

Collision Geometry Is Linear

Diophantine Certificates for Rational Signal Machines and Conservative Particle Dynamics

Event-sparse chambers, canonical quadratic and quartic certificates, two routes to complete chronology, a conservative universal frontend, sparse lattice certificates, conserved-mass thresholds, and mass four in every dimension

AI-assisted research report prepared for Vladimir Reshetnikov from twelve manuscripts (ProveIt’s incoming reports: batch 78, manuscripts 07 and 11; batch 79, manuscripts 16 and 19; batch 80, manuscripts 10, 02, 05 and 03; batch 82, manuscripts 19, 06, 17 and 01), 2 October 2026

Abstract

A rational signal machine computes by the collisions of finitely many kinds of particles moving at fixed rational speeds on a line. Its first two Parts print two independent manuscripts that prove one theorem by two routes: once a complete finite collision history is fixed, its realizations are exactly the natural zeros of a sum of squares of integer affine residuals, of degree two, with exactly one witness tuple for each realization. In both, typed endpoint conditions on maximal adjacency intervals exclude unrecorded earlier collisions, omitted simultaneous inputs and omitted simultaneous sites, and causal depth controls the denominators of the event coordinates.

Part I (source 07) eliminates every event coordinate. The realizing initial-gap vectors form a relatively open rational polyhedral cone cut out by at most $3(n-1) + (3m+1)E$ linear conditions, and the only witnesses are the slacks of the strict rows. The chamber geometry yields small integral realizations, an NP upper bound for bounded synthesis, a robustness–codimension criterion, a canonical first-hit quartic for a fixed event budget, exact quasi-polynomial seed counts, an exactly solved first-collision tournament and a four-speed accumulating clock with exponentially growing denominators. Part II (source 11) keeps batch times and event positions as natural witnesses on a fixed rational grid, adds halting semantics, proves that the witness matrix has full column rank (so the certificate is strictly convex in its witnesses), gives linear witness and residual ledgers and a full-layer reference construction, and bounds the canonical witnesses by $60^K \max(1, M_0)$ for the four-speed interface of Durand-Lose’s universal machine.

Parts III and IV (sources 12 and 13, from batch 79) certify conservative dynamics at a fixed horizon. Part III writes out a reversible number-conserving universal signal machine (Morita’s universal Turing machine through Durand-Lose’s stacks) with 18 live signals, whose event dynamics is an 18-dimensional integer-linear map with a canonical quadratic step certificate; Part IV compiles one-dimensional number-conserving lattice dynamics into a quartic sum of squares with a unique natural witness and proves decidability at total mass at most two.

Parts V and VI (sources 14–17, batch 80) find the least conserved mass of an undecidable observation. With many weight-one symbols, a fixed reversible automaton simulates a universal reversible two-counter machine at total mass three, also for exact pairs (Part V); with at most one, masses three and four stay decidable, so integer-state number-conserving automata need five particles (Part VI). Source 18 and Part VII (sources 19–21, batch 82) add timed quartics and mass-four decidability on every $\mathbb{Z}^d$.

None of the seven Parts compiles unbounded halting into one fixed-arity polynomial or improves the repository’s 75- and 87-operation bounds; Parts I and II do not transcribe the published universal machine, and Part III transcribes a different one. Re-derived devices are printed as pointers or second routes. Part I’s convex-quadratic barrier is a special case of Part XI of the collection’s report *canonical-diophantine-certificates*. The report is AI-assisted and unrefereed; nothing in it is formalized in Lean or Rocq.

Status. Conventional proofs and finite exact-arithmetic checks; not kernel-checked. Section 1 compares the two routes, introduces Parts III–VII, and states what is not claimed; Appendix E gives the provenance.

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1 Scope, the two routes, and conventions

[write] This section was written when the two manuscripts were merged. Part I and Part II print the two sources; text there without a [write] marker is the source’s own. Sections 1.7–1.11 were added on 2 October 2026, when two manuscripts of batch 79 became Parts III and IV; where they make a sentence of the earlier subsections stale, a dated note says so. Sections 1.12–1.16 were added later the same day, when four manuscripts of batch 80 became Parts V and VI, again with dated notes where earlier sentences became stale. Sections 1.17–1.21 were added on 3 October 2026, when four manuscripts of batch 82 became source 18 of Part VI and Part VII, likewise with dated notes.

1.1 The two sources

This report is built from two AI-assisted research manuscripts that arrived in one delivery as batch 78 of ProveIt’s incoming reports (manuscripts 07 and 11; Appendix E gives the full provenance). They are called *source 07* and *source 11* after their batch-78 manuscript numbers, which are also the file prefixes of their shipped programs and data. Neither cites the other, and a text-overlap check at placement found almost no shared wording (at most 0.5 per cent of either text’s eight-word sequences occur in the other). They nevertheless prove the same theorem, by the same mechanism, in two different witness formats (Section 1.2).

- **Part I (source 07, the base), *Collision geometry is linear*.** Arbitrary rational speeds, of either sign. Every event time and position is eliminated as a rational linear form in the initial gaps, so a complete chronological skeleton becomes a homogeneous rational chamber (Theorem 5.3) and a convex quadratic with unique slack witnesses (Theorem 6.1). The chamber geometry then gives denominator and height bounds, small integral realizations, an NP upper bound, robustness and codimension, finite-horizon semilinearity, a canonical first-hit quartic, exact Ehrhart counts, two exact examples and a convex-quadratic barrier.
- **Part II (source 11), *Direct Diophantine certificates*.** Speeds and positions normalized to natural numbers by an external Galilean change of coordinates. Batch times and event positions stay in the certificate as natural witnesses on the grid $B = D^K$, and the certificate has prefix and halting versions (Theorem 18.3). Its additions are the halting semantics with terminal rays, full column rank of the witness matrix and hence strict convexity in the witnesses (Lemma 22.1), an explicit witness and residual ledger, a full-layer reference construction with exact counts (Section 23), the witness-height bound for Durand-Lose’s four-speed interface (Theorem 25.2) and a positive-witness translation.

Source 07 is the base because it assumes less (signed rational speeds, no normalization interface) and proves more about the chamber; source 11 is printed in full as the second route together with its additions. Each Part keeps its source’s definitions, letters and numbering of statements. ([write] Added 2 October 2026: two further Parts, from batch 79, follow Part II; Section 1.7 describes them. They do not prove the common theorem of this subsection and the next again.) ([write] Added 2 October 2026, batch 80: two more Parts, from four manuscripts of batch 80, follow Part IV, so that the report now prints eight sources in six Parts; Section 1.12 describes them.) ([write] Added 3 October 2026, batch 82: a third source joins Part VI and a seventh Part follows it, so that the report now prints twelve sources in seven Parts; Section 1.17 describes them.) The common theorem is stated in both Parts, because its two statements have different inputs and witnesses; where a lemma of Part II coincides with one of Part I, a [write] note at the lemma says so, and source 11’s proof is kept as a marked second route.

1.2 One theorem by two routes

The common theorem. Fix a rational signal machine, an initial label word and a finite chronological collision history (a *skeleton* in Part I, a *schema* in Part II). There is an effective sum of squares of integer affine residuals, in the input coordinates and natural witnesses, whose natural zeros are exactly the realizations of that history, with exactly one witness tuple for each realization; its degree is at most two. This is Theorems 5.3 and 6.1 of Part I and Theorem 18.3 of Part II (for prefix semantics).

The common mechanism. Both sources record each pair of adjacent signals once, over the maximal interval during which it stays adjacent, and impose a *typed* condition at each end of that interval: equality exactly when both segments are born (or die) at the same collision there, and strict positivity otherwise, including for a pair that survives the last listed batch. Affineness of the gap then excludes every interior meeting (Part I, Lemma 5.2; Part II, Lemma 21.1), and the strict conditions at a batch boundary make the declared blocks exactly the collision sites (the soundness part of the proof of Theorem 5.3; Lemma 21.2). The two sources then prove the same auxiliary statements: the lifetime count $(n - 1) + \sum_e (q_e + 1)$ ((5.2) and (18.2), where Part II writes $|\text{out}(e)|$ for q_e), causal-depth denominators dividing D^h (Theorem 7.1, Lemma 19.1), and semilinearity of the inputs accepted under a fixed event bound (Theorem 10.1, Proposition 26.1). Table 1 compares the two formats.

Table 1: The two routes compared.

	Part I (source 07)	Part II (source 11)
Speeds	arbitrary rationals, of either sign; time rescaled by $\tau = t/L$ to make them integers (Section 3)	natural numbers, after the external change of coordinates (19.1) (a Galilean shift and two scalings)
Input	the initial gap vector g , translation removed; real in the chamber theorem, positive integers in the certificate	the absolute initial positions $\mathbf{p}$, natural numbers in increasing order
Event times and positions	eliminated: rational linear forms in g , computed data (Section 4)	witnesses: natural T_i and X_e , scaled to the grid $B = D^K$ (18.1)
Witnesses	one slack $z_j = G_j g - 1$ per retained strict row; s in all	batch times T_i , time slacks d_i , event positions X_e , input-order slacks u_j , endpoint slacks $h^\pm$; $W \leq E + 2K + 2L + n_-$
Size	at most $3(n - 1) + (3m + 1)E$ raw conditions (5.3); a row may be dense in g	$R \leq I + K + 2L + n_- \leq (r + 2s + 3)E + 3n_-$ residuals, each of bounded width for a fixed machine (18.3)–(18.4)
Denominators	D^h at an event of causal depth h , for the coefficients of its forms	D^h for coordinates of depth h ; the compiler clears with D^K , a multiple of D^H
Convexity	convex, Hessian $2A^T A \succeq 0$; unique slack by (6.3)	strictly convex in the witnesses for fixed input: full column rank over $\mathbb{R}$, even for unrealizable schemas (Lemma 22.1)
Semantics	prefixes: exactly the first B batches; nothing after them	prefixes, and halting realizations certified by terminal rays
Heights	row coefficients at most $2d(1+V)(DK)^h$ (Proposition 7.3); a realizing seed of size at most $d^{d/2}A^d$ (Theorem 8.1)	every canonical input and witness at most $60^K \max(1, M_0)$ for speeds in $\{0, 2, 3, 4\}$ (Theorem 25.2)

	Part I (source 07)	Part II (source 11)
Shuttle	four-speed clock, speeds 0, 2, -1, -2; denominator exactly $2 \cdot 3^k$ at event $2k$ for the seed (1, 2); Zeno time $S = u + v$ (Section 12)	bounded shuttle, speeds 3, 2, 4, 0; denominator exactly 3^{k-1} at the k th left event; Zeno time 2 (Proposition 25.3)
Missed simultaneity	two independent sites, word P, N, P, N , gaps (2, 5, 2) (Section 11.3)	hidden collision at the final cut, word d, a, d, a at (0, 4, 10, 14) (Section 24.2)
Only here	chamber geometry, small realizations, real = rational = integer realizability, NP bound, robustness and codimension, first-hit quartic, Ehrhart counts, the tournament, the convex-quadratic barrier	halting semantics, strict convexity, the full-layer construction and its exact counts, the four-speed height bound, positive witnesses

Normalizations. The two definitions of D agree. Part I multiplies all speeds by an integer L and leaves their signs; Part II subtracts a common speed c and multiplies by an integer m . A common shift does not change speed differences, so with $m = L$ both take D to be the least common multiple of the same nonzero integer differences. Part II needs nonnegative speeds and an origin because its inputs are natural absolute positions; Part I needs neither because its input is the gap vector. Part I’s integer seeds come from homogeneity (scaling a rational chamber point); Part II fixes the scale outside the witness tuple and warns that a scale witness or variable denominators would not preserve its canonical quadratic claim without a new argument (Section 19.1).

Two shuttles, one mechanism. Part I’s clock and Part II’s shuttle are both a particle reflected between two converging walls, each round trip shrinking the remaining time to the accumulation by a factor three, so that the reduced denominators grow exponentially in the number of events while their bit lengths grow linearly. They are different machines: Part I’s has a stationary left wall and a right wall of speed -1 , and its every finite prefix has the whole positive quadrant as chamber (robust chronology with exponentially growing precision); Part II’s lives entirely in $0 < t < 2$, $0 < x < 6$ and is stated for its natural-speed interface. Neither source gives the accumulation point a continuation. Source 07 credits the accumulation phenomenon to Becker et al. [6].

Two missed simultaneities, one failure mode. In both examples a proposed batch omits a second, spatially separate collision that happens at the same time, and in both the omitted pair is an unchanged adjacent pair whose gap reaches zero exactly at the last listed boundary. The strict upper endpoint of a surviving pair rejects it: in Part I through the chamber row that must be positive there, in Part II through the residual $G(T_1) - h^+ - 1 = -h^+ - 1$. Part II adds the observation that weaker conditions (two nonnegative endpoint gaps with positive sum) would accept the false schema.

Two meanings of “sparse”. In Part I “event-sparse” counts geometric conditions: the bound (5.3) is linear in n and E , but a propagated intercept may depend on every initial gap, so the rows can be dense and storing them can take $O(dR_\sigma)$ coefficients (the paragraph after Theorem 5.4). In Part II the sparse certificate also has residuals of bounded width, because event coordinates are witnesses; its coefficients then contain the grid constant $B = D^K$. *Neither* size statement is a constant arithmetic-operation count for an unbounded computation, as both sources say.

1.3 Notions and letters

Each Part keeps its source’s letters. Many letters carry different meanings in the two Parts; Table 2 lists every letter used in both, so that a statement of one Part is never read with the other Part’s meaning. Inside a Part the meaning is always the one in that Part’s column. No symbol was renamed and no normalization was changed: source 11’s macro for the natural numbers is called `\NN` in this file and still prints $\mathbb{N}$, and Part I prints $\mathbb{N}_0$; both denote $\{0, 1, 2, \dots\}$. ([write] The letters of Parts III and IV are listed in Section 1.8, and those of Parts V and VI in Section 1.13.)

Table 2: Letters used in both Parts.

Letter	Part I (source 07)	Part II (source 11)
$\mathbb{N}_0, \mathbb{N}$	$\mathbb{N}_0 = \{0, 1, 2, \dots\}$	$\mathbb{N} = \{0, 1, 2, \dots\}$ (the same set)
A	coefficient bound (7.2); a matrix in proofs	witness coefficient matrix (Lemma 22.1); birth-position expression A_σ ; start coordinates $A_{i,j}$
B	number of batches; a nonsingular matrix in the proof of Theorem 8.1; a label of the tournament	the grid modulus $B = D^K$ (18.1), <i>not</i> a number of batches; a label of the shuttle
C	the chamber $\mathcal{C}$; the count $C(H)$; a label	matrix of input coefficients; end coordinates $C_{i,j}$
D	lcm of nonzero speed differences (7.1)	the same lcm (18.1) (Section 1.2)
d	$d = n - 1$	time slacks d_i ; a label
E	number of collision sites; $\mathbf{E}$ the equality matrix	number of events (a site of Part I)
G	$\mathbf{G}$ the strict-row matrix	the gap function $G_{\sigma\tau}(T)$ (20.5)
g	the initial gap vector	number of blocks g_i of a slab (Section 23); a gap in proofs
H, h	h causal depth; H the side of a counting box	h a depth bound, H the maximal depth; $h, h^\pm$ endpoint slacks
K	$K = 2 + 4V$ (Lemma 7.2); number of first-hit chambers	number of batches, <i>not</i> a number of chambers
L	integer clearing the speeds; a label of the clock	number of recorded lifetimes; a label of the shuttle
M	the label set	largest position M_j ; $\mathcal{M}$ the machine
m	number of distinct speeds	the speed multiplier of (19.1); number of generators of a linear set
n	number of initial signals	the same; $n_- = \max(n - 1, 0)$
p, q	p_e, q_e incoming and outgoing counts at site e	initial positions p_j ; the position scale q
R	raw row count R_σ ; a label	residual count; a label
r, s	numbers of retained equality and strict rows	largest incoming arity and largest fanout; residuals r_a
S	the accumulation time $u + v$ of the clock	a label; $\mathcal{S}$ the schema
T	the site budget	scaled batch times T_i
u, v	initial gaps of the examples; $v(a)$ a speed	input-order slacks u_j ; $u(\ell)$ an original speed, $v(\ell)$ a normalized speed
V	$\max(1, \max v(a))$	the maximal speed (remark after Theorem 25.2)
W	$W = 3u + 2v$ of the clock	number of witnesses
z, y	z the slack witnesses	$\mathbf{y}$ the witnesses; $\mathbf{z}$ a test vector

Three words need the same care.

- **Skeleton** (Part I) and **schema** (Part II) carry the same information: an ordered list of batches, each a set of disjoint contiguous blocks of the live list. A skeleton lists only the collision blocks (at least two members); a schema partitions the whole live list and marks singletons as persisting signals. A *site* of Part I is an *event* of Part II.
- **Realization.** Part I’s realization and Part II’s prefix realization are the same notion: exactly the listed batches as the first batches, nothing asserted afterwards. Part II’s *halting* realization additionally excludes every future collision. The “halting relation” of Part I’s Section 13 is the undecidable halting of a universal machine, a different thing.
- **Convex.** Part I’s certificate is convex (positive semidefinite Hessian) jointly in seed and slacks; Part II’s is *strictly* convex in the witnesses with the input held fixed, and Part II states that joint strict convexity is neither needed nor generally true.

1.4 Relation to the repository and to the literature

Neither source cites the related material below; this subsection supplies the cross-references (pins and dates in Appendix E). Reports of the collection are cited by label name, since their labels do not resolve in this file.

- **Report *canonical-diophantine-certificates, Part XI* [11]**, “Order-free self-assembly and a positivity-restricted degree threshold”. Its Lemma `cdc:of:lem:affinezero`, Theorem `cdc:of:thm:cubicsemilinear` and Theorem `cdc:of:thm:classification` ($D_2^+(k) = D_3^+(k) = \text{SL}(k)$) prove, for polynomials of degree at most three that are nonnegative only on the real nonnegative orthant, that the full natural zero set is a finite union of natural solution sets of rational affine systems, hence semilinear and effectively so. Part I’s Theorem 13.1 and Corollary 13.2 are the globally nonnegative quadratic special case; they are printed with source 07’s direct proof, as a second route, and credited in Remark 13.3. No novelty is claimed for them. The same Part’s Theorem `cdc:of:thm:singlefoldsl` (a single-fold orthant-nonnegative quadratic for every semilinear set) bears on Part I’s first-hit quartic; see Remark 10.3. The bounded-search argument at the end of Part I’s Section 13 is the standard one of Proposition `cdc:bd:prop:nobound` there, for a different statement (site counts of accepting runs).
- **Report *liveness-beyond-halting* [12]**. Its Part V prints a question of its source 11, “Optimal degree under stronger geometric restrictions”, which asks for the minimal degree and dimension of a witness-free, binary-branching local polynomial that is globally nonnegative on all real inputs, and remarks that the globally nonnegative quadratic case “has a much more rigid affine geometry”. Part I’s Theorem 13.1 (and, more generally, Part XI of [11]) states that geometry: such a quadratic has a rational affine zero set. This report does not settle that degree question.
- **The Hilbert-tenth-problem programme has already reviewed both manuscripts.** The review of the delivery of 2 October 2026 [13], with its detailed signal-machine part [14], replayed both suites from the original archives, read both articles in full and reports “PASS for both reports’ stated finite-schema, exact-domain theorems and generated examples”, with no mandatory correction. It also proves and checks two reductions that neither source claims: deleting source 11’s identically zero same-birth rows and its same-death rows (differences of

retained flight rows), and erasing the initial-order slacks when $K > 0$ or the certificate is halting, which takes the worked annihilation from six residuals and five witnesses to four and four (a bijection of natural zero sets, not of nonnegative rational ones, with a rational counterexample); and deleting redundant strict rows, with computed slacks, from the five exported packets of source 07. Those reductions are the review’s, are not re-proved here, and do not change any theorem of either Part. This report’s intake is therefore not a first review; it adds the merge, the cross-references and the credit of Remark 13.3.

- **The programme’s audits of finite-history models.** Source 07 cites the reversible finite-history audit [2] for its distinction between cheap local relations and a complete computation certificate; that file is unchanged from source 07’s pin to the write. A sibling audit [15] notes that “finite particles can in principle store unbounded values in their separation” and that “a new construction would have to establish that interface”. Durand-Lose’s halting universal signal machine [3] uses a finite initial configuration and a halting event (Part II, Section 17), so signal machines are a candidate for such an interface, and both Parts compile its finite collision histories in principle; but neither transcribes its universal rule table or its input loader (Part I, Section 14; Part II, Question 1), so this report does not establish that interface. ([write] Added 2 October 2026: for signal machines Part III now establishes it, with a different universal machine and a prepared rational loader; see Section 1.9.)
- **The repository’s universal bounds.** The project README [1] still states the established bounds of 75 operations for a complete certificate and 87 for one universal polynomial. Neither Part improves or bears on them; both say so. ([write] Nor do Parts III–VI.)
- **Other reports of the category.** No result of *probabilistic-quantum-and-continuous-computation* or *group-theoretic-substrates* is re-proved here. Apart from the programme’s review described above, nothing in the repository treated signal machines before this report.

1.5 Status and what is not claimed

The report is AI-assisted and unrefereed, and **nothing in it is formalized in Lean or Rocq**. Both sources give conventional proofs together with finite exact-arithmetic checks that are regression evidence, not proofs. Every limitation stated by either source is kept in its Part; collected here, neither Part claims

- historical priority, or exhaustive novelty, for its constructions (Part I, Section 2; Part II, Section 17); Part II adds that a general linearizability of signal-machine evolution is already in Durand-Lose’s linear real-register simulation [4];
- one fixed-arity polynomial for unbounded halting, a single-fold or finite-fold Diophantine representation of c.e. sets, or a fixed-universal arithmetic-operation bound: the skeleton or schema, the machine and the input length are compiler data, and the row and residual counts are not operation counts;
- that the published universal machine (13 meta-signals, 21 non-blank rules) or its input loader has been transcribed, executed or certified;
- any semantics at or after an accumulation point; the clock and the shuttle are tests of finite prefixes, and the clock is not a claim to have discovered accumulation;
- NP-hardness (Corollary 8.3 is an upper bound), an efficient enumeration of first-hit chambers or of their volumes, or an optimized exhaustive enumerator in the code;

- that the denominator-growth examples are lower bounds for arbitrary Diophantine representations: Part II’s is a lower bound for common-grid encodings;
- that integer scaling of a rational seed preserves a prescribed physical length or deadline;
- that the finite checks are exhaustive, or that the shared simulator of Part II’s two compilers is independently verified;
- any Lean or Rocq verification; neither source’s code is a verified compiler.

The report itself claims no novelty for Part I’s convex-quadratic barrier (Remark 13.3) and none for the reductions of the programme’s review. ([write] The non-claims of Parts III and IV are collected in Section 1.10, those of Parts V and VI in Section 1.15, and those of source 18 and Part VII in Section 1.20.)

1.6 Research questions

Both agendas are printed in full: the nine research directions of Part I (Section 15) and the five questions of Part II (Section 27.1), each in its Part and with its source’s wording. Four of Part II’s questions coincide with, or overlap, directions of Part I; Table 3 pairs them, so that the two formulations are read together. None of them is answered by the other source, by the repository, or by the programme’s review; the one partial overlap is that Part II’s Question 3 begins with causal-depth clearing, which both sources already prove (a [write] note there re-scopes it). ([write] Added 2 October 2026: this sentence describes the report before batch 79. Part III now answers Part I’s direction 15.2, except for an ordinary-integer loader, and Parts III and IV answer Part II’s Question 2 in part; Section 1.11 and dated notes at the questions give the details.) ([write] Added 2 October 2026, batch 80: Parts V and VI answer two of Part IV’s questions and the remark after its mass-two theorem; Section 1.16 gives the details.)

Table 3: Concordance of the research questions.

Part I (source 07)	Part II (source 11)	Relation
15.2 An audited universal signal-machine frontend	Q1 Instantiate the universal table and input map	the same task; [write] answered by Part III for a different universal machine, except the ordinary-integer loader; the 13-label table of Q1 is still not transcribed
15.1 Formalizing the chronology theorem	Q5 Formalize geometric completeness	the same target (endpoint lemma and sweep invariant; typed endpoints and atomic replacement)
15.3 Sharper denominator invariants; 15.4 From few rows to small arithmetic circuits	Q3 Sharpen the denominator and cost ledger	overlapping: prime-valuation denominators, and a literal gate ledger
15.5 Avoiding exhaustive first-hit enumeration	Q4 Compress histories without losing canonicity	overlapping: both separate bounded compression from fixed-arity unbounded compression
15.6, 15.7, 15.8, 15.9	—	Part I only
—	Q2 Exploit conservative bounded populations carefully	Part II only; [write] answered in part, at a fixed horizon, by Parts III and IV

Part I (source 07)	Part II (source 11)	Relation
—	—	[write] Part IV’s five questions (Section 48); Part III states none
—	—	[write] Part IV’s fourth and fifth questions are answered, in substance and for the weighted three-mass automaton respectively, by Parts V and VI (Section 1.16); its first stays open
—	—	[write] Part V’s questions (Sections 63 and 73; source 14’s third is answered by source 15, both sources’ fourth in part by Part VI) and Part VI’s (Sections 81 and 92; source 16’s four-versus-five question is answered by source 17)
—	—	[write] Source 17’s seventh question is answered for a fixed input by source 18; its second, fourth and sixth are touched by sources 18, 20 and 21, and its first, third and fifth stay open (Section 1.21)
—	—	[write] The questions of sources 18–21 (Sections 100, 117, 127 and 140); source 20’s first is answered for binary rules by source 21, the others are open

1.7 Parts III and IV: conservative frontends (batch 79)

[write] This subsection and the four that follow were written on 2 October 2026, when two manuscripts of batch 79 of ProveIt’s incoming reports were added to the report as Parts III and IV. They are called *source 12* and *source 13* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 07 and 11. Their batch-79 manuscript numbers are different: **source 12 is batch-79 manuscript 16, and source 13 is batch-79 manuscript 19**. Appendix E gives the full provenance. Neither manuscript cites Parts I and II, the other new manuscript, or any repository file.

- **Part III (source 12), *Conservative signal machines as Diophantine frontends*.** A fixed reversible, number-conserving universal signal machine $\mathcal{U}_{18}$, written out completely: 114 meta-signals, 445 explicit injective two-to-two collision rules, speeds $\{-12, -9, -4, 0, 4, 9, 12\}$ and exactly 18 live signals on every encoded input (Theorem 29.1). Its source is Morita’s 15-state, 6-symbol reversible universal Turing machine, compiled through Durand-Lose’s 2012 conservative rational stacks; the input of an explicitly halting cyclic-tag program is loaded into two rational distances. At event cuts the complete simultaneous-event dynamics of any fixed-population, equal-cardinality machine is a partial piecewise homogeneous integer-linear

map with the pivot-scaled gap matrices $N_j = c_j I - c_j e_j^T$, $N_j^2 = c_j N_j$, plus one mode coordinate (Sections 34–35). A selector/copy construction then gives one explicit quadratic polynomial with a unique natural witness for each legal step, and for each fixed horizon (Theorem 36.1); for $\mathcal{U}_{18}$ the step polynomial has exactly 4,196,998 auxiliary variables over a closed family of 49,700 modes and 80,501 branches (Appendix H). A ten-signal binary version is proved to exist but not printed (Theorem 29.2).

- **Part IV (source 13), *Sparse Diophantine certificates for finite mass lattice dynamics*.** The discrete counterpart: one-dimensional partitioned number-conserving cellular automata with three lanes. A conserved mass M bounds the number of unit records but not the gaps between them, and a fixed horizon T of any supplied total conservative table compiles into one quartic sum of squares with exactly one natural zero, paying for collision detection, simultaneous redistribution, sorting, table lookup and empty space, with exact witness and residual ledgers (Theorem 38.1) and an option that is exact on the nonnegative real orthant. A source interface to Morita and Imai’s reversible conservative automaton corrects a printed sign, adds a reachable-legality hypothesis that an explicit false observer shows to be necessary, pays for a loader and a first-pulse observer, and works a fully specified, nonuniversal doubling example. Decidability results bound what any such frontend can do: under a bounded orbit diameter, and without it at total mass at most two (Theorem 44.1); at mass at most two the relation is effectively semilinear (Theorem 45.1).

How they relate to Parts I and II. Neither new Part proves the common theorem of Section 1.2 again; both certify a step, or a fixed horizon, of a *deterministic* dynamics rather than the realizations of a prescribed collision history. Part III meets Parts I and II in one lemma: its complete event selection (Lemma 34.1) is their typed-endpoint argument for a single batch, and its uniform lift is their common-denominator device ([write] notes at both places). Part III is also the instantiated universal frontend that Part I’s research direction 15.2 asks for, though for a different universal machine than the one named there (Section 1.11). Part IV shares no lemma with Parts I–III; it answers, for lattice dynamics, the conservative-population question of Part II.

Two Morita sources, no shared data. Both new Parts draw their universal source from Morita, but from different constructions: Part III from Morita’s 2008 reversible universal Turing machine UTM(15,6), through Durand-Lose’s signal-machine stacks; Part IV from Morita and Imai’s 2001 number-conserving reversible partitioned automaton, built on Morita’s 1996 reversible two-counter machine. They share no table, program or data file. Part III prints its universal machine in full; Part IV deliberately leaves its safe universal source as an existence construction and prints only a nonuniversal example.

1.8 Notions and letters of Parts III and IV

[write] Parts III and IV keep their sources’ letters and numbering of statements, by section, like Parts I and II; no symbol was renamed and no normalization changed. Both sources’ macro for the natural numbers is written `\NN` in this file, as for source 11, and prints $\mathbb{N} = \{0, 1, 2, \dots\}$; Part I’s $\mathbb{N}_0$ is the same set. The table lists every letter of Parts III and IV that has another meaning in the other new Part or in Parts I–II (Table 2 gives the meanings in Parts I and II in full). Inside a Part the meaning is always that Part’s.

Letter	Part III (source 12)	Part IV (source 13)	Parts I–II
A	auxiliary-variable count $A = B(d + 1) + T + W$; loader numerators A_L, A_R ; the projection A_j	—	coefficient bound; witness matrix
B	number of branches of the step map, <i>not</i> batches	the control baseline $B = m + 16$ of the source interface	batches (I); grid modulus D^K (II)
b	base $\ell + 1$; selectors b_r ; the blank $\mathbf{b}$	comparison flag b ; output masses b_i ; baselines b_i	—
C	the set C_ℓ of stack codes	centre lane C ; number C of congruence atoms	chamber (I); input matrix (II)
c	gadget speed c ; speed-difference vector c , $c_i = v_i - v_{i+1}$; pivot speed c_j	output masses c_i ; a configuration c	—
D	lcm of the nonzero speed differences, <i>the same</i> D as in Parts I–II	the diameter of the initial support	lcm of speed differences
d	dimension of a branch system; a move direction	comparison slack d ; a modulus	$n - 1$ (I); time slacks (II)
E	number of equality guards; escape tokens $E_{q,a}$	residual count E_Φ of the Presburger compiler	sites (I); events (II)
F	the partial step map on $\mathbb{N}^d$	the global cellular-automaton map	—
g	rational gap vector	the local table $g : \{0, \dots, K\}^3 \rightarrow \{0, \dots, K\}^3$; an initial gap	gap vector (I); blocks (II)
H	halt labels $H_{q,a}$	the witness-height bound (38.4)	box side (I); depth (II)
h	integer gap vector h_k , loader h_0	comparison flags h_i ; a time	depth (I); depth, end-point slacks (II)
K	the horizon (number of steps)	the alphabet cap, $q = K + 1$; witness count K_Φ	$2 + 4V$, chambers (I); batches (II)
L	a guard form $L(X)$; the move direction L	left lane L ; an affine atom L ; the term count $\mathcal{L}$	speed-clearing integer (I); lifetimes (II)
M	uniform-lift matrices M_j ; branch matrices $\widehat{M}_r$	the conserved mass	labels (I); largest position (II)
m	the padded word length of the loader	number of source states	distinct speeds (I); speed multiplier (II)
n	the population (live signals)	a counter value; counters n_0, n_1	initial signals
p	timestamp numerators p_k ; productions p_i	the period of a type cycle	in-counts (I); positions (II)
Q	the loader scale ℓb^m , or a clearing denominator	a finite cellular-automaton alphabet	—
q	mode code; Morita's states q_i	$q = K + 1$; source states q_i ; quotients $q^\pm$	out-counts (I); position scale (II)
R	the layout half-width $\ell + 2$, <i>and</i> the residual count $1 + 2d + E + T + W$	right lane R ; residual count; the right walker's position $R(t)$	raw rows (I); residuals (II)
r	branch index; the length of a run in (33.1)	ranks r_i ; the radius r ; a table entry	retained rows (I); arity (II)
S	the span $S = \mathbf{1}^\top h$; the source alphabet	number of allowed rows of a restricted list	accumulation time (I)
s	guard slacks $s_{r,L}$; Morita's target state	table selectors $s_{i,\tau}$; comparison slacks	strict rows (I); fanout (II)
T	number of strict guards, <i>not</i> a time	the horizon (number of lattice steps)	site budget (I); batch times (II)
u	mode coordinate $u = qS$	incoming masses u_i	gaps (I); slacks (II)
V	a bound on speed differences	witness count V	speed bound (both)

Letter	Part III (source 12)	Part IV (source 13)	Parts I–II
W	number of weak guards	—	$3u + 2v$ (I); witnesses (II)
w	a written symbol; the input word	incoming masses w_i ; the weight function	—
z	packet witnesses $\mathbf{z}$	channel index of a record	slacks (I); test vector (II)

Several words need the same care.

- **Horizon.** Part III’s horizon is K steps of the event map, Part IV’s is T lattice steps; Part II’s K is a number of collision batches of a prescribed schema. In Parts III and IV the horizon is a compiler parameter that fixes the number of variables; it is never a free unknown of one polynomial.
- **Batch.** Part III’s “signal batches” (Section 37) are the collision batches of Parts I–II, counted along a run. Its *mode* is an ordered label word, and a *germ* is a post-event state whose zero gaps separate.
- **Source.** “Source 12” and “source 13” name manuscripts; inside Parts III and IV “the source” is the simulated machine (Morita’s Turing machine, Morita’s two-counter machine).
- **Conservative.** Part III’s rules conserve the number of signals and a positive energy; Part IV’s local tables conserve the total lane mass. Both are reversible in the cases printed, but neither certificate theorem needs reversibility.
- **Sparse.** Part III’s “sparse scaled projections” have at most $2(n - 2)$ nonzero entries; Part IV’s “sparse” certificate keeps M records however large the gaps between them. Neither is Part I’s event-sparse row count, and none of them is a fixed-arity statement.
- **Loader.** Part III’s loader writes a prepared tape into two rational distances at a cost linear in the tape length (Section 33); Part IV’s uniform loader leaves two counter positions as free natural parameters and pays $4 + 3M(M - 1)$ witnesses and residuals (42.2). Neither converts an ordinary binary integer input.

1.9 Relation of Parts III and IV to the repository

[write] Neither new source cites the material below; these cross-references are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers or second routes.** Part III’s complete event selection (Lemma 34.1) is the one-batch case of Part I’s Theorem 5.3, and its uniform lift is the denominator device of Theorem 7.1 and Part II’s grid. Part IV’s comparison gadget is Lemma `cdc:mem:lem:compare` and its sorting comparator the two-field record comparator (`cdc:mem:prop:comparator-cost`) of Part V of *canonical-diophantine-certificates* [11]; Part IV’s horizon-free quartic (Corollary 46.1) is a degree-four second route to the single-fold degree-two representation of semilinear sets of that report’s Part XI (`cdc:of:thm:singlefolds1`); and the finite-search step of Proposition 43.2 is

`cdc:bd:prop:nobound`. Each place carries a [write] note; no novelty is claimed for any of them.

- **The degree-two orthant format.** Part III’s step polynomial is literally of the form $\sum A_i^2 + \sum B_j C_j$ of the spine of *quadratic-orthant-certificates* (`qoc:eq:spine`), nonnegative on the real orthant; its existence also follows from `cdc:of:thm:singlefoldsl`, since a step relation with finitely many linear branches is semilinear; source 12’s contribution is the explicit construction and its ledger. Its selector/copy construction is Balas’s disjunctive formulation [26, 27, 28], credited at the place. (Added 2 October 2026, batch 79: that report’s Parts V and VI use the same selector device with the selector added to the right factor of each product, which makes their nonnegative real fibres natural; the note after Theorem 36.1 compares the two gates and shows that the real fibre of Part III’s packet is natural at every natural input $X \neq 0$, and at $X = 0$ as well under the theorem’s disjoint homogeneous branch hypothesis; the weak gates alone do not force one-hot selection at zero.) Part IV’s certificates are sums of squares of quadratic residuals, of degree four; its orthant-exact option restores exactness on the nonnegative real orthant with norm-one residuals.
- **The finite-input interface asked for by a programme audit.** The block-automaton audit [15] notes that finite particles could store unbounded values in their separations and that a new construction would have to establish that interface. For signal machines, Part III establishes it: a fixed machine with 18 live signals, finite support, stack values stored in two distances, and observable halt and null outcomes with an escape of eight crossings. It does so with a prepared rational loader, not from an ordinary integer, and not for billiard-ball or block automata, which remain as the audit leaves them. (This supersedes, for signal machines, the last sentence of the corresponding item of Section 1.4.)
- **Other reports.** Part IV’s mass-two theorem is related only loosely to Part I’s rigidity theorem (both end in semilinearity, for different reasons). Part IV bears on the question “Local certificates on adaptive support geometry” of Part XVI of *canonical-diophantine-certificates* without answering it (note at Part IV’s questions). The fixed-horizon form of Part III is the same kind of statement as the trace theorem `pqc:pm:thm:trace` of *probabilistic-quantum-and-continuous-computation*, for different dynamics.
- **The programme’s reviews.** The Hilbert-tenth-problem programme reviewed the earlier edition of source 12 (batch-79 manuscript 07) and wrote the patch that the corrected edition printed here applies [29], and then reviewed the corrected edition itself, with a natural-domain projection of its packet [31]. It reviewed source 13 completely and wrote a constructor patch [30], which was not applied here; the corrected code edition of source 13 (batch 80) contains an equivalent repair of its own and is now shipped, and the programme’s correction audit [36] confirms it. It later reduced two of source 13’s ledgers [32, 33]. Its review index [34] lists all of these as complete, and its placement audit [35] authenticates every file of both sources placed by `a7ae02511` against the delivered archives (five of source 13’s have since been replaced by the corrected edition’s bytes). Sections 37 and 47 give the details. No review changes a theorem of this report, and the reductions are the programme’s, not re-proved here.
- **The repository’s universal bounds.** The 75- and 87-operation bounds stated in the project README [1] are untouched by Parts III and IV, which say so themselves (no fixed-arity polynomial for unbounded histories). The programme’s review of source 12’s earlier edition records the same conclusion.
- **Formal status.** Nothing in Parts III and IV is formalized in Lean or Rocq; the Hilbert-tenth-

problem Lean development has no signal-machine, cellular-automaton or Presburger-compiler module that states any of their results. One classical ingredient that Part IV imports is formalized elsewhere in the repository: Cooper’s quantifier elimination for Presburger arithmetic, used in Section 45, is the Lean development `Logic/PresburgerArithmetic` (for instance `PresburgerArithmetic.Formula.holds_iff_quantifierEliminate` and the decision procedure `PresburgerArithmetic.Formula.presburgerArithmetic_decidable`). That formalizes the imported step only, over the integers, and none of Part IV’s own statements.

1.10 Status and what Parts III and IV do not claim

[write] Both new sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; both are AI-assisted and unrefereed. Every limitation they state is kept in its Part. Collected here, they do not claim

- priority, or an exhaustive literature search, for the precise synthesis (Part III), the sparse constructions or the mass-two theorem (Part IV); a new universal computational mechanism or new classical universality; the stack, selector/copy and circuit-to-sum-of-squares devices as new;
- a fixed-arity polynomial for an unknown, unbounded horizon or for arbitrary halting histories; the horizon fixes the number of variables in both Parts, and “there exists some finite horizon” is an unbounded union that needs a separate compression theorem;
- for Part III: that the ten-signal binary interpreter is printed or instantiated; optimality of 18 or ten live signals, of the loader or of the dimension; injectivity of the integer event map; that every path of the 49,700-mode graph is realizable from one input (it is a mode-only overapproximation), or that the graph bounds the length of a universal run; a flat monomial expansion (the trillion-term count is derived, not emitted); a minimal gate count; any semantics for malformed rational configurations or at accumulation points; that the finite replays prove universality, which is imported from Morita’s theorem; that the polynomial is convex or a sum of squares over all reals;
- for Part IV: a printed universal source table or its state count; faithful interpretation of arbitrary counter pairs; intrinsic universality, an efficient simulation, exact-target undecidability, or global halting (the pulse is an anchored observation); a minimum universal mass, or that three particles suffice; any statement over unrestricted reals; that the displayed counts are lower bounds; an arbitrary-rule quantifier eliminator; and anything about arbitrary c.e. sets from the decidable mass-two subclass ([write] added 2 October 2026, batch 80: these remain true of Part IV; Parts V and VI now prove exact-target undecidability and the sufficiency of three units for a weighted automaton with many weight-one symbols, and decidability at masses three and four with a single one; Section 1.16);
- a proof-assistant verification; neither source’s code is a verified compiler.

1.11 Research questions answered or touched by Parts III and IV

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its relation column.

- **Part I’s direction 15.2** (an audited universal signal-machine frontend) is *answered by Part III*,

except for the ordinary-integer loader: Part III instantiates a published finite-input universal construction with its exact labels, speeds, all collision rules, forbidden (not transparent) undefined encounters, input encoding and marked acceptance, audits simultaneous and triple collisions, and proves the correspondence of its loader with the simulated machine. Two differences from the direction as worded remain. The machine is Morita’s UTM(15,6) through Durand-Lose’s 2012 stacks, not the 13-type construction of [3] named as a candidate; and the input is a prepared cyclic-tag tape loaded as two rational distances, so the equivalence with “an ordinary integer program input” would still need a digit-extraction and base-conversion loader, whose cost Part III leaves explicit (Section 33).

- **Part II’s Question 1** (transcribe the 13-label, 21-rule machine) stays open as stated: Part III transcribes a different universal machine.
- **Part II’s Question 2** (exploit conservative bounded populations) is *answered in part, at a fixed horizon*, by both new Parts. Part III instantiates every finite transition branch of a reversible conservative universal signal machine, pays for strict guards and for its loader, and excludes accumulation on encoded runs; Part IV does the same for lattice dynamics, where conserved mass bounds the records but not the gaps. Neither encodes unbounded histories in a fixed arity, which the question also mentions.
- **Part II’s Questions 3 and 4** are not answered; Part III’s literal ledger is a variable-and-residual count, not the gate ledger asked for.
- **Part IV’s own five questions** (Section 48) are open; Part III states no separate questions. ([write] Added 2 October 2026, batch 80: the fourth and fifth are now answered, in substance and for the weighted three-mass automaton respectively, by Parts V and VI; Section 1.16.)

1.12 Parts V and VI: conserved-mass thresholds (batch 80)

[write] This subsection and the four that follow were written on 2 October 2026, when four manuscripts of batch 80 of ProveIt’s incoming reports were added to the report as Parts V and VI. They are called *sources 14–17* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 13. Their batch-80 manuscript numbers are different: **source 14 is batch-80 manuscript 10, source 15 is manuscript 02, source 16 is manuscript 05 and source 17 is manuscript 03**. A fifth archive of the batch, manuscript 06, is the superseded original of manuscript 05 and is not printed. Appendix E gives the full provenance. None of the four manuscripts names a repository commit; source 16 cites Part IV (source 13) by its title, for its post-elimination compiler, and sources 15 and 17 cite sources 14 and 16 as their “companion” reports. All four are dated 3 October 2026.

Source	Batch 80	Archive and title	Printed as
14	manuscript 10	<code>Three_Mass_Reversible_Computatio n.zip</code> ; <i>Three mass units in a reversible weighted cellular automaton</i>	Part V, Sections 49–63
15	manuscript 02	<code>Exact_Targets_Three_Mass_Units.zip</code> ; <i>Exact targets with three conserved mass units</i> , a declared extension of source 14	Part V, Sections 64–73
16	manuscript 05	<code>Single_Unit_Three_Mass_Decidability (1).zip</code> ; <i>Three mass units with a single unit symbol</i> , the attribution revision of manuscript 06	Part VI, Sections 74–82
17	manuscript 03	<code>Four_Mass_Decidability_Package.zip</code> ; <i>Four mass units and exact reachability</i> , which builds on source 16	Part VI, Sections 83–92
—	manuscript 06	<code>Single_Unit_Three_Mass_Decidability. zip</code> ; the original of manuscript 05	not printed (superseded)

- **Part V (sources 14 and 15), *Three conserved mass units suffice*.** The model is a power-set alphabet 2^T of Boolean channel types, the weight of a symbol being its cardinality, so that many distinct symbols have weight one. Source 14 compiles every separated reversible two-counter machine into a globally reversible, mass-conserving cellular automaton of forward and inverse radius at most four whose canonical configurations have total mass exactly three and gap $12 \cdot 2^a 3^b$, and in which a clean one-cell symbol $\text{Ready}(q_f)$ occurs exactly when the source halts (Theorem 49.1); radius one follows by four-site spatial blocking at the same clock, or by a four-phase dilation at four times the time. With Part IV’s mass-two theorem this makes three the least mass with fixed-rule undecidable pattern occurrence in that model (Theorem 49.2). Source 14 also gives exact type and collision-row counts, exact microtimes, and a quadratic natural-number certificate at a fixed source horizon with a unique witness (Theorem 58.1). Source 15 wraps a fresh-entry source so that it runs forward, then its logical inverse, and then relabels its state: for one fixed reversible cellular automaton, exact reachability $\exists t A^t(x_N) = y_N$ between two computable configurations of mass three and the same support is undecidable (Theorems 64.1 and 67.1); it adds an exact first-target clock and a compact certificate whose natural zero fibres are in affine bijection with those of the full wrapped certificate (Theorem 70.2).
- **Part VI (sources 16 and 17), *A single unit symbol*.** With at most one weight-one symbol, as in an ordinary integer-state number-conserving automaton, source 16 proves that the timed evolution relation at total mass at most three is effectively Presburger, uniformly in all positions and the time for each fixed rule (Theorem 74.1); exact reachability and pattern occurrence are decidable, and horizon-free canonical quartics follow. Source 17 proves that exact reachability and pattern occurrence remain decidable at mass four (Theorem 83.1), but that the timed relation is no longer Presburger there: a binary radius-six shuttle hits an anchored pattern exactly at the times $k^2 + (2d - 11)k$ (Proposition 89.2), with a one-witness quartic for that timing. For integer-state number-conserving automata the numerical particle lower bound is therefore five (Corollaries 79.1 and 83.2). ([write] Added 3 October 2026: batch 82 adds a third source to Part VI and a Part VII, which extends the decidability at mass four to every dimension; Section 1.17.)

The thresholds proved in this report, by model (“undecidable” means fixed-rule undecidable for a computable family of initial configurations):

Model	Mass at most 2	Mass 3	Mass 4	Mass 5
Weighted alphabet, several weight-one symbols	decidable (Part IV, Theorem 44.1)	undecidable pattern occurrence and exact-pair reachability, fixed reversible rule (Part V)	not treated	not treated
At most one weight-one symbol (includes integer-state NCCA)	decidable (Part IV)	decidable, timed relation Presburger (Part VI, source 16)	decidable, timed relation not Presburger (Part VI, source 17)	universal at large radius by the Alhazov–Imai abstract [24]; reversibility not asserted, full paper not checked
[write] The same, on $\mathbb{Z}^d$ for every finite d (added 3 October 2026)	decidable (Part VII, source 19)	decidable, fixed-input timed Presburger (source 19, Lemma 104.1)	decidable (Theorem 101.1); timed relation not Presburger (source 17’s shuttle embeds linewise)	source 19 infers universality by embedding the one-dimensional binary five-particle construction of batch-82 Reports 16 and 26 linewise (Corollary 101.2); Report 16 is Part II of five-particle-binary-automata, Report 26 is placed there but not yet written, and this report does not establish it

How they relate to Parts I–IV. Both new Parts work in Part IV’s weighted model (Section 43: finite alphabet, finite radius, positive integer weights, a unique zero-weight vacuum, conservation on finite configurations), which allows several symbols of weight one. Part IV’s Theorem 44.1 is the lower half of Part V’s threshold, and source 14 proves it again (a pointer replaces that proof, Section 59). Part VI restricts the same model to at most one weight-one symbol, so its mass-two statements are special cases of Part IV’s; source 16 reruns Part IV’s semilinearity architecture one mass higher, with a new no-split lemma, and reuses Part IV’s post-elimination compiler by name. The two new Parts do not contradict each other: Part V keeps its finite control in weight-one controller types and changes its gap multiplicatively, neither of which a single stateless unit symbol permits, as sources 14 and 16 both say. Parts V and VI share no lemma with Parts I–III, except that the certificate gate of sources 14 and 15 is the selector device of Part III and of *quadratic-orthant-certificates* (Section 1.14).

Morita and Imai again. Part IV and Part V both rest on Morita and Imai’s 2001 paper [19], in different ways: Part IV interfaces with their number-conserving reversible partitioned automaton itself (and corrects a sign in its rule (8.2)), while Part V uses only their reversible two-counter normal form (Definitions 3.1–3.3, Proposition 3.4 and, for source 15, Lemma 3.5). No table, program

or data file is shared, except that source 14 ships a byte copy of one source-13 receipt, which is not shipped twice.

1.13 Notions and letters of Parts V and VI

[write] Parts V and VI keep their sources' letters and their numbering of statements, by section; equations are numbered by section (sources 16 and 17 tagged theirs (1), (2), ...). The four sources' macro for the natural numbers is written `\NN` in this file; it prints $\mathbb{N} = \{0, 1, 2, \dots\}$, as each source defines it. No other symbol was renamed and no normalization changed. Inside Part V, source 15 restates source 14's interface in source 14's letters; inside Part VI, source 17 restates source 16's stationary frame in source 16's letters. The table lists the letters of Parts V and VI that carry another meaning in the other new Part or in Part IV (the table of Section 1.8 gives Part IV's meanings in full). Inside a Part the meaning is always that Part's.

Letter	Part V (sources 14, 15)	Part VI (sources 16, 17)	Part IV
R	the stationary right-marker channel type R ; in source 15 also a source program R , with R_{cl} its wrapped program	the radius R of F	right lane; residual count
r	residue $N \bmod 6$; clock phase in $\mathbb{Z}_{144}$; the radius in the mass-two summaries; number of input coordinates in source 15's ledger	phase index $0 \leq r < p$	the radius; ranks
S	static shuttle types S_j ; the channel shift in $F = S \circ \Pi$; state-code sums S_t	the padded comoving radius $S = \max(1, R + \delta)$	number of allowed rows
L	left-controller types $L_{j,x}$	the left marker coordinate (source 17)	left lane
u	controller velocity $u = d - p$; quotient offsets u_{tj}	the weight-one symbol u	incoming masses
d	the divisor in a factor p/d ; the number of increment or decrement instructions	a drift; packet widths d_r ; the initial right-marker coordinate d of the binary shuttle	comparison slack; modulus
D	$D = 6qm$ data triples; tick durations D_j	drift of a mass-two profile (source 16); marker gap $D = R - L$ (source 17)	diameter
N	the gap integer, $N = 2^a 3^b$ or raw $N = C 2^a 3^b$, with $n = 12N$	numbers N_2, N_3 of weight-two and weight-three labels (source 16); the threshold $N = 4B + 12S$ (source 17)	—
B	number of residue-expanded branches	Boolean gate count of source 16's quartic (Part IV's G); seed-prefix bound (source 17)	control baseline
C	completed pair-support count; raw cofactor C (source 15: c); new-value forms C_j ; a loader bound	number of congruence atoms (source 16); support constant C_1 and a time coefficient (source 17)	centre lane; congruence atoms
E	selector sums E_t	residual count of the quartic (source 16, Part IV's E_Φ); first-hit predicate E_q ; seed library $\mathcal{E}$ (source 17)	residual count E_Φ

Letter	Part V (sources 14, 15)	Part VI (sources 16, 17)	Part IV
G	finite gates G_I, G_q, G_H	the comoving automaton $G = \tau_{-\delta} F$	gate count
H	incoming-instruction table H ; the terminal state H of the wrapped source (source 15)	span predicate H_q (source 16)	height bound
K	loader selectors $K = (A + 1)(C + 1)$; $K_{\text{types}}, K_{\text{prescribed}}, K_{\text{completed}}$	witness count of the quartic (Part IV's K_Φ)	alphabet cap; K_Φ
M	a source machine (F_M, A_M)	total mass $M(c)$; the gap-residue modulus (source 17)	the conserved mass
P	prescribed pair-row count; the certificate P_h	de Bruijn potential (source 17)	—
p	the prime 2 or 3, factors p/d ; a period	a period	period of a type cycle
q	number of states; a source state	a normalized pair shape (source 16); a finite mode (source 17)	$q = K + 1$
T	translation T_a ; physical times T_h, T_* , T_{cl}	times in Proposition 77.1; section times T_n (source 17)	the horizon
τ	a source-step duration	translation τ_a	—
α	the singleton type map	label tuples α, β (source 16); a positive rational slope (source 17)	singleton type map
δ	a tail translation	the displacement of the isolated unit	a tail translation
h	the source horizon	first core time (source 16); edge duration (source 17)	a time; flags

Several words need the same care.

- **Mass, unit, particle.** In Part V a symbol is a set of Boolean channels and its mass is the number of channels, so a canonical configuration has *three mass units in two occupied cells*. In an ordinary integer-state number-conserving automaton (NCCA) the state is its own mass, and a “particle” of the literature is one unit of numerical mass. The tempting false reading of Part V as “three particles suffice for an NCCA” is refuted by Part VI: with a single weight-one symbol, three and four units are decidable.
- **Source and companion.** “Source 14” to “source 17” name manuscripts; inside Part V “the source” is the simulated reversible two-counter machine. Source 15’s “companion construction” is source 14, source 16’s “companion report” is Part IV (source 13), and source 17’s “companion” is source 16; bracketed notes say so where they occur.
- **Halt and target.** Source 14 observes the clean symbol $\text{Ready}(q_f)$ *somewhere*; source 15 asks for equality of the *whole* configuration with an input-dependent target y_N ; Part VI’s observations are exact targets and finite patterns, anchored or anywhere. Part IV’s pulse is an anchored first-pulse observation. None is global halting.
- **Timed and untimed.** Source 16’s relation includes the time and is uniform in all positions; source 17’s mass-four description forgets the time and is for a fixed input, and the binary shuttle shows that the timed relation cannot be Presburger.

- **Horizon.** Sources 14 and 15 fix an external source horizon h (source steps, not CA ticks); sources 16 and 17 need no horizon. None of them is a fixed-arity polynomial for unbounded halting.

1.14 Relation of Parts V and VI to the repository

[write] The four sources cite none of the material below except Part IV, which source 16 cites by title; these cross-references are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers.** Source 14’s Section 59 proves Part IV’s Theorem 44.1 again by the same argument; its proof is replaced by a pointer, and so are its bounded-diameter proposition (Proposition 43.1) and its finite-ring paragraph (Part IV’s Section 43); its two-mass test program is source 13’s audit minus one line, and its receipt is byte-identical to source 13’s, which is not shipped twice. Source 15’s summary of the mass-two theorem is a pointer too, and its compiler interface restates source 14’s results in its own words (Section 65). Source 16’s Lemma 75.2 and Sections 76–79 follow the architecture of Part IV’s Sections 44–45, one mass higher, and its quartic is Part IV’s compiler (Section 46); source 17’s Section 84 restates source 16. Each place carries a bracketed note; no novelty is claimed for any of them. New are source 16’s no-split lemma (Lemma 75.1), its mass-three core and uniform first-entry composition, and all of source 17 after its Section 84.
- **The certificate gate.** The product $(E_t - e_{tj})(e_{tj} + u_{tj})$ of sources 14 and 15 is the inactive-product gate of Lemma `qoc:rn:lem:gates` in Part VI of *quadratic-orthant-certificates*, and a form of Part III’s selector/copy device (note after Theorem 36.1); source 14’s natural-domain counterexample is consistent with that report’s real-orthant one-hot lemma: one-hot selection survives over the nonnegative reals, divisibility does not.
- **canonical-diophantine-certificates, Part XI [11].** For source 16’s semilinear relation, that Part’s Theorem `cdc:of:thm:singlefolds1` gives an orthant-nonnegative single-fold representation of degree two, as it does for Part IV’s (note after Corollary 80.1). Its Theorem `cdc:of:thm:classification` ($D_2^+ = D_3^+ = \text{SL}$) shows that source 17’s non-semilinear hit set has no orthant-nonnegative representation of degree at most three, so the degree four of source 17’s one-witness quartic is the least possible in that class (note after (90.1)).
- **The programme’s reviews.** The Hilbert-tenth-problem programme reviewed all five archives of this cluster before this write. Its review of sources 14 and 15 [45] reports “PASS within the stated model and interfaces”, with no repair, after both complete author replays and independent checks (360 literal core cases, 2,400 natural assignments, 45 compact/full zero-fibre correspondences, 7,968 microscopic cleanup ticks); it checked Morita and Imai’s Proposition 3.4 and Lemma 3.5 in the primary paper (pp. 245–246) and records the off-zero and rational limitations of the compact certificate as documented limitations, not defects. Its review of sources 16 and 17 and of manuscript 06 [46] reports “PASS within the stated mathematical and executable scopes”, with no correction; it confirms that manuscript 05 changes only the attribution of manuscript 06 (the mathematical body is byte-identical), reconstructs all 8,192 local windows of the binary shuttle, checked Kong’s thesis, could not certify the Japanese text of the KAKEN report or obtain the full Alhazov–Imai paper, and notes the five-witness refinement of source 16’s quartic. Its index [47] lists the reviews, and its placement audit [48] authenticates the 116 files placed by 345a9e44e. Those reviews concern the archives and the placement; this write, which assembles the text, has not been

reviewed by the programme. ([write] Added 3 October 2026: the programme has since reviewed this write too, in `review_signal_batch80_typesetting_ef114b0bb.md` with the independent challenge `review_signal_batch80_quantifiers_ef114b0bb.md`, both in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit `f787251a2`). They account for every formal statement, proof and display of sources 14–17 (71 statements and proofs retained and 4 printed as pointers; 98 displays retained and 7 declared deduplications) and confirm the degree-four minimality note after Proposition 89.2 in its stated class. Their patch `signal_batch80_quantifiers_ef114b0bb.patch` corrected two summaries written for this merge, and both corrections were applied on 3 October 2026: the last cell of the threshold table above had called the Alhazov–Imai construction “not reversible”, whereas its abstract merely does not assert reversibility; and the README’s summary of Part V had one fixed automaton simulate every separated reversible two-counter machine, whereas Theorem 49.1 compiles each machine M into its own automaton F_M and the one fixed rule of Theorem 49.2 comes from fixing a universal source. The abstract’s sentence on Part V, which had a fixed automaton simulate reversible two-counter machines, now says, with source 14’s own abstract, that it simulates a universal one.) Sections 61, 72, 82 and 92 give the details.

- **The repository’s universal bounds.** The 75- and 87-operation bounds of the project README [1] are untouched by Parts V and VI. Sources 14 and 15 give degree-two certificates for each external horizon only, and sources 16 and 17 give horizon-free quartics only for decidable classes; all four say so, and the programme’s reviews record the same conclusion.
- **Formal status.** Nothing in Parts V and VI is formalized in Lean or Rocq; the Hilbert-tenth-problem Lean development has no cellular-automaton module. Cooper’s quantifier elimination, which source 16 imports, is the Lean development `Logic/PresburgerArithmetic` (Section 1.9); that formalizes the imported step only.
- **The literature.** Sources 14 and 15 credit Brandt, Portmann and Uitto [37] (three finite-state agents, reflected-shuttle arithmetic, prime exponents in one gap), Dobrev et al. [38], the abstract of Dobrev, Královič and Pardubská [39] (three synchronous agents, full paper not obtained), Durand-Lose [5], Morita and Imai’s partitioned automaton of mass $m + 19$ [19], and Bennett’s uncomputation [40]. Sources 16 and 17 credit Kong’s 2021 thesis [41] (binary three-particle nonuniversality, with the four-particle case open in 2021), Imai’s 2021 KAKEN report [42] (decidability at most three particles, the problem left implicit), the Alhazov–Imai abstract [24] (five particles, not asserted reversible) and a 2021 RIMS session [43, 44]. Source 14 does not cite Kong or Imai’s report, which concern the numerical model of Part VI. No source claims priority, and this report claims none.

1.15 Status and what Parts V and VI do not claim

[write] The four sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; all four are AI-assisted and unrefereed. Every limitation they state is kept in its Part. Collected here, they do not claim

- for source 14: an expanded universal source table, a numerical universal alphabet or row count, or a frozen universal transition table (universality is imported from Morita and Imai’s Proposition 3.4); anchored halt observation or intrinsic universality; an efficiency bound (the gap is exponential in the counters); a three-numerical-particle theorem for integer-state

NCCA; real or rational exactness of the certificate; a fixed-arity polynomial for an unknown horizon, or a finite-fold or single-fold representation; an exponentiation gadget (the loader is a bounded table); that the syntactic slot counts are lower bounds; novelty of the radius reductions; optimal overhead, a smallest alphabet, priority or literature-wide novelty; global halting, freezing or an eventual fixed point;

- for source 15: one target independent of the input; that the target is stationary or reached once; microscopic reversal of the forward paths; a bound on the overhead of the fresh-entry normalization; a map of the whole natural orthant, an identity off the zero sets, or rational or real exactness of the lift; novelty or priority for forward-and-reverse computation; a numerical universal alphabet;
- for source 16: one formula with the rule as a parameter; anything about intrinsic universality, signed masses, infinite alphabets, nonuniform or time-dependent rules, or active zero-weight backgrounds; a universal bound on the formula-dependent arity K , efficient preprocessing or optimal arity; a general rule-to-formula or quantifier-elimination implementation (the code validates the orbit accelerator); real or rational fibre exactness; a finite-fold conclusion for c.e. sets; novelty with respect to Kong’s thesis and Imai’s report;
- for source 17: an efficiency bound; infinite backgrounds, signed weights, infinite alphabets, active zero-weight backgrounds, several unit labels, or time-dependent or nonuniform rules; a five-particle upper bound for binary automata; an implemented rule-to-section compiler or a formalization; reversibility of the binary shuttle (it is not injective); a general fixed-arity single-fold quartic compiler; uniform timed Presburger definability at mass four; priority, or any statement on the present status of the four-particle question;
- a proof-assistant verification; none of the sources’ code is a verified compiler.

The report claims no novelty for the results that the four sources prove again (Section 1.14).

1.16 Research questions answered or touched by Parts V and VI

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its last rows.

- **Part IV’s remark after Theorem 44.1** (“It does not show that three suffices, or identify a least universal mass”) is *answered, model by model*. With several weight-one symbols three suffices, for pattern occurrence (Theorem 49.2) and for exact pairs (Theorem 64.1); with a single weight-one symbol neither three nor four does (Theorems 74.1 and 83.1), so that the least universal numerical mass is at least five, and exactly five if the Alhazov–Imai construction holds as its abstract states (its full paper has not been checked, and it is not asserted reversible). ([write] Added 3 October 2026, batch 82: the lower bound five holds in every finite dimension (Part VII, Theorem 101.1); source 19 states an inherited sharp four-versus-five boundary in every dimension, conditional on batch-82 Reports 16 and 26, which this report does not print: Report 16 is Part II of **five-particle-binary-automata**, whose Part I (Reports 14 and 15) proves the binary one-dimensional upper bound five, and Report 26 is placed there but not yet written.)
- **Part IV’s fourth question** (“Resolve small-particle model comparisons”) is *answered in substance*: the number of weight-one symbols is the decisive convention (Section 1.13). Still open: the full Alhazov–Imai construction, a reversible numerical threshold, a binary upper

bound, and the least number of weight-one symbols that permits a universal three-mass construction.

- **Part IV’s fifth question** (“Keep exact targets separate from halt signals”) is *answered for the three-mass weighted automaton* by source 15, whose output normalization is inverse execution from a fresh entry state; it is not answered for Part IV’s own Morita–Imai pulse interface.
- **Part IV’s non-claims** “exact-target undecidability, ... a minimum universal mass, or that three particles suffice” (Section 1.10) remain true of Part IV; Parts V and VI supply those statements in the models stated above.
- **Part IV’s first question** (“Print one safe universal source”) is *not* answered: sources 14 and 15 also leave the universal source as an existence construction (their first questions).
- **Source 14’s third question** (exact finite targets need explicit treatment of the final counters) is answered by source 15; its anchored halt observer stays open. **Source 14’s and source 15’s fourth questions** (the integer-state mass convention) are answered in part by Part VI (at least five units for undecidable observations); the sharp reversible threshold stays open. **Source 16’s “four-versus-five boundary”** is answered by source 17 within its class.
- All other questions of Parts V and VI (Sections 63, 73 and 92, and source 16’s two directions in Section 81) are open. ([write] Added 3 October 2026: source 17’s seventh question is now answered for a fixed input by source 18, and its second, fourth and sixth are touched by sources 18, 20 and 21; Section 1.21.)

1.17 Source 18 and Part VII: mass at most four in every dimension (batch 82)

[write] This subsection and the four that follow were written on 3 October 2026, when four manuscripts of batch 82 of ProveIt’s incoming reports were added to the report: one as a third source of Part VI, the other three as Part VII. They are called *sources 18–21* after the file prefixes of their shipped programs and data, which continue this report’s local sequence after 17. Their batch-82 manuscript numbers are different, and so are the report numbers of the research pipeline that wrote them: **source 18 is batch-82 manuscript 19 (the pipeline’s addendum 13), source 19 is manuscript 06 (Report 29), source 20 is manuscript 17 (Report 30) and source 21 is manuscript 01 (Report 31)**; source 17 is the pipeline’s Report 12. All four arrived in commit db37d18c8 and are dated 3 October 2026; Appendix E gives the full provenance. None names a repository commit.

Source	Batch 82; pipeline	Archive and title	Printed as
18	manuscript 19; addendum 13	<code>Timed_Four_Mass_Quartic_Certificates.zip</code> ; <i>Canonical quartic certificates for timed four mass orbits</i> , a declared addendum to source 17	Part VI, Sections 93–100
19	manuscript 06; Report 29	<code>Dimension_Independent_Particle_Thresholds_Package.zip</code> ; <i>Dimension independent particle thresholds</i>	Part VII, Sections 101–117, Appendices I–J
20	manuscript 17; Report 30	<code>Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip</code> ; <i>Sparse orbit geometry and exact counting</i>	Part VII, Sections 118–128, Appendix K
21	manuscript 01; Report 31	<code>Binary_Planar_Four_Particle_Shuttle_Package.zip</code> ; <i>A binary planar four particle shuttle</i>	Part VII, Sections 129–140

Citations among them, all pinned by SHA-256 and verified at placement: source 18 cites source 17; source 19 cites sources 17 and 18 and batch-82 Reports 16 and 26; source 20 cites source 19; source 21 cites sources 17, 18 and 20. Reports 16 and 26 belong to the binary five-particle line of the same batch and were placed in the collection’s report `five-particle-binary-automata` (commits `bca6383e9` and `7d2b1b245`); Report 16 is printed there as Part II, and Report 26 is not yet written. Neither is printed here.

- **Source 18 (Part VI), *timed quartics for a fixed input*.** Conditional on source 17’s effective normal form (Proposition 87.1), the complete timed orbit of a fixed input of mass at most four is a disjoint union of half-open charts with at most two natural parameters, affine domains and quadratic outputs (Lemma 94.1), and a canonical quartic with $B + K + M$ natural witnesses has exactly one witness tuple at each complete timed configuration (Theorem 93.1, Proposition 96.1). For source 17’s binary shuttle it writes the two charts explicitly and specializes the certificate to four witnesses and seven squares, exact even for nonnegative real witnesses and uniform in the gap (Corollary 99.1). It answers source 17’s last question for a fixed input (Question 7).
- **Source 19 (Part VII), *every dimension*.** For a conservative cellular automaton on $\mathbb{Z}^d$ with a finite positively weighted alphabet, a unique vacuum and at most one weight-one symbol, exact and translated reachability and anchored and translated pattern occurrence are decidable from every input of mass at most four, uniformly in d and without reversibility (Theorem 101.1); its appendix lifts source 18’s timed quartics to $\mathbb{Z}^d$ (Theorem I.1). Its sharp four-versus-five boundary (Corollary 101.2) is conditional on Reports 16 and 26.
- **Source 20, *orbit geometry and counting*.** For a fixed input of mass at most four the stationary-frame visited set is semilinear, lies in finitely many translates of one rational plane of rank at most two, and has eventually quasipolynomial box counts of sharp degree $\text{rank}_{\mathbb{Q}}\{E, \nu\}$ (Theorem 118.2); an expanding shuttle discovers sites at the rate $T^{r/2}$ and has piecewise quadratic first arrivals (Theorem 118.3); a nonzero drift can destroy semilinearity and quasipolynomiality (Theorem 118.4), as an explicit weighted radius-six planar automaton shows with the exact count $3N - 3\sqrt{N} + O(1)$ (Theorem 125.1); canonical first-visit quartics follow (Theorem K.1).
- **Source 21, *a binary planar shuttle*.** An ordinary binary, quiescent, conservative radius-six automaton on $\mathbb{Z}^2$, obtained by lifting two outputs of source 17’s binary shuttle by one row,

has a four-particle orbit with source 17’s clock (Theorem 132.1), visited set a wedge minus two lines (Theorem 133.1), exact stationary and drifted counts $4N - 4\sqrt{N} + O_k(1)$ (Theorems 134.1 and 136.2), Boolean degree exactly 45 with exactly 78 essential cells (Theorem 137.1), and a one-step quartic with 614 unique auxiliaries (Theorem 138.1).

The threshold table of Section 1.12 gains a row for every finite dimension ([write], 3 October 2026). Its entry at mass five records only what is true now: the inherited five-particle upper bound in every dimension rests on source 19’s linewise embedding and on Reports 16 and 26; Report 16 is Part II of **five-particle-binary-automata**, Report 26 is placed there but not yet written, and the programme’s reviews of both leave published theorems as explicit dependencies (note after Corollary 101.2).

1.18 Notions and letters of source 18 and Part VII

[write] The four sources keep their letters and their numbering of statements, by section; equations are numbered by section. Their macro for the natural numbers is written `\NN` here and prints $\mathbb{N} = \{0, 1, 2, \dots\}$: source 18 wrote $\mathbb{N}$, and sources 19–21 wrote $\mathbb{Z}_{\geq 0}$ for the same set (a change of rendering only). Sources 19–21’s maximum norm $\|\cdot\|_\infty$ is typeset by a renamed macro, because Part IV’s `\norm` prints $\mathcal{N}(\cdot)$; no other symbol was renamed and no normalization changed. The table lists the letters that carry different meanings among the new sources or against Part VI; inside a source the meaning is always that source’s.

Letter	Source 18	Part VII (sources 19, 20, 21)	Part VI (sources 16, 17)
d	the initial right-marker coordinate of the binary shuttle, $d \geq 7$, with $x = d - 7$	the dimension of $\mathbb{Z}^d$ (sources 19, 20); the current gap $d = k + n$ (source 21)	the initial right-marker coordinate (source 17)
k	k_i , the number of parameters of chart i	a period count (source 19); the initial gap, $k \geq 3$ (source 20) or $k \geq 7$ (source 21)	the witness of (90.1); the index of (89.3)
S	the common squared part; transition starts $S_\ell(n)$	the padded comoving radius (source 19, as in Part VI); the visited set S , S_F (source 20); an occupied set $S \subseteq \mathbb{Z}^2$ (source 21)	the padded comoving radius
B	the number of charts	the seed-prefix bound B , a marker B and the chart count B_{ch} (source 19); a clause count B , the count B_n , the function $B(N)$ (source 20); a rule output B and $B_k(N)$ (source 21)	the seed-prefix bound (source 17)
C	the certificates C_a , C_{prod} , C_{SOS} , C_d	a time coefficient and the quartic $C(y; w)$ (source 19); the box counts $C(N)$, $C_F(N)$ and a congruence count C (source 20); $C_{G,k}$, $C_{F,k}$ (source 21)	a time coefficient (source 17)
E	the selector sum	the anchor-update vector $E \in \mathbb{Z}^d$ (sources 19, 20); the selector sum (source 19’s appendix); a head state of weight two (source 20); the eastward pair rule (source 21)	the anchor update (source 17)
W	—	the ordinary threshold (source 19); a witness count and a head state (source 20); the westward pair rule (source 21)	the threshold W (source 17)

Letter	Source 18	Part VII (sources 19, 20, 21)	Part VI (sources 16, 17)
N	the witness count	the threshold (source 19); a box radius and a shared cycle variable (source 20); a box radius (source 21)	the threshold (source 17)
τ	transition durations $\tau_\ell(n)$; translation	translation τ_v (sources 19, 20); first visit time $\tau(y)$ (source 20); unit vertical translation (source 21)	translation τ_a
D	the marker gap D_n	the drift vector of a packet (source 19); the discovery count $D(K)$, drifts D_i , a denominator (source 20)	the marker gap $D = R - L$ (source 17)
R	the relation R_a ; residuals $R_0, R_{ir}^\geq, R_q^{\text{out}}$	the radius (source 19); the inverse $R(M)$ (source 20); a rule and $R_{G,k}, R_{F,k}$ (source 21)	the radius
M	the inequality count	$M = 10^6$, M_0 and an inequality count (source 19); a level in an inverse and a number of square lifts (source 20); a level (source 21)	the gap-residue modulus (source 17)
H	the equality count	a Cramer vector and an equality count (source 19); the inverse $H(M)$ and an equality count (source 20); the halo $H(A + a)$ (source 21)	a span predicate (source 16)
P	chart outputs P_i	a packet profile (source 19); a polynomial $P(y; w)$ (source 20); a polynomial and a predicate $P_{A,a}$ (source 21)	the de Bruijn potential (source 17)
α	coefficients α_ℓ of flight lengths	a rational slope (source 19); the leading coefficient of $D(K)$ (source 20); $2k - 10$ (source 21)	a rational slope (source 17)
λ, ν	—	a Bezout functional with $\lambda(\nu) = 1$ for the primitive flight direction ν (sources 19, 20)	—
V	—	the discovery count $V(T)$ (source 20); the visited set V_k (source 21)	—

Several readings need the same care.

- **The gap and the dimension.** In sources 17 and 18 the letter d is the initial right-marker coordinate of the binary shuttle; in sources 19 and 20 it is the dimension. Source 21 calls the initial gap $k \geq 7$; its k is source 17's and source 18's d , and its $d = k + n$ is source 18's $r = d + n$. The tempting reading of source 18's " $d \geq 7$ " as a dimension bound is false.
- **Particles and mass.** In source 21 every occupied site has weight one, so "four particles" is mass four. Source 20's explicit automata have mass four in three occupied sites: two unit markers and one head of weight two. Its letters E, W are head states; source 21's E, W, R, L are names of rules of a binary automaton.
- **Timed and untimed, fixed and uniform.** Source 18's relation keeps the time and the complete configuration; source 20's first-visit certificates forget the time and keep one witness per site. Every certificate of the four sources is for one fixed rule and one fixed input, with an arity that depends on the input; source 16's uniform Presburger description (Theorem 74.1) has no analogue here at mass four.

- **One step and one orbit.** Source 21’s 614-auxiliary quartic certifies one local transition of its rule; source 18’s and source 20’s quartics certify one orbit.

1.19 Relation of source 18 and Part VII to the repository

[write] The four sources cite only Part VI’s source 17, each other and the batch-82 Reports 16 and 26; the cross-references below are added. Results of other reports are cited by label name.

- **Re-derived results, printed as pointers or second routes.** Source 18 restates source 17’s binary rule and its conservation certificate (a pointer replaces that subsection, Section 98), its return quartic (90.1) and its normal form (notes in Sections 94 and 99), and ships three of source 17’s files byte for byte (shipped once). Its compiler is a second route to Part I’s Theorem 10.2, with Part V’s inactive gate and Lemma `qoc:rn:lem:gates` of *quadratic-orthant-certificates* (note in Section 96). Source 19 proves source 17’s Theorem 83.1 again at $d = 1$ by the same architecture while extending it to $\mathbb{Z}^d$ (notes from Section 101 on); its Lemma 104.1 is a fixed-input, $\mathbb{Z}^d$ form of source 16’s Theorem 74.1; its mass-two case meets Part IV’s Theorem 44.1; its appendix restates source 18’s chart partition and compiler (printed in full, with notes). Source 20’s canonical membership quartic is a second route to Part IV’s compiler (Section 46) and to Theorem `cdc:of:thm:singlefolds1` of *canonical-diophantine-certificates* (note in Appendix K); its Lemma 122.1 is classical. Source 21’s orbit theorem is the planar lift of source 17’s Proposition 89.2, and its one-step quartic uses Part IV’s device of gate residuals. Each place carries a bracketed note, and no novelty is claimed for any of them.
- **New in the four sources.** Source 18’s complete-orbit chart partition, constant-term lifting, ledgers, selector elimination, signed external pairs and the four-witness binary certificate; source 19’s actual-site contact, exact contact rays, finite nonparallel switch set, compact triples that miss, the $\mathbb{Z}^d$ dispatch and drift cutoff, and the $\mathbb{Z}^d$ chart refinements; all of source 20’s geometry, counting, explicit automata and first-visit certificates beyond the membership compiler; source 21’s binary planar rule with its full-shift conservation proof, wedge, counts, exact dependency set and one-step quartic.
- **Degree four is least for source 18’s binary certificates.** The note after (90.1) transfers. The complete timed relation of the binary shuttle meets the semilinear condition $(x_0, x_1, x_2) = (0, 3, 4)$ exactly at the return times $T(n)$ of Section 98, which form a set that is not semilinear (Proposition 89.2); so the relation is not semilinear, and by Theorem `cdc:of:thm:classification` it has no natural-witness representation of degree at most three by a polynomial nonnegative on the real nonnegative orthant. This observation is [write], made at this merge; the programme has not reviewed it.
- **The programme’s reviews.** The Hilbert’s-tenth programme reviewed three of the four sources on the day of their placement, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`. Source 21: `review_binary_planar_four_particle31.md` (commit 16f50dcc6) [56], “PASS within the stated scope; no error found”, with an independent reconstruction of the whole local quartic from the four printed rules (Section 139). Source 19: `review_dimension_independent_four_mass.md` (commit 7b547f686) [58]: the proof of Theorem 101.1 passes, a second reviewer found no gap, and the inherited five-particle upper bound is not reproved (Section 116). Source 20: a scoped intake, `review_original_frame_first_hit30_intake.md` (commit c875bad40) [59], which finds no defect in the whole-orbit-injectivity step of Corollary K.4 but is not a review of

the compiler, and names source 19’s timed companion proof as the next audit; a follow-up (`original_frame_first_hit30_fixture.md` with its independent review `review_original_frame_first_hit30_fixture.md`, commit `5b4c091e6`) [60] then read that proof in full, found it sound as a conditional statement, and emitted a complete 35-operation first-hit quartic for one drifted orbit of source 20’s weighted planar automaton (note in Appendix K). Source 18 has not been reviewed: the programme’s archive inventory `reviewed_report_archive_replay.md` lists its archive as relocation only. The programme has also reviewed Reports 16 and 26, on which source 19’s corollary rests (commits `69730d3e8` [61] and `fb7e3cb47` [57]; note after Corollary 101.2). The batch-80 review of source 17 [46], whose normal form sources 18–20 import, still applies to that import. This write has not been reviewed by the programme.

- **The repository’s universal bounds.** The project README’s fixed-universal operation bounds [1] are untouched: all quartics of the four sources are for one fixed rule and one fixed input, with arity depending on the input, or for one local transition (source 21); the programme’s review of source 21 says the same for it.
- **Formal status.** Nothing in source 18 or Part VII is formalized in Lean or Rocq, and the Hilbert-tenth-problem Lean development has no cellular-automaton module. The Presburger decision step that source 19 applies to its explicit formulas is Cooper’s quantifier elimination, formalized in `Logic/PresburgerArithmetic` (Section 1.9); that formalizes the imported step only. Woods’ counting theorems, which source 20 uses, are not formalized in the repository.

1.20 Status and what source 18 and Part VII do not claim

[write] The four sources give conventional proofs and finite exact-arithmetic checks, which are regression evidence, not proofs; all four are AI-assisted and unrefereed. Every limitation they state is kept in its place. Collected here, they do not claim

- for source 18: a proof of the imported normal form (the theorem is conditional on it); an implemented rule-to-chart preprocessing, a decision of the semantic bijectivity of a chart family, or a complexity bound; a bound uniform over inputs, uniform timed Presburger definability, or a single-fold or finite-fold Diophantine conclusion, including for MRDP; real or rational exactness of the generic compiler (only the four-witness binary certificate is exact for nonnegative real witnesses, and unrestricted real witnesses give spurious zeros); uniqueness after forgetting the time; witness minimality; novelty, priority or a literature status;
- for source 19: uniform timed Presburger definability; a useful complexity bound or small libraries; a general rule compiler or an end-to-end simulation of every input; reversibility; the inherited five-particle upper bound, which it does not reconstruct (Corollary 101.2 is conditional); several unit labels, negative or zero weights, infinite alphabets, an active vacuum, infinite scenery or time- or position-dependent rules; untimed finite-foldness, a single-fold MRDP theorem, a uniform arity bound or real or rational exactness for its appendix; a formalization; novelty or priority;
- for source 20: a binary or reversible realization; new theorems about Presburger arithmetic (its counting steps apply Woods’ theorems); practical algorithms or size bounds; an exhaustive asymptotic taxonomy in the original frame; a general original-frame first-hit certificate for sites or patterns under drift; a single-fold MRDP theorem, a uniform arity bound or real or rational exactness; an efficient general compiler, a formalization or a universality threshold; novelty with respect to the pebble-automaton literature;

- for source 21: reversibility or universality of either rule; a smallest mass, radius, number of essential cells, Boolean degree or auxiliary count; a classification of malformed orbits; an unbounded-time or finite-fold representation from its local quartic; a real-witness conclusion inherited from source 20;
- a proof-assistant verification; none of the sources' code is a verified compiler.

The report claims no novelty for what the four sources prove again (Section 1.19).

1.21 Research questions answered or touched by source 18 and Part VII

[write] Dated notes at the questions themselves record the following; Table 3 carries the same information in its last rows.

- **Source 17's seventh question** (Question 7, a fixed-input timed quartic compiler from disjoint phases) is *answered for a fixed input* by source 18, conditional on Proposition 87.1 and with natural witnesses; source 19's Appendix I lifts the answer to $\mathbb{Z}^d$. The rule-to-chart preprocessing is not implemented.
- **Source 17's second and fourth questions** (uniform untimed spatial definability; a stronger normal form for original-frame timed relations) are *touched*: for a fixed input the visited set is semilinear with quasipolynomial counts (source 20), the complete timed orbit is a disjoint union of quadratic charts (source 18), and with drift original-frame site sets can be nonsemilinear (Theorem 125.1). The uniform versions stay open.
- **Source 17's sixth question** (minimal radius for the binary timing example) is *touched* by source 21: its planar analogue has exact radius six and exactly 78 essential cells, with no minimality claimed; the one-dimensional question stays open.
- **Source 17's first, third and fifth questions** stay open; source 19 adds literature (Section 115), shows by an example that naive geometric bounds fail (Section 107), and leaves several unit labels open (its Question 4).
- **The numerical particle threshold** of Section 1.16 (at least five units, exactly five if the Alhazov–Imai construction holds) *extends to every finite dimension* for the lower bound (Theorem 101.1); the sharp upper bound in every dimension is conditional on Reports 16 and 26 (Corollary 101.2).
- **Source 20's first question** (binary or reversible realizations) is *answered for binary rules* by source 21; reversible realizations stay open.
- **Source 19's fifth question** (optimizing the timed certificates) is touched by source 18's selector elimination and four-witness certificate, without a general answer. All other questions of the four sources (Sections 100, 117, 127 and 140) are open.

Part I

Collision geometry is linear (source 07)

[write] Part I prints batch-78 manuscript 07, *Collision Geometry Is Linear: Event-Sparse Quadratic Diophantine Certificates for Rational Signal Machines*, “research manuscript prepared with ChatGPT

for Vladimir Reshetnikov’s **ProveIt** research program”, dated 2 October 2026. Its labels carry the prefix `smc:cg::`; its sections are Sections 2–16 here and its two appendices are Appendices A and B. Its abstract and keywords follow.

Abstract (source 07). A rational signal machine computes by the collisions of finitely many types of particles moving at fixed rational velocities on a line. Its continuous geometry seems to require nonlinear arithmetic certificates. We show that, once a complete chronological collision skeleton is fixed, the geometry is instead governed by homogeneous rational linear equations and strict linear inequalities in the initial gaps. A sweep construction using maximal adjacency intervals proves that these conditions exclude both unrecorded earlier events and omitted simultaneous collisions. For n initial signals, m distinct velocities, and E collision sites, at most $3(n - 1) + (3m + 1)E$ raw linear conditions suffice.

After positive denominator clearing, one globally nonnegative convex quadratic polynomial certifies the skeleton over the nonnegative integers. Every accepted integer seed has exactly one slack-witness tuple. A finite disjoint first-hit family then yields one quartic with unique selector and slack witnesses. We also prove causal-depth denominator and coefficient bounds, a small integral realization theorem, a robustness–codimension criterion, and a finite-horizon quasi-polynomial counting theorem. Explicit examples include a first-collision tournament with an exact counting formula and inverse, and a four-speed accumulating clock whose denominator size grows exponentially in causal depth. Finally, we prove that globally nonnegative quadratic certificates cannot be compressed into a single fixed-arity universal halting representation without leaving their decidable linear zero-set class. ([write] This last result is a special case of Part XI of the collection’s report *canonical-diophantine-certificates* [11]; see Remark 13.3.)

These are manuscript-level results with executable exact-arithmetic checks, not proof-assistant-certified theorems. The generic compiler and regression suite accompany the article. No priority claim, resolution of finite-fold MRDP, or improvement to an existing fixed-universal operation bound is made.

Keywords. Diophantine representation; signal machines; collision chronology; rational polyhedral cones; Presburger arithmetic; Ehrhart quasi-polynomials; Zeno accumulation; exact computation.

2 The question and the contribution

2.1 A concrete gap between universality and arithmetic

The MRDP theorem guarantees that a computably enumerable acceptance relation has a Diophantine representation. That existence theorem does not itself give a transparent arithmetic description of a particular computational substrate. A useful substrate-specific compiler should explain what its witnesses mean, why they cannot describe a counterfeit computation, and which resource bound is actually being measured. The distinction is especially important when the substrate has a geometric, asynchronous, or apparently continuous dynamics. For background on the classical theorem and a machine-based formalization, see Matiyasevich [9] and Bayer et al. [10].

The **ProveIt** Hilbert-tenth-problem material motivates precisely this separation. Its reversible finite-history audit distinguishes cheap local relations from a complete finite-input computation certificate, and points to additional obligations concerning wiring, initialization, boundaries, and halting observations [1, 2]. This article develops an analogous complete *geometric frontend* for a different substrate: one-dimensional rational signal machines. It does not attempt another operation-count reduction for the repository’s universal polynomial.

[write] The repository has since reviewed this manuscript, and the related manuscript printed as

Part II, in its Hilbert-tenth-problem research tree (Section 1.4). The reversible-history audit cited here is unchanged from this source’s pin to the write.

Signal machines are not merely collections of unrelated finite circuits. Durand-Lose gives a halting universal construction with 13 signal types and 21 collision rules, obtained through cyclic tag systems [3]. This established universality is the reason to study the substrate. We do not reproduce or independently certify that universal machine’s full rule table and input loader here. Instead we prove a generic finite-history compiler whose interface applies to the rational fixed-speed model.

The central question addressed is the following.

Can an explicitly proposed finite collision history be represented arithmetically without introducing arbitrary event coordinates, without overlooking an earlier collision, and without splitting a genuinely simultaneous event into a false sequence?

Our answer is affirmative. In fact, the required geometric residuals can be linear rather than quadratic, and their sum of squares can therefore be quadratic rather than quartic. The essential work is not squaring equations; it is showing that a small set of order constraints certifies the *complete* chronology.

2.2 Results and their logical scope

For a fixed initial label word and a fixed discrete skeleton σ , we construct integer matrices $\mathbf{E}_\sigma, \mathbf{G}_\sigma$ such that the real initial-gap vectors realizing that skeleton are exactly

$$\mathcal{C}_\sigma = \{g \in \mathbb{R}^{n-1} : \mathbf{E}_\sigma g = 0, \mathbf{G}_\sigma g > 0\}. \quad (2.1)$$

Initial gap positivity is included among the strict rows. Every event time and position is a rational linear form in g . Section 5 proves the event-sparse bound

$$R_\sigma \leq 3(n-1) + (3m+1)E \quad (2.2)$$

for the number of raw linear residuals and inequalities.

The corresponding natural-number certificate is

$$P_\sigma(g, z) = \sum_i (\mathbf{E}_{\sigma,i} g)^2 + \sum_j (\mathbf{G}_{\sigma,j} g - 1 - z_j)^2. \quad (2.3)$$

Its degree is at most two, and its entire auxiliary tuple is forced to be $z_j = \mathbf{G}_{\sigma,j} g - 1$. This uniqueness concerns a fixed externally supplied skeleton, not a fixed universal MRDP polynomial.

The same linearization gives several additional conclusions. A collision at causal depth h has coordinate denominators dividing D^h after integer speed normalization, where D is the least common multiple of the nonzero speed differences. Every nonempty chamber has a small positive integer point. A chamber’s synchronization equalities determine its exact codimension and its perturbation behavior. For a fixed event budget, first-acceptance seeds form a finite disjoint union of such chambers, and counting them in growing integer boxes gives exact quasi-polynomials.

These conclusions solve specific representation, realizability, precision, and counting questions formulated here. They are not presented as the settlement of a named longstanding conjecture. The literature review establishes the underlying signal-machine model and related accumulation results; it does not establish historical priority for every combination of lemmas in this article. The proofs are intended to stand on their own and to be suitable for further independent review.

3 Rational signal machines and chronological skeletons

3.1 Particles, velocities, and collision rules

Definition 3.1 (Machine). A rational signal machine consists of a finite label set M , a velocity map $v : M \rightarrow \mathbb{Q}$, and a deterministic rule for every admissible incoming collision set under consideration. An incoming set contains at least two labels with pairwise distinct velocities. Its output is a possibly empty set of labels with pairwise distinct velocities. A signal of label a moves along a straight line of slope $v(a)$ until it participates in a collision.

We use a transparent default rule when an explicit rule is absent: the incoming labels reappear as outgoing labels, ordered by increasing velocity. Such a crossing still counts as a collision. Alternatively, one may specify all relevant rules explicitly and reject a skeleton requiring an undefined rule. The attached implementation supports both conventions.

At a collision all incoming signals are consumed and the prescribed outputs are born at that space-time point. Newly born outputs are ordered by increasing velocity in the right-hand time limit. This makes them separate rather than immediately collide again. At each positive time, all spatially distinct collisions occurring at that time are processed together.

This is the fixed-speed straight-line model used in abstract geometrical computation [3]. Our arithmetic interface restricts initial configurations to finitely many signals at distinct positions and considers ordinary finite prefixes. An infinite sequence accumulating at a finite time has no automatically assigned continuation in this article.

3.2 Initial coordinates and normalization

Fix $n \geq 2$ initial labels $a_0, \dots, a_{n-1}$, in increasing spatial order, and write $d = n - 1$. Translation invariance lets us put

$$x_0 = 0, \quad x_i = g_1 + \dots + g_i \quad (1 \leq i \leq d), \quad g_i > 0. \quad (3.1)$$

Repeated labels at distinct positions are permitted. We call g the *seed geometry*; the label word is separate discrete input.

The principal chamber theorem allows real g . The Diophantine theorem uses integer g . A rational gap vector can be multiplied by a positive common denominator to become integral. Scaling all positions by $c > 0$ scales all event times by c without changing any collision rule or chronological skeleton. Thus integral seeds suffice for existence of a rational realization, but they need not preserve a separately specified physical length or time bound.

When all velocities are rational, choose an integer $L > 0$ such that $Lv(a)$ is integral for every label. In the time coordinate $\tau = t/L$, all speeds are integers $Lv(a)$. Denominator bounds below are stated in this normalized time coordinate. An arbitrary common translation in velocity is not needed.

3.3 Batches and stable signal identities

Definition 3.2 (Chronological skeleton). A finite skeleton is an ordered list of nonempty batches. Each batch is an ordered list of pairwise disjoint contiguous blocks in the current live signal list, each block having at least two members. The blocks are listed from left to right. Each block is a

collision site, with its rule and outputs determined by the incoming labels. New signal identities are assigned by batch, then by site, then by increasing output velocity.

The initial identities are $0, \dots, n-1$. A consumed identity cannot be used again; a noninitial identity must have been generated by an earlier batch. Within a single batch, outputs cannot serve as inputs of another site. These are discrete well-formedness conditions, not arithmetic constraints left for a witness to evade.

Let B be the number of batches and E the total number of collision sites. At site e , write p_e and q_e for the numbers of incoming and outgoing signals. Let m be the number of distinct velocities. Then

$$2 \leq p_e \leq m, \quad 0 \leq q_e \leq m, \quad B \leq E. \quad (3.2)$$

The quantities B , E , and causal depth are different. Many sites may share a batch; many batches may contain events unrelated to a given causal chain.

Definition 3.3 (Realization). A positive seed g realizes a skeleton σ when its actual execution has exactly the specified batches as its first B batches, with exactly the specified incoming blocks and outputs at every listed site. In particular, no earlier event and no simultaneous site or extra incoming signal may be omitted. Nothing is required about events after the final listed batch.

The empty skeleton is realized by every positive seed. It states only that no initial portion has yet been requested; it does not state that the machine never collides. A malformed skeleton is assigned the empty realization set. One can encode a statically rejected skeleton by the constant polynomial 1 if a total polynomial-producing interface is desired.

4 Eliminating every event coordinate

4.1 Intercept propagation

For each signal segment s , let v_s be its fixed velocity and write its supporting line as

$$x_s(t) = b_s + v_s t. \quad (4.1)$$

Its intercept b_s need not be its birth position: the birth may occur at a positive time. Initially, the b_s are the partial sums in (3.1), hence homogeneous integer linear forms in g .

Suppose an event contains two inputs s, r with $v_s \neq v_r$. Their formal meeting time and position are

$$t_e = \frac{b_r - b_s}{v_s - v_r}, \quad x_e = b_s + v_s t_e. \quad (4.2)$$

For an outgoing label of velocity w , its intercept is

$$b_{\text{out}} = x_e - w t_e. \quad (4.3)$$

The denominator $v_s - v_r$ is fixed by the machine, not a variable. Consequently these formulas preserve rational linearity.

Lemma 4.1 (Coordinate elimination). *For every well-formed finite skeleton, all formally reconstructed intercepts, event times, and event positions are homogeneous rational linear forms in the initial gap vector. An event with more than two incoming signals requires only additional homogeneous rational linear equalities.*

Proof. Induct on the ordered batches. Initial intercepts are linear. Equations (4.2) and (4.3) use only addition and multiplication or division by fixed rational constants. An additional incoming signal u requires $b_u + v_u t_e = x_e$, again a homogeneous linear equality. Every new intercept is then available for subsequent batches. No output of the current batch is used by another event in that batch, so this induction is acyclic. $\square$

All event coordinates are therefore *computed data*. They need not appear as free Diophantine witnesses. A certificate containing unconstrained coordinate variables would obscure both the low degree and the uniqueness that result from this elimination.

4.2 Local incidence does not imply a real computation

The incidence equations alone are insufficient. Three signals starting at $0, u, u + v$ with respective speeds $2, 0, -1$ formally give the first two a meeting time $u/2$. If $u = 3$ and $v = 1$, however, the second and third signals collide already at time 1, before $u/2 = 3/2$. An equation witnessing the later formal intersection cannot establish that the two original segments are still alive there.

There is also a boundary failure. When $u = 2v$, all three signals meet at once. A purported binary event containing only the first two signals has perfectly consistent incidence coordinates but the wrong incoming set. A compiler must distinguish strict separation from weak separation at precisely these boundaries.

For events in the same batch, choose the leftmost site's formal time as t_k . Equate all other site times in that batch to t_k , and require

$$t_k - t_{k-1} > 0, \quad t_0 = 0. \quad (4.4)$$

These conditions are necessary but still do not exclude an unlisted earlier collision. The missing ingredient is a complete spatial-order certificate.

5 A sparse certificate for complete collision chronology

5.1 Maximal adjacency intervals

Between collisions the live signals have a strict spatial order. The skeleton tells us how that order is updated: replace each incoming block by its output block, in increasing velocity order. This replacement changes only a bounded number of adjacent pairs.

Definition 5.1 (Adjacency interval). A maximal adjacency interval records an ordered pair (s, r) , left signal first, that remains adjacent in the live list from one batch boundary to a later batch boundary. The interval begins when the pair first becomes adjacent and ends when it ceases to be adjacent or when the requested prefix ends. An unchanged pair is not closed and reopened merely because an unrelated event occurs elsewhere.

Intervals created at the final batch have zero duration and need not be checked: their output order is already determined by the distinct velocities. Every recorded positive-duration interval has an affine gap

$$\Delta_{sr}(t) = b_r - b_s + (v_r - v_s)t. \quad (5.1)$$

At its initial boundary this gap is prescribed to be zero exactly when both segments are born at the same collision there. Otherwise it is prescribed to be strictly positive. At its final boundary

it is prescribed to be zero exactly when both segments are consumed at the same collision there. Otherwise it is strictly positive.

Both endpoints cannot be prescribed zero for a positive-duration interval. Indeed, two distinct straight-line segments cannot separate after a common birth and then meet again without one undergoing an intervening collision. Such a proposal is rejected statically. Each surviving interval contributes exactly two linear conditions before redundant zero rows are removed.

Lemma 5.2 (Endpoint lemma). *Let $a < b$ and let f be affine on $[a, b]$. If $f(a), f(b) \geq 0$ and at least one of them is positive, then $f(t) > 0$ for every $a < t < b$.*

Proof. For $t = (1 - \lambda)a + \lambda b$ with $0 < \lambda < 1$, $f(t) = (1 - \lambda)f(a) + \lambda f(b)$ is a positive convex combination of nonnegative numbers, at least one of which is positive. $\square$

[write] Part II proves the same statement, with terminal rays added, as Lemma 21.1, and the boundary half of the soundness argument below as Lemma 21.2.

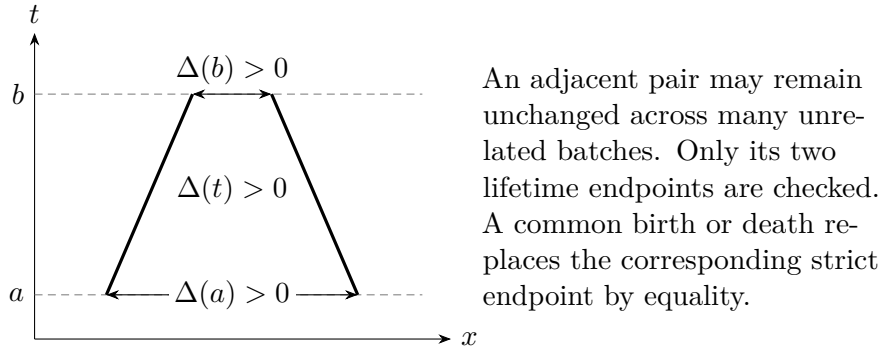


Figure 1: The affine gap between adjacent straight segments cannot become zero in the interior if its permitted endpoint conditions hold.

5.2 Soundness and completeness

Theorem 5.3 (Exact chronological chamber). *Fix a rational signal machine, an initial label word, and a finite skeleton σ . There is an effective construction of rational homogeneous linear equalities and strict inequalities whose solution set is exactly the set of real seed geometries realizing σ . All simultaneous collisions and all extra incoming signals are accounted for. Every nonempty solution set is a relatively open rational polyhedral cone in the kernel of its equality matrix.*

Proof. First reject malformed identities, overlapping or noncontiguous input blocks, undefined rules under the nontransparent convention, and incoming blocks whose velocities fail to decrease strictly from left to right. The latter condition is necessary because immediately before a multiple meeting, two incoming lines s, r ordered left to right have gap $(v_s - v_r)(t_e - t) > 0$ and hence $v_s > v_r$.

Propagate every intercept and event coordinate using (4.2)–(4.3). Add the additional incoming-incidence equalities, equal-time conditions within each batch, (4.4), initial gap positivity, and all adjacency-interval endpoint conditions described above. Every resulting expression is a homogeneous rational linear form in g .

For soundness, suppose all these conditions hold. On each positive-duration adjacency interval, the endpoint lemma gives strict separation in its interior. Hence no adjacent pair meets at an

unlisted intermediate time. A meeting of nonadjacent signals would force an adjacent meeting: at the first failure of strict spatial order, at least one consecutive gap is zero. Thus no collision of any arity can occur between the listed batches.

At a batch boundary, the incidence equalities put the signals of each specified block at its reconstructed common site. Every adjacent gap not belonging to that block is strictly positive there. Consequently no additional incoming signal can join the site. The same strict conditions separate the sites of distinct blocks. They also exclude an omitted disjoint collision elsewhere in the configuration: an omitted site would have a zero gap at a boundary where strict positivity is imposed, or would lie in the interior of an unchanged adjacency interval, where positivity has just been proved.

The deterministic rule is therefore applicable to exactly the listed incoming set at each site. Its outgoing order is increasing in velocity, so outputs separate in the right-hand time limit. Static identity validation ensures that every segment has a legitimate birth and is consumed at most once. Induction on batches now proves that the reconstructed execution is exactly the actual initial prefix.

For completeness, consider an actual execution realizing σ . The coordinate propagation agrees with its genuine intersection coordinates. The extra incidences and batch-time equalities hold, batch times strictly increase, and each endpoint gap is zero precisely at a common birth or common death of the pair. Every other endpoint gap is positive. A pair of straight segments with a common birth and a later common death cannot occur. Thus the execution satisfies every generated condition.

Finally, a finite intersection of homogeneous equality subspaces and open linear half-spaces is a relatively open convex cone in that subspace. If nonempty, its strict inequalities leave a neighborhood in the equality subspace around each point. All coefficients are rational. $\square$

5.3 The event-sparse bound

Theorem 5.4 (Linear number of chronology conditions). *Let J be the number of positive-duration maximal adjacency intervals used by the construction. Then*

$$J \leq (n - 1) + \sum_{e=1}^E (q_e + 1) \leq (n - 1) + (m + 1)E, \quad (5.2)$$

$$R_\sigma \leq 3(n - 1) + (3m + 1)E. \quad (5.3)$$

These bounds apply before deduplication or removal of identically zero equalities.

Proof. Initially there are $n - 1$ adjacent pairs. Replacing a collision block by q_e outputs creates at most $q_e - 1$ internal output pairs and two external pairs, for a total at most $q_e + 1$ when $q_e \geq 1$. If $q_e = 0$, it creates at most one bridge pair between the surviving neighbors. At a boundary or when neighboring blocks change simultaneously, fewer pairs may be created. Existing unchanged pairs need no new interval. This proves (5.2).

There are $d = n - 1$ initial gap inequalities, B positive batch-time increments, $\sum_e (p_e - 2)$ additional incoming-incidence equalities, $E - B$ equalities synchronizing the sites within batches,

and $2J$ endpoint conditions. Therefore

$$\begin{aligned}
R_\sigma &\leq d + B + \sum_e (p_e - 2) + (E - B) + 2J \\
&= d + \sum_e p_e - E + 2J \\
&\leq d + (m - 1)E + 2d + 2(m + 1)E \\
&= 3d + (3m + 1)E.
\end{aligned}$$

□

[write] The first inequality of (5.2) is Part II's Lemma 20.1; Part II's own size ledger (18.3) counts witnesses and residuals in its event-coordinate format.

The word *sparse* here refers to the number of geometric conditions, not to the number of nonzero coefficients in each eliminated linear form. A deeply propagated intercept may depend on every initial gap. Storing all rows densely can take $O(dR_\sigma)$ coefficient slots; compiling them may have additional bookkeeping cost. Neither (5.3) nor degree two is a constant arithmetic-operation bound for an unbounded computation.

An implementation may scan the live list at each batch for simplicity. That does not change the mathematical row bound. A linked-list implementation can update adjacencies locally, while the attached small implementation prioritizes clarity and exact arithmetic over this engineering optimization.

6 Quadratic Diophantine certificates with unique slacks

6.1 Positive denominator clearing

Every nonzero rational row can be multiplied by a positive common denominator and then divided by the positive greatest common divisor of its integer coefficients. For equality rows the sign may subsequently be normalized; for strict inequality rows it may not. Negating a strict row would reverse its meaning.

Delete identically zero equality rows. An identically zero strict row makes the chamber empty. Remove duplicate rows of the same kind. Denote the resulting integer matrices by $\mathbf{E}$ and $\mathbf{G}$, with r equality rows and s strict rows. Their matrix presentation is a complete, explicit description of the polynomial, even when its expanded quadratic has many monomials.

Theorem 6.1 (Unique-slack quadratic representation). *For a nonrejected skeleton, define*

$$P(g, z) = \|\mathbf{E}g\|_2^2 + \|\mathbf{G}g - \mathbf{1} - z\|_2^2, \quad (g, z) \in \mathbb{N}_0^{d+s}. \quad (6.1)$$

Then P is an integer polynomial of total degree at most two, globally nonnegative and convex on $\mathbb{R}^{d+s}$. Moreover,

$$\exists z \in \mathbb{N}_0^s \ P(g, z) = 0 \iff g \in \mathbb{Z}^d \cap \mathcal{C}_\sigma. \quad (6.2)$$

For every accepted g , the auxiliary tuple is unique:

$$z = \mathbf{G}g - \mathbf{1}. \quad (6.3)$$

Proof. Each summand is the square of an affine linear form with integer coefficients. Thus P is a quadratic sum of squares and has Hessian $2A^\top A \succeq 0$, where A is the combined coefficient matrix of these affine forms. Over the reals, and hence over the integers, a sum of squares vanishes precisely when every squared form vanishes. Its zeros therefore require $\mathbf{E}g = 0$ and $\mathbf{G}_j g = 1 + z_j \geq 1$ for every strict row. Because these expressions are integers, this is equivalent to $\mathbf{G}g > 0$. The chamber theorem proves (6.2). The same equations force (6.3), which is a nonnegative integer tuple whenever g is accepted. $\square$

[write] Theorems 5.3 and 6.1 together are the common theorem of the two Parts (Section 1.2). Part II proves it again as Theorem 18.3, with batch times and event positions kept as natural witnesses, adds halting semantics, and strengthens uniqueness to full column rank of the witness matrix (Lemma 22.1).

All event coordinates are reconstructed from g and are absent from the auxiliary tuple. This avoids arbitrary rational encodings with multiple numerator–denominator representatives. If the seed itself is given as a rational geometry in a different interface, one must separately choose and validate an encoding or a scale; the uniqueness theorem should not be silently transferred to a noncanonical rational encoding.

6.2 What uniqueness does not assert

The polynomial depends on the externally supplied skeleton. Its number of rows and slack variables is not fixed independently of the requested history. Even though each member of this family has unique auxiliary witnesses, placing an arbitrary unbounded skeleton inside one fixed Diophantine equation is a different task. An unspecified application of MRDP to a skeleton verifier does not preserve this uniqueness for free.

The present result is therefore not a proof that all computably enumerable sets admit single-fold or finite-fold Diophantine representations. Its stronger-than-usual feature is narrower and fully explicit: for each fixed collision skeleton, projection of *all* natural zeros of (6.1) onto the initial gaps is a bijection onto the actual integer realizations. There are no additional zeros corresponding to incomplete local histories.

7 Causal depth controls denominators and coefficient size

7.1 A denominator invariant

Work in the integer-speed normalization of Section 3, and set

$$V = \max(1, \max_{a \in M} |v(a)|), \quad D = \text{lcm}\{|v(a) - v(b)| : v(a) \neq v(b)\}. \quad (7.1)$$

The least common multiple of an empty set is taken as 1; in that case no collision is possible from a simple initial configuration.

Assign depth zero to initial segments. The depth of an event is one plus the maximum depth of its incoming segments. Every output segment inherits the depth of its birth event. This is causal depth, not the chronological index of the batch.

Theorem 7.1 (Causal denominator bound). *At an event of depth h , each coefficient of its time and position forms belongs to $D^{-h}\mathbb{Z}$. The same holds for every outgoing intercept. At an integer seed, the event coordinates therefore have reduced denominators dividing D^h .*

Proof. Initial intercept coefficients are integral. Suppose an event has depth h . Its incoming intercepts have denominators dividing powers of D no larger than D^{h-1} . Put them over the common denominator D^h . Formula (4.2) divides their difference by a nonzero integer velocity difference dividing D . Its time coefficients thus belong to $D^{-h}\mathbb{Z}$. The position and all output intercepts use only addition and multiplication by integer velocities, preserving that lattice. Evaluation at integer gaps preserves the same denominator divisibility. $\square$

[write] Part II proves the same divisibility for event coordinates as Lemma 19.1, with its own proof.

Only primes already present in the speed differences can appear in these coordinate denominators at an integer seed. At a rational seed, primes in its initial common denominator may also occur. Parallel collisions elsewhere do not increase the denominator exponent along an unrelated causal chain.

This is an upper bound, not an assertion that all factors of D^h occur. Cancellations and special velocity configurations can make the actual precision requirement much smaller. Section 12 exhibits an explicit case in which exponential denominator magnitude is nevertheless unavoidable.

7.2 An explicit coefficient bound

For a rational linear form, use the ℓ^1 norm of its coefficient vector. Put $K = 2 + 4V$. Initial intercepts have norm at most d .

Lemma 7.2 (Growth of propagated forms). *Every intercept of depth at most h has coefficient norm at most dK^h . Every event time and event position of depth at most h satisfies the same bound.*

Proof. Assume the input intercept norms at a depth- h event are at most $M = dK^{h-1}$. A nonzero integer speed difference has absolute value at least one, so (4.2) gives $\|t_e\|_1 \leq 2M$. Consequently

$$\|x_e\|_1 \leq (1 + 2V)M, \quad \|b_{\text{out}}\|_1 \leq (1 + 4V)M.$$

Each factor is at most K . Induction proves the claim; forms of smaller depth obey the larger bound as well. $\square$

Proposition 7.3 (Integer row height). *If the maximum event depth is h , each primitive integer chamber row can be chosen with every coefficient of absolute value at most*

$$A = 2d(1 + V)(DK)^h. \tag{7.2}$$

In particular, for a fixed machine, row coefficient bit lengths are $O(h + \log d)$.

Proof. An endpoint gap is a difference of two intercepts plus a velocity difference times a batch time. Its coefficient norm is at most $2dK^h + 2VdK^h = 2d(1 + V)K^h$. Additional incidence residuals and differences of event times satisfy the same bound. Initial gap rows also do. All these coefficients have denominators dividing D^h by Theorem 7.1. Multiplication by D^h bounds every resulting integer coefficient by (7.2). Passing to a primitive integer row can only decrease this bound. $\square$

The exponent in (7.2) bounds coefficient *magnitude*; its logarithm is linear in depth. Exact rational arithmetic can therefore remain polynomial in the externally bounded history length even when ordinary fixed-precision floating-point simulation becomes unreliable. This is not an assertion that the total number of events before acceptance has a computable bound in the initial input length.

8 Small integral realizations and finite-horizon synthesis

Theorem 8.1 (Small positive integer seed). *Let $\mathcal{C} = \{g \in \mathbb{R}^d : \mathbf{E}g = 0, \mathbf{G}g > 0\}$ be a nonempty chamber whose strict rows include $g_i > 0$. Suppose all matrix coefficients are integers of absolute value at most $A \geq 1$. Then $\mathcal{C}$ contains a positive integer point satisfying*

$$\max_i g_i \leq d^{d/2} A^d. \quad (8.1)$$

Thus every realizable collision skeleton has a realization with a seed encoding of polynomial bit length in the explicit skeleton and initial label word, for a fixed machine.

Proof. Homogeneity permits a positive scaling of any point in $\mathcal{C}$ so that every strict expression is at least 1. Hence the closed rational polyhedron

$$Q = \{g : \mathbf{E}g = 0, \mathbf{G}g \geq \mathbf{1}\}$$

is nonempty. It lies in $[1, \infty)^d$. Minimizing $\sum_i g_i$ over Q attains an optimum: intersecting with a nonempty objective sublevel gives a compact set. The optimal face is a nonempty compact polytope and has a vertex; that point is also a vertex of Q .

At such a vertex choose d independent active constraints, including sufficient equality rows. They give a nonsingular integer matrix B and a right-hand side c with entries in $\{0, 1\}$. The vertex is $B^{-1}c$. Let $\delta = |\det B|$, a positive integer. Cramer's rule shows that $g = \delta B^{-1}c$ is integral. It is positive, satisfies the homogeneous equalities, and has every strict row at least $\delta \geq 1$.

Each coordinate of g is, up to sign, a determinant in which one column of B has been replaced by c . Hadamard's inequality bounds its magnitude by $d^{d/2} A^d$; using the sharper bound on the replaced column is unnecessary. This proves (8.1). Combining it with (7.2) gives the stated bit-length conclusion. $\square$

Corollary 8.2 (Real, rational, and integer realizability coincide). *A fixed skeleton is realizable by a positive real seed if and only if it is realizable by a positive rational seed, if and only if it is realizable by a positive integer seed. Realizability is decidable by rational linear feasibility, using $\mathbf{E}g = 0, \mathbf{G}g \geq \mathbf{1}$.*

The equivalence is specific to homogeneous rational chambers. It should not be transferred to arbitrary nonlinear geometry. Nor does the scaled integer seed necessarily satisfy extra nonhomogeneous requirements such as an absolute prescribed distance or a fixed deadline.

Corollary 8.3 (An NP upper bound for bounded synthesis). *Fix a rational signal machine and a distinguished output label. Given an initial label word of length n and a unary bound T , the problem of whether some positive seed produces the distinguished label within a complete prefix containing at most T collision sites is in NP.*

Proof. A skeleton of at most T sites has at most T batches and at most $n + mT$ created signal identities. Since m is fixed, its discrete description has polynomial length in $n + T$. Guess such a skeleton, check its identities and rules, and check that the distinguished label appears in its prefix.

If it is realizable, Theorem 8.1 supplies a positive integer seed with polynomial bit length: $d = n - 1$, $h \leq T$, and $\log A = O(T + \log n)$ for the fixed machine. Guess that seed as well. Propagating rational forms and verifying every chamber row then takes a polynomial number of operations on polynomial-length integers. This is a polynomial-time verifier. No search over exponentially long real numbers is required. $\square$

Only an upper bound is claimed. There is no NP-hardness result here. The budget counts total sites, including all sites in the accepting batch, not elapsed physical time. A physical-time bound can contain arbitrarily many events and even an accumulation, so it is not interchangeable with T .

9 Robustness and the rank of synchronization constraints

The chamber description distinguishes two different sources of numerical difficulty. A trace can be sensitive because it demands an exact equality among initial gaps, or its exact coordinates can require large denominators even while its combinatorial pattern is robust. The latter phenomenon will occur in the accumulating clock.

Theorem 9.1 (Certified perturbation radius and codimension). *Let $\mathcal{C} = \{g : \mathbf{E}g = 0, \mathbf{G}g > 0\}$ be nonempty, with zero strict rows removed, and let $g^* \in \mathcal{C}$. Define*

$$\rho(g^*) = \min_j \frac{\mathbf{G}_j g^*}{\|\mathbf{G}_j\|_1} > 0. \quad (9.1)$$

Every perturbation $\eta \in \ker \mathbf{E}$ with $\|\eta\|_\infty < \rho(g^)$ preserves the skeleton. The chamber has relative dimension $d - \text{rank } \mathbf{E}$. It has nonempty ambient interior if and only if $\text{rank } \mathbf{E} = 0$; otherwise it has ambient Lebesgue measure zero.*

Proof. The equality constraints persist for $\eta \in \ker \mathbf{E}$. For every strict row, Hölder’s elementary norm inequality gives

$$\mathbf{G}_j(g^* + \eta) \geq \mathbf{G}_j g^* - \|\mathbf{G}_j\|_1 \|\eta\|_\infty > 0.$$

Hence the perturbed point stays in the chamber. This proves relative openness in the full equality kernel, whose dimension is $d - \text{rank } \mathbf{E}$. A proper linear subspace has empty interior and measure zero in $\mathbb{R}^d$. Conversely, when the rank is zero, the finitely many strict inequalities define an open neighborhood of each accepted point. $\square$

The radius in (9.1) is a certified lower bound, not necessarily the largest possible radius in the equality subspace. Restricting perturbations to that subspace can increase the true distance to a forbidden boundary.

One must also distinguish genuinely independent synchronization equations from identities already forced by previous dynamics. Some formal coincidences yield an identically zero row and impose no additional codimension. Therefore the statement “every multiple collision is fragile” is too strong. What matters is the rank of the *nontrivial* conditions on the input geometry. When a loader or problem specification already restricts the admissible inputs to a subspace, robustness should be measured relative to that allowed input space.

The theorem provides a useful diagnostic for a proposed geometric computer. A simulator can report not merely a diagram, but also an exact equality matrix, a synchronization rank, and a rational certified tolerance at an integer seed. Those objects separate an intended robust mechanism from an accidental coincidence created by a particular numeric sample.

10 Finite-horizon acceptance, semilinearity, and counting

10.1 Disjoint first-acceptance chambers

Fix a machine, an initial label word, a marked accepting label, and a site budget T . Acceptance means the first appearance of that label in a batch such that the total number of processed sites through that entire batch is at most T . If the initial word already contains the marked label, every seed accepts with zero events.

There are only finitely many candidate skeletons with at most T sites. Indeed, at every stage there are at most $n + mT$ identities; the rule table and possible contiguous input blocks are finite. Enumerate those skeletons that end at the first accepting batch and reject all statically invalid ones. Distinct nonempty chambers in this first-acceptance list are disjoint: for a fixed seed, deterministic complete chronology has just one first accepting prefix.

Theorem 10.1 (Finite-horizon semilinearity). *The positive integer seed geometries accepted within site budget T form an effectively constructible semilinear set. More specifically, they are a finite disjoint union of the integer points of rational homogeneous chambers of the form (2.1).*

Proof. The enumeration above is finite and effective. For each skeleton, Theorem 5.3 gives a complete linear characterization. On integer points, a strict integer inequality is equivalent to being at least 1, so each chamber is definable in Presburger arithmetic. Finite union remains Presburger-definable. The effective equivalence with semilinear sets is the classical theorem of Ginsburg and Spanier [7]. $\square$

This use of the classical equivalence is not a new proof of that theorem. The new substrate-specific input to it is the exact finite union of collision chambers, including chronology and simultaneous-event boundaries.

[write] Part II proves the same statement for its fixed schemas and event bounds, prefix and halting, as Proposition 26.1.

10.2 A single quartic for an unknown bounded first-hit skeleton

The skeleton need not remain externally supplied when the event budget is fixed. Enumerate the nonrejected first-acceptance skeletons $\sigma_1, \dots, \sigma_K$ and let their matrices be $\mathbf{E}_k, \mathbf{G}_k$. Empty chambers may remain in this list without harm. Introduce natural selectors $b_1, \dots, b_K$ and a separate slack vector z_k for each chamber. Define

$$\begin{aligned} Q_T(g, b, z) = & \left(\sum_{k=1}^K b_k - 1 \right)^2 + \sum_{k=1}^K \sum_i (b_k \mathbf{E}_{k,i} g)^2 \\ & + \sum_{k=1}^K \sum_j (b_k (\mathbf{G}_{k,j} g - 1) - z_{k,j})^2. \end{aligned} \quad (10.1)$$

For an empty family, use $Q_T = 1$.

Theorem 10.2 (Canonical first-hit quartic). *Equation (10.1) is a globally nonnegative integer polynomial of degree at most four. Its natural zeros project bijectively onto the integer seeds accepted within the fixed event budget. No Boolean equations for the selectors and no additional inactive-slack constraints are needed.*

Proof. At a natural zero, $\sum_k b_k = 1$ forces exactly one selector to be 1 and all others to be 0. For the selected chamber, the remaining equations are precisely $E_k g = 0$ and $G_k g - \mathbf{1} = z_k \geq 0$. For an unselected chamber, each slack residual becomes $-z_{k,j}$, forcing all of its slacks to zero. Thus the polynomial accepts exactly the union of the chamber integer points.

Those chambers are disjoint by canonical first-acceptance chronology. Therefore an accepted seed determines exactly one selected skeleton, its active slacks are uniquely determined, and every inactive slack is zero. Every residual has degree at most two before squaring, proving the quartic bound. The construction is a sum of squares and hence globally nonnegative, but it is not in general convex. $\square$

The same construction works for any finite disjoint family of chambers. If chambers overlap, it still represents their union, but a seed in several chambers has several selector witnesses. First-hit canonicalization is therefore mathematically important, not cosmetic.

Remark 10.3 ([write] Relation to a single-fold quadratic of another report). The accepted set of Theorem 10.2 is a finite union of chamber integer points, hence semilinear (Theorem 10.1). The collection's report *canonical-diophantine-certificates* [11] proves in its Part XI (Theorem `cdc:of:thm:singlefolds1`) that every semilinear set has a representation by an integer polynomial of degree at most two that is nonnegative on the real nonnegative orthant, with exactly one natural witness for each member, effectively from a Presburger formula. So, if only orthant nonnegativity is required, the first-hit set has a single-fold quadratic certificate. Theorem 10.2 gives something different: an explicit construction from the chamber matrices, globally nonnegative on all real inputs, whose selector and slacks have the chamber meaning. Within the globally nonnegative class a quadratic has a rational affine zero set (Theorem 13.1), and the selector products of (10.1) leave that class, as the paragraph after Corollary 13.2 says. Whether a globally nonnegative quadratic suffices for a given first-hit family is not decided here.

This theorem removes external skeleton choice for a *bounded* problem; it does not remove the bound. The number of candidate skeletons, and hence the number of selectors and slacks, may be very large and grows with T . The attached implementation provides the matrix-based quartic constructor, and checks a two-chamber first-hit example. It does not include an optimized exhaustive enumerator for all skeletons of an arbitrary event budget.

10.3 Exact quasi-polynomials rather than merely leading asymptotics

Let $A_T(H)$ count the accepted seeds in $\{1, \dots, H\}^d$, with $H \geq 1$. For a single chamber put

$$\Pi_\sigma = \mathcal{C}_\sigma \cap (0, 1]^d.$$

Its closure is a bounded rational polytope in its equality subspace. The original set is obtained from that closed polytope by excluding finitely many rational faces where one or more required strict expressions vanish. The upper box facets remain included.

Theorem 10.4 (Exact finite-horizon counting). *For fixed machine, initial word, accepting label, and event budget T , $A_T(H)$ is an exact quasi-polynomial in H for all positive integers H . It has degree at most d . Its ordinary generating function is rational and has poles only at roots of unity. Furthermore,*

$$\lim_{H \rightarrow \infty} \frac{A_T(H)}{H^d} = \sum_{\sigma} \text{vol}_d(\Pi_\sigma) \in \mathbb{Q}, \quad (10.2)$$

where the sum ranges over the first-acceptance chambers, and lower-dimensional terms have zero d -dimensional volume.

Proof. Homogeneity gives $\mathcal{C}_\sigma \cap (0, H]^d = H\Pi_\sigma$. For a nonempty chamber, the closure of Π_σ is obtained by replacing its strict homogeneous inequalities by weak ones. To see equality with this weak closure, take a small strictly feasible point with every coordinate less than 1 and form convex combinations with any weakly feasible point. Those combinations converge to the latter while preserving all strict conditions and the box.

A forbidden equality $G_j g = 0$ cuts out a face of the weak closure, since $G_j g \geq 0$ throughout it. Inclusion–exclusion over these faces expresses the lattice-point count in $H\Pi_\sigma$ as a finite integer combination of counts in dilates of closed rational polytopes. Ehrhart’s rational-polytope theorem makes each term a quasi-polynomial; see Beck and Robins [8]. This also covers lower-dimensional faces. Disjointness of first-acceptance chambers permits summing these exact counts.

A quasi-polynomial of degree at most d and period p has a rational generating function whose denominator divides a power of $1 - z^p$. This proves the generating-function assertion. The normalized leading coefficient of a full-dimensional rational polytope count is its volume; removing proper faces does not change it. Equivalently, the same limit follows from Riemann sums for a bounded polyhedral set. Rational polytopes have rational volume by a rational triangulation and determinant formulas. Summing over the disjoint chambers proves (10.2). $\square$

Thus a finite-horizon seed-counting problem has a particularly rigid asymptotic form: a polynomial in H with periodic coefficient functions, not an arbitrary irregular error term. Its leading density is rational. If the initial gaps are instead sampled independently and uniformly from $(0, 1]$, the probability of finite-horizon acceptance is exactly the same rational volume sum.

The theorem does not promise an efficient procedure for enumerating all chambers or computing all their volumes. Their number can grow rapidly with T . It also does not justify interchanging an unbounded-time limit with the box-size limit. Removing the event bound can leave an infinite union of chambers, outside the finite quasi-polynomial argument.

Because each integer seed in one chamber has one slack tuple, its chamber count is also the number of complete natural zeros of its quadratic certificate with $1 \leq g_i \leq H$, even when no separate bound is placed on z . Across different skeletons this statement uses a tagged disjoint family of certificates, not a single unqualified polynomial.

11 An exact first-collision tournament

11.1 Three chambers and a complete polynomial

Consider labels A, B, C with respective speeds $2, 0, -1$, initially at $0, u, u + v$, where $u, v > 0$. Introduce two stationary output labels H, R and rules

$$\{A, B\} \longrightarrow \{H\}, \quad \{B, C\} \longrightarrow \{R\}, \quad \{A, B, C\} \longrightarrow \{R\}.$$

Use transparent rules otherwise. The question here is whether H is produced at the *first* collision batch.

The possible first events are exactly

event	condition	time
AB	$u < 2v$	$u/2$
BC	$u > 2v$	v
ABC	$u = 2v$	$v = u/2$.

(11.1)

For instance, the formal AC meeting time is $(u + v)/3$. If $u < 2v$, it is later than $u/2$; if $u > 2v$, it is later than v . It cannot be a different first event. The equality case is the triple collision.

The complete accepting chamber has the particularly small certificate

$$P_{AB}(u, v, z_1, z_2, z_3) = (u - 1 - z_1)^2 + (v - 1 - z_2)^2 + (2v - u - 1 - z_3)^2. \quad (11.2)$$

All five variables range over $\mathbb{N}_0$. At $u = v = 1$, for example, the unique auxiliary tuple is $(0, 0, 0)$. At $(u, v) = (3, 1)$, the third strict condition fails, correctly rejecting the incidence-only counterfeit. At $(2, 1)$ it also fails, correctly rejecting an omitted simultaneous third input.

For the triple-collision skeleton the compiler instead produces

$$P_{ABC}(u, v, z_1, z_2) = (u - 2v)^2 + (u - 1 - z_1)^2 + (v - 1 - z_2)^2. \quad (11.3)$$

Its equality rank is one. The two binary-event chambers have rank zero and are open in the positive quadrant; the triple-collision chamber is one-dimensional and requires exact synchronization.

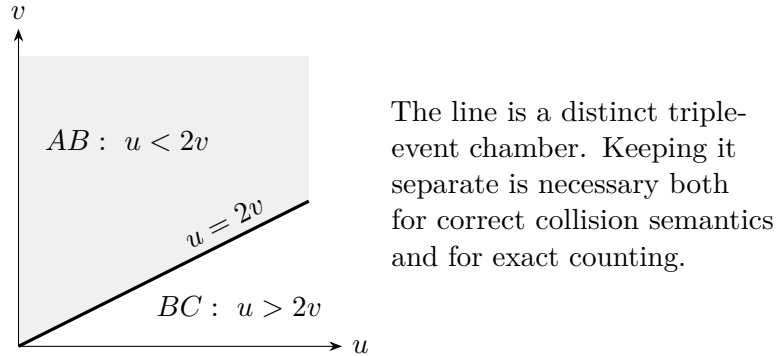


Figure 2: The first-event geometry is partitioned into two open cones and a synchronization ray. The coordinate axes are excluded.

11.2 Counting, generating function, and inversion

Let $C(H)$ count the accepting gap pairs with $1 \leq u, v \leq H$, and set $C(0) = 0$. Direct summation gives

$$\begin{aligned}
 C(H) &= \sum_{v=1}^H \min(H, 2v - 1) \\
 &= \begin{cases} 3m^2, & H = 2m, \\ 3m^2 + 3m + 1, & H = 2m + 1, \end{cases} \\
 &= \left\lfloor \frac{3H^2 + 1}{4} \right\rfloor = \frac{3H^2}{4} + \frac{1 - (-1)^H}{8}.
 \end{aligned} \quad (11.4)$$

For $H = 2m$, split the sum after $v = m$; the odd-number sum contributes m^2 and the remaining m terms contribute $2m^2$. The odd case follows by the same split after $m + 1$.

The first values are

$$0, 1, 3, 7, 12, 19, 27, 37, 48, 61, 75, 91, 108, 127, 147, 169, \dots$$

No assertion about an OEIS identification or novelty of this elementary sequence is needed. It is included as an exactly solvable counting test for the compiler and as an illustration of the general quasi-polynomial theorem. Its ordinary generating function is

$$\sum_{H \geq 0} C(H)z^H = \frac{z(1+z+z^2)}{(1-z)^3(1+z)}. \quad (11.5)$$

This follows from the standard geometric-series derivatives applied to (11.4), or by expanding the displayed rational function.

There is also an exact inverse threshold. For every positive integer y ,

$$\min\{H \geq 0 : C(H) \geq y\} = \left\lceil \sqrt{\frac{4y-1}{3}} \right\rceil. \quad (11.6)$$

Indeed, because y is integral, $\lfloor (3H^2 + 1)/4 \rfloor \geq y$ is equivalent to $3H^2 + 1 \geq 4y$. The smooth expression inside the ceiling has the expansion

$$\sqrt{\frac{4y-1}{3}} = \frac{2}{\sqrt{3}}y^{1/2} - \frac{1}{4\sqrt{3}}y^{-1/2} - \frac{1}{64\sqrt{3}}y^{-3/2} - \frac{1}{512\sqrt{3}}y^{-5/2} + O(y^{-7/2}). \quad (11.7)$$

The binomial series supplies all further orders. The ceiling in (11.6) is essential: omitting it would incorrectly replace an integer inverse's bounded rounding oscillation by a decaying error.

11.3 Independent simultaneous sites

Take the transparent two-speed machine with $v(P) = 1$, $v(N) = -1$, and initial word P, N, P, N . With gaps $(2, 5, 2)$, the two disjoint pairs $(0, 1)$ and $(2, 3)$ collide simultaneously at time 1. A correct first batch must contain both sites. A proposed batch containing only $(0, 1)$ is rejected by the unchanged adjacency interval of $(2, 3)$, whose gap would vanish at the final boundary where positivity is required.

For general positive gaps (g_1, g_2, g_3) , this simultaneous first-batch skeleton has the single nontrivial equality $g_1 = g_3$. Changing the last gap from 2 to 3 destroys that equality. This example tests a different failure mode from the omitted third input of a multiple collision: the missing event is spatially separate, not an extra signal at the same site.

[write] Part II's hidden collision at the final cut (Section 24.2) is another instance of this failure mode, rejected by the same strict endpoint of a surviving pair (Section 1.2).

12 A four-speed clock with exponentially growing denominators

12.1 Rules and a closed-form execution

Consider the four labels

$$v(L) = 0, \quad v(P) = 2, \quad v(R) = -1, \quad v(N) = -2$$

with rules

$$\{P, R\} \longrightarrow \{N, R\}, \quad \{L, N\} \longrightarrow \{L, P\}, \quad (12.1)$$

and transparent defaults. Initially put L at 0, P at u , and R at $u + v$, with $u, v > 0$. The left boundary is stationary, the right boundary moves left at speed 1, and a particle alternately travels right at speed 2 and left at speed 2, reflecting at the boundaries via the two rules.

Write $S = u + v$ and $W = 3u + 2v$. The collision times are

$$t_{2k+1} = S - \frac{W}{3^{k+1}}, \quad k \geq 0, \quad (12.2)$$

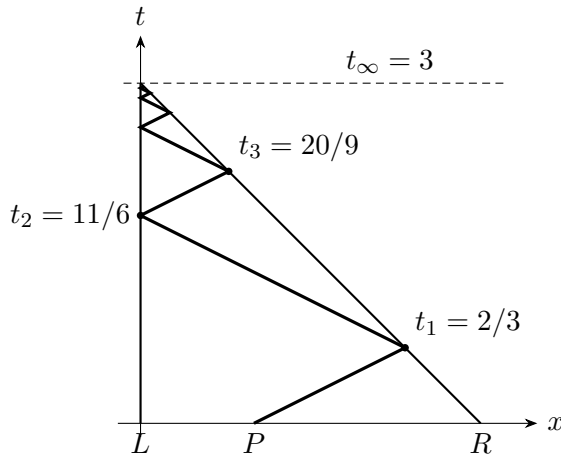
$$t_{2k} = S - \frac{W}{2 \cdot 3^k}, \quad k \geq 1. \quad (12.3)$$

Odd collisions occur on the right boundary at position $S - t_{2k+1}$; even collisions occur at position zero.

Proposition 12.1 (Clock correctness and common chamber). *For every $u, v > 0$, equations (12.2)–(12.3) describe the actual alternating execution. The collision times strictly increase to S . Every finite alternating prefix has the whole positive (u, v) quadrant as its seed chamber.*

Proof. The first rightward meeting occurs at $t_1 = v/3$, which equals $S - W/3$. At a right-boundary collision at time t , the remaining boundary distance is $S - t$. The new left-moving particle takes $(S - t)/2$ to reach zero, leaving time-to-boundary-meeting $(S - t)/2$. From there the right-moving particle closes the remaining gap at relative speed 3, leaving two thirds of that remaining time when it next hits the right boundary. Hence a full right-to-right bounce multiplies $S - t$ by $1/3$, with the intermediate left collision having half the previous remaining time. This proves the formulas inductively.

The first time is positive. Every subsequent difference is positive, and all times are less than S . Between collisions the moving particle lies strictly between the two boundaries. The boundaries themselves meet only at S , not at any finite indexed event. There is therefore no unlisted collision in any finite prefix. The labels alternate exactly as specified by (12.1), independently of the positive gaps. This proves both correctness and the common-chamber statement. $\square$



Initial gaps (1, 2). Each full bounce shrinks the remaining time by a factor of 3. Every finite part is covered by the compiler; the accumulation point itself is not an extra finite event.

Figure 3: A four-speed accumulating clock. The drawing displays only a finite number of segments; the formulas specify all finite prefixes.

12.2 A denominator lower bound without fragile synchronization

For the particular integer seed $(u, v) = (1, 2)$, $S = 3$ and $W = 7$, giving

$$\frac{2}{3}, \frac{11}{6}, \frac{20}{9}, \frac{47}{18}, \frac{74}{27}, \frac{155}{54}, \frac{236}{81}, \frac{479}{162}, \dots$$

At event $2k$, the reduced denominator is exactly $2 \cdot 3^k$:

$$t_{2k} = \frac{6 \cdot 3^k - 7}{2 \cdot 3^k},$$

and the numerator is odd and is not divisible by 3. The causal depth of event j is j , since the reflected particle links every event to its predecessor. Thus denominator magnitude is exponential in causal depth, while its bit length is linear in depth.

Here the integer speed differences have least common multiple $D = 12$, so Theorem 7.1 correctly gives a divisible upper bound 12^j . The example does not make that upper bound sharp in its base, but it rules out a depth-independent denominator bound or a polynomial-magnitude bound for general rational signal machines.

At the same time, every finite prefix has equality rank zero and remains valid throughout the positive quadrant. Growing exact precision is therefore not evidence of a fragile collision pattern. Robust symbolic chronology and large rational denominators are compatible.

Accumulations in signal machines are established phenomena. Becker et al. [6] distinguish the behavior of small speed sets and show, in particular, that four speeds permit accumulation whereas the three-speed case with rational speed and initial-distance ratios does not. The clock is an explicit test case and a denominator-growth illustration, not a claim to have discovered accumulation. Our compiler verifies every ordinary finite prefix before S ; it assigns no transition or computational power to the limiting point at S .

[write] Part II's bounded shuttle (Proposition 25.3) uses the same mechanism on a different four-speed machine, with reduced denominators exactly 3^{k-1} (Section 1.2).

13 Why a fixed universal convex quadratic cannot follow

A natural question is whether the quadratic frontend can be compressed into one fixed polynomial with a fixed number of witnesses for arbitrary unbounded computations. Within the globally nonnegative quadratic class, there is an elementary obstruction stronger than a failure of the present construction.

Theorem 13.1 (Linear zero-set rigidity). *Let $Q(w)$ be a quadratic polynomial with rational coefficients that is nonnegative for every $w \in \mathbb{R}^N$. Its real zero set is either empty or a rational affine subspace. The projection of its nonnegative integer zeros onto any subset of coordinates is an effectively semilinear, decidable set.*

Proof. Write

$$Q(w) = w^T A w + 2b^T w + c$$

with A rational and symmetric. Global nonnegativity implies $A \succeq 0$. It also implies $b \perp \ker A$: otherwise the restriction to an appropriate kernel direction would be a nonconstant affine function unbounded below. Thus $b \in \text{range } A$ and the consistent rational system $A w_0 = -b$ has a rational solution.

Completing the square gives

$$Q(w) = (w - w_0)^\top A(w - w_0) + \mu, \quad \mu = c + b^\top w_0 \geq 0.$$

If $\mu > 0$, there are no zeros. If $\mu = 0$, positive semidefiniteness gives $Q(w) = 0$ exactly when $A(w - w_0) = 0$, equivalently $Aw = -b$. This is a rational affine subspace.

Clearing denominators gives finitely many integer linear equations. Intersecting with $\mathbb{N}_0^N$ and projecting coordinates is an existential Presburger definition. Effective semilinearity and decidability follow from Ginsburg–Spanier and Presburger arithmetic [7]. $\square$

Corollary 13.2 (Convex-quadratic compression barrier). *An undecidable halting relation cannot be represented by one fixed globally nonnegative rational quadratic with a fixed finite number of natural parameters and witnesses, even after a computable input-loading map.*

Proof. The projected zero relation would be decidable by Theorem 13.1. Its preimage under a computable input-loading map would then be decidable as well, contradicting the assumed undecidability. $\square$

This theorem applies to global nonnegativity on *all real variables*, not merely nonnegativity on a special natural-number input slice. It does not classify arbitrary quadratic equations, arbitrary nonconvex polynomials, or cubic Diophantine representations. For example, $x - y^2 = 0$ is a quadratic equation with a nonlinear zero set, but its polynomial is not globally nonnegative and is outside the theorem.

The quartic first-hit construction in (10.1) leaves the convex-quadratic class by multiplying selectors and input-dependent linear forms. Its finite-horizon relation remains decidable for other reasons. A truly unbounded compiler must additionally encode an unbounded amount of discrete chronology into finitely many integer variables. That compression, not the local collision geometry, is where the universal Diophantine problem re-enters.

Remark 13.3 ([write] Credit). Theorem 13.1 and Corollary 13.2 are a special case of results already in Part XI of the collection’s report *canonical-diophantine-certificates* [11], which source 07 does not cite. There, Lemma `cdc:of:lem:affinezero` shows that a real polynomial p of degree at most three that is nonnegative on an open convex set O and vanishes at $a \in O$ vanishes on O exactly along $a + \ker \nabla^2 p(a)$; with $O = \mathbb{R}^N$ this is the affine zero set of Theorem 13.1. Theorem `cdc:of:thm:cubicsemilinear` shows that the full natural zero set of a *rational* polynomial of degree at most three that is nonnegative only on the real nonnegative orthant is a finite union of natural solution sets of rational affine systems, and Theorem `cdc:of:thm:classification` that the coordinate projections of such zero sets are exactly the semilinear sets, $D_2^+(k) = D_3^+(k) = \text{SL}(k)$, effectively. Those results assume less (nonnegativity on the orthant only, degree up to three, no convexity or sum-of-squares hypothesis) and give the same decidable, semilinear conclusion; that report itself notes that its theorem is stronger than the observation that a globally nonnegative polynomial cannot have odd highest degree. Corollary 13.2 therefore also holds for orthant-nonnegative polynomials of degree at most three. No novelty is claimed here for either statement. Source 07’s proof by completing the square is kept as a direct second route for the globally nonnegative quadratic case, and its clause “even after a computable input-loading map” is its own: a computable preimage of a decidable set is decidable.

13.1 The classical unbounded existence result remains available

For a finite rational initial configuration, an exact event-driven simulator can find the next collision batch by computing all positive pairwise meeting times and taking the smallest. It can then apply all rules at the sites of that batch. Repeating this procedure semidecides the appearance of a marked signal in an ordinary finite prefix. An infinite accumulating run simply causes the simulator to execute indefinitely when no earlier acceptance has occurred; no post-accumulation convention is needed for semidecidability.

Consequently the pre-accumulation finite-prefix acceptance relation, with machine and rational input encoded by natural numbers, is computably enumerable. MRDP supplies a fixed-arity Diophantine representation of that encoded relation [9, 10]. This last existence step is classical. Neither MRDP nor the present frontend automatically supplies unique witnesses, an explicit small expanded universal polynomial, or the repository’s particular operation ledger.

For a fixed universal machine with an undecidable marked-halting interface, there can be no total computable function of the encoded seed that bounds the number of sites needed by every accepting run. Such a function, followed by exact bounded simulation, would decide halting. This explains why the bounded chamber theorems and the universal existence theorem coexist without contradiction.

[write] The last argument is the standard bounded-search argument, used for other statements in Proposition `cdc:bd:prop:nobound` of [11] and elsewhere in the collection.

14 Implementation, reproducible checks, and trust boundaries

14.1 What the attached code computes

The accompanying standard-library Python implementation uses exact `Fraction` arithmetic for every geometric calculation. The essential interfaces are:

```
Machine(speeds, rules, transparent_default=True)
simulate(machine, initial_labels, gaps, max_batches=20)
compile_skeleton(machine, initial_labels, skeleton)
UnionCertificate(tuple_of_chamber_certificates)
```

A skeleton is a tuple of batches; each batch is a tuple of input-ID tuples. For example, `((0, 1), (2, 3)),` denotes one batch with two independent simultaneous sites. Fresh identities are generated canonically by increasing outgoing velocity.

The simulator uses an all-pairs minimum-time calculation. The symbolic compiler uses intercept propagation and adjacency intervals. Their different chronology mechanisms provide a useful cross-check: the regression suite compares exact numerical prefixes against the regions predicted by the symbolic compiler on many other initial seeds.

A compiled `Certificate` contains the complete integer matrices, exact rational event forms, causal depths, interval count, and raw row ledger. Its `witnesses(gaps)` method returns the unique natural slack tuple, while `evaluate(gaps, witnesses)` evaluates the full polynomial. The `UnionCertificate` implements (10.1). Its constructor does not prove that a supplied family is globally disjoint; that is a mathematical precondition for uniqueness. At a supplied seed, its witness routine rejects multiple accepting chambers rather than silently claiming a unique answer.

The machine constructor copies and freezes the speed and rule tables. This prevents a caller’s

later dictionary mutation from changing the meaning of an already constructed machine. Input validation rejects unknown labels, repeated-speed outputs, malformed groups, use of dead signals, noncontiguous blocks, overlapping sites, and inappropriate incoming speed orders. These checks are part of the semantic contract.

The exported JSON matrices specify full polynomials, not just sample zeros. For example, `tournament_left.json` contains all rows of (11.2), while `first_hit_quartic_union.json` supplies every matrix and the exact selector formula for a two-chamber union. Expanding these expressions into monomials is optional and provides no extra mathematical information.

14.2 Run and reproduce

From the extracted archive’s root directory, run

```
python code/run_checks.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Python 3.10 or newer and a standard LaTeX installation are sufficient. No Python third-party dependencies are needed. The test command overwrites the generated JSON data with deterministic results, including `data/checks.json`.

[write] These are the delivered names. In this report the programs are shipped as `code/07-collision-geometry-run_checks.py` and `code/07-collision-geometry-signal_certificates.py` (with the delivered Makefile as `code/07-collision-geometry-Makefile`), and the seven JSON files of `data/` as `data/07-collision-geometry-*.json`. The test program imports `signal_certificates` by its delivered name and rewrites all seven JSON files in place, relative to its own directory’s parent; it must therefore be run on a copy with the delivered layout (see the README). It was so run at the write, under Python 3.14.4 on Windows: it passed with 34,560 assertions, and the seven regenerated files equal the shipped ones apart from line endings. The delivered article PDF is not shipped; this report’s PDF replaces it.

The completed run accompanying this manuscript passed **34,560 exact assertions**. Its principal coverage is summarized in Table 4. These are finite regression tests, not exhaustive verification of all machines or all input geometries.

Check family	Instances
Tournament seeds $(u, v) \in \{1, \dots, 40\}^2$	1,600
Exact seed-count boxes $H = 0, \dots, 80$	81
Exact inverse thresholds $y = 1, \dots, 2000$	2,000
Four-speed clock batches at $(1, 2)$	24
Other positive clock seeds with the same 24-batch skeleton	25
Pseudorandom rule tables (fixed random seed 20261002)	6
Distinct nonempty prefix skeletons obtained from those tables	24
Independent seed–skeleton chronology comparisons	1,536
Two-chamber quartic-union seed checks	1,600

Table 4: Finite exact test coverage. Individual assertions also check witness mutations, denominators, malformed inputs, and table immutability.

The pseudorandom cases include creation, annihilation, and transparent rules. The checked skeletons are generated from exact sample executions and then tested against other seed geometries.

The suite also specifically rejects an earlier omitted collision, an omitted third input, and an omitted independent simultaneous site. It tests the clock’s exact closed-form times and denominator divisibility. For the quartic union, it checks selector validity and that inactive slacks are forced to zero.

14.3 An auditable resource ledger

Table 5 separates retained conditions from raw generated rows. The generic bound is deliberately loose; redundancy elimination can be substantial. The 24-batch clock is a useful example: its entire seed chamber reduces to only three retained strict rows, even though its geometric coordinates become increasingly complicated. One of those rows is redundant with positive gaps, but the implementation does not attempt full linear redundancy minimization.

Certificate	E	B	r	s	Raw rows	Bound
Tournament AB	1	1	0	3	7	16
Tournament BC	1	1	0	3	7	16
Tournament ABC	1	1	1	2	8	16
Two simultaneous sites	2	1	1	4	11	23
Clock, 24 batches	24	24	0	3	122	318

Table 5: Exact compiler ledger: r is the retained equality count and s the retained strict-row count. The rightmost column is $3(n - 1) + (3m + 1)E$, not an arithmetic-operation count.

14.4 What has and has not been verified

The prose proofs establish general claims by mathematical argument. The code and data test finite instances with exact arithmetic. No Lean or other proof-assistant verification of this article is included, and the implementation is not a formally verified compiler. The universal 13-type machine’s complete published rule table and finite-input loader have not been reimplemented in the package. The code supplies a generic machine interface and the specific worked examples only.

The repository was inspected on 2 October 2026. The observed main-branch head was

`f1edb38f93aa0a362a34b531a600909f6e74608a`.

The relevant reversible-history audit was fetched as blob

`cb31d0a108b9a19fc33da6fa89a73b604e15e260`.

No repository-wide build was performed and no repository files were changed. The current README and the older audit have different historical operation ledgers; none is used as a theorem in the present proofs. The new row counts are not comparable to a fixed-universal arithmetic count without a separate input and history compression analysis.

[write] The pin `f1edb38f9` is an ancestor of this report’s placement commit. The audit blob `cb31d0a10` is still the current `Computability/HilbertTenthProblem/Papers/1980/REVERSIBLE_FINITE_HISTORY_MODELS.md` at the write; the project README [1] has gained text since the pin and still states the 75- and 87-operation bounds. The repository’s own review of this package [14] is described in Section 1.4.

15 Further research questions and a concrete development path

[write] Table 3 pairs these directions with the questions of Part II. The first two coincide with Part II's Questions 5 and 1.

15.1 Formalizing the chronology theorem

The most direct formalization target is the endpoint lemma followed by the sweep invariant: the live list is in strict spatial order on every open interval between batches, and its equal-position blocks at each boundary are exactly the declared sites. Stable IDs make consumption and birth finite-map properties. A Lean development should keep the real semantic statement separate from rational coefficient computation, then prove that evaluation commutes with the intercept compiler. Finally, the sum-of-squares zero lemma and natural slack uniqueness can be linked to the repository's Diophantine infrastructure.

A practical first milestone is a small checker for exported matrices and rational event forms. A second is a proof-producing compiler rather than a trusted Python translator. Tests that compare two unverified programs should remain auxiliary evidence, not be promoted to kernel verification.

15.2 An audited universal signal-machine frontend

Instantiate a published finite-input universal construction with its exact labels, velocities, all required collision rules, transparent defaults, input encoding, and marked acceptance convention. The 13-type construction of Durand-Lose [3] is a concrete candidate. The task is not merely to copy a picture: simultaneous three-input cases, finite workspace supply, and the loader's correspondence with the simulated machine must be audited. The eventual theorem should state an exact equivalence between an ordinary integer program input and the existence of a compiled accepting prefix.

This would turn the present generic geometry theorem into a fully explicit end-to-end substrate result. A fixed-arity universal polynomial would still require a separate outer compression step and its own complete resource ledger.

[write] **Added 2 October 2026 (batch 79).** This direction is answered by Part III, except for the ordinary-integer loader. Part III instantiates a published finite-input universal construction literally: Morita's 15-state, 6-symbol reversible universal Turing machine compiled through Durand-Lose's 2012 conservative stacks, with 114 labels, their speeds, all 445 collision rules, undefined encounters forbidden rather than completed, the input encoding, and marked acceptance distinguished from null termination (Theorem 29.1); simultaneous and triple collisions are treated exactly (Lemma 34.1; a triple collision is undefined), the workspace is a fixed population of 18 signals, and the loader's correspondence with the simulated machine is proved (Sections 32–33). The machine is not the 13-type construction named above, and the input is a prepared cyclic-tag tape loaded as two rational distances: the exact equivalence with an *ordinary integer* program input asked for here still needs a digit-extraction and base-conversion loader, whose cost Part III states but does not pay (Section 33). As the paragraph above says, no fixed-arity universal polynomial follows; Part III's certificates are for fixed horizons.

15.3 Sharper denominator invariants

The factor D^h treats all speed differences uniformly. Can one use a velocity-labelled dependency graph to compute exact prime-valuation bounds or a smaller denominator ideal for each event? Which

speed sets force uniformly bounded denominators on all integral seeds, and which permit exponential growth? The clock supplies one lower-bound mechanism, while the small-speed accumulation results [6] suggest that the interaction between arithmetic precision and global dynamics is subtler than a least-common-multiple bound alone.

A particularly concrete project is to propagate prime-valuation vectors through (4.2) and detect systematic cancellation at each rule. Any claimed lower bound must use reduced denominators, not merely the unreduced fractions produced by a particular implementation.

15.4 From few rows to small arithmetic circuits

The event-sparse theorem controls row count but permits dense coefficient vectors. Can common subexpressions in intercept forms give a substantially smaller arithmetic circuit than dense matrix evaluation? An appropriate cost model must distinguish fixed machine numerals, input-dependent arithmetic, equality tests, and the conversion from a residual system to one polynomial. The original intercept dependency graph already provides a sharing structure; the challenge is to preserve its certified semantic meaning while optimizing its arithmetic schedule.

A useful result would be a proved tradeoff among witness count, maximum degree, coefficient height, and literal addition/multiplication count. No one-dimensional comparison such as “quadratic beats a degree-169 polynomial” is meaningful when arity and unbounded-history coding differ.

15.5 Avoiding exhaustive first-hit enumeration

The quartic in (10.1) is explicit but can contain an enormous number of chambers. Can a local selector circuit encode a bounded unknown skeleton with polynomially many variables while retaining canonical first-hit uniqueness, including all simultaneous events? A promising direction is to encode changes of adjacency directly rather than a rectangular array of all particle positions. The main obstruction is that a locally plausible selection must still describe one globally ordered and complete execution.

Even a polynomial-size bounded compiler would not solve fixed-arity unbounded compression. Keeping those two improvements distinct will make future operation claims interpretable.

15.6 Robust universality and minimal synchronization rank

For a given computational simulation, determine the rank of its required input-geometry equalities. Can the same computation be realized on a full-dimensional chamber, or does its interface genuinely require exact ratios among certain gaps? The answer must be relative to the legal input space rather than to coordinates the loader never permits to vary.

One possible invariant is the minimum independent synchronization rank among equivalent signal-machine implementations with a fixed speed set. The chamber theorem makes that invariant concrete for each finite trace, but extending it uniformly over a universal family needs a precise notion of simulation and allowed perturbation.

15.7 Counting complexity and asymptotic structure

Determine when first-hit chamber counts can be computed without enumerating all skeletons. Are there useful machine classes for which a recurrence on adjacency types produces the Ehrhart

generating function directly? When do quasi-polynomial periods collapse, and which restrictions imply polynomial counts? The exact inverse in the tournament example is simple; in higher degree, inverse asymptotics must retain residue-class corrections and rounding effects rather than replacing them by an unjustified smooth error.

For unbounded acceptance, investigate convergence of finite-horizon rational densities. The increasing accepted sets give monotonicity, but a rational limit is not automatic. Quantitative tail estimates would be needed for a useful approximation theorem and for interchanging input-box and time limits.

15.8 Algebraic velocities and nonlinear motion

With velocities in a fixed algebraic number field, formal event coordinates remain linear over that field. However, positivity is no longer a rational half-space condition in the same elementary form, and integer realization by scaling is not automatic. A number-field sign certificate might extend part of the method, but the rational-cone, Presburger, and Ehrhart conclusions would require new arguments or additional hypotheses.

If signals accelerate or their velocities depend on variable parameters, (4.2) ceases to divide only by a fixed constant. This identifies a precise boundary of the present linearization. Studying which broader motion laws still admit low-degree complete chronological certificates is a more informative problem than assuming every geometric model behaves like the fixed-speed case.

15.9 Accumulation limits without fictitious infinite witnesses

The clock gives a finitely described accumulation time, but a generic finite prefix certificate cannot assert all events of an infinite run. A separate notion of a certified self-similar tail could perhaps prove a limit formula or a no-collision property beyond a given recursive pattern. Such a notion must specify the dynamics and limit semantics explicitly. It must not silently treat an infinite sequence of successful finite checks as one ordinary existential Diophantine witness.

16 Conclusion

For rational fixed-speed signal machines, the continuous coordinates of a finite computation are not the difficult nonlinear part. Once a discrete chronological skeleton is fixed, they are rational linear forms in the initial gaps. The nontrivial certificate obligation is completeness of chronology: excluding unlisted earlier meetings and accounting for every signal and every site at a simultaneous event. Maximal adjacency intervals solve that obligation with a linear number of endpoint conditions.

The resulting exact chamber description yields quadratic natural-number certificates with unique slack tuples, and finite disjoint first-hit unions yield quartics with unique selector and slack tuples. It also yields useful results about denominator growth, small integral realizations, perturbation stability, and seed-counting quasi-polynomials. The executable examples test these claims against independent exact simulation and expose the boundary cases that incidence-only encodings miss.

The unresolved universal task is sharply separated: finitely encode an unbounded discrete skeleton together with the ordinary input interface. The convex-quadratic rigidity theorem explains why this cannot be achieved by merely hiding more geometry inside one fixed globally nonnegative quadratic. ([write] The rigidity theorem is a special case of Part XI of *canonical-diophantine-certificates*; Remark 13.3.) The present contribution is a complete and auditable geometric frontend,

together with a clear account of which further arithmetic and formalization steps remain.

Part II

Direct Diophantine certificates (source 11)

[write] Part II prints batch-78 manuscript 11, *Direct Diophantine Certificates for Rational Signal Machines: Finite collision schemas with unique quadratic witnesses*, “research article and exact arithmetic replay”, dated 2 October 2026; its document metadata name the author as “Research report prepared with OpenAI”. Its labels carry the prefix `smc:sd:`; its sections are Sections 17–28 here and its two appendices are Appendices C and D. It proves the theorem of Part I by a second route (Section 1.2); it is printed in full, and its lemmas that coincide with lemmas of Part I carry [write] notes. Its abstract and keywords follow.

Abstract (source 11). A prescribed finite collision history of a rational signal machine admits an explicit Diophantine certificate that retains the machine’s geometry. After external normalization, we compile each finite schema into a sum of squares of integer affine residuals. Natural zeros are exactly its realizations, and every admissible input has exactly one witness tuple. The witness coefficient matrix has full column rank, so the polynomial is strictly convex in the witnesses. A sparse construction uses event coordinates, batch times, and typed endpoints of maximal adjacency lifetimes; for bounded collision arity and fanout its size is linear in the initial support and number of events. The endpoint types enforce complete collisions, simultaneous events, and the absence of a hidden collision at the final cut. We prove denominator divisibility and an explicit witness-height bound for the four-speed interface of a known universal signal machine. The construction is a uniform compiler for externally supplied finite schemas, not a fixed-arity representation of unbounded halting. Exact standard-library replays and fully worked examples accompany the proofs.

Keywords. Diophantine representation; signal machine; abstract geometrical computation; finite trace; unique witness; convex quadratic; semilinear set.

17 Purpose and relation to prior work

Turing-complete substrates can store and transform information in very different ways. Signal machines use a finite collection of particle types moving continuously on a line and interacting at collisions. A direct arithmetization of their geometry should explain both the prescribed events and the absence of all other events. Merely writing down line intersections leaves the latter obligation unresolved.

This article establishes that obligation for each fixed finite collision schema. Its contribution is an explicit integer certificate, including denominator control, canonical witnesses, sparse size bounds, and exact endpoint conditions. No general claim that signal-machine evolution is newly linearizable is intended. Durand-Lose’s linear real-register simulation already computes the next delay, updates gaps, and processes simultaneous events by finitely many linear branches [4, Section 4]. Our arithmetic constraints begin after a particular finite branch history has been fixed.

The substrate itself is already Turing universal. Durand-Lose gives a halting construction with 13 meta-signals and 21 *non-blank* collision rules; its four speeds are $-2, 0, 1, 2$ [3, Theorem 5 and Figure 4]. The construction uses a finite initial configuration and is separate from the paper’s weaker,

non-halting cellular-automaton simulation with an infinite background. After a Galilean shift the speeds become 0, 2, 3, 4. The rational normalized placements used in the halting arrangement can be scaled to integer positions, as proved generally below. We have not transcribed or executed that universal collision table. The accompanying replay uses small exact test machines.

The results are deliberately bounded in a precise sense. The machine, input length and labels, schema, and number of batches are compiler data. The polynomial's arity grows with the schema. Nothing here packs unbounded execution into one fixed-arity quadratic, resolves the single-fold Diophantine representation problem, improves a universal-polynomial arithmetic-operation bound, or formalizes the proof in a proof assistant. A targeted literature check supports the stated relation to earlier work; it does not establish exhaustive novelty.

[write] Source 11 does not know source 07, printed as Part I, which proves the same theorem with all event coordinates eliminated and allows signed rational speeds; Section 1.2 compares the two routes. The repository has since reviewed this manuscript in its Hilbert-tenth-problem research tree (Section 1.4).

18 Model and exact theorem

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Inputs and witnesses are natural numbers unless explicitly stated otherwise.

Definition 18.1 (Normalized signal machine). A machine $\mathcal{M}$ consists of a finite set of labels, a fixed speed $v(\ell) \in \mathbb{N}$ for each label, and a deterministic collision table. A legal incoming set has at least two labels with pairwise distinct speeds. Its outgoing set also has pairwise distinct speeds and may be empty. An initial configuration consists of labels $\ell_0, \dots, \ell_{n-1}$ at positions

$$0 \leq p_0 < p_1 < \dots < p_{n-1}, \quad p_j \in \mathbb{N}.$$

Between collisions a signal follows a straight line of its label's speed. At a collision every incident signal is consumed and the designated outgoing set is emitted at that point.

If an incoming set has no rule, the collision is forbidden, unless the model explicitly completes the table by a transparent rule. Such a rule consumes the incident segments and emits the same labels, ordered by speed. The theorem works for either convention after the machine is fixed. The replay scripts use identity completion, so an omitted rule in their input is transparent, not forbidden.

We distinguish labels from *segment identities*. Each initial signal has its own identity, and each emitted signal receives a fresh one, even if a transparent rule reproduces the same label. A segment has one birth and at most one death. Initial signals are born at time zero. This distinction is essential for endpoint typing.

Definition 18.2 (Finite collision schema). Fix the machine and the initial label list. A schema $\mathcal{S}$ specifies K nonempty chronological batches. In batch i , a partition of the ordered pre-batch live list into consecutive blocks designates each nonsingleton block as an event and each singleton as a persisting signal. Every event uses a legal rule, and its incoming speeds strictly decrease from left to right. Its outputs are ordered by strictly increasing speed. All replacements in a batch are performed atomically. Segment births precede their deaths by at least one batch.

The schema fixes no event coordinates or durations. Its events within a batch are ordered from left to right by their incoming blocks. Zero-output events are retained as events. A schema with

$K = 0$ has no events. No initial collision markers, infinite initial support, or accumulation events are included.

A *prefix realization* has exactly these first K nonempty collision batches and ends immediately after batch K . The $K = 0$ prefix is the initial configuration alone. A *halting realization* additionally has no future collision. Halting does not require all signals to disappear or stop moving. There is no arbitrary terminal observation time.

Let E be the number of events, $I = \sum_e |\text{in}(e)|$ the number of incoming incidences, r the largest incoming arity, and s the largest fanout. Set $n_- = \max(n - 1, 0)$ and

$$D = \text{lcm}\{|v - w| : v, w \text{ are machine speeds, } v \neq w\}, \quad B = D^K, \quad (18.1)$$

with the empty least common multiple equal to one.

[write] Theorem 18.3 is, for prefix semantics, the theorem of Part I's Theorems 5.3 and 6.1, in a different witness format: natural absolute positions as input, and batch times, event positions and slacks as witnesses on the grid B . Its halting version, the full column rank and the size ledger (18.3)–(18.4) are not in Part I. Source 11's proof is printed in full as the second route.

Theorem 18.3 (Sparse canonical certificate). *There is an effective compiler which, for every such fixed schema, produces integer affine residuals $r_1, \dots, r_R$ in the input coordinates $\mathbf{p}$ and witnesses $\mathbf{y} \in \mathbb{N}^W$, and a polynomial*

$$Ps(\mathbf{p}, \mathbf{y}) = \sum_{a=1}^R r_a(\mathbf{p}, \mathbf{y})^2.$$

For prefix semantics, a natural zero exists exactly when $\mathbf{p}$ realizes the schema. For halting semantics the same statement holds after the terminal conditions below are included. For every admissible input the witness is unique. The witness coefficient matrix has full column rank; consequently, if $W > 0$, the Hessian in the witness variables is positive definite.

If L denotes the number of recorded adjacency lifetimes, including terminal rays when present, then

$$L \leq n_- + \sum_e (|\text{out}(e)| + 1), \quad (18.2)$$

$$W \leq E + 2K + 2L + n_-, \quad R \leq I + K + 2L + n_-. \quad (18.3)$$

In particular,

$$W \leq (2s + 5)E + 3n_-, \quad R \leq (r + 2s + 3)E + 3n_-. \quad (18.4)$$

The polynomial has degree at most two, and degree exactly two whenever its witness vector is nonempty.

Syntactically illegal schemas are rejected before compilation. For the halting version, a final live order with a negative adjacent relative velocity is likewise rejected as impossible. One can instead output the constant polynomial one for these cases; the stated resource ledger concerns the structurally admitted construction. At $K = 0$, the prefix compiler may use only input-order checks. For $n \leq 1$ it has no witnesses and returns zero. This harmless degenerate case is not a strictly convex function on a positive-dimensional space.

19 Rational normalization and denominator control

19.1 Normalization is an external interface

Suppose the original speeds are rational numbers $u(\ell)$. Choose a rational $c \leq \min_{\ell} u(\ell)$ and a positive integer m such that every $m(u(\ell) - c)$ is an integer. For a fixed rational input choose a rational translation a and positive integer q so that every $q(x_j + a)$ is a nonnegative integer. The invertible coordinate change

$$X = q(x + a - ct), \quad T = \frac{q}{m}t \quad (19.1)$$

preserves simultaneous incidence, spatial order at each time, positive time direction, and the collision rules. Its speeds are

$$\frac{dX}{dT} = m(u(\ell) - c) \in \mathbb{N},$$

and its initial coordinates are natural numbers. Thus every finite rational configuration can be put into the theorem's interface.

The constants a, c, m, q are chosen *outside* the existential witness tuple. For a varying input family with fixed denominator and shift, the natural input $p_j = q(x_j + a)$ is the declared coordinate encoding. Allowing an arbitrary additional scale as a witness produces many proportional encodings; allowing raw variable numerators and denominators introduces products when denominators are cleared. Neither change preserves the present canonical, jointly quadratic claim without a new argument. Inputs at arbitrary real coordinates are outside this interface.

[write] Part I needs no Galilean shift: it allows signed integer speeds after the time rescaling $\tau = t/L$ (Section 3) and takes the gap vector as input. With $m = L$ the two normalizations give the same speed differences, hence the same D (Section 1.2).

19.2 Causal depth and a fixed grid

Assign depth zero to initial births. The depth of event e is one plus the maximum depth of the birth events of its incoming segments, with initial births contributing zero. Every birth lies in an earlier batch, so the maximum event depth H satisfies $H \leq K$.

Lemma 19.1 (Denominator divisibility). ([write] *The same as Part I, Theorem 7.1, there stated for the coefficient forms in the initial gaps; source 11's proof is kept as a second route because it works with event-coordinate witnesses.*) *Every event time and position of depth at most h has reduced denominator dividing D^h . In particular D^H , and hence $B = D^K$, clears every event coordinate and time. It also clears the position of any live segment sampled at an event time.*

Proof. Initial birth positions and times are integers. Suppose an event of depth h has incoming lines

$$x = b_1 + v_1(t - a_1), \quad x = b_2 + v_2(t - a_2), \quad v_1 \neq v_2.$$

All birth coordinates have denominators dividing D^{h-1} by induction. The intersection time is

$$t = \frac{b_2 - v_2 a_2 - b_1 + v_1 a_1}{v_1 - v_2}. \quad (19.2)$$

The numerator's denominator divides D^{h-1} , while $|v_1 - v_2|$ divides D . Thus the reduced denominator of t divides D^h . Substitution into either line gives the same assertion for its position. A multi-signal

event is pinned by any two incoming speeds; the remaining incidences are additional equalities. Finally a sampled position is an integer-linear combination of a birth position, birth time and sample time, all cleared by D^H . $\square$

This lemma bounds denominator divisibility, not numerator magnitude. Its practical advantage is that B is a computable constant of the schema. The compiler can use the sharper D^H if desired; the supplied implementation uses D^K .

20 The sparse construction

20.1 Geometry variables and flight equations

Use a natural variable T_i for the scaled time of batch i , with $T_0 = 0$, and a natural variable X_e for the scaled position of each event. Introduce $d_i \in \mathbb{N}$ and require

$$T_i - T_{i-1} - d_i - 1 = 0 \quad (1 \leq i \leq K). \quad (20.1)$$

Thus each batch occurs at a strictly later time on the B -grid.

For a segment σ of speed v_σ , let $b(\sigma)$ be its birth batch. Define its scaled birth-position expression A_σ to be Bp_j for initial segment j , and X_f if it is emitted at event f . Its position at any relevant batch time is the affine expression

$$Z_\sigma(T_i) = A_\sigma + v_\sigma(T_i - T_{b(\sigma)}). \quad (20.2)$$

If σ is consumed by event e in batch i , impose the flight residual

$$X_e - Z_\sigma(T_i) = 0. \quad (20.3)$$

There are I such residuals. All coefficients are integers because the speeds and B are fixed integers. No product of two unknowns appears.

Input order is encoded, rather than silently assumed, by natural slacks u_j and residuals

$$p_{j+1} - p_j - u_j - 1 = 0 \quad (0 \leq j < n - 1). \quad (20.4)$$

An explicitly restricted ordered-input domain permits these residuals and witnesses to be omitted. We keep them in every count.

20.2 Maximal adjacency lifetimes

Maintain the ordered live list combinatorially. A consecutive ordered pair of segment identities has a lifetime beginning at the first boundary at which it becomes adjacent and ending at the boundary at which it ceases to be adjacent. A pair that remains adjacent across several batches has one lifetime, not a new copy in every slab. Perform all replacements in a simultaneous batch before creating new pairs; pairs appearing only during an arbitrary sequential processing order are never recorded.

For a prefix, close each still-active positive-duration lifetime at the final cut T_K . A pair newly born at batch K has no positive-duration finite lifetime and is omitted. For halting semantics, still-active pairs receive infinite terminal lifetimes instead, including pairs born at the final batch. The initial $K = 0$ prefix needs no finite lifetimes.

For a pair (σ, τ) , left to right, its scaled gap is the affine function

$$G_{\sigma\tau}(T) = Z_\tau(T) - Z_\sigma(T), \quad (20.5)$$

whose slope is the fixed relative velocity $v_\tau - v_\sigma$.

Lemma 20.1 (Lifetime count). ([write] *The same as the first inequality of (5.2) in Part I, Theorem 5.4; source 11's proof is kept as a second route because it works with event-coordinate witnesses.*) *The number L of recorded lifetimes satisfies (18.2).*

Proof. There are at most n_- initial adjacent pairs. Replacing one consecutive event block by k outputs creates at most $k + 1$ new pairs: at most $k - 1$ internal output pairs and two outer-neighbor pairs when $k \geq 1$, or at most one bridge pair when $k = 0$. Boundary events can create fewer. A sequential accounting of the disjoint replacements in a batch therefore counts at most $\sum_e (|\text{out}(e)| + 1)$ creations. Some counted pairs may be transient and disappear within the same batch; atomic processing removes those and cannot increase the count. Every lifetime has exactly one creation, including any terminal lifetime. Closing a lifetime does not create an additional one. $\square$

20.3 Typed endpoint constraints

For a finite lifetime from batch a to batch $b > a$, impose two constraints:

1. At the lower endpoint, impose $G(T_a) = 0$ if and only if both segments were emitted by the *same event* in batch a . Otherwise impose $G(T_a) = h^- + 1$ with a fresh $h^- \in \mathbb{N}$.
2. At the upper endpoint, impose $G(T_b) = 0$ if and only if both segments are consumed by the *same event* in batch b . Otherwise impose $G(T_b) = h^+ + 1$ with a fresh $h^+ \in \mathbb{N}$.

The “if and only if” here specifies which residual the compiler emits from the schema; it is not a new logical connective in the polynomial. Permitted equality means an equality residual, not an arbitrary nonnegative gap. In particular, a surviving pair at the final cut receives a *strict* upper gap.

For a terminal ray starting at batch a , require its relative velocity to be nonnegative, a check on machine constants. Impose a zero initial gap only for two same-event outputs at that starting boundary; impose $G(T_a) = h + 1$ otherwise. Same-event outputs have strictly increasing speeds, so a zero-gap terminal pair automatically has strictly positive relative velocity. A terminal pair with zero relative velocity must therefore start with positive gap.

Every stated condition is either an integer affine residual or a finite compile-time structural check. Define P_S by squaring and adding the residuals (20.1), (20.3), (20.4), and all endpoint residuals.

21 Proof of geometric correctness

21.1 Why endpoint types suffice

Lemma 21.1 (Positive interior gap). ([write] *The same as Part I, Lemma 5.2 (the endpoint lemma) together with the static rejection of a common birth and a common death that precedes it; source 11's proof is kept as a second route because it works with event-coordinate witnesses, and it adds terminal rays.*) *Every finite lifetime satisfying its endpoint residuals has strictly positive gap in its open time interval. Every admitted terminal lifetime has strictly positive gap at every time after its start.*

Proof. A finite gap is affine. Its endpoint values are nonnegative. If either is strictly positive, every proper convex combination of the endpoint values is strictly positive. If both are zero, the endpoint types say that the pair shares its birth event at the lower boundary and its death event at the upper boundary. Birth ordering gives $v_\sigma < v_\tau$, so

$$G(T_b) = G(T_a) + (v_\tau - v_\sigma)(T_b - T_a) > 0,$$

a contradiction. Thus at least one endpoint is strict, and no extra endpoint-sum residual is needed. On a terminal ray the nonnegative slope preserves a strict initial gap; a zero initial gap has strictly positive slope and becomes positive immediately. $\square$

It is important that this argument uses segment identities and the exact start and end boundaries. Merely allowing two nonnegative endpoint gaps, even with positive sum, does not exclude an unrecorded meeting at the upper endpoint.

Lemma 21.2 (Complete boundaries). *At every batch satisfying the residuals, each designated incoming block meets at its event point, distinct designated events have distinct ordered positions, and no persisting signal or signal of another event also meets that point. The post-batch order is valid; its only zero gaps are between outputs of the same event.*

Proof. Before the batch, every adjacent pair belongs to a lifetime that has existed for a positive duration. For a pair in one event block, the flight equations place both segments at that event. Any pair across two distinct pre-batch blocks is not consumed by the same event. If its lifetime ends at this boundary, the strict upper endpoint gives a positive gap. If it persists across the boundary and this is an interior point of its lifetime, Lemma 21.1 gives a positive gap. If the boundary is the final cut, a surviving pair has a strict upper endpoint, or lies on an admitted terminal ray. These cases exhaust the pre-batch neighboring pairs.

All positions in an incoming block are equal by the flight equations, whereas successive blocks are strictly separated. Hence the designated blocks are precisely the maximal equality classes at that time. In particular a third incident signal cannot be omitted, two designated events cannot coincide, and a completely erased event still has a position in the strict block order. Replacing each event block at its own point by the prescribed speed-ordered outputs therefore preserves weak spatial order. Across distinct surviving blocks the gap stays strict; within one output fan the gap is zero and the increasing speeds separate it immediately. If erased blocks disappear, the new bridge gap spans one or more former strict block separations, so is positive. The replacements are atomic and introduce no additional intermediate configuration. $\square$

This proof also explains why final-born prefix pairs can be omitted. Their boundary order is already certified by the incoming block partition and flight equations. Immediately after that boundary, a common output fan separates and all other gaps remain positive for a sufficiently small interval. A prefix does not assert the absence of collisions later in that interval's future.

21.2 Soundness and completeness

Proof of the existence assertion in Theorem 18.3. For soundness, suppose $P_S(\mathbf{p}, \mathbf{y}) = 0$ with natural inputs and witnesses. Each square is nonnegative, so every residual is zero. The input packet gives strict initial order. Time residuals give strictly increasing batch times. Divide all scaled times and event positions by B and draw the prescribed segments using (20.2).

On each open slab, the combinatorial live list is fixed and every neighboring pair is in an active lifetime. Lemma 21.1 excludes all adjacent meetings. A meeting of nonadjacent signals would force an adjacent equality in the ordered list at or before that meeting, so there are no other meetings either. At the next batch, Lemma 21.2 proves that the designated events are exactly all collisions and contain every incident signal. Their legal rules are the fixed schema rules. Induction across the K batches gives precisely the prescribed prefix. For halting semantics every final adjacent gap remains positive for all future time by the terminal-ray argument, so no future collision occurs.

For completeness, start with an actual realization. Lemma 19.1 puts every event time, event position and endpoint sample on the B -grid. Assign their scaled values to T_i and X_e , and set $d_i = B(t_i - t_{i-1}) - 1$. All are natural. Input gaps determine natural u_j . Each lower or upper endpoint at which the schema prescribes equality is an actual common birth or death point. Every other endpoint is strictly separated by the definition of a complete collision history; its positive scaled integer gap minus one supplies the unique natural slack. An actual halting configuration has nonnegative adjacent relative velocities, since a negative one would eventually cause a meeting, and its terminal initial gaps have exactly the required types. All residuals vanish. $\square$

The argument covers simultaneous multi-signal collisions, disjoint events in the same batch, annihilation, transparent collisions and empty final configurations. Its dependence on complete blocks and typed boundaries is substantive, not a cosmetic choice of encoding.

22 Canonical witnesses and strict convexity

Uniqueness of integer witnesses alone would not prove strict convexity. We establish the stronger linear-algebra statement directly.

Lemma 22.1 (Full column rank). *Write the residual vector as*

$$\mathbf{r}(\mathbf{p}, \mathbf{y}) = A\mathbf{y} + C\mathbf{p} + \mathbf{c}.$$

For every structurally admitted schema, the matrix A has trivial kernel over $\mathbb{R}$, even if the schema has no realization for the chosen input.

Proof. Let $A\Delta\mathbf{y} = 0$ and hold the input fixed. Initial birth positions have zero variation. Induct on batches. Assume all earlier times and event positions have zero variation. Choose any event e in batch i and two of its incoming segments σ, τ with distinct speeds. The homogeneous flight equations are

$$\Delta X_e - v_\sigma \Delta T_i = 0, \quad \Delta X_e - v_\tau \Delta T_i = 0,$$

because both segments were born in earlier batches. Their difference gives $(v_\sigma - v_\tau)\Delta T_i = 0$, hence $\Delta T_i = 0$ and $\Delta X_e = 0$. Every other event in the batch then has zero position variation from any one of its flight equations. Equation (20.1) gives $\Delta d_i = 0$. Induction fixes all geometry variables and time slacks.

Each remaining witness is an input-order or endpoint slack with coefficient -1 in its own defining residual. With the geometry and input fixed, that residual forces its variation to be zero. Thus $\Delta\mathbf{y} = 0$. $\square$

Corollary 22.2 (Canonical zero and Hessian). *For fixed input there is at most one real solution of the residual system, and hence at most one natural zero of P_S . The Hessian in witness coordinates is $2A^\top A$, positive definite when $W > 0$.*

Proof. Two solutions differ by a vector in $\ker A$, which is zero by Lemma 22.1. Moreover $\mathbf{z}^\top(2A^\top A)\mathbf{z} = 2\|A\mathbf{z}\|^2 > 0$ for every nonzero $\mathbf{z}$. Existence for admissible inputs was proved above. $\square$

Strict convexity is asserted in the witness variables with the input held fixed. Joint strict convexity in input and witness coordinates is neither needed nor generally true. The compiler introduces no witness for a schedule choice, denominator choice or arbitrary final observation time. Adding any of those could destroy the canonical property.

Proof of the size assertions in Theorem 18.3. Geometry uses E event positions, K batch times and K time slacks. A finite lifetime needs at most two slacks and two residuals; a terminal lifetime needs at most one of each. Input order uses n_- slacks and residuals. Flight and time residual counts are I and K . This proves (18.3). Lemma 20.1, $I \leq rE$, $K \leq E$ and $|\text{out}(e)| \leq s$ give (18.4). Squaring affine residuals gives degree at most two. If $W > 0$, full column rank forces a nonzero quadratic part. $\square$

The ledger counts variables and affine residuals. With a fixed machine, every residual has bounded width: an endpoint gap uses only the birth positions and times of two segments, one endpoint time, and possibly a slack. A linear-size arithmetic circuit follows if integer constants are treated as available coefficients. This does not give a literal gate ledger under a model that charges separately for constructing large constants, input loading, multiplication conventions or natural-to-positive conversion. In particular, no improvement to previously studied 75-operation or 87-operation constructions is asserted here.

23 A full-layer reference construction

A denser construction is useful for cross-checking. In slab i , let N_i be the number of pre-batch signals and g_i the number of blocks, including singleton through-signals. Introduce natural scaled end coordinates $C_{i,j}$, a duration slack d_i , and one gap slack between successive blocks. The scaled start coordinate $A_{i,j}$ is Bp_j initially or an alias of an earlier endpoint. Require

$$\begin{aligned} C_{i,j} - A_{i,j} - v_j(d_i + 1) &= 0 && \text{for every live signal,} \\ C_{i,j} - C_{i,\text{first}(b)} &= 0 && \text{inside each collision block } b, \\ C_{i,\text{first}(b')} - C_{i,\text{first}(b)} - h - 1 &= 0 && \text{for successive blocks } b, b'. \end{aligned}$$

All blocks, including zero-output collisions, participate in the last row. The exact counts, with input-order checks, are

$$W = n_- + \sum_i (N_i + g_i), \quad R = n_- + \sum_i (2N_i - 1). \quad (23.1)$$

Indeed each slab uses $N_i + 1 + (g_i - 1)$ witnesses, and $N_i + (N_i - g_i) + (g_i - 1)$ residuals.

The two boundary lists have nonnegative adjacent gaps, with strict separation across ending blocks. A pair with zero gap at its starting boundary is a same-event output pair with increasing speeds; it cannot meet again at the ending boundary after positive time. Affineness therefore excludes all interior collisions. The full ending partition supplies completeness at each boundary. Conversely the denominator lemma gives a witness from every realization. The same two-incoming-speed induction fixes each duration, then all endpoints and gap slacks, and proves full witness rank. A

halting certificate additionally checks that the final live speeds are nondecreasing; the final boundary order is already certified.

This reference compiler stores coordinates for every live signal at every layer, even if that signal survives unchanged through many layers. The sparse compiler avoids these repetitions by storing only event coordinates and maximal adjacency lifetimes. Both implementations share an exact simulator and small arithmetic utilities; agreement between them is useful evidence, but is not independent formal verification of the common simulator.

24 Worked examples

24.1 One annihilation with its complete quadratic

Consider labels R, S with speeds $2, 0$ and the rule $\{R, S\} \mapsto \emptyset$. Let the initial positions be $p_0 < p_1$, and specify one event. Here $D = B = 2$. Write T, X, d for its scaled time, position and duration slack, u for input order and h for the lower adjacency slack. The compiler gives

$$\begin{aligned} P = & (p_1 - p_0 - u - 1)^2 + (T - d - 1)^2 \\ & + (X - 2p_0 - 2T)^2 + (X - 2p_1)^2 \\ & + (2p_1 - 2p_0 - h - 1)^2 + (2p_1 - 2p_0 - 2T)^2. \end{aligned} \quad (24.1)$$

The last row is the zero upper gap of the two segments consumed by the same event. Its redundancy with flight equations is harmless. Every $p_0 < p_1$ has the unique witness

$$T = p_1 - p_0, \quad X = 2p_1, \quad d = u = p_1 - p_0 - 1, \quad h = 2(p_1 - p_0) - 1.$$

For $(p_0, p_1) = (0, 1)$ the tuple (T, X, d, u, h) is $(1, 2, 0, 0, 1)$, corresponding to physical time $1/2$ and position 1 . There are five witnesses and six residuals. The final state is empty, so the prefix is also halting. Changing any one witness while fixing the others cannot keep all residuals zero.

24.2 A hidden collision exactly at the final cut

Use the four-speed interface $\{0, 2, 3, 4\}$, with labels a, b, c, d respectively, and the annihilation rule $\{d, a\} \mapsto \emptyset$. Start with

$$(\ell_0, \ell_1, \ell_2, \ell_3) = (d, a, d, a), \quad (p_0, p_1, p_2, p_3) = (0, 4, 10, 14).$$

There are two simultaneous collisions at physical time 1 , at positions 4 and 14 . Now submit a false one-batch schema containing only the first event and marking the last two signals as through-signals. Since $B = 12$, the first event's flight equations force $T_1 = 12$ and $X_0 = 48$.

The omitted pair's gap is

$$G(T) = 12(14 - 10) - 4T = 48 - 4T.$$

It is positive before the cut, has lower value 48 , and reaches zero at the cut. Thus conditions of the form $G(0) \geq 0$, $G(T_1) \geq 0$, and $G(0) + G(T_1) > 0$ would accept its endpoints. Our upper endpoint is strict because the pair survives rather than dying in one designated event. Its residual becomes

$$G(T_1) - h^+ - 1 = -h^+ - 1,$$

which cannot vanish for $h^+ \in \mathbb{N}$. The false schema has no certificate. The correct simultaneous schema consumes both pairs atomically and has event positions 48 and 168 on the scaled grid.

[write] Part I's independent simultaneous sites (Section 11.3) are another instance of this failure mode: an omitted, spatially separate simultaneous collision, rejected by the strict endpoint of the surviving pair (Section 1.2).

24.3 An outgoing fan at a finite cut

Replace the annihilation rule in the preceding setup by $\{d, a\} \mapsto \{b, c\}$. Each collision emits speeds 2 and 3. At the final cut the two members of each fan have zero gap and immediately separate; they need no zero-duration finite lifetime. In a halting certificate they instead receive terminal rays with zero starting gap and positive relative velocity. The gap between the two different fans is strict at the cut. Its relative velocity is $2 - 3 = -1$, so that terminal ray is rejected: the two fans will interact later. This separates valid finite-prefix geometry from the stronger no-future-collision requirement.

25 Witness height for the four-speed interface

Let the normalized machine speeds lie in $\{0, 2, 3, 4\}$, and let M_j be the largest live position immediately after batch j , with $M_0 = \max_j p_j$ and $\max \emptyset = 0$. Let t_j be elapsed physical time, with $t_0 = 0$.

Lemma 25.1 (Geometric growth). *Whenever batch $j + 1$ exists, its delay satisfies $\delta_{j+1} \leq M_j$. All positions at that boundary, including erased event positions, are at most $5M_j$. Consequently*

$$M_j \leq 5^j M_0, \quad t_j \leq M_0 \frac{5^j - 1}{4}.$$

Proof. A next collision must be approached by an adjacent closing pair. The pair has a strictly positive gap g and positive integer closing speed $c \geq 1$. Because all live positions are nonnegative, $g \leq M_j$; its meeting delay is $g/c \leq M_j$. The earliest collision delay is no larger. During this interval every signal advances by at most $4M_j$, so every pre-collision position is at most $5M_j$. Outputs remain at those event positions and deletion cannot increase the maximum. Iterate this bound and sum $\delta_i \leq 5^{i-1} M_0$ to obtain the stated geometric series. $\square$

The argument concerns an existing next batch. Empty or singleton configurations have none. Nor does a valid configuration whose maximum position is zero: any coincident outgoing fan has strictly increasing speeds and separates. This avoids dividing by a nonexistent positive gap.

Theorem 25.2 (Canonical height). *For $B = 12^K$, every canonical input coordinate and witness of the sparse or full-layer compiler is at most*

$$60^K \max(1, M_0). \tag{25.1}$$

Each such integer therefore has bit length $O(K + \log(M_0 + 1))$.

Proof. For the full speed set, the nonzero absolute differences are 1, 2, 3, 4, whose least common multiple is 12. The same 12 is a valid clearing modulus for a subset. Up to batch K , every live sample and every event position is at most $5^K M_0$, including positions subsequently erased. Multiplication

by 12^K bounds them by $60^K M_0$. Every endpoint gap is the nonnegative difference of two positions in this range, so its slack is smaller. Input-order slacks are at most M_0 . The time bound is

$$12^K t_j \leq 12^K M_0 \frac{5^K - 1}{4} \leq 60^K M_0,$$

and the scaled duration minus one is no larger than the corresponding scaled time. The empty and all-zero cases are covered by $\max(1, M_0)$; if $K = 0$, all remaining slacks are initial-gap slacks bounded by M_0 . Taking binary logarithms proves the bit bound. $\square$

This bounds actual zero witnesses, not arbitrary trial assignments. Degree two alone would supply no such height guarantee. Likewise, the bound is exponential in batch count K as a numerical magnitude while linear in K as a bit length. For a fixed machine with maximal speed V , the same argument gives $M_j \leq (1 + V)^j M_0$; the four-speed constant above is the concrete interface relevant here.

[write] Part I bounds the chamber's row coefficients (Proposition 7.3) and the size of *some* realizing seed (Theorem 8.1); Theorem 25.2 bounds the canonical witnesses of every realizing input. The two bounds measure different things and neither implies the other.

25.1 Exponential denominator growth can be necessary

Proposition 25.3 (A bounded geometric shuttle). *There is a fixed four-label machine with integer initial positions whose $2k$ -batch prefix contains a time of reduced denominator 3^{k-1} , although every event lies in $0 < t < 2$ and $0 < x < 6$. Thus a common integer grid for exact event geometry may require exponentially large scale in the number of batches.*

Proof. Give labels L, R, F, B speeds 3, 2, 4, 0, respectively, and use rules

$$\{F, R\} \mapsto \{B, R\}, \quad \{L, B\} \mapsto \{L, F\}.$$

Start with L, F, R at positions 0, 1, 2. The two walls follow $x_L(t) = 3t$ and $x_R(t) = 2 + 2t$, so their separation is $2 - t$ until they meet at $t = 2$. The first right collision occurs at $t_{R,1} = 1/2$ and the first left collision at $t_{L,1} = 1$. At a left collision at time $a < 2$, the outgoing speed-four shuttle catches the right wall after $(2 - a)/2$, giving right time $r = 1 + a/2$. Its speed-zero replacement is then caught by the left wall after $(2 - r)/3$, giving the next left time

$$a' = r + \frac{2 - r}{3} = \frac{4}{3} + \frac{a}{3}.$$

These positive delays place the shuttle strictly between the walls between alternating collisions; there are no other earlier meetings. The recurrence and first collision give

$$t_{L,k} = 2 - 3^{1-k}, \quad t_{R,k} = 2 - \frac{3}{2} 3^{1-k} \quad (k \geq 1).$$

The first six batch times are $1/2, 1, 3/2, 5/3, 11/6, 17/9$. At the k th left event the numerator of $(2 \cdot 3^{k-1} - 1)/3^{k-1}$ is coprime to three, so the reduced denominator is exactly 3^{k-1} . Event positions lie on a wall before time two and hence are below six. Every finite prefix is valid. $\square$

This example supplies a lower bound for common-grid geometry encodings: after $2k$ batches a clearing scale is divisible by 3^{k-1} , and its scaled last time requires $\Omega(k)$ bits. It makes the linear

event-count dependence in the witness-bit bound qualitatively sharp for this type of encoding; it is not a lower bound on arbitrary Diophantine representations. The collision times approach a Zeno accumulation at $t = 2$. We assert nothing about a continuation at or beyond that accumulation, which is outside the finite-schema theorem. The replay checks 64 batches with exact fractions.

[write] Part I's four-speed clock (Section 12, Proposition 12.1) is the same shuttle mechanism on a different machine, with speeds $0, 2, -1, -2$ and reduced denominators exactly $2 \cdot 3^k$ at event $2k$ for the seed $(1, 2)$. The worked-example receipt shipped as `data/11-signal-certificates-expected-worked_examples_receipt.json` records the last left denominator of the 64-batch prefix as $617673396283947 = 3^{31}$, which is 3^{k-1} for $k = 32$.

26 Fixed-schema arithmetic and the unbounded boundary

Proposition 26.1 (Semilinear feasible inputs). ([write] *The same as Part I, Theorem 10.1, for prefix semantics; source 11's proof is kept as a second route because it works with event-coordinate witnesses, and it adds the halting case.*) *For a fixed normalized machine, fixed initial label list and fixed schema, the set of natural input positions admitting a prefix certificate is semilinear. The same holds for the halting certificate. For a fixed total-event bound, the union over all schemas with that bound is also semilinear, provided the initial length and labels are fixed.*

Proof. A zero of a sum of squares is exactly a common zero of its affine residuals. Existence of natural witnesses is therefore an existential formula using only integer constants, addition, order and equality. Such Presburger-definable sets are semilinear [7, Theorem 1.3]. Terminal slope checks depend only on the fixed schema; they either remove it or leave the same argument.

For a total-event bound E_0 , at most $n + sE_0$ segment identities can have been created. A finite label set and bounded finite list admit finitely many consecutive partitions, legal rule choices and batch assignments with at most E_0 events. Thus there are finitely many schemas. Finite unions of semilinear sets are semilinear. $\square$

More concretely, a semilinear set is a finite union of sets $\mathbf{a} + \sum_{j=1}^m \mathbb{N}\mathbf{b}_j$. The proposition describes a strong boundary: a fixed externally supplied finite schema reduces to integer linear feasibility despite its quadratic packaging. Strict convexity does not import undecidability into that fixed finite system.

Enumerating all finite schemas gives an enumerable family of quadratics covering finite-event halting, with varying arities and coefficients. In a deterministic machine an actually halting run has a definite finite sequence of complete nonempty batches. That fact does not encode its arbitrary length into one fixed tuple. A single fixed-arity representation of unbounded halting requires an additional history-packing construction, with its own decoder, arithmetic costs and witness-multiplicity proof. Generic Diophantine representability results do not automatically preserve this compiler's quadratic degree, sparsity, or single witness. Nor does the construction certify continuation through a Zeno accumulation; no finite schema contains infinitely many events.

[write] For a fixed event budget, Part I's Theorem 10.2 packs the finite union over all first-hit skeletons into one quartic with unique selector and slack witnesses; it does not remove the budget either.

26.1 Strictly positive witnesses

If a convention requires witnesses in $\mathbb{Z}_{>0}$, replace every y_j by $Y_j - 1$ and define

$$Q_S(\mathbf{p}, \mathbf{Y}) = P_S(\mathbf{p}, \mathbf{Y} - \mathbf{1}).$$

This is an integer polynomial of the same degree. The coordinatewise translation is a bijection from positive witness tuples to natural witness tuples and preserves uniqueness and the Hessian. The input-coordinate convention remains unchanged unless it is translated separately and explicitly. This substitution is exact but is not an assertion that conversion is free in an arithmetic-operation ledger.

27 Verification evidence and research questions

The package supplies two exact-arithmetic compilers, a rational event simulator, and deterministic test suites. Arithmetic uses Python integers and `fractions.Fraction`; no floating-point solver or Diophantine oracle is involved. Table 6 summarizes reproduced checks. Counts of certificates include both prefix and halting variants when both are generated; they are not counts of distinct universal-machine programs.

Check	Full layers	Sparse	Additional
Certificates	122	444	878
Exact full-rank tests	30	40	12
Single-coordinate mutations	4 229	10 102	—
One-layer candidate schedules	832	832	—
Terminal certificates	—	198	372
Height checks	—	444	878

Table 6: Finite implementation evidence. The sparse suite includes 1 114 layers, 1 186 events, 3 982 lifetimes and a dedicated hidden-final-collision regression. The additional suite covers 506 fixtures and 2 152 local growth inequalities.

The full-layer suite additionally enumerates all 625 natural tuples in a small witness box for a two-signal example and finds its unique zero. The sparse suite’s original height test uses a factor-two safety margin; the additional suite checks the sharper constant-one bound (25.1). A release-specific example check verifies the complete annihilation polynomial, the omitted-final-collision obstruction, empty inputs, positive-witness translation, simultaneous events, output fans and the denominator-growth shuttle. Appendix C describes how to reproduce every reported receipt.

These checks can catch coefficient, endpoint and bookkeeping defects. They do not prove an infinite theorem, establish exhaustive novelty, or replace an independent formalization. The mathematical proofs above establish the stated scope.

[write] The counts in Table 6 equal the four expected receipts shipped as `data/11-signal-certificates-expected-*_receipt.json`. The replay was run at the write on a copy with the delivered layout, under Python 3.14.4 on Windows: all four receipts and both coefficient-example files matched. The repository’s review [14] replayed it as well and adds independent finite checks (Section 1.4).

27.1 Questions with concrete next steps

[write] Table 3 pairs these questions with the research directions of Part I: Questions 1 and 5 coincide with Sections 15.2 and 15.1, and Questions 3 and 4 overlap Sections 15.3, 15.4 and 15.5.

1. **Instantiate the universal table and input map.** Transcribe the 13-label, 21-non-blank-rule machine with an explicit convention for every blank collision. Verify the cyclic-tag input encoding and rational halting gadget, then compile and replay complete finite examples. This would connect the present implementation to the cited universal machine rather than only to its speed interface. ([write] Added 2 October 2026: still open as stated. Part III instantiates a different universal machine, Morita’s UTM(15,6) through Durand-Lose’s 2012 stacks; the 13-label table has not been transcribed.)
2. **Exploit conservative bounded populations carefully.** A reversible conservative rational construction is already Turing universal [5]. Its positive energy assignment bounds the number of simultaneously live signals for fixed total energy, but not the number of collisions. A direct unbounded representation would need to instantiate its finite transition branches, encode unbounded histories of rational distances, and pay for strict guards and input loading. Conservation alone neither bounds runtime nor excludes accumulation. ([write] Added 2 October 2026: answered in part, at a fixed horizon. Part III instantiates every finite transition branch of a reversible conservative universal signal machine with 18 live signals, pays for the strict guards and the loader, and excludes accumulation on encoded runs (Section 32.4); Part IV does the same for conservative lattice dynamics. Neither encodes unbounded histories of rational distances in a fixed arity.)
3. **Sharpen the denominator and cost ledger.** Causal depth can replace batch count in B . Further denominator cancellation may depend on the actual branch structure. A useful next result would make such improvements canonical and give a literal arithmetic-gate ledger, including the construction of coefficients and any conversion of input conventions. ([write] Causal-depth clearing itself is proved: Lemma 19.1 shows that D^H suffices, and Part I, Theorem 7.1, gives D^h event by event. What remains open is a canonical choice compatible with the witness format, and the gate ledger.)
4. **Compress histories without losing canonicity.** The sparse certificate removes repeated geometry, not the unbounded number of events. Can a restricted family of repeated collision motifs be packed with a unique decoder witness and explicit degree, height and operation bounds? Any answer should separate verification of a supplied history from existence of an outcome.
5. **Formalize geometric completeness.** The most valuable proof-assistant target is the typed-endpoint soundness theorem, including atomic simultaneous replacement and disappearing event blocks. The replay data provide regression fixtures, but the formal objects must model the real semantics, not only the executable’s output.

28 Conclusion

A finite signal-machine schema admits a direct, canonical Diophantine encoding because its collision geometry can be expressed by affine integer constraints on a predetermined rational grid. Sparse adjacency lifetimes make the certificate linear in event count and initial support. Correct endpoint types are what turn line-incidence equations into a complete execution certificate. The resulting

quadratic is strictly convex in its witness variables, and the four-speed case has an explicit linear witness-bit bound. The remaining unbounded problem is to encode arbitrary schema length without losing the quantitative and multiplicity properties that make this finite construction useful.

Part III

Conservative signal machines as Diophantine frontends (source 12)

[write] Part III prints batch-79 manuscript 16, *Conservative signal machines as Diophantine frontends*, an AI-assisted research note dated 2 October 2026, from the archive `Conservative_Signal_Frontend_Corrected.zip`. That archive is the corrected edition of the batch’s manuscript 07, which has the same article and is not printed separately (Section 37 and Appendix E). It is called source 12 after the prefix of its shipped files. Its labels carry the prefix `smc:cs::`; its sections are Sections 29–37 here, and its three appendices are Appendices F–H, after the provenance appendix of the report. Its statements keep their numbering by section; equations are numbered by section (the source numbered them consecutively). Source 12 was written without Parts I and II and names no repository commit; [write] notes at Lemma 34.1, in Section 35 and after Theorem 36.1 relate it to Parts I–II and to the collection, and Section 1.7 describes the Part. Its abstract follows.

We instantiate a fixed reversible, number-conserving universal signal machine with 114 meta-signals, 445 explicitly enumerated binary collision rules, and exactly 18 live signals on every encoded input. The source is Morita’s 15-state, 6-symbol reversible universal Turing machine, compiled through the stack construction of Durand-Lose. Rational finite-support tape encodings occupy a fixed spatial interval; an exact replay checks the published 184-step example at every Turing-machine read. For any fixed-population, equal-cardinality signal machine, complete simultaneous-event dynamics admits the integer branch normal form $N_j = c_j I - c e_j^T$. These gap matrices satisfy $N_j^2 = c_j N_j$ and have coefficients of absolute value at most 24 in the literal machine. Encoding finite label control in one additional coordinate yields an 18-dimensional partial piecewise homogeneous integer-linear map for the literal machine. A separate binary-alphabet existence construction gives dimension and population ten. Finally, a direct branch-copy construction produces a nonnegative quadratic polynomial with a unique natural witness for each legal step, and canonical quadratic certificates for every fixed finite horizon. No fixed-arity compression of unbounded computation is claimed. The release contains the complete collision table, exact arithmetic code, receipts, and reproducible article source.

29 Results and scope

Signal machines are an unusually direct esoteric substrate for arithmetic descriptions of computation: a finite set of point particles moves at fixed velocities, and only collisions change their labels. Their geometry is continuous, but every finite collision history starting from rational data is rational. The issue is to expose that arithmetic without silently assuming generic position, constant-time tape loading, or a fixed number of witnesses for an unbounded history.

The universal computational mechanisms here are prior work. Durand-Lose supplies reversible conservative stacks and a reversible Turing-machine compiler [16]; Morita supplies the particular universal transition table [17]. Event simulation by linear real-register operations is also established background [4]. This note makes their composition literal, derives a population-dimensional integer

normal form with full tie semantics, and gives its canonical quadratic finite-step certificate. Priority of this precise synthesis has not been established by an exhaustive literature search.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A *meta-signal* is a label equipped with a speed; a *live signal* is one instance at one position. A fixed finite set of labels is compatible with arbitrarily long computations. Population and alphabet size are different resources.

Theorem 29.1 (Literal universal frontend). *There is a fixed partial signal machine $\mathcal{U}_{18}$ whose complete finite data are included with this note, with the following properties.*

- (i) *It has 114 meta-signals and 445 collision rules. Every rule has exactly two inputs and two outputs, with distinct speeds on each side. Distinct input sets have distinct output sets.*
- (ii) *Its speeds are $\{-12, -9, -4, 0, 4, 9, 12\}$. The least common multiple of all positive speed differences is $D = 65520$.*
- (iii) *Each well-formed finite input to Morita’s explicitly halting cyclic-tag interpreter is loaded into exactly 18 rational-position signals. Before halt these signals remain in $[-8, 8]$. They simulate its reversible Turing machine exactly. The signal table does not depend on the input program or word.*
- (iv) *Designated terminal pairs produce a halt label and an escape token. The token undergoes exactly eight further crossings on these encoded configurations, then no further collision occurs. Infinite encoded computations have unbounded elapsed time.*
- (v) *At event cuts, 17 nonnegative integer gaps and one integer mode coordinate define an exact partial piecewise homogeneous integer-linear map on $\mathbb{N}^{18}$. Finite reachability of its designated accepting modes is universal under the explicit input recoding described below.*

Here “universal” imports the halting-preserving simulation theorem for Morita’s source machine; the finite replay is an implementation check, not a replacement proof of universality. The source distinguishes its phase-zero halt from exhaustion of the current word. Only $(q_1, \mathbf{b})$ is accepting; $(q_2, \$)$ is nonaccepting null termination. Both outcomes are observable separately in $\mathcal{U}_{18}$. Unspecified source instructions and collisions remain undefined.

Theorem 29.2 (Binary existence bound). *There exists a fixed reversible number-conserving universal signal machine with ten live signals on every effectively encoded input. It uses speeds $\{-6, -3, -2, 0, 2, 3, 6\}$ and has a nine-gap integer lift using the current positive pivot speed c_j as its scale multiplier. Including label control gives a partial piecewise homogeneous integer-linear map on $\mathbb{N}^{10}$ with universal finite accepting reachability. Each gap branch is a rank-eight scaled projection with at most 16 nonzero entries, all of absolute value at most 12. A uniform alternative multiplier is $D = 360$.*

Theorem 29.2 is an existence-by-compiler statement using established binary reversible Turing-machine simulation [18, 17]. This release does not print or instantiate the resulting universal binary interpreter. The fully literal result is Theorem 29.1. Neither bound is claimed optimal.

Theorem 29.3 (Canonical quadratic finite-step certificate). *For either fixed integer map, one can effectively construct a polynomial of total degree at most two which vanishes over natural arguments precisely on legal one-step input/output pairs. Its full auxiliary witness is unique whenever it exists. For each externally fixed horizon K , the same construction gives a quadratic certificate for K steps with size linear in K , the machine being fixed.*

For the literal machine, the release enumerates a forward-closed family of 49,700 modes and 80,501 branches, and supplies the complete numeric mode/event data, an exact sparse coefficient exporter, and a factored polynomial circuit. The one-step packet has exactly 4,196,998 auxiliary variables. It is not expanded into a flat monomial list; that representation is much larger. The binary existence result remains specified by a compiler. The horizon K determines the arity of the certificate. It is not another free variable in a single fixed-arity quadratic polynomial.

30 Primary sources and the literal interpreter

30.1 The source machine and its terminal convention

Morita's Theorem 3.3 and Table 5 specify a 15-state, 6-symbol reversible universal Turing machine with 62 quintuples [17]. Its states are $q_0, \dots, q_{14}$ and its tape alphabet is

$$S = (\mathbf{b}, \mathbf{Y}, \mathbf{N}, *, \$, 1),$$

with $\mathbf{b}$ the blank. A quintuple (q, a, b, d, s) means: in state q scanning a , write b , move one cell in direction $d \in \{-1, +1\}$, and enter s . The complete table is reproduced as mathematical data in Appendix F and in the release JSON.

Two cells have designated terminal meaning:

$$(q_1, \mathbf{b}) = \text{halt}, \quad (q_2, \$) = \text{null}.$$

The other 26 empty cells are unspecified. In particular, q_1 and q_2 are not unconditional halting states: they also have ordinary transitions. Treating the entire states as final would change the computation.

For each target state s , all incoming transitions have one common direction d_s , and their written symbols are distinct. These local conditions imply reversibility: the target state fixes the previous head location; the symbol there identifies the unique predecessor transition, prior state, and overwritten symbol. All 62 transitions were checked against these conditions.

30.2 The finite input language

The source interpreter uses the explicitly halting cyclic-tag convention of Morita's Definition 3.3. A program has $k \geq 1$ cyclic productions

$$(\text{halt}, p_1, \dots, p_{k-1}), \quad p_i \in \{\mathbf{Y}, \mathbf{N}\}^*.$$

An empty current word gives the outcome `null`. Reading $\mathbf{N}$ deletes it and advances the phase. Reading $\mathbf{Y}$ in a nonzero phase deletes it, appends that phase's production, and advances the phase. Reading $\mathbf{Y}$ in phase zero halts. The simulation theorem applies to this convention; no unproved identification with every other cyclic-tag convention is used.

Let $\text{rev}(p)$ denote reversal. At phase zero the finite program region is

$$P = \text{rev}(p_{k-1}) * \dots * \text{rev}(p_1) * \mathbf{b}. \quad (30.1)$$

The tape is $P\$w$, surrounded by infinite blank tails, and the head initially scans the first cell of w , or a blank if w is empty, in state q_0 . Equivalently, start with a delimiter for every production

including halt, then replace the rightmost delimiter by the blank phase marker. Other initial phases replace the appropriate delimiter counted from the right. There is finite nonblank support on both sides of the head; there is no nonblank periodic background.

For (halt, YN, YYN) and $w = \text{NYY}$, the source Figure 28 starts with

$$\text{NYY} * \text{NY} * \text{b\$NYY}.$$

The printed prose following Table 5 reverses the letters of p_1 relative to the example and figure. We use the general reversal convention and the figure, which agree with the full 184-step table replay. The release documents this discrepancy rather than silently changing the sample.

30.3 What is imported and what is checked

Durand-Lose’s 2012 Sections 3–4 give the stack construction and reversible-TM translation used here. The later shrinking construction in that paper is not used. The 2006 result also establishes reversible conservative universality, but omits some compiler detail and is not an enumerated fixed universal collision table [5]. The 2007 linear-BSS comparison already treats next-event arithmetic [4].

The literal collision table is a new finite instantiation of these ingredients, not a claim to a new universal computational mechanism. Its structural checks, source-tape comparison, and integer event identities are exact. They do not constitute proof-assistant formalization or an exhaustive audit of every source reduction.

31 Finite-support stacks with a fixed number of particles

31.1 Exact rational encoding

For an alphabet of $\ell \geq 2$ symbols, number the digits $1, \dots, \ell$, with digit 1 blank, and put $b = \ell + 1$. Encode a finite digit word followed by infinitely many blanks by

$$\sigma(d_1 \cdots d_m) = \sum_{i=1}^m \frac{d_i}{b^i} + \frac{1}{\ell b^m}. \quad (31.1)$$

Let C_ℓ be the set of all such values. Trailing blank digits give the same eventually blank tape, as they should. Delimiters in a source input language determine any required finite-word boundary.

Lemma 31.1 (Stack arithmetic). *$C_\ell \subset [1/\ell, 1)$ is closed under*

$$\text{push}_v(\sigma) = \frac{v + \sigma}{b}, \quad \text{pop}(\sigma) = (v, b\sigma - v), \quad v = \lfloor b\sigma \rfloor.$$

The pop digit belongs to $\{1, \dots, \ell\}$, the remainder belongs to C_ℓ , and no admissible value lies on a digit boundary v/b .

Proof. Equation (31.1) is the base- b expansion of an infinite digit stream in $\{1, \dots, \ell\}$ eventually equal to 1. Its minimum is the all-1 stream, $1/\ell$; its value is strictly below the all- ℓ stream, which has value 1. Removing or prepending a digit leaves an eventually-1 stream. Thus $b\sigma = v + \rho$ with $\rho \in [1/\ell, 1)$, proving the floor formula and excluding boundaries. $\square$

This is an invariant on the encoded domain. Arbitrary rationals in the same interval are not substitutes: for example $2/3$ is a forbidden binary boundary. The gadget has exact rational dynamics, not a rounding convention for malformed stack values.

31.2 The complete parametric collision gadget

There are stationary marks $m_0, \dots, m_\ell$ at positions $0, \dots, \ell$, one stationary memory signal at σ , and one operation token arriving from the right. Choose a positive speed c and put $a = \ell/(\ell + 2)$. The labels and speeds are

Labels	Speeds
m_i, mem	0
$\text{ml}, \text{sl}_v, \text{gl}, \text{vl}_v$	$-c$
$\text{mr}, \text{sr}_v, \text{ack}, \text{vr}_v$	c
$\text{catch}, \text{fix}, \text{gr}, \text{set}$	$c/3, -ac, ac, -c/3$

The nontransparent rules are, for $1 \leq v \leq \ell$,

$$\{m_0, \text{ml}\} \mapsto \{m_0, \text{mr}\}, \quad \{\text{mem}, \text{sl}_v\} \mapsto \{\text{ml}, \text{sr}_v\}, \quad (31.2)$$

$$\{\text{sr}_v, m_v\} \mapsto \{m_v, \text{catch}\}, \quad \{\text{mr}, \text{catch}\} \mapsto \{\text{ml}, \text{fix}\}, \quad (31.3)$$

$$\{\text{mr}, \text{fix}\} \mapsto \{\text{mem}, \text{ack}\}, \quad \{\text{mem}, \text{gl}\} \mapsto \{\text{ml}, \text{gr}\}, \quad (31.4)$$

$$\{\text{mr}, \text{gr}\} \mapsto \{\text{ml}, \text{set}\}, \quad \{m_v, \text{set}\} \mapsto \{\text{vl}_v, m_v\}, \quad (31.5)$$

$$\{\text{mr}, \text{vl}_v\} \mapsto \{\text{mem}, \text{vr}_v\}. \quad (31.6)$$

For every $1 \leq i \leq \ell$, add identity crossings of m_i with

$$\text{ml}, \text{mr}, \text{fix}, \text{ack}, \text{gl}, \text{gr}, \text{sl}_v, \text{vr}_v \quad (1 \leq v \leq \ell),$$

and with sr_v, vl_v only when $i < v$. All other encounters are undefined. This last clause is essential: completing every unspecified pair by an identity rule can create an output-set collision with a nontransparent rule and destroy reversibility.

There are $5\ell + 10$ labels and

$$5 + 4\ell + 6\ell + 2\ell^2 + \ell(\ell - 1) = 3\ell^2 + 9\ell + 5 \quad (31.7)$$

rules. Thus one binary module has 20 labels and 35 rules, while one six-symbol module has 40 labels and 167 rules. The gadget uses only $\ell + 3$ live signals, including its operation token.

Every displayed rule is two-to-two, with distinct input and output speeds. Its output set determines which displayed case and, where applicable, which v occurred. The transparent outputs are the corresponding untouched inputs, and the listed transparent cases exclude the nontransparent outputs. This proves local rule injectivity; the release also checks it mechanically.

31.3 Geometric proof of push and pop

For push, take time zero at the encounter of **mem** with **sl_v** at $x = \sigma$. The memory moves left at speed c , reflects at zero, then moves right at speed c . Meanwhile **sr_v** goes right to mark v and produces a token of speed $c/3$. The reflected memory and that token meet at

$$x_0 = v + \sigma.$$

They produce leftward memory of speed c and a **fix** token of speed ac . After the memory reflects once more, their encounter has coordinate

$$x_0 \frac{1-a}{1+a} = \frac{v+\sigma}{\ell+1}.$$

The memory becomes stationary there and the acknowledgement leaves to the right. This is exactly push.

For pop, at the first memory/get encounter the memory moves left and reflects while the get token moves right at speed ac . Their meeting coordinate is

$$\sigma \frac{1+a}{1-a} = (\ell+1)\sigma = v + \rho, \quad \rho \in C_\ell.$$

The new set token moves left at speed $c/3$, so its first lower integer mark is exactly v . That mark turns it into the leftward value token vl_v of speed c . The reflected memory meets this token at ρ , becomes stationary, and returns vr_v to the right.

All additional marker crossings are precisely those in the transparent list. The values $\sigma, \rho \in [1/\ell, 1)$ keep the critical meetings away from integer marks. Each construction has a fixed finite number of moving segments and finitely many marker crossings, hence a uniform finite collision bound for fixed ℓ . Its largest local coordinate is less than $\ell + 1$. The memory and operation token remain the same two live resources throughout.

32 Reversible control and the fixed population

32.1 Alternating write and move form

Let a reversible Turing machine have disjoint write states Q_W and move states Q_M , with partial maps

$$\delta_W : Q_W \times S \longrightarrow Q_M \times S, \quad \delta_M : Q_M \longrightarrow Q_W \times \{L, R\}.$$

Require δ_W to be injective and the projection of δ_M onto Q_W to be injective. Thus each target write state has a unique preceding move and a fixed incoming direction. Require also that every move-state in the image of δ_W is in the domain of δ_M , and every used noninitial write or terminal state has an incoming move. The initial read side must agree with any existing incoming direction. A designated terminal *pair* (q, a) has no writing transition; other undefined pairs remain forbidden. The compiler may alternatively be given states whose every symbol is terminal, but this stronger condition is not imposed on Morita's table.

For a reversible quintuple table with nonzero movements, replace each transition (q, a, b, d_s, s) by the write $(q, a) \mapsto (r_s, b)$ followed by $r_s \mapsto (s, d_s)$, using one fresh move state for each target s . The local reversibility conditions from Section 30 give exactly the two injectivity requirements. Morita's table yields 15 move states and 62 writing transitions. Its q_0 has incoming direction right, so the initial read can use the right module consistently even though q_0 has predecessors.

32.2 Layout and tape invariant

Put $R = \ell + 2$. The left module has marks at $-R + i$ and memory at $-R + \sigma_L$; the right module is its reflected, disjointly labelled copy, with marks at $R - i$ and memory at $R - \sigma_R$. Put the current controller state at zero. There is one travelling operation/value token shared by the modules. The population is

$$2((\ell + 1) \text{ marks} + 1 \text{ memory}) + 1 \text{ controller} + 1 \text{ token} = 2\ell + 6. \quad (32.1)$$

In particular, $\ell = 6$ gives 18 and $\ell = 2$ gives ten. The available state labels depend on the fixed interpreter, but the number of live controller signals is always one.

At a controller read, the arriving value token holds the scanned symbol; the two stationary memories hold the cells strictly left and strictly right of the head, nearest cell first. Suppose the current write sends (q, a) to (r, b) and the next move is $r \mapsto (s, d)$. The controller sends a push- b token to the side opposite d . On acknowledgement it changes to s and sends a pop request to side d . The returned value is the new scanned symbol. This preserves the tape invariant and performs exactly one source transition.

Distinct writing rules cannot share an output unless they share both (r, b) , which injectivity forbids. Distinct acknowledgement rules cannot share an output unless they share the target s , also forbidden. The writing and acknowledgement output phases have disjoint controller-state sets. Gadget outputs contain no controller state. These observations prove injectivity of the compiled control rules and their separation from the gadget rules.

32.3 Terminal exits and conservation

For each designated terminal pair (q, a) , replace its arriving controller/value pair by a fresh stationary label $H_{q,a}$ and a fresh token $E_{q,a}$ moving right at maximum speed. Add transparent crossings of this escape label with every stationary marker and memory label. All these rows are two-to-two and their fresh labels preserve output injectivity.

At that time both memories and all marks are stationary. On the right there are $\ell + 1$ marks and one memory, all at distinct positions. Hence exactly $\ell + 2$ further crossings take the token to the rightmost mark; thereafter it has no future encounter. This is eight crossings for the literal machine and four for the binary construction. The larger bound of all $2\ell + 4$ stationary module signals is also immediate but unnecessary.

A terminal mode is recorded by the presence of $H_{q,a}$. Only the specified accepting subset is a reachability target: in the literal instance this is $H_{q_1,b}$, while $H_{q_2,s}$ records nonaccepting termination. Such a cut can precede the last escape crossing. If a collision-free cut is desired, additionally require the token to be rightmost and every adjacent velocity difference nonpositive. For an accepting label the two acceptance cuts differ by a fixed finite suffix. Acceptance is never a Zeno-limit convention.

Giving every signal energy one makes every rule conserve both number and this positive energy. Local reversibility refers to the partial collision table and finite well-defined flow; no uniqueness assertion about continuation through singular accumulation points is needed.

32.4 No accumulation on encoded runs

All active stack geometry remains outside $(-1, 1)$, apart from the controller token travelling between the center and a gadget. A center-to-gadget-to-center exchange covers distance at least two at speed at most $c_{\max}$, so takes at least $2/c_{\max}$ time. Each exchange contains only uniformly finitely many collisions, by Section 31.3. Thus an infinite encoded computation requires infinitely many exchanges and unbounded time. A finite number of source steps has finitely many events and finite duration. This excludes Zeno acceleration on the encoded domain without making a claim about every malformed rational configuration of the partial machine.

The outer marks remain at $-R, R$. At every event cut on an encoded run, including the final escape crossing, the span is exactly $2R$. The token later flies beyond R , but those later non-event observation times are not states of the event map.

32.5 The ten-signal existence specialization

Morita, Shirasaki and Gono give a simulation by a one-tape two-symbol reversible Turing machine [18]; the blank is included in that two-symbol alphabet [17]. Fix one universal ordinary interpreter and apply that conversion. If its reversible description uses separate write/move quadruples, writing becomes a zero-move quintuple and a movement becomes one identity-writing quintuple per symbol. Alphabet size does not change.

Factor transitions by their target state as above. A zero-move entry into s can be replaced by a right move into a fresh write state z_s , an identity write into a fresh move state t_s , and a left move into s . Both new maps satisfy the same injectivity conditions. The detour restores the head and tape. Initial entry is chosen consistently with the existing incoming direction, using the two orientations of the loader below; if there is no incoming move either entry can be chosen. Halting can be designated by the finitely many undefined terminal pairs. Applying the binary gadget proves Theorem 29.2; this is a compiled-existence argument, not a second printed universal table.

33 The exact input loader and its cost

33.1 Both finite tape halves

At the initial head position, list the cells strictly to its left in decreasing index order to form w_L . List the scanned cell and the cells to its right in increasing order to form w_R . Include every nonblank cell; omitted tails are blank. This works whether finite data lie on one side or both sides of the head, as happens in Morita's program/data layout.

Let $m = \max(|w_L|, |w_R|)$, append blank digits to the shorter word, and define

$$Q = \ell b^m, \quad A_\epsilon = \ell \sum_{i=1}^m d_{\epsilon,i} b^{m-i} + 1 \quad (\epsilon \in \{L, R\}).$$

Then $\sigma_\epsilon = A_\epsilon/Q$. Place the initial pop token at $x = 1$ pointing toward the right module. All initial labels and their order are fixed. Only two memory coordinates vary.

The ordered gap numerators, with a run of r equal entries denoted $Q^{[r]}$, are

$$h_0 = (A_L, Q - A_L, Q^{[\ell-1]}, 2Q, Q, Q, Q^{[\ell-1]}, Q - A_R, A_R). \quad (33.1)$$

There are $2\ell + 5$ entries, whose sum is $2(\ell + 2)Q$. For the binary case this is

$$h_0 = (A_L, Q - A_L, Q, 2Q, Q, Q, Q, Q - A_R, A_R). \quad (33.2)$$

For the literal machine it gives 17 gaps and span $16Q$. The initial mode coordinate, introduced below, is its fixed mode code times this sum.

If the initial state instead has left incoming direction, include the scanned cell in w_L and exclude it from w_R . Place a leftward pop token at -1 . The binary gap vector becomes

$$(A_L, Q - A_L, Q, Q, Q, 2Q, Q, Q - A_R, A_R).$$

For general ℓ , the vector is

$$(A_L, Q - A_L, Q^{[\ell-1]}, Q, Q, 2Q, Q^{[\ell-1]}, Q - A_R, A_R).$$

These are two fixed entry conventions, each with a fixed initial label order. The literal six-symbol interpreter uses right entry.

33.2 Arithmetic and representation cost

The rational loader uses $O(m)$ multiply-by- b and add-digit operations on $O(m)$ -bit integers for fixed alphabet. This is linear in the prepared tape-word length in an arithmetic-operation model; it is not a constant-cost conversion from a raw base-two integer or a claim of linear bit complexity with arbitrary multiplication primitives. Ordinary binary input integers require digit extraction, source-interpreter recoding, length/power construction, and base conversion. Those costs must be paid either in preprocessing or by a separate arithmetic representation theorem.

The source program in (30.1) has length linear in its explicit production description, and its two tape halves can be obtained effectively. The separate universal reduction into that source language is imported from the cited simulation theorem, not counted as free computation.

Keeping rational positions gives a fixed physical box. Scaling all positions and times by Q instead gives integer initial positions with the same fixed speed and rule tables, at the price of input-dependent physical scale. These two conventions have the same live population and event ordering.

34 The complete simultaneous-event normal form

The next results do not depend on stacks or Turing machines. Fix any finite partial signal machine with rational speeds and fixed population $n \geq 2$, whose rules preserve the cardinality of each collision block and whose input and output sets each have pairwise distinct speeds. Rescale the time unit once so all speeds are integers.

34.1 Post-event germs and first-collision guards

A state at an event cut is an ordered label word $a_1, \dots, a_n$ and a gap vector $g \in \mathbb{Q}_{\geq 0}^{n-1}$. Put $v_i = v(a_i)$ and $c_i = v_i - v_{i+1}$. We require

$$g_i = 0 \implies v_i < v_{i+1}. \quad (34.1)$$

Thus signals emitted at the same point are sorted by strictly increasing outgoing speed. This is a post-event *germ*, not an arbitrary coincident snapshot. It prevents zero-time recollision and equal-speed overlap. Strictly ordered initial slices satisfy the condition automatically.

If every $c_i \leq 0$, there is no future collision. Otherwise each closing edge has $g_i > 0$ by (34.1); its putative encounter time is g_i/c_i . Let J be the nonempty set attaining the smallest such time and fix the canonical pivot $j = \min J$. For a fixed mode and candidate J , the exact branch guards are

$$c_j > 0, \quad g_j > 0, \quad (34.2)$$

$$c_j g_i - c_i g_j = 0 \quad (i \in J), \quad (34.3)$$

$$c_j g_i - c_i g_j > 0 \quad (i \notin J). \quad (34.4)$$

The first condition is a compile-time test on speeds. These guards imply that every $i \in J$ is closing; equality would otherwise contradict nonnegativity and the germ condition.

Lemma 34.1 (Complete event selection). *Under the germ condition and guards (34.2)–(34.4), the next positive collision time is $\tau = g_j/c_j$. Exactly the adjacent edges in J have zero gap then. The maximal connected components of J are the complete collision blocks.*

Proof. Every gap evolves affinely as $g_i - c_i t$. At $t = \tau$, multiplication by $c_j > 0$ gives exactly the expressions in (34.3)–(34.4). The endpoint gaps are nonnegative, zero precisely for $i \in J$. An initially positive gap remains positive for $0 < t < \tau$, as a convex combination of its positive initial value and nonnegative endpoint. An initially zero gap has $c_i < 0$ and is strictly positive for every $t > 0$. Hence no collision is earlier. Zero endpoint gaps join exactly the contiguous signals occupying the same endpoint position. $\square$

[write] Lemma 34.1 is the one-batch case of the chamber theorem of Part I, Theorem 5.3, with its typed endpoints (Lemma 5.2); Part II proves the same step as Lemmas 21.1 and 21.2. The guards (34.3)–(34.4) are exactly Part I’s equality and strict rows for one batch, written after multiplication by the positive pivot speed, and a batch with several components of J is a batch of several sites in Part I’s terms. The one addition is the germ condition (34.1): Parts I and II start from strictly ordered positions, while an event cut here may carry zero gaps between diverging outputs, and the proof’s sentence on initially zero gaps disposes of them. Source 12 was written without these Parts; its proof is kept as a second route. What is new in this Part is the use made of the lemma: the mode-indexed integer map of Section 35, the pivot-scaled lift, and the literal universal machine to which they are applied.

For every nontrivial collision block, its exact incoming set must be a defined rule input. Apply all such rules atomically and sort each output by speed. An undefined block has no branch. Thus a triple arrival cannot be replaced by a preferred pairwise rule, and simultaneous collisions at distinct locations cannot be processed with an accidental temporal priority. The literal table has only two-input rules, so a genuine triple collision is undefined.

34.2 Why no coordinate permutation is needed

The endpoint gap vector before label replacement is

$$\bar{g} = A_j g, \quad A_j = I - \frac{c e_j^\top}{c_j}. \quad (34.5)$$

Each collision replaces k signals at one position by exactly k signals at that position. Therefore the ordered multiset of positions is unchanged by replacement. Reordering outgoing labels at a coincident point changes the mode, but not the gap vector. Consequently $g' = \bar{g}$: equation (34.5) is the entire branch map even when several collision blocks occur simultaneously.

The matrix is a rank-one *correction* to the identity, not a rank-one matrix. Equal-cardinality at each block is important. Merely conserving the total population across a collection of events would not by itself imply this coordinate identity.

35 Integer lifting and internalized finite control

35.1 The pivot-scaled integer lift

For a branch with canonical pivot j , the rational projection from (34.5) has only the denominator c_j . Multiplying by that very denominator gives

$$h_{k+1} = N_j h_k, \quad N_j = c_j I - c e_j^\top \in \mathbb{Z}^{(n-1) \times (n-1)}. \quad (35.1)$$

Choose Q clearing the initial gap denominators and let

$$\lambda_0 = Q, \quad \lambda_{k+1} = c_{j_k} \lambda_k, \quad h_k = \lambda_k g_k. \quad (35.2)$$

Here c_{j_k} is the positive closing speed of the uniquely selected branch at step k . All guards are homogeneous, so they apply unchanged to h_k . The multiplier is determined by the branch, and no extra scale coordinate, division, gcd reduction, or denominator witness is needed in the integer map.

Conversely, a legal integer orbit determines its branch sequence, hence all λ_k . Set $g_k = h_k / \lambda_k$ and reconstruct the event delay as

$$\tau_k = \frac{h_{k,j_k}}{\lambda_k c_{j_k}}.$$

Lemma 34.1 and induction reconstruct exactly the rational geometry, up to global translation. This proves finite-orbit equivalence in both directions, including simultaneous events.

Proposition 35.1 (Sparse scaled projections). *For every branch with $c_j > 0$,*

$$A_j^2 = A_j, \quad N_j^2 = c_j N_j, \quad \text{rank } N_j = n - 2.$$

Row j vanishes. Every other row has a diagonal entry c_j and possibly one further entry in column j . There are at most $2(n - 2)$ nonzero entries. If all speed differences have absolute value at most V , every matrix entry has absolute value at most V .

Proof. Since $e_j^\top c = c_j$, multiplication expands to $A_j^2 = A_j$. Its kernel is exactly the span of the nonzero vector c : $A_j c = 0$, while $A_j x = 0$ implies $x = c x_j / c_j$. Its rank is $(n - 1) - 1$. Since $N_j = c_j A_j$, its square and rank follow. For $i \neq j$ the only possible nonzero row entries are c_j in column i and $-c_i$ in column j , proving the sparse and coefficient bounds. $\square$

For the literal machine, the 17-dimensional gap matrices have rank 16, at most 32 nonzero entries, and entries of absolute value at most 24. The binary existence construction has nine-dimensional gap matrices of rank eight, at most 16 nonzero entries, and entries of absolute value at most 12. These bounds do not depend on the source interpreter's number of states.

35.2 A uniform alternative for comparison

It is also valid to use one fixed multiplier

$$D = \text{lcm}\{|v - v'| : v \neq v' \text{ are available integer speeds}\},$$

with $D = 1$ if the set is empty. This gives $\tilde{h}_k = Q D^k g_k$ and

$$\tilde{h}_{k+1} = M_j \tilde{h}_k, \quad M_j = D I - \frac{D}{c_j} c e_j^\top, \quad M_j^2 = D M_j. \quad (35.3)$$

The literal machine has $D = 65520$, with coefficient bound $8D = 524160$ because the minimum positive speed difference is 3. The binary speeds give $D = 360$ and coefficient bound 4320. This alternative is exact and useful for cross-checking the replay, but is unnecessary for a pure integer reachability frontend: the pivot-scaled matrices are much smaller. Both conventions have the same homogeneous guards and mode sequence.

[write] The multiplier D is the least common multiple of the nonzero speed differences used in Parts I and II ((7.1); (18.1)), and the uniform lift is their device: after k events every gap is an

integer once multiplied by QD^k , which is the event-by-event denominator bound of Theorem 7.1 (with the event count bounding the causal depth) and the grid $B = D^K$ of Part II on which Lemma 19.1 places the event coordinates. The pivot-scaled lift (35.2) sharpens that device for one branch sequence: it multiplies by the actual pivot speed c_j instead of D and needs no extra coordinate, and on the primary example it ends with a largest gap numerator of 39,320 bits against 179,449 under the uniform QD^k lift (Section 37).

35.3 Control uses one coordinate

There are finitely many ordered n -label modes; enumerate them injectively by positive integers q . This can be done lexicographically, including modes later discarded as invalid. Put

$$S = \mathbf{1}^\top h, \quad u = qS, \quad X = (h, u) \in \mathbb{N}^n. \quad (35.4)$$

Require $S > 0$. This mode encoding is used on the positive-span invariant domain of the compiled machines. An arbitrary signal machine can collapse its entire population to one point; at such a zero-span cut this particular coordinate does not distinguish outgoing modes and needs a separate convention. The generic gap theorem itself does not require positive span. In fact, positive span is invariant under every legal step of either compiled table, not only its encoded runs: zero output span would mean that all n particles form one complete collision block, requiring an n -input rule, whereas these tables have only two-input rules and $n > 2$. Each branch for mode q includes the homogeneous equality $u - qS = 0$, applies (35.1), and sets

$$u' = q'\mathbf{1}^\top N_j h, \quad (35.5)$$

where q' is the uniquely determined outgoing mode. These equations define a homogeneous integer-linear matrix $\widehat{M}_r$ on all n coordinates. Its last input column is zero; the mode is read through its guard. The small coefficient bounds 24 and 12 apply to the gap block; the added mode row contains the fixed, potentially large codes q' .

The guard also includes the germ conditions. On natural gap variables these are particularly simple: whenever $c_i \geq 0$, require $h_i > 0$; when $c_i < 0$, no extra constraint is needed beyond $h_i \geq 0$. The remaining guards are the first-collision equalities and inequalities. They are homogeneous rational polyhedral-cone conditions, with strict positivity interpreted on natural arguments as value at least one.

For the literal release, give the 114 labels their zero-based lexicographic indices $\iota(a)$ and use the explicit code

$$q(a_1, \dots, a_{18}) = 1 + \sum_{i=1}^{18} \iota(a_i) 114^{18-i}.$$

The initial and final collision-free modes have codes

$$\begin{aligned} q_{\text{initial}} &= 473631179350110573467332809270905830, \\ q_{\text{accept}} &= 473631179350110518412162422097252939, \\ q_{\text{null}} &= 473631179350110521664110943253890124. \end{aligned}$$

Their exact label orders and decimal codes are included in the machine JSON. Thus $u_0 = 16q_{\text{initial}}Q$, while the final accepting target is the single equality $u = q_{\text{accept}} \sum h$ on admissible post-event germs. The null code is a different nonaccepting target. The code values are serialized as strings to avoid loss of precision in JSON consumers using floating-point numbers.

Distinct mode codes cannot overlap when $S > 0$. For a fixed valid mode and gap vector, the set of minimizing edges is unique and its pivot is fixed. The branch guards are consequently disjoint. Enumerating every mode and every nonempty subset of edges, discarding impossible or undefined cases, is a complete finite branch compiler. This proves the pure n -coordinate claims of Theorems 29.1 and 29.2.

35.4 Limits of the normal form

No injectivity of the integer event map is asserted. Reversing a signal flow, quotienting global translation, forgetting elapsed time, taking an event section, and changing the common scale are different operations. The original partial collision table is reversible; that does not automatically make (35.1) invertible.

For a general configuration, summing the pivot-scaled update gives

$$\mathbf{1}^\top N_j h = c_j \mathbf{1}^\top h - (v_1 - v_n) h_j.$$

It is not valid in general to discard the second term. On encoded runs, the extreme markers are stationary at the input of every event, including the last escape crossing. Their fixed event-cut span gives

$$S_k = \sum_i h_{k,i} = 2(\ell + 2)\lambda_k, \quad S_{k+1} = c_{j_k} S_k.$$

At the last cut the token may be the rightmost outgoing germ, but there is no subsequent event requiring stationary extrema. For all these encoded cuts the geometry and time can be decoded directly from the integer state:

$$g_k = \frac{2(\ell + 2)}{S_k} h_k, \quad \tau_k = \frac{2(\ell + 2)h_{k,j_k}}{S_k c_{j_k}}.$$

For the literal machine $S_k \leq 16Q 24^k$, and for the binary specialization $S_k \leq 8Q 12^k$. Every gap is bounded by S_k , and $u_k \leq q_{\max} S_k$. Their bit length is $O(k + \log Q)$ for the fixed machine. Fixed coordinate dimension does not make integer-height growth free.

Exact points and accepting modes are different targets. In the literal mode closure of Appendix H, every active branch has stationary extreme labels, and every pivot speed is at least 3. Thus $S' \geq 3S$. Given two exact positive-span integer states $X_{\text{in}}, X_{\text{out}}$ in this domain, point-to-point reachability is decidable by deterministic simulation for at most $\lfloor \log_3(S_{\text{out}}/S_{\text{in}}) \rfloor$ steps when the ratio is at least one; a smaller target span rules out a positive-length run. For the fixed machine this costs polynomial time in the binary endpoint length. The universal question is reaching the fixed accepting mode or cone with unbounded output span, not reaching one prescribed integer point. No endpoint-size bound for acceptance is supplied.

Optional exact elapsed time. One additional natural coordinate suffices if the absolute collision timestamp is wanted. Starting at time zero, let $p_k = \lambda_k t_k$. The same branch updates it by

$$p_{k+1} = c_{j_k} p_k + h_{k,j_k}, \quad p_0 = 0.$$

Indeed, multiplying $t_{k+1} = t_k + h_{k,j_k}/(\lambda_k c_{j_k})$ by $\lambda_{k+1} = c_{j_k} \lambda_k$ gives exactly this integer-linear row. On literal encoded runs the timestamp is $t_k = 16p_k/S_k$. The timed frontend has 19 coordinates,

or 11 in the binary specialization; the extra row still has coefficients bounded by 24, respectively 12. Its finite-horizon packet remains quadratic. This optional coordinate is separate from the 18-dimensional untimed reachability theorem.

The dimension retains every physical collision and is not an optimized arithmetic state dimension. For comparison, at completed Turing-machine reads a standard two-stack macro-description can use (A_L, A_R, Q, u) , with u coding the finite state and scanned-symbol pair times Q . A right move writing w and popping v from the right can use

$$Q' = 7Q, \quad A'_L = A_L + wQ, \quad A'_R = 49A_R - 7vQ,$$

with digit guard $vQ < 7A_R < (v+1)Q$; left moves are symmetric. This coarser quotient skips internal collisions and is distinct from the sparse projection frontend. Elementary stack arithmetic and the selector/copy technique are used as standard algebraic mechanisms, not claimed as new inventions.

[write] The selector/copy construction of Section 36, in which one selector per branch and one copy of the input per branch represent a disjoint union of polyhedral pieces, is the disjunctive (convex-hull) formulation of a union of polyhedra due to Balas [26, 27], here in homogeneous form with the selectors entering the strict guards; Jeroslow and Lowe [28] characterized the sets that such formulations represent with integer variables. Source 12 cites neither; the credit is added here, and source 12 claims no novelty for the technique.

36 A direct canonical quadratic Diophantine packet

36.1 Finite linear branch systems

Let F be a partial map on $\mathbb{N}^d$ with finitely many disjoint branches $r = 1, \dots, B$. Branch r has update $Y = \widehat{M}_r X$, where $\widehat{M}_r$ has integer entries, and finitely many homogeneous integer-linear guards of three kinds:

$$L(X) = 0, \quad L(X) > 0, \quad L(X) \geq 0.$$

The event map above has this form with $d = n$. Naturality of the input coordinates is part of the domain; its mode, germ, nonzero-span, first-collision, tie, and collision-legality conditions are included in the branch description. Collision legality is a finite compile-time filter.

For every branch introduce a selector $b_r \in \mathbb{N}$, a local input copy $X_r \in \mathbb{N}^d$, and one slack $s_{r,L} \in \mathbb{N}$ for each strict or weak inequality. Form the following affine residuals:

$$\sum_{r=1}^B b_r - 1, \tag{36.1}$$

$$X_i - \sum_r X_{r,i} \tag{36.2} \quad (1 \leq i \leq d),$$

$$Y_i - \sum_r (\widehat{M}_r X_r)_i \tag{36.3} \quad (1 \leq i \leq d),$$

$$L(X_r) \tag{36.4} \quad \text{for each equality guard,}$$

$$L(X_r) - b_r - s_{r,L} \tag{36.5} \quad \text{for each strict guard,}$$

$$L(X_r) - s_{r,L} \tag{36.6} \quad \text{for each weak guard.}$$

Let $\mathcal{R}$ be this list and define the explicit polynomial

$$P(X, Y; \mathbf{z}) = \sum_{R \in \mathcal{R}} R^2 + \sum_{r=1}^B \left(\sum_{h \neq r} b_h \right) \left(\sum_{i=1}^d X_{r,i} \right). \quad (36.7)$$

All sums here are finite and their coefficients are directly computable from the branch table. The second group consists of nonnegative complementarity products on natural arguments. No appeal to MRDP is involved.

Theorem 36.1 (Exactness and canonical witness). *For $X, Y \in \mathbb{N}^d$, there exists $\mathbf{z} \in \mathbb{N}^A$ with $P(X, Y; \mathbf{z}) = 0$ if and only if $Y = F(X)$. Whenever such a tuple exists, it is unique. The polynomial has total degree at most two and is nonnegative on every natural assignment.*

Proof. Every square and every displayed complementarity product is nonnegative on natural arguments. A zero therefore makes all terms vanish. Equation (36.1) forces exactly one selector, say b_{r_*} , to be 1 and all the others to be zero. For every $r \neq r_*$, the first factor of its complementarity term is 1, so all coordinates of X_r are zero. Equation (36.2) then gives $X_{r_*} = X$.

Homogeneity makes every inactive guard form zero; equations (36.5) and (36.6) consequently force its slack to zero. On the active branch an equality guard vanishes, a strict guard has value $1 + s_{r_*,L} \geq 1$, and a weak guard has value $s_{r_*,L} \geq 0$. Thus X lies on that branch and (36.3) gives $Y = \widehat{M}_{r_*} X$.

Conversely, for a legal step set its unique branch selector to 1, set its local copy to X , set its strict slack to $L(X) - 1$ and weak slack to $L(X)$, and set every inactive quantity to zero. These are natural values and every term vanishes. Disjoint branch guards fix the active selector, and all copies and slacks are then forced, proving uniqueness. The residuals are affine and the additional products are quadratic, proving the degree bound. $\square$

The product must use $\sum_{h \neq r} b_h$, as written. Replacing it by $1 - b_r$ would lose global nonnegativity on arbitrary natural assignments before the selector equation has been enforced. The polynomial is not asserted to be convex or a sum of squares over all real arguments.

[write] Two remarks place Theorem 36.1 in the collection. (i) Source 12 states nonnegativity on natural arguments only, but P is nonnegative on the whole real nonnegative orthant: the residuals enter squared, and each complementarity product $(\sum_{h \neq r} b_h)(\sum_i X_{r,i})$ is a product of two affine forms with nonnegative coefficients. So (36.7) has exactly the form $\sum_i A_i^2 + \sum_j B_j C_j$ of the spine of *quadratic-orthant-certificates* (its equation `qoc:eq:spine`), the format of that report's three Parts. This is a statement about the sign of P only; no claim is made here that the nonnegative real zeros of P are natural. (ii) The one-step graph $\{(X, F(X))\}$ of a homogeneous-guarded piecewise-linear map on $\mathbb{N}^d$ with finitely many branches is a semilinear set, so the *existence* of a single-fold representation of degree two, nonnegative on the real orthant, also follows from Theorem `cdc:of:thm:singlefolds1` of Part XI of *canonical-diophantine-certificates*, via Presburger quantifier elimination. The construction here is explicit, needs no elimination, and carries the exact ledger (36.8)–(36.9); that is what this Part contributes, not the existence statement. The fixed-horizon sum (36.10) is of the same “one natural zero, fixed horizon” kind as the trace polynomial of Theorem `pqc:pm:thm:trace` in *probabilistic-quantum-and-continuous-computation*, which handles a different dynamics with a clamp gadget; that network is total, so its trace polynomial has exactly one natural zero, whereas the event map here is partial and the natural fibre of (36.10) is empty or a singleton, “one natural zero” meaning at most one. (The text after the semicolon was added on 2 October 2026, after finding 3 of the review cited at the end of the next paragraph.)

[write] **Added 2 October 2026 (batch 79): the strong selector gates of *quadratic-orthant-certificates*.** Part V of that report (its source 16, batch-79 manuscript 09, written in commit 11abe5008) uses the same selector device for the branches of literal counter programs, with one change: its products $(\sum_{k \neq r} e_{j,k})(e_{j,r} + \sum_i x_{j,r,i})$ (its equation `qoc:um:eq:polynomial`) add the selector itself to the right factor. Two positive selectors would then make some product positive, so the gate forces one-hot selection over the nonnegative reals, and for natural inputs the nonnegative real zero fibre of that polynomial is its natural fibre (its Theorem `qoc:um:thm:poly`); its source 17 isolates the gate as Lemma `qoc:rn:lem:gates` and uses it again for reset-net traces in its Part VI. The products of (36.7) lack that term, and source 12 claims exactness over natural witnesses only. An elementary observation, not made by either source, shows how far the present gates reach. At a nonnegative real zero of (36.7), if two selectors were positive, every first factor $\sum_{h \neq r} b_h$ would be positive, so every local copy X_r would vanish and $X = 0$ by (36.2). Hence, for natural $X \neq 0$ and natural Y , exactly one selector equals one, every inactive copy and slack vanishes, the active slacks are the integers $L(X) - 1$ and $L(X)$, and the nonnegative real fibre is the natural one. At $X = 0$ all local copies vanish, and (36.5) forces $b_r = 0$ for every branch with a strict guard. Every branch with only homogeneous equality and weak guards contains zero; disjointness therefore allows at most one such branch. The selector equation then makes the whole packet's nonnegative real fibre natural at zero as well (or empty). The weak gates alone do not enforce one-hot selection at zero. Adding b_r to each right factor, as in that report, would enforce it independently of branch disjointness, without changing the counts (36.8)–(36.9). Neither report re-proves a theorem of the other. (Corrected in place on 2 October 2026 following the Hilbert's-tenth programme's review `review_batch79_j2_bbc67d225.md` of `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit 37a829e0b, finding 4): this note had ended by calling fractional selectors at $X = 0$ an exception of the packet; they are possible for the weak gates in isolation only, and the sentences on $X = 0$ above replace that ending. For the actual event branches, whose positive-span and positive-pivot guards exclude zero, the case $X = 0$ does not arise.)

36.2 The exact witness and residual ledger

Let E, T, W be, respectively, the total counts of equality, strict, and weak guards across all B branches. There are

$$A = B(d + 1) + T + W \quad \text{auxiliary natural variables,} \quad (36.8)$$

$$R = 1 + 2d + E + T + W \quad \text{squared affine residuals,} \quad (36.9)$$

$$B \quad \text{unsquared nonnegative quadratic products.}$$

The variables are exactly B selectors, Bd local coordinates, and $T + W$ slacks. For the literal frontend, replace d by 18, so $A = 19B + T + W$ and $R = 37 + E + T + W$. For the binary existence frontend, $A = 11B + T + W$ and $R = 21 + E + T + W$.

This is a variable-and-residual ledger, not a minimal arithmetic-gate count. Expanding a squared linear form may create many monomials. The release gives the literal machine's numeric ledger and a complete factored coefficient description in Appendix H. No minimization or flat monomial expansion is claimed.

36.3 Fixed horizons and terminal reachability

For an externally fixed $K \geq 1$, introduce intermediate states $X_1, \dots, X_{K-1}$ and a separate packet witness $\mathbf{z}_k$ for each step. Define

$$P_K(X_0, X_K; \mathbf{Z}) = \sum_{k=0}^{K-1} P(X_k, X_{k+1}; \mathbf{z}_k). \quad (36.10)$$

There are exactly $KA + (K - 1)d$ witness variables with the two endpoints supplied as parameters, KR squared affine residuals, and KB complementarity products. The polynomial is quadratic and nonnegative on natural arguments. By Theorem 36.1, its natural fiber is empty or a singleton: determinism fixes every intermediate state, and the step witnesses are canonical.

A designated accepting mode is a finite disjunction of homogeneous linear mode equalities, optionally with the linear collision-free condition. Its finite disjunction can be handled by the same selector/copy construction. If the intended question is *first* acceptance at step K , disallow accepting input modes in the earlier branch packets. Nonaccepting null modes are never promoted to accepting ones. Reaching H and finishing the escape suffix are separate predicates, as discussed above.

The statement “there exists some finite K ” is an unbounded union of these fixed-horizon relations. A uniform fixed-arity Diophantine representation of that union would need an additional sequence or iteration compression theorem. None is supplied here. Universality of the dynamics and existence of a direct quadratic *step* polynomial do not remove that requirement.

37 Exact replay and the release contract

37.1 Literal machine counts

For $\ell = 6$, the two stack copies contribute 80 meta-signals and 334 rules. The controller has 15 write-state labels and 15 move-state labels. The 62 source transitions give 62 writing rules and the 15 target states give 15 acknowledgement rules. Two designated terminal pairs contribute four fresh labels, two terminal rules, and 32 transparent escape rules, one per terminal pair and each of the 16 stationary module labels. Hence

$$80 + 30 + 4 = 114, \quad 334 + 62 + 15 + 2 + 32 = 445.$$

The exported table lists every label with its integer speed and every input/output pair. Undefined encounters are explicitly forbidden; there is no implicit identity completion.

37.2 Replays and their meaning

The Python replay uses standard-library integer arithmetic and exact fractions. The core stack suite checks 1,120 operations over alphabets 2, 3 and 4, totaling 24,028 collisions and 34 simultaneous batches. A small compiled binary source machine is tested on all 63 binary words of length at most five, totaling 5,045 batches. These tests cover gadget behavior, conservation, local injectivity, complete earliest-event guards, simultaneous processing, and the rational/integer branch identity.

The source-machine audit checks all 184 transitions of Morita’s published example in both directions. The Figure 28 snapshots at times 0, 6, 59, 178 and 184 match state, head, and full displayed tape window. A separate bounded differential suite covers 2,430 source cyclic-tag cases,

7,687 macro checks and 169,019 Turing microsteps; 40 cases remain active at its 12-macrostep bound. A q_0 state alone is not a reliable macro boundary, so that test also compares the restored program, phase marker, and active word.

For the literal signal machine, each controller read is checked against an independent ordinary tape: controller state, returned symbol, and both complete rational half-stack values must all agree. The tested cases are

Program and word	TM steps	Signal batches	Outcome
(halt, YN, YYN), NYY	184	11214	halt
Same program, Y	2	154	halt
Same program, empty word	1	106	null
(halt), N	11	722	null
(halt, ε), NY	25	1602	null
Total	223	13798	

All these signal batches are single-collision batches, and each has an eight-crossing escape suffix. The primary example's final exact clock is

$$\frac{142256492930823919}{199397583417606}.$$

An additional event-timing implementation computes all pairwise intersections and groups coincident endpoints independently of the adjacent-gap selector. It matches all 11,214 primary-example batches and verifies the cumulative QD^k integer lift at every batch; the final largest numerator has 179,449 bits. Its generic suite checks 8,092 valid germs, including 1,184 simultaneous-event batches, 352 triple-or-larger events, and 3,430 initially zero-gap germs. A missing triple rule is explicitly rejected. There are also 1,296 two-half loader cases and 32 two-entry-side compiled runs totaling 2,574 batches. The pivot-scaled suite independently checks all 13,798 literal batches and 7,282 generic events, including 4,708 events whose total span changes. The primary example's final largest gap numerator has 39,320 bits under pivot scaling, compared with 179,449 under uniform scaling. It also checks exact fixed-span geometry and delay decoding, the production adaptive-lift helper, and the optional timestamp numerator at every event; the primary example's final time numerator has 39,329 bits. These are checks of the same rule data, not an independent proof of universality.

These finite comparisons substantiate the literal implementation. The proof for all inputs is the stack geometry, control invariant, and imported source universality theorem, not extrapolation from the tested cases.

37.3 Reproduction and boundaries

The release has four parts: the article source and PDF; original exact replay code; complete mathematical machine data; and public-safe replay receipts with provenance links. The run script recreates the source table and its checks, the generic stack checks, and the literal signal table and replays. A separate generic event-branch compiler and arithmetic packet demonstration exercise exact canonical witnesses and residual counts on a fully enumerated finite branch system. A separate closure compiler enumerates a forward-closed mode family for the literal initial mode and records all 80,501 numeric event branches, as described in Appendix H. This avoids enumeration of all 114^{18} possible label words.

No third-party source PDFs, source-page images, private working notes, or third-party software are bundled. Source PDFs can be obtained from the linked primary locations; their SHA-256 digests are recorded for comparison. The Python programs use only the standard library. Rebuilding the article additionally requires a standard LaTeX installation.

The strongest literal deliverable is the 18-live machine. The smaller ten-live result is derived. The general event theorem includes exact ties. The direct Diophantine result concerns a step and every fixed finite horizon. No fixed-arity polynomial for arbitrary halting histories, no fastest loader, and no optimal signal-count claim is made.

[write] **The release in this report.** “The release” is the delivered archive. Its programs are shipped as `code/12-conservative-signal-*` and its records as `data/12-conservative-signal-*`, with the delivered sub-directories `code/`, `numeric/`, `correction/`, `receipts/` and `data/` flattened into these two directories, and its `CORRECTION.md`, `SOURCE-PROVENANCE.md` and `numeric/PROOF_AND_LEDGER.md` at the report root under the same prefix; the README of this report maps every name. The delivered article source and PDF and the delivered README are not shipped: this Part prints the article, Appendix F prints its Morita table, and this report’s README replaces the delivered one. The two compact closure records `numeric/MODES.jsonl.gz` (1,054,196 bytes) and `numeric/EVENTS.jsonl.gz` (580,458 bytes) are *not* shipped either: they are regenerated by the shipped compiler in about a minute, with identical decompressed content, and the README gives the command (“Reconstructing the excluded data”); the machine file, the ledger and the circuit file named in Appendix H are shipped, and the verbose branch and residual-row files were already omitted from the delivered archive, as that appendix says. The programs need the delivered layout and must be run on a copy (README).

[write] **The corrected archive and the repository’s review.** This Part prints the corrected edition of the archive (batch 79 manuscript 16). An earlier edition (manuscript 07 of the same batch) has the same article, data and receipts, and differs only in the generic packet evaluator `code/quadratic_packet.py`. The repository’s Hilbert-tenth-problem programme reviewed that earlier edition [29]: it passed the stated construction, independently enumerated the complete outgoing successor set of all 49,700 modes and all 80,501 branches (the least mode-only closure), checked that the trillion-term monomial count of Appendix H cannot be reduced by cancellation, and found genuine input-validation defects in the generic evaluator (a padded output (1,999), a floating endpoint and a surplus slack were accepted). It wrote the repair `conservative_signal_packet_domains.patch` [29]. The corrected edition applies exactly that patch and says so in its `CORRECTION.md` (shipped as `12-conservative-signal-CORRECTION.md`); the shipped evaluator `code/12-conservative-signal-quadratic_packet.py` is the patched one, and the correction receipts are shipped. The repair does not change Theorem 36.1, the literal construction or any ledger, and it does not make arbitrary caller-supplied overlapping branch families disjoint. The programme then reviewed the corrected edition itself [31]: it reread the article and the proof notes, compared the two archives member by member, passed all ten original author entryptoints and the three correction runs, rechecked one explicit integral event for every branch, and found no new defect on the valid domain. That review also proves a reduction that source 12 does not claim, a natural-domain guard projection: deleting the germ rows, the span row and the first-event rows of unselected edges with negative closing speed (18 rows and slacks per branch) is a bijection of complete *natural* zero sets, though not of nonnegative real ones, and reduces the literal one-step packet from 4,196,998 to 2,747,980 witnesses and from 2,762,961 to 1,313,943 squared residuals, with an explicit evaluation schedule 7,489,424 operations shorter. The projection is the review’s, is not re-proved here, and changes no theorem of this Part. Neither review is a proof-assistant verification, and neither finds a new universal operation bound.

Part IV

Sparse Diophantine certificates for finite mass lattice dynamics (source 13)

[write] Part IV prints batch-79 manuscript 19, *Sparse Diophantine certificates for finite mass lattice dynamics*, “Research report prepared with AI assistance for Vladimir”, dated 2 October 2026, from the archive `Sparse_Lattice_Diophantine_Certificates.zip`. It is called source 13 after the prefix of its shipped files. The delivered article is a main file that reads five further files; they are printed here in the delivered order, without its separate table of contents. Its labels carry the prefix `smc:sl:`; its sections are Sections 38–48 here, and it has no appendices. Its statements keep their numbering by section; equations are numbered by section (the source numbered them consecutively). Source 13 names no repository commit and cites no repository file. Its comparison and sorting gadgets and its post-elimination compiler re-derive devices of *canonical-diophantine-certificates*; [write] notes at those places say so, and Section 1.7 describes the Part. Its abstract follows.

Finite conserved mass bounds the number of occupied unit records without bounding the distances between them. We exploit this distinction to compile a fixed finite horizon of one-dimensional partitioned lattice dynamics into one quartic sum of squares with a unique natural witness. The construction explicitly pays for collision detection, simultaneous redistribution, sorting, finite table lookup, and empty space. We give exact witness and residual ledgers, a coefficient-level reproducible implementation, and an optional strengthening whose nonnegative-real fiber is also exact. A source-critical interface to the reversible conservative construction of Morita and Imai includes a necessary sign correction and a reachable-legality hypothesis; a fully specified doubling example is separated from the existence of a fixed safe universal source. Complementary proofs show decidability under bounded orbit diameter and, without a diameter promise, at total mass at most two. At mass at most two, a stronger effective semilinearity argument yields horizon-free canonical quartics of fixed rule-dependent arity. No fixed-arity representation of unbounded universal computation, new classical universality result, or minimum universal mass is claimed.

38 Results and model

The geometric resource in this report is an unbounded integer gap, rather than an unbounded number of occupied cells. A conserved positive integer mass M permits a canonical description by M repeated unit records, even when the occupied positions are very far apart. The goal is to verify every lattice microstep directly on those records, including collisions and the assertion that no output has been omitted in empty space.

Three levels of claim must be kept separate. First, the main certificate theorem applies to *any supplied total conservative finite table*. Second, the corrected Morita–Imai compiler provides explicit reversible examples and an interface from suitably legal two-counter computations. Third, a fixed safe universal source exists by classical simulation constructions, but the release does not print a particular universal source machine and its numerical state count. Its fully printed worked machine computes doubling and is not universal.

The sparse construction and its exact ledgers are the arithmetic contribution considered here. The circuit-to-sum-of-squares device and classical reversible computational universality are not new claims. Dense fixed-box or ring tableaux already record bounded cellular computations; a direct sparse infinite-line argument additionally has to justify grouping, ordering, and empty space.

Priority of the particular constructions and the mass-two theorem below has not been established by an exhaustive literature review.

[write] The dense fixed-box tableau is, for example, the finite cellular-automaton compiler of Part VIII of *canonical-diophantine-certificates* (its Section `cdc:mem:sec:substrates`), whose one-hot tableau occupies a fixed box; this Part is the sparse infinite-line counterpart for one-dimensional partitioned conservative dynamics. Source 13 was written without this report and without that one.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Fix $K \geq 1$, put $q = K + 1$, and supply every row of a function

$$g : \{0, \dots, K\}^3 \longrightarrow \{0, \dots, K\}^3, \quad r + c + l = g_L(r, c, l) + g_C(r, c, l) + g_R(r, c, l).$$

A configuration c gives a triple $(L_c(x), C_c(x), R_c(x))$ at each $x \in \mathbb{Z}$. One update is

$$F(c)(x) = g(R_c(x - 1), C_c(x), L_c(x + 1)). \quad (38.1)$$

Thus the *input* order of the table is (R, C, L) and its *output* order is (L, C, R) . In physical terms the right channel streams right, the left channel streams left, and the center stays in place; g then redistributes the incoming mass. The input-output order matters even when the numerical table happens to be the identity.

Conservation forces $g(0, 0, 0) = (0, 0, 0)$. Finite support stays finite, and the sum of all channel values remains M . If g is a permutation, then F is globally bijective: apply g^{-1} at each site and reverse the two lane shifts. Both operations preserve the zero background and finite support. Reversibility is useful for the intended substrate, but is unnecessary for the forward certificate theorem.

38.1 Canonical records and fixed parameters

Let the valid initial configuration be nonempty, with minimum occupied position o and diameter D . Replace each channel mass by that many copies of a record (x, z) , where $z = 0, 1, 2$ means left, center, right. Sort lexicographically, and shift the physical position by $T - o$, where $T \geq 0$ is the externally fixed horizon. Thus a record at physical site p has

$$x = p - o + T.$$

The initial row has exactly M records. At time $t \leq T$, every position lies between $T - t$ and $D + T + t$. Repeated records are allowed and have no individual labels. The sorted repeated-unit row therefore encodes the physical configuration uniquely. There is neither zero padding nor a hidden list of empty sites.

The initial data, its translation, the local table, and the horizon are compiler parameters, not quantified witnesses. Later we explicitly pay for a loader when two initial counter positions are allowed to remain free natural parameters. The vacuum $M = 0$ is a separate trivial constant orbit.

Theorem 38.1 (Sparse finite-horizon certificate). *For a supplied total conservative g , valid nonempty initial configuration of mass M , and fixed horizon T , there is an explicitly generated integer polynomial*

$$P(\mathbf{z}) = \sum_{j=1}^R \rho_j(\mathbf{z})^2, \quad \deg \rho_j \leq 2,$$

with exactly one zero in $\mathbb{N}^V$. Its row decoder gives precisely $c, F(c), \dots, F^T(c)$. The preferred full-table construction uses exactly

$$V = T[M(q^3 + 14) + 5M(M - 1)], \quad (38.2)$$

$$R = T[17M + 5M(M - 1)]. \quad (38.3)$$

Every witness in this unique solution is at most

$$H = 3(D + 2T) + M + 3K + 3. \quad (38.4)$$

Consequently the tuple has at most $V[\log_2(H + 1)]$ bits under fixed-width coordinate encoding. A prescribed endpoint of mass M can be imposed with $2M$ additional affine residuals and no witnesses, leaving zero or one natural solution.

For $T = 0$ the witness tuple is empty, and is the unique element of $\mathbb{N}^0$. The empty sum is the zero constant. A mismatched-mass endpoint can instead be rejected by adding the constant residual 1. The degree bound is at most four, not a claim that every small instance has degree exactly four. Dimension grows with T and M ; the theorem is not a fixed-arity equation with an unknown unbounded horizon.

39 The canonical arithmetic layer

39.1 Comparison and equality without ambiguous ties

For already established nonnegative integral expressions x, y , define $\text{CMP}(x, y)$ by fresh natural witnesses b, d and residuals

$$b(b - 1), \quad y - x - (2b - 1)d - b + 1. \quad (39.1)$$

If $b = 1$, the second equation says $d = y - x$, possible exactly when $y \geq x$. If $b = 0$, it says $d = x - y - 1$, possible exactly when $x > y$. This proves unique solvability, including ties, with

$$b = \mathbf{1}_{\{y \geq x\}}, \quad d = \begin{cases} y - x, & y \geq x, \\ x - y - 1, & x > y. \end{cases}$$

The inputs can be affine expressions; there is no charge for an unnecessary alias wire. To test equality, use two comparisons, $\text{CMP}(x, y)$ and $\text{CMP}(x + 1, y)$, and take the affine difference of their flags:

$$e(x, y) = b_{\text{weak}} - b_{\text{strict}} = \mathbf{1}_{\{x=y\}}. \quad (39.2)$$

This costs four witnesses and four residuals, with no additional equality-variable witness.

[write] The comparison gadget (39.1), with the same two residuals and the same case analysis, is Lemma `cdc:mem:lem:compare` of Part V of *canonical-diophantine-certificates*. Source 13 does not cite it; the gadget is printed here as delivered and no novelty is claimed for it.

39.2 Streaming and all simultaneous collision groups

Suppose the current row is (x_i, z_i) , $0 \leq i < M$, in sorted order. Each x_i, z_i is affine in earlier variables. Introduce natural L_i, C_i, R_i with residuals

$$L_i + C_i + R_i - 1, \quad z_i - C_i - 2R_i. \quad (39.3)$$

For valid integral $z_i \in \{0, 1, 2\}$, these force the unique one-hot channel vector. A streamed position y_i is forced by

$$y_i + L_i - x_i - R_i = 0. \quad (39.4)$$

It is nonnegative at every layer $t < T$, because $x_i \geq T - t \geq 1$. This is genuine line transport; there is no clipping at zero and no wraparound.

For each unordered pair $i < j$, compute $e_{ij} = \text{EQ}(y_i, y_j)$; set $e_{ji} = e_{ij}$ and $e_{ii} = 1$ as aliases and constants. The whole collision partition is determined at once. Introduce three incoming mass variables with residuals

$$u_i - \sum_j e_{ij} R_j, \quad v_i - \sum_j e_{ij} C_j, \quad w_i - \sum_j e_{ij} L_j. \quad (39.5)$$

Their order is (R, C, L) , as required by (38.1). Equal destinations have identical incoming triples. Each component is at most K : a particular incoming lane is a translated copy of one old lane at one site. That statement uses validity of the previous row, not a guessed bound on the number of simultaneous records.

39.3 A single table-row selector simplex

Index every input table row by $\tau = (r, c, l) \in \{0, \dots, K\}^3$. For particle i , introduce q^3 natural selectors $s_{i,\tau}$, three output masses a_i, b_i, c_i , and the seven affine residuals

$$\begin{aligned} & \sum_{\tau} s_{i,\tau} - 1, \\ u_i - \sum_{\tau} r s_{i,\tau}, \quad v_i - \sum_{\tau} c s_{i,\tau}, \quad w_i - \sum_{\tau} l s_{i,\tau}, \\ a_i - \sum_{\tau} g_L(\tau) s_{i,\tau}, \quad b_i - \sum_{\tau} g_C(\tau) s_{i,\tau}, \quad c_i - \sum_{\tau} g_R(\tau) s_{i,\tau}. \end{aligned} \quad (39.6)$$

A natural vector summing to one has exactly one nonzero entry, equal to one. The three input moment equations select precisely the row (u_i, v_i, w_i) , because all input triples are distinct and the table is total. The output equations then force $g(u_i, v_i, w_i)$. This single simplex avoids factoring the lookup into three separate selector vectors and all their products. Its small residual count does not make the q^3 selectors or the long coefficient sums free.

39.4 Rank redistribution

Within each destination group use the old record order to define its one-based rank

$$r_i = \sum_{j \leq i} e_{ij}.$$

This is an affine expression, not a fresh witness. Compute two flags by comparisons

$$h_i = \text{CMP}(r_i, a_i), \quad k_i = \text{CMP}(r_i, a_i + b_i),$$

where the notation denotes the returned flag. The next channel is the affine expression

$$z'_i = 2 - h_i - k_i. \quad (39.7)$$

Ranks within a group are exactly $1, \dots, u_i + v_i + w_i$. Conservation makes this number $a_i + b_i + c_i$. The first a_i ranks have $(h_i, k_i) = (1, 1)$ and channel zero; the next b_i have $(0, 1)$ and channel one; the last c_i have $(0, 0)$ and channel two. Thus the output multiset at the collision site is exact. The intermediate rank convention only chooses identical physical units, and no persistent identity survives.

39.5 A fixed sorting network

Sort the M records (y_i, z'_i) by a fixed bubble sorting network of $\binom{M}{2}$ adjacent compare-exchanges. For inputs $(x, z), (y, w)$ compare the keys $3x + z$ and $3y + w$ using (39.1), giving b, d . Introduce four output witnesses and residuals

$$\begin{aligned} x_{\min} &= y - b(x - y), & z_{\min} &= w - b(z - w), \\ x_{\max} &= x - y + x_{\min}, & z_{\max} &= z - w + z_{\min}. \end{aligned} \quad (39.8)$$

For $b = 1$ the left input is the smaller or tied key; for $b = 0$ the right is smaller. Ties retain the left input, but equal keys are already the same physical record. Since channels lie in $\{0, 1, 2\}$, key order is lexicographic order. Every comparator costs six witnesses and six residuals, including its comparison. The network outputs are the next row by alias, without copy variables. All equations in this layer are of degree at most two.

[write] This compare-exchange is the two-field case of the record comparator of the same Part V (its equations `cdc:mem:eq:record-low` and `cdc:mem:eq:record-high`, with the cost count of Proposition `cdc:mem:prop:comparator-cost`); it is printed as delivered, as a re-derivation of that device.

40 Correctness and exact paid costs

Proof of Theorem 38.1. Assume inductively that the current row is the canonical true row and every earlier auxiliary witness is uniquely fixed. The channel equations and comparison lemma uniquely fix streaming and the entire equality partition. Equations (39.5) fix the incoming mass. The single natural simplex forces the unique table row and its output. Conservation and rank redistribution then give exactly the outgoing multiset. The sorting network uniquely produces the canonical next row.

The empty-space assertion is indispensable. Every nonzero incoming site appears among the y_i , since it has at least one streamed unit. A site absent from that list has incoming triple $(0, 0, 0)$ and hence zero output. A listed site has positive incoming total, which conservation keeps positive. Therefore there is no hidden output outside the represented sites, no omitted occupied site, and no ghost site introduced by an unconstrained witness.

This proves soundness and uniqueness by induction through T layers. Conversely, start with the actual physical history, expand each configuration into repeated unit records, and evaluate the stated one-hot vectors, comparisons, masses, ranks, and sorting gates. They provide nonnegative integral values satisfying every residual. Thus the unique natural solution exists. Counting records of channel z at position x and undoing the shift by $T - o$ is the decoder. The deterministic gate evaluation is its inverse on the fixed-input, fixed-horizon solution set.

If g is reversible, the final physical configuration also determines all preceding physical configurations by F^{-1} . This does not assert reversibility of the auxiliary arithmetic circuit, nor any property of arbitrary real solutions.

For the height bound, positions stay in $[0, D + 2T]$. Sorting keys are at most $3(D + 2T) + 2$; comparison slacks are bounded by the appropriate key difference, group rank, or local mass. Incoming and outgoing lane masses are at most K , ranks at most M , and selectors at most one. Bound (38.4) covers all these quantities with slack. Endpoint equations only select or reject the already unique row. $\square$

40.1 Scalar ledger

The per-layer cost is as follows. The lookup row excludes the separately listed incoming masses.

Block	Witnesses	Residuals
Channel decoding	$3M$	$2M$
Streaming	M	M
Pairwise destination equality	$4\binom{M}{2}$	$4\binom{M}{2}$
Incoming lane masses	$3M$	$3M$
Full row-selector lookup	$M(q^3 + 3)$	$7M$
Rank comparisons	$4M$	$4M$
Sorting network	$6\binom{M}{2}$	$6\binom{M}{2}$
Total	$M(q^3 + 14) + 5M(M - 1)$	$17M + 5M(M - 1)$

The exact counts include slacks, Boolean comparison flags, sorting outputs, and repeated simultaneous-collision work. Constants and affine aliases are not witnesses. The local table itself has q^3 input rows and three output entries per row; its construction, range/conservation checks, storage, and any permutation test are separate explicit costs. Expanding weighted input into M unit records requires $\Omega(M)$ output work and storage. The size bounds are polynomial in the numerical mass M , not in its binary bit length. For a fixed source substrate, $M = m + 19$ is constant while gaps and their bit lengths may grow.

Let $\mathcal{L}$ be the total number of nonzero monomial terms in the residual list, and $\mathcal{O}$ the sum of the degrees of those monomials, including repeated variable occurrences. The release declares the following literal evaluator: initialize a monomial with its signed integer coefficient, multiply by each listed variable, add its value into a residual accumulator initially zero, square each residual, and add those squares starting at zero. Under exactly this convention,

$$\#\text{multiplications} = \mathcal{O} + R, \quad \#\text{additions} = \mathcal{L} + R. \quad (40.1)$$

Multiplication by a coefficient is already included by the initialization convention; negation signs reside in signed coefficients. Even a final zero residual is squared by this evaluator. The exporter supplies these exact counts and the largest residual-coefficient bit length for each instance.

There are $O(T[M^2 + Mq^3])$ such arithmetic operations. This is scalar arithmetic, not bit complexity. Witness height, coefficient size, polynomial-description length, arithmetic cost, and witness dimension are different resources. The statement concerns a coefficient-explicit residual-list sum of squares. Fully expanding and collecting the quartic polynomial can create many more terms; no equality with the residual-list cost is asserted.

40.2 Restricted row lists

A useful optional backend replaces all q^3 rows in (39.6) by a supplied list of S distinct allowed input rows, each conserving mass. It has

$$V_S = T[M(S + 14) + 5M(M - 1)], \quad R_S = R.$$

The same proof certifies exactly histories whose every occupied incoming triple belongs to that list. If some required row is absent, its input moments cannot be matched by a natural one-hot selector, and the certificate is unsatisfiable. This backend does *not* certify all histories of an arbitrary total

extension merely because the listed rows extend to one. Coverage of the particular history or a valid-trajectory invariant is a separate proof obligation. The worked source example uses such a coverage theorem; the generic theorem always uses the supplied full table.

40.3 The domain of uniqueness and an orthant-exact option

The default simplex equations rely on natural integrality. They need not be unique even over the nonnegative real orthant. For a concrete full-table example let $K = 2$, $M = 1$, $T = 1$, and $g(r, c, l) = (r, c, l)$ numerically, with the stated input/output orders. Start with one right-channel unit. Its incoming triple is $(1, 0, 0)$ and its output is one left unit. Replace the table selector at $(1, 0, 0)$ by half at $(0, 0, 0)$ and half at $(2, 0, 0)$. All three input moments and all output moments remain unchanged. Every other witness can keep its canonical value, so the original residuals still vanish, but the selector is fractional. The release checks this with exact rational arithmetic.

For any nonnegative simplex $s_1 + \dots + s_d = 1$, add the one residual

$$\sum_{j=1}^d s_j^2 - 1. \quad (40.2)$$

If it vanishes, then

$$0 = \left(\sum_j s_j\right)^2 - \sum_j s_j^2 = 2 \sum_{i < j} s_i s_j.$$

Every summand is nonnegative; at most one selector is nonzero, and that selector equals one. Adding (40.2) to each channel simplex and each table simplex therefore restores the inductive proof over $\mathbb{R}_{\geq 0}$ for fixed natural input data. Comparisons have Boolean flags already, and their nonnegative slacks are uniquely forced once the inputs are fixed integers. All subsequent witnesses remain the canonical natural values.

Corollary 40.1 (Exact nonnegative-real fiber). *The strengthened full-table or restricted-row compiler has the same witness count and exactly $2MT$ extra residuals. Its nonnegative-real solution set is the same singleton or empty set as its natural solution set. Residual degree stays at most two and polynomial degree at most four.*

This says nothing about the unrestricted real domain, where negative selectors invalidate the nonnegativity argument. An earlier factorized lookup remains an independently tested baseline: three q -selectors, q^2 pair products, and q^3 triple products give

$$V_{\text{fact}} = T[M(q^3 + q^2 + 3q + 14) + 5M(M - 1)],$$

$$R_{\text{fact}} = T[M(q^3 + q^2 + 19) + 5M(M - 1)].$$

Its orthant strengthening costs $4MT$ residuals. These are baseline counts, not the preferred single-row-simplex ledger in (38.2)–(38.3).

41 The reversible conservative source interface

Morita and Imai [19] construct a reversible number-conserving partitioned automaton from a reversible two-counter machine. The following source audit fixes the exact contract used here. It distinguishes the published theorem, a correction forced by its encoding, and an additional reachable-legality argument needed when unspecified local rows are completed arbitrarily.

41.1 Source syntax and encoding

A transition is (i, j, op, k) , with source state q_i , target q_k , counter $j \in \{0, 1\}$, and operation increment, decrement, no-op, zero test, or positive test. There are $m \geq 2$ states, q_0 has no incoming transition, and q_{m-1} is designated final with no outgoing transition. Distinct transitions with the same source may only be complementary zero/positive tests on the same counter; the same restriction applies to distinct transitions with the same target. This is the paper's syntactic deterministic and reversible model, a stronger specification than a vague assertion of injectivity for some other instruction language.

Put

$$a_i = i + 10, \quad B = m + 16, \quad b_i = B - a_i - \gamma_i,$$

where $\gamma_i = 2, 4, 6, 7$ for increment counter zero, decrement counter zero, increment counter one, decrement counter one respectively, and zero otherwise. The alphabet cap is $K = m + 18$ and the number of ordinary cell states is $(m + 19)^3$.

The encoding $\phi(q_i, n_0, n_1)$ places (a_i, b_i, γ_i) at the origin. Counter j contributes marker mass 2^j at position n_j , normally in the center lane. If $n_j = 0$ and q_i increments that counter, its marker is instead added to the right lane at the origin. Contributions at the same site add. Hence

$$M = B + 1 + 2 = m + 19 = q, \tag{41.1}$$

independently of counter magnitude. The support occupies at most three sites, all in $\{0, n_0, n_1\}$. Coincident markers have total three, retaining their separate zero/positive information through distinct weights. Input size is paid through distances and their bit lengths, not a variable number of particles.

41.2 The printed sign defect and the repaired local injection

On printed page 249, rule (8.2) has an output-center $+\gamma_k$. When a no-op enters a state incrementing counter h , the input and repaired output are

$$\begin{aligned} & (a_i, B - a_i + c2^h + c'2^{1-h}, 0) \\ & \longmapsto (a_k, B - a_k - \gamma_k + c'2^{1-h}, \gamma_k + c2^h), \quad c, c' \in \{0, 1\}. \end{aligned} \tag{41.2}$$

The printed plus would increase the local total by $2\gamma_k$, namely four or twelve. The minus conserves mass and gives exactly the declared target encoding. This is a necessary correction to that printed formula; no author-issued erratum is asserted. The target's increment counter h must also be read from q_k , rather than borrowed from the no-op quadruple's semantically irrelevant counter tag. Opposite-tag cases are included in the executable audit.

[write] The repository's Hilbert-tenth-problem programme checked this sign against the primary paper (printed page 249 of [19]) in its review of source 13 [30]: subtracting the target opcode is forced by mass conservation and the target encoding, and the target's increment counter, not the no-op tag, controls the marker routing. Like source 13, the review records a mathematical correction to a published formula, not an author erratum. No report or README of the repository made a claim that this correction refutes.

For clarity, the structural proof behind finite row checks is as follows. The ordinary signal rules route signal mass or transfer a marker between center and signal; waiting rows are identities. After (41.2), state-update rows preserve B plus the zero-marker mask. Startup and final rows

exchange a unit signal plus a reservoir $m + 15$ with control mass B . For nonfinal i , the smallest baseline obeys

$$b_i \geq m + 6 - (m - 2) - 7 = 1;$$

the final baseline is seven. Added marker mass is at most three, all center values are at most $m + 18$, and state signals are at most $m + 9$. Thus all coordinates stay in range.

Routine signal domains differ from control domains by their nonzero lane pattern and by signal values below or above ten. At a fixed control state its unique operation type chooses the rule family. Zero masks are distinct; complementary test branches have disjoint masks. The no-outgoing convention at the final state removes domain conflicts there. This proves the function property.

For injection, different target state signals have different left components. At a fixed target, its center/right pair recovers the zero-mask bits: an increment target carries its own zero marker in the right signal and leaves the other bit in the center; otherwise both bits remain centered. A target with a movement or no-op predecessor has only that incoming transition. If it has two incoming tests, they test the same counter with complementary outcomes and hence disjoint masks. The *whole output triple* distinguishes them, even when the center alone does not. Waiting outputs have right component zero, whereas a transition into an operation state has a positive right signal. Startup targets q_0 , which has no incoming source transition, and the final unit signal with its large reservoir is distinct from an ordinary travelling unit signal. The remaining routine signal output patterns are disjoint by lane and signal type. Together these observations establish a mass-preserving partial injection.

Lemma 41.1 (Completion within mass fibers). *Every finite mass-preserving partial injection on the local triple alphabet extends effectively to a mass-preserving permutation. This applies to the repaired table.*

Proof. For each s , let Ω_s be the triples with coordinate sum s . Used inputs and used outputs have equal cardinality within Ω_s . Bijection of their unused complements, for example by pairing lexicographically sorted lists, completes this fiber. Their union is the required permutation. The unique weight-zero triple is fixed. Lane transport followed by the local permutation is a global reversible conservative CA. $\square$

Simply assigning identity to every unspecified input is generally wrong. The specified row $(2, 1, 0) \mapsto (0, 0, 3)$ can leave $(0, 0, 3)$ unspecified as an input. Its identity image would duplicate an already used output. Also, Lemma 41.1 is about a partitioned/block local permutation; it is not a completion theorem for arbitrary partial ordinary-neighborhood CA rules.

41.3 Exact source microtimes and the pulse observer

For a *legal* source step, the repaired specified rows, without using any completion row, carry $\phi(I)$ to $\phi(J)$ in the following number of microsteps:

Source operation on a counter of value n	Microsteps
Increment, including $n = 0$	$2n + 2$
Decrement, with $n > 0$	$2n$
No-op or legal zero/positive test	2

Here is the geometric reason for the clocks. The state signal travels from the origin to site -1 and back, while the control waits for its operation signal. An increment signal traverses the distance to its marker, carries that marker one step farther right, deposits it, and returns to the origin, a total of $2n + 2$ steps. When $n = 0$, the marker is already in the initial right signal, giving the same two-step formula. A decrement signal reaches its positive-position marker, reflects while carrying it one step left, deposits it, and returns in $2n$ steps. At $n = 1$ the returning marker joins the origin's zero mask directly. The other counter's marker is transparent to these routes, including coincident marker positions, because the rule families carry its distinct mass unchanged. Tests and no-ops only require the state signal's two-step return. The state-update rows then redistribute the zero mask into exactly the target encoding. These routes also explain why no constant slowdown is promised.

Let t_h be the sum of these times up to first entry into $q_f = q_{m-1}$. From $\phi(q_f, n_0, n_1)$ the state signal is in the right lane at -1 at $t_h + 1$; the final rule creates a left unit at the origin at $t_h + 2$; and

$$L_{t_h+3}(-1) = 1. \quad (41.3)$$

That unit subsequently moves left forever. Before final entry, the specified simulation trajectory never has this left-unit signal. After its departure, the fixed centers are

$$C(0) = m + 15 + \mathbf{1}_{\{n_0=0\}} + 2\mathbf{1}_{\{n_1=0\}}, \quad C(x) = \mathbf{1}_{\{x=n_0\}} + 2\mathbf{1}_{\{x=n_1\}} \quad (x > 0).$$

All right channels are zero. The results remain readable while information escapes spatially. This is an anchored pulse, not an absorbing globally halted configuration. The complete successful orbit has unbounded diameter.

The equivalence between this observer and source halting requires that *every reachable nonfinal source configuration on the input under discussion has a legal next instruction*. This is called reachable legality below. It is not asserted for every arbitrary counter pair at every internal state.

41.4 An explicit false observer on an unsafe source

Take $m = 2$ and the sole source instruction “decrement counter zero from q_0 to q_1 .” On counters $(0, 0)$ that instruction is illegal, so the designated final state is not reached. Its encoded initial CA state is $(10, 7, 4)$ at site zero, of mass 21. At time one the nonzero sites are

$$-1 : (0, 0, 10), \quad 0 : (0, 7, 0), \quad 1 : (0, 0, 4).$$

The next origin input $(10, 7, 0)$ is unspecified. Both this input and the output $(1, 16, 0)$ are unused, and both have mass 17, so adding

$$(10, 7, 0) \mapsto (1, 16, 0)$$

is compatible with Lemma 41.1. At time two the configuration is $0 : (1, 16, 0)$ and $2 : (0, 0, 4)$; at time three it has a left unit at -1 , falsely satisfying the proposed observer. The release supplies the entire resulting 9,261-row permutation and a replay through time eight.

Thus an arbitrary partial reversible source and an arbitrary conservative permutation completion do not provide a halting equivalence on every encoded pair. Missing test branches and unnamed dead states have the same logical risk. This example does not affect the generic theorem, which always certifies the actual supplied CA dynamics, and does not refute a safe universal-source construction.

42 Safe universal sources and the paid loader

42.1 The existence argument and its exact scope

Morita's 1996 theorems [20] give designated-final-state simulation equivalences. The following preservation of reachable legality is an explicit invariant argument through those constructions, rather than an attribution of a separate totality theorem to that paper.

Start with a fixed universal deterministic Turing machine whose transition is defined at every nonhalting state/symbol pair. On its computably valid program/input encodings, choose a complete guarded counter simulation: decrement is entered only after positivity, every appropriate test branch exists, and each arithmetic routine terminates on its finite inputs. A fresh initial state and a terminal designated final state give the required endpoint conventions. This yields a safe irreversible counter simulator. Splitting high-indegree target states through finite no-op merge chains, as in Morita's preparatory reduction, preserves legal progress: one legal step only acquires a finite chain of unconditional steps.

The reversible history compiler adds a history counter H and scratch counter W . At an original-state boundary the invariant is

$$(\text{simulated counters correct}), \quad H \in \mathbb{N}, \quad W = 0.$$

An already reversible incoming relation is copied. For an overlapping incoming pair, the compiler takes the original legal operation and then records which predecessor was used. It transfers H to W through positivity-guarded decrements and reconstructs $2H$ or $2H + 1$ in the history counter while emptying W . Each decrement is legal by its positive guard; required positive back-edge tests follow increments; each transfer loop terminates by decreasing its source counter. The boundary invariant is restored at the target. Thus a legal step expands into a finite nonempty legal path, without a new nonfinal dead end. Erasing history is unnecessary for this existence argument.

The reversible multi-counter machine is then represented by two counters

$$(N, S) = \left(\prod_{i=1}^k p_i^{v_i}, 0 \right), \quad N \geq 1, \quad (42.1)$$

where the p_i are distinct fixed primes. This is a constrained encoding of all data, history, and scratch counters, not an arbitrary pair in $\mathbb{N}^2$. The prime compiler has the following legal-progress invariants.

- An increment of counter i transfers N to scratch through guarded decrements and reconstructs $p_i N$ while consuming scratch. Both loops terminate and end with scratch zero
- A legal decrement has $v_i > 0$, so $p_i \mid N$. After transfer, each positive scratch batch has at least p_i units. Its p_i successive decrements are legal even where only the first is explicitly guarded. Consuming all batches gives $(N/p_i, 0)$
- A test computes quotient and residue by a guarded finite-state division loop. Divisibility by p_i is exactly $v_i > 0$; nonzero residue is exactly $v_i = 0$. The appropriate source branch is selected, and quotient and residue restore the original N through terminating positively guarded loops
- A no-op is copied. If a source has only one test exit, the omitted exit is never needed on its legal encoded trajectory

The original state, prime encoding, and scratch-zero invariant hold again at each boundary. Fresh internal states preserve the source's required reversible incoming syntax and initial/final conventions. Induction proves that every reachable nonfinal primitive two-counter state on a valid input has a successor. If the original machine never halts, each of its steps expands into a nonempty finite legal path, so no stuck intermediate configuration can masquerade as a halt.

Fixing the universal program fixes the resulting finite two-counter source and hence, by the corrected local construction and a chosen completion, one finite reversible conservative CA. Its valid program/data encodings vary only through the initial counter positions. Eventual anchored pulse occurrence is therefore computably enumerable hard on this computable valid-input family; direct simulation makes it computably enumerable. This supplies the usual undecidable local-observation problem for a fixed finite-mass substrate.

The release does not materialize a chosen universal source table or give its numerical m . It does not claim faithful source interpretation for all $(n_0, n_1) \in \mathbb{N}^2$, intrinsic CA universality, or an efficient simulation. In particular, the often cited 96-state construction using an ultimately periodic infinite background is not a finite-mass substitute. Exact-target reachability is also a different question: the final counter positions need not be computable in advance, and the escaping halt signal is not a specified complete target. Transferring exact-target undecidability would require an additional normal form or compute/uncompute argument.

42.2 A uniform initial loader with no hidden input oracle

For fixed source data m , initial opcode γ_0 , and fixed T , leave $n_0, n_1 \in \mathbb{N}$ as free polynomial parameters. At position T form ten left records, $m + 6 - \gamma_0$ center records, and γ_0 right records. Add one counter-zero record at $n_0 + T$ and two counter-one records at $n_1 + T$. A marker's channel is right precisely when its position parameter is zero and q_0 increments that counter; otherwise it is center.

Two comparisons $\text{CMP}(n_j, 0)$ determine the zero flags, costing four witnesses and four residuals. The initial list is sorted by the same network as before. Its exact loader cost is

$$V_{\text{load}} = R_{\text{load}} = 4 + 3M(M - 1). \quad (42.2)$$

Even unused zero tests are included in this declared ledger. The exported residual coefficients are independent of the supplied values of n_0, n_1 ; only those parameter values and the canonical witnesses vary. Bound (38.4) remains valid with $D = \max(n_0, n_1)$, including loader comparison slacks.

42.3 Paying for the exact first pulse

For $T \geq 1$, site -1 is shifted to $T - 1$. For each $t = 1, \dots, T$ and each record, compute $\text{EQ}(x_{t,i}, T - 1)$. Reuse the channel one-hot vectors already present at $t < T$, and add $3M$ channel witnesses with $2M$ residuals for the final row. Add one residual at each time,

$$\sum_i e(x_{t,i}, T - 1) L_{t,i} - \mathbf{1}_{\{t=T\}}. \quad (42.3)$$

This forces detector left mass zero at every earlier observed time and one at T . The initial detector is blank because ϕ is supported on nonnegative sites. The additional cost is exactly

$$V_{\text{obs}} = 4MT + 3M, \quad (42.4)$$

$$R_{\text{obs}} = 4MT + 2M + T. \quad (42.5)$$

For the orthant-exact variant add M final decoder norm residuals, on top of the core $2MT$ extra residuals. No observation witnesses are needed for a fixed endpoint beyond the core row witnesses, but that is a different predicate.

These guards mean *zero before and one now*. For an arbitrary CA they are not the generic first occurrence of the value one, because earlier detector masses greater than one are also excluded. On valid source simulations they match the exact unique pulse in (41.3). Combining (38.2), (38.3), (42.2), (42.4), and (42.5) gives the full paid uniform-input first-pulse certificate. Safe-source legality concerns the semantic interpretation of that predicate, not the arithmetic soundness of the actual-CA certificate.

42.4 The fully specified doubling example

The worked source has seven states and the following complete transition list; all other source transitions are absent:

Source	Counter	Operation	Target
q_0	1	zero test	q_1
q_1	0	zero test	q_6
q_1	0	positive test	q_2
q_2	0	decrement	q_3
q_3	1	increment	q_4
q_4	1	increment	q_5
q_5	1	positive test	q_1

On $(q_0, n, 0)$ every reachable nonfinal instruction is legal: the first test sees zero, each decrement follows a positive counter-zero test, and the last positive test follows two increments. The machine reaches $(q_6, 0, 2n)$ after $5n + 2$ source steps. Summing the exact microclocks gives

$$t_h = 5n^2 + 7n + 4, \quad t_{\text{pulse}} = 5n^2 + 7n + 7. \quad (42.6)$$

Indeed iteration $k = 0, \dots, n - 1$ takes a positive test, a decrement of $n - k$, increments at $2k$ and $2k + 1$, and a final test, totalling $2n + 6k + 10$ microsteps. The initial and final zero tests add four; summing yields (42.6).

Here $K = 25$, $M = q = 26$. The corrected specified local table has 97 distinct rows, and its lexicographic mass-fiber completion has all $26^3 = 17,576$ rows. For $n = 2$, twelve source steps reach $\phi(q_6, 0, 4)$ at time 38, create a left unit at the origin at time 40, and produce the detector pulse at time 41. The full table and trajectory checks are shipped. This explicit example verifies the interface and its costs without representing a universal-machine claim.

43 Geometric barriers and translating periodicity

The next results use a more general model than the triple-channel compiler. Let Q be a finite alphabet, 0 a quiescent state, and $F : Q^{\mathbb{Z}} \rightarrow Q^{\mathbb{Z}}$ a deterministic translation-invariant CA of radius r . Let

$$w : Q \rightarrow \mathbb{N}, \quad w(0) = 0, \quad w(a) \geq 1 \quad (a \neq 0),$$

conserve total weight on every finite-support configuration. Thus the vacuum is the unique zero-weight symbol. A mass- M configuration has at most M occupied sites. Distinct states may have the same positive weight; this matters when comparing particle bounds in numerical-state models.

Zero-weight active symbols, signed weights allowing cancellation, and a nonzero infinite periodic background are outside these hypotheses. Quiescence and locality preserve finite support at each finite time, but do not bound its position or its diameter uniformly in time. A finite-support configuration on $\mathbb{Z}$ is not a finite-ring configuration.

Write $(\mathcal{T}_a x)(i) = x(i - a)$ for translation to the right. For nonempty x , let $\ell(x) = \min \text{supp}(x)$ and $\mathcal{N}(x) = \mathcal{T}_{-\ell(x)} x$ be its normalized shape. The vacuum is handled separately.

Proposition 43.1 (Deciding a diameter bound and its tail). *Given F , a finite configuration x , and an integer $D \geq 0$, one can decide whether $\text{diam}(F^t x) \leq D$ for every $t \geq 0$. Under this full-orbit bound, exact finite-configuration reachability and finite-word occurrence, anchored or at any translate, are decidable. Without the bound promise, the same procedure decides occurrence before the first escape and otherwise reports the escape, without concluding anything about later unrestricted occurrence.*

Proof. There are finitely many normalized nonempty shapes of diameter at most D , at most $(|Q| - 1)|Q|^D$. With mass M a sharper bound is

$$\sum_{k=1}^{\min(M, D+1)} \binom{D}{k-1} (|Q| - 1)^k. \quad (43.1)$$

Position zero is occupied, the remaining $k - 1$ positions are chosen in $\{1, \dots, D\}$, and all occupied symbols are nonzero. An exact-weight count is the coefficient of z^M in

$$\left(\sum_{a \neq 0} z^{w(a)} \right) \left(\sum_{a \in Q} z^{w(a)} \right)^D.$$

Simulate until the diameter exceeds D or a normalized shape repeats. One event must happen in finitely many steps. If

$$\mathcal{N}(F^i x) = \mathcal{N}(F^j x), \quad i < j, \quad p = j - i, \quad \delta = \ell(F^j x) - \ell(F^i x),$$

then determinism and translation equivariance give, for every $0 \leq s < p$ and $k \geq 0$,

$$F^{i+s+kp} x = \mathcal{T}_{k\delta} F^{i+s} x. \quad (43.2)$$

Consequently no future shape escapes. An escape disproves the diameter bound, while a repeat proves it.

For exact target y , inspect the finite prefix and then each phase $z = F^{i+s} x$. It can reach y exactly when its normalized shape agrees and

$$\ell(y) - \ell(z) = k\delta \quad \text{for an integer } k \geq 0.$$

This is a divisibility and sign test; when $\delta = 0$ it is simply equality of offsets.

For a full word on a fixed interval, each occupied phase site moves as $u + k\delta$. Required nonzero symbols give position equalities and symbol tests; required zeros exclude the moving occupied sites from their positions. The resulting finite Boolean combination of affine equalities and inequalities in

one integer $k \geq 0$ is decidable, including an all-zero word. For occurrence anywhere, translation of the entire phase does not change the answer, so a finite prefix and one list of phases suffice. A local predicate allowing several alphabet symbols is a finite union of these tests.

If an escape is encountered before an observed hit, it only disproves the bound. It does not reject a target that might appear after the escape. This is why the unrestricted-reachability conclusion requires the stated full-orbit promise. $\square$

If F is injective on finite configurations, a normalized repeat has no true transient. From $F^j x = \mathcal{T}_\delta F^i x$ and equivariance, cancel F^i to obtain $F^{j-i} x = \mathcal{T}_\delta x$. Thus a wholly bounded-diameter reversible orbit is translating periodic from time zero. If the existence of some diameter bound is promised but no value is supplied, simply simulate normalized shapes until a repeat; the promise ensures termination. Without that promise this is only a semidecision of boundedness.

These arguments are elementary single-orbit consequences of finite normalized shape space. Classical number-conserving CA dynamics already studies bounded-diameter classes and periodic or generalized-shift behavior; see Durand et al. [21] and Formenti–Grange [22]. No new classification theorem is claimed here.

Proposition 43.2 (No computable complete diameter cutoff). *Suppose a fixed F and computable finite input encodings $e \mapsto x_e$ reduce an undecidable predicate to eventual occurrence of an effective finite local observation, possibly anchored. There is no total computable $B(e)$ such that, whenever the observation first occurs at h , every configuration through time h has diameter at most $B(e)$.*

Proof. If such B existed, simulate until a hit, an escape from the bound, or a normalized repeat. A hit answers yes. Escape before a hit answers no by the assumed completeness of B . A normalized repeat permits the exact tail decision in Proposition 43.1, including anchored observations. One of these events occurs in finitely many bounded-width shapes, contradicting undecidability. $\square$

This does not forbid sparse witnesses with large gaps, unbounded integer values, or an externally varying horizon. It forbids a computable complete geometric cutoff through successful runs. A computable complete time or witness-length cutoff would similarly turn the relevant finite search into a decision procedure.

[write] In *canonical-diophantine-certificates* the finite-search step of this proof is the standard argument of Proposition `cdc:bd:prop:nobound` (also used at the end of Part I’s Section 13); the statement, a cutoff on orbit *diameter*, is source 13’s.

43.1 Finite rings and global halting

On a specified n -cell ring, every mass sector is finite, so orbit exploration decides reachability. Its cardinality is

$$[z^M] \left(\sum_{a \in Q} z^{w(a)} \right)^n \leq \sum_{k \leq M} \binom{n}{k} (|Q| - 1)^k.$$

For triple-channel cap K it is $[z^M](1 + \dots + z^K)^{3n}$, at most $\binom{3n+M-1}{M}$. Fixed M makes the bound polynomial in the *numerical* ring length, not necessarily in its binary encoding length. A reversible ring orbit is a cycle from its start.

A line computation from support diameter D_0 is reproduced through a specified horizon T by a ring whose periodic copies have disjoint radius- rT influence intervals; $n > D_0 + 2rT$ suffices. The zero padding and ring length depend on T . This cannot replace the infinite line for all times at once.

An injective dynamics cannot enter an existing fixed point or finite cycle from outside: the cycle already supplies the unique predecessor of each of its points. A reversible simulator therefore reports a halt by an observation while retaining information elsewhere. Stabilization of a finite window, or a departing pulse, must not be called global CA halting.

44 Decidability at total mass at most two

Theorem 44.1 (Two-unit-mass decision theorem). *Under the positive-weight, unique-vacuum hypotheses of Section 43, exact finite-configuration reachability and finite-word occurrence, anchored or anywhere, are decidable from every finite-support initial configuration of total mass at most two. Reversibility is not required. For a fixed CA F , these problems have an algorithm polynomial in the input size when finite configurations are sparse lists of nonzero coordinate/state pairs with binary coordinates, and a pattern is an explicit full finite word, whose length counts toward input size. An anchored word has a binary anchor coordinate.*

The theorem rules out universality by these observation problems at masses zero, one, and two within this model. It does not show that three suffices, or identify a least universal mass. The proof is given in detail because the absolute-position and long-excursion issues are easy to miss. ([write] Added 2 October 2026, batch 80: Part V shows that three suffices in this model when there are many weight-one symbols (Theorem 49.2, and for exact pairs Theorem 64.1); Part VI shows that with at most one weight-one symbol masses three and four remain decidable (Theorems 74.1 and 83.1). Source 14 proves the present theorem again; Section 59.)

44.1 Single particles are finite-state walkers

Let $Q_j = \{a \in Q : w(a) = j\}$. From a singleton of type $a \in Q_1$ at position u , positivity and conservation force the next state to be a single weight-one site. Locality puts it within r of u . Thus computable maps

$$\alpha : Q_1 \rightarrow Q_1, \quad d : Q_1 \rightarrow [-r, r] \cap \mathbb{Z}$$

give its next type $\alpha(a)$ and displacement $d(a)$. A type trajectory has a finite transient and cycle. For two hypothetical independent walkers use the product type map on Q_1^2 . Its transient μ and period p satisfy $\mu + p \leq |Q_1|^2$. At each cycle phase $0 \leq s < p$ their positions are

$$L(\mu + s + kp) = L_s + k\Delta_L, \quad R(\mu + s + kp) = R_s + k\Delta_R, \quad k \geq 0. \quad (44.1)$$

Types are fixed within a phase. The displacements over a full product cycle are constants. Reversibility would make the singleton type map a permutation, but no part of the proof needs that simplification.

44.2 Exact acceleration to the next encounter

A mass-two configuration is *near* if its diameter is at most $2r$. Every other such configuration consists of two weight-one sites $L < R$ with gap greater than $2r$. No radius- r neighborhood sees both. Quiescence then implies that the next update is exactly the superposition of the isolated updates, as long as the configuration starts far.

Each step changes the gap by at most $2r$. Starting with gap greater than $2r$, the walkers cannot cross without entering the near region. For each affine phase of (44.1), solve

$$R_s - L_s + k(\Delta_R - \Delta_L) \leq 2r, \quad k \geq 0. \quad (44.2)$$

If the drift is negative, the first admissible k is the appropriate integer ceiling, truncated below by zero. For zero or positive drift, the phase either already satisfies the inequality at $k = 0$ or never does. Inspect the finite transient too and choose the least resulting physical time. This gives the exact first encounter or proves no encounter occurs.

At that first near time the independently calculated positions are still correct, because the preceding update began far. Stop independent motion there and use the actual CA rule for subsequent behavior. Hypothetical crossings or negative signed gaps after that first time are never used as real dynamics. This boundary is essential for collision correctness.

44.3 Only finitely many encounter shapes

There are exactly

$$N_{\text{near}} = |Q_2| + 2r|Q_1|^2 \quad (44.3)$$

normalized mass-two near shapes: one weight-two site, or an ordered pair of weight-one types at separation $1, \dots, 2r$. Record each near shape together with its time and absolute offset. From a previously unseen near configuration, perform one actual CA step. If the result is still near, continue directly. If it is far, accelerate to its first subsequent near encounter by (44.2), or conclude that this is a terminal forever-far segment.

A weight-two singleton may split, but its next diameter is at most $2r$ and is included. A near pair may merge into one weight-two site. Simultaneous collision, exchange, crossing, and coincident outputs are all handled by the actual CA update rather than a proposed particle collision law. A step starting near has diameter at most $4r$, so every newly departing far excursion starts with gap at most $4r$.

After at most $N_{\text{near}} + 1$ near visits, either a normalized near shape repeats or a terminal forever-far segment has been found. Every finite excursion is computed exactly. A repeated near shape gives the translating periodic tail (43.2). This proves termination of the compressed orbit construction. If $r = 0$, there are no separated-pair interactions, the near-pair term in (44.3) is zero, and each occupied site evolves locally; the same singleton and phase arithmetic handles the case directly.

Mass zero is the stationary vacuum. Mass one is the finite-state walker already described. These complete the orbit descriptions needed for Theorem 44.1.

44.4 Testing every intermediate time

Skipping an excursion without checking its intermediate configurations would not decide reachability. On a phase of a far segment restrict the integer k in (44.1) to its exact interval, which may have no upper bound. The two types a_s, b_s are fixed.

For an exact target with two weight-one sites (u, a) and (v, b) , $u < v$, require the type matches and solve

$$L_s + k\Delta_L = u, \quad R_s + k\Delta_R = v$$

for a common admissible integer k . Incompatible mass or occupied-site counts reject the phase. A weight-two singleton is checked at actual near states.

For a full word on an anchored interval $I = [A, B]$, there are only four possibilities relevant to a far segment:

1. More than two nonzero symbols, or a weight-two nonzero symbol, cannot match
2. Two nonzero symbols require the two matching type and position equations
3. One nonzero symbol at u requires a matching walker there and the other walker strictly outside I
4. An all-zero word requires both walkers strictly outside I

Each case is a finite union of intersections of linear equalities and inequalities in one integer k . Solve them by interval intersection, ceiling/floor operations, and divisibility. This also treats the mass-one walker by omitting the second position.

For occurrence at an arbitrary translate, anchor any nonzero word at a matching occupied site. Two nonzero word symbols require their specified relative gap and types. With one nonzero symbol the other particle must lie outside the corresponding translated interval. An all-zero finite word always occurs sufficiently far from finite support. Again the tests reduce to one-variable affine arithmetic. An empty word, if allowed, occurs everywhere by convention.

A later translated copy of a repeated near loop may hit an anchored target that the first loop missed. Therefore do not stop at a failed first copy. Expand the finite loop into its phases and apply (43.2) to every phase, including all intermediate excursion times; the offset arithmetic then searches every later copy. Unanchored queries need only one copy. This establishes correctness of every stated observation decision.

44.5 Why the fixed-rule algorithm is polynomial

Fix F . Alphabet size, radius, product-type transients and periods, and the number of near shapes are constants. Only the initial far gap may be arbitrarily large. Formula (44.2) computes its first encounter, if any, using binary arithmetic; that time is at most $O(g + 1)$ with a rule-dependent constant, but the algorithm does not simulate g steps. The resulting coordinates have input-linear bit length.

Every subsequent returning excursion begins with gap at most $4r$. During the constant-length product-type transient and one cycle, gap changes by a rule-dependent constant. If cycle drift is nonnegative and this inspected prefix has no encounter, none will occur. If drift is negative, it is an integer at most -1 , so the remaining number of cycles to encounter is also bounded by a rule-dependent constant. Thus every returning post-encounter excursion has constant duration for fixed F .

There are constantly many distinct near visits before a repeat or forever-far tail. Accordingly the compressed orbit has constantly many affine-phase pieces, and every repeating near loop has constant length. The target tests use numbers of input-linear bit length and a fixed number of arithmetic variables. Scanning the explicitly supplied word and performing those tests is polynomial time. No optimal exponent or uniform complexity bound in the description size of F is claimed. Binary encoding of a gigantic gap is crucially different from listing every empty site in that gap.

This proof resembles classical one-counter reachability: after quotienting translation, a separated pair has one unbounded gap, finite internal control, and exceptional dynamics only near a fixed threshold. The direct argument additionally retains absolute displacement for anchored observations. Haase et al. [23] provides related one-counter context; it is not cited as stating this precise CA

theorem. A targeted search did not establish priority of the exact statement.

44.6 What particle bounds from the literature do and do not say

Alhazov and Imai’s 2016 paper on particle complexity [24] is directly relevant. Its verified abstract describes finite number-conserving CA with nonnegative integer states and finite total numerical sum, and announces a universal construction with five particles and a sufficiently large finite radius. It also announces a radius-one construction based on two-register machines. The abstract does not assert reversibility. These are finite-mass infinite-line claims, not finite-ring universality.

Only the abstract-level model and upper-bound statements are relied on here; the full construction and any optimality claim have not been verified for this report. In a numerical-state NCCA with weight equal to the state value, there is only one weight-one state. The broader weighted-alphabet model of Theorem 44.1 allows multiple internal types of weight one. A small-mass construction in that broader model cannot automatically be quoted as a numerical-state particle bound. Neither the five-particle literature statement nor the mass-two decision theorem establishes a three-particle reversible universal model or a minimum universal mass.

45 Horizon-free semilinearity at mass at most two

The low-mass decision argument supports a stronger Diophantine conclusion. Unlike the generic sparse certificate, the following result has no supplied horizon and has fixed witness arity after the rule and its arithmetic description are fixed. It applies to a decidable subclass, so it creates no fixed-arity statement for arbitrary universal computation.

Theorem 45.1 (Effective timed and untimed semilinearity). *Fix a CA satisfying the hypotheses of Theorem 44.1, and fix the ordered nonzero labels of an initial configuration and a target, each of mass at most two. The numerical-coordinate relation*

$$\mathcal{R}_F(\mathbf{x}, \mathbf{y}, T) \iff T \in \mathbb{N} \text{ and } F^T(x) = y$$

is effectively Presburger-definable. Its projection $\exists T \mathcal{R}_F$ is also effectively Presburger-definable. Equivalently these are effective semilinear coordinate relations, with signed coordinates interpreted by orthants or the canonical natural encoding below. Taking a finite union allows the labels to vary over their finite allowed cases.

Here Presburger arithmetic has addition, order, integer constants, and quantification, but no multiplication of two unbounded variables. Semilinearity concerns the numerical coordinates, not a language of their binary digit strings. The proof must show why the initial unbounded gap never becomes a variable period or a variable coefficient in a later phase.

45.1 A linear guard for the entire independent prefix

Fix the initial ordered weight-one types at positions $u < v$, with $g = v - u > H_0 = 2r$. Let λ_i, ρ_i be the cumulative independent displacements at time i , starting at zero. For the product-type transient μ and period p , put $i_s = \mu + s$ and write

$$\begin{aligned} t &= i_s + pk, \\ L(t) &= u + \lambda_{i_s} + D_L k, & R(t) &= v + \rho_{i_s} + D_R k, & k &\geq 0. \end{aligned} \tag{45.1}$$

All displayed coefficients except u, v, k are fixed by the rule and initial types. Set $c_i = \rho_i - \lambda_i$ and $D_* = D_R - D_L$. The gap at phase s is $g + c_{i_s} + D_*k$.

Define $\text{Safe}(g, T)$ to mean that every gap at $0 \leq h < T$ exceeds H_0 . This ensures that actual motion agrees with the independent motion at time T , including a possible first near time. It is already a Presburger statement by quantifying h and expanding its finite phase alternatives. A quantifier-free finite guard in the chosen phase is also explicit.

If $T = i < \mu$, require $g + c_h \geq H_0 + 1$ for $0 \leq h < i$. If $T = \mu + s + pk$, first impose those inequalities for all $h < \mu$. For a cycle phase $j < s$, earlier occurrences have indices $0, \dots, k$; for $j \geq s$ they have indices $0, \dots, k-1$, an empty set if $k = 0$. Minima of these arithmetic progressions give the remaining guards:

Drift	$j < s$	$j \geq s$
$D_* \geq 0$	$g + c_{i_j} \geq H_0 + 1$	$k = 0$ or $g + c_{i_j} \geq H_0 + 1$
$D_* < 0$	$g + c_{i_j} + D_*k \geq H_0 + 1$	$k = 0$ or $g + c_{i_j} + D_*(k-1) \geq H_0 + 1$

These are Boolean combinations of constant-coefficient affine inequalities. At time zero all earlier-index sets are empty, as required. Conjoin the applicable guard, the time equation, the current type matches, and target-coordinate equations from (45.1). A finite disjunction over phases describes every target time before or at first near entry, as well as all times on a forever-free initial flight.

45.2 First entry with the absolute position preserved

For $r \geq 1$ the first near entry from an initially far pair has two distinct weight-one sites and gap $e \in \{1, \dots, H_0\}$. A step from gap at least $2r + 1$ leaves gap at least one, so a merger or crossing cannot skip this boundary. There are finitely many entry shapes E .

For a transient entry time $\tau = i$, impose its Safe guard, phase labels, and

$$g + c_i = e, \quad \xi = u + \lambda_i, \quad \tau = i.$$

For a periodic entry at $\tau = \mu + s + pk_0$, impose its Safe guard, $k_0 \geq 0$, phase labels, and

$$g + c_{i_s} + D_*k_0 = e, \quad \xi = u + \lambda_{i_s} + D_Lk_0, \quad \tau = \mu + s + pk_0. \quad (45.2)$$

Here ξ is the absolute leftmost entry coordinate. The guard enforces *earliest* entry, rather than an arbitrary later meeting of hypothetical walkers. Every actual first entry satisfies one such case; conversely the guard and bounded displacement prove that each satisfying case is the real first entry. Thus the full relation between initial coordinates, entry shape, time, and absolute offset is Presburger.

45.3 Finite rule-fixed orbit templates after entry

For each normalized near shape E at offset zero, preprocess its orbit by the near/far algorithm of Theorem 44.1. There are only N_{near} such shapes, each fixed by the rule. Every returning excursion has rule-dependent constant duration; preprocess until a normalized near repeat or a terminal nonreturning free flight. Expand the finite returning flights and one complete recurrent loop when present.

This gives a finite collection of templates, each with fixed occupied-site count, fixed ordered labels, and integer constants $\theta, \pi, \eta_j, \delta_j$, representing

$$\text{elapsed time} = \theta + \pi n, \quad \text{site } j = \eta_j + \delta_j n. \quad (45.3)$$

A listed isolated time fixes $n = 0$; an infinite template has $n \geq 0$ and $\pi > 0$. A translating loop has the same displacement for every particle in a phase. A terminal free flight has the finite-state phase templates already used above, possibly with different left and right drifts. The occupied-site order holds on every represented time, by translation for a loop and by nonreturn for a free tail.

Starting E at offset ξ adds ξ to all positions. Compose (45.3) with (45.2):

$$T = \tau + \theta + \pi n, \quad y_j = \xi + \eta_j + \delta_j n.$$

Substitution gives affine expressions in the unbounded variables k_0, n . There is no product $k_0 n$: the entry shape has been chosen from a *finite* set, so every subsequent period and drift is a rule-fixed constant, independent of the initial gap. A finite union over entry cases and templates therefore remains Presburger. Targets before or at entry were already covered; these compositions cover every later target. Overlap of descriptions is harmless for definability.

An initially near input selects one of the same finite templates directly, with its initial offset and $\tau = 0$. A mass-two singleton can split and a pair can merge, so one discards mismatched *template* site counts rather than assuming the occupied-site count is conserved. A total-mass mismatch is impossible. Vacuum and mass-one cases have stationary or finite-state affine descriptions. At radius zero, each occupied coordinate is fixed and its label follows a finite map, giving a finite transient and periodic label condition. These arguments prove the timed part of Theorem 45.1; existentially quantifying T proves the untimed part.

45.4 Effective elimination and canonical signed input

Standard effective Presburger quantifier elimination converts the constructed formula into a quantifier-free Boolean combination of affine inequalities and fixed-modulus congruences; see Cooper [25]. The effective equivalence with semilinear sets is classical, as in Ginsburg–Spanier [7]. These standard results are used for the final arithmetic normal form, rather than assumed as a CA-specific reachability theorem.

Encode each signed coordinate a by natural inputs (a^+, a^-) with

$$a = a^+ - a^-, \quad a^+ = 0 \text{ or } a^- = 0.$$

This gives exactly one representation of each integer. Include this domain condition and strict ordering of occupied coordinates in the predicate before elimination. With at most two initial and two target sites, there are at most eight natural coordinate inputs, and at most nine external natural inputs when T is retained. For untimed reachability, eliminate T itself before the next compiler. Leaving it as a witness would generally destroy uniqueness when a target is visited at multiple times.

46 Canonical fixed-arity quartics for the low-mass relation

Here is a complete deterministic compiler from a fixed quantifier-free Presburger predicate $\Phi(\mathbf{z})$ on natural inputs. Normalize its atoms to $L(\mathbf{z}) \geq 0$ or $L(\mathbf{z}) \equiv 0 \pmod{d}$, where L is affine with integer coefficients and $d \geq 1$ is fixed. Equality becomes two inequalities and strict inequality a shifted

weak inequality. Express the Boolean structure by AND, OR, and NOT gates. Let I, C, G be the actual numbers of inequality atoms, congruence atoms, and Boolean gates in the chosen circuit, respectively. Identical nodes may be shared or repeated; the counts refer to the circuit actually used.

46.1 Inequalities and congruences have unique arithmetic witnesses

For $L \geq 0$ introduce natural b, s with residuals

$$b(b-1), \quad L - (2b-1)s - b + 1.$$

They force $(b, s) = (1, L)$ when $L \geq 0$, and $(b, s) = (0, -L-1)$ otherwise. Thus both witnesses exist uniquely for every input, and b is the truth bit even when the atom is false.

For $L \equiv 0 \pmod{d}$ introduce six natural witnesses q^+, q^-, a, h, b, s and five residuals

$$\begin{aligned} q^+q^-, \quad L - d(q^+ - q^-) - a, \\ a + h - (d-1), \quad b(b-1), \quad a - (2b-1)s - b. \end{aligned} \tag{46.1}$$

The first three impose the unique signed Euclidean quotient $q = q^+ - q^-$ and remainder $0 \leq a < d$. The product zero condition gives the unique sign-separated quotient representation; $h = d - 1 - a$. The last two force $b = s = 0$ if $a = 0$, and $b = 1, s = a - 1$ if $a > 0$. Therefore the congruence truth bit is the affine expression $1 - b$. All witnesses are uniquely fixed, including negative quotients and modulus one.

46.2 Boolean gates and the exact ledger

For each Boolean gate introduce one natural output z_g and its corresponding residual

$$\begin{aligned} \text{AND : } z_g - uv, \\ \text{OR : } z_g - u - v + uv, \\ \text{NOT : } z_g - 1 + u. \end{aligned}$$

Inputs are preceding outputs, atom truth bits, or constants zero and one. They are already Boolean, so each gate output is uniquely its correct Boolean value. Add one residual $o - 1$ for the final output o . Summing the squares of all residuals gives a polynomial of degree at most four with exactly

$$K_\Phi = 2I + 6C + G \quad \text{natural witnesses,} \quad E_\Phi = 2I + 5C + G + 1 \quad \text{residuals.} \tag{46.2}$$

Every input has exactly one complete assignment to the atom and gate witnesses. The final residual accepts it exactly when Φ is true. Thus the natural fiber is a singleton for true inputs and empty for false inputs. This does not rely on a product of branch equations with unconstrained inactive witnesses.

Corollary 46.1 (Horizon-free canonical low-mass certificates). *For each fixed CA and ordered label case in Theorem 45.1, timed and untimed exact reachability admit effectively constructible integer polynomials of degree at most four with zero or one natural auxiliary witness tuple. Their witness arity is the rule- and compiler-dependent constant K_Φ in (46.2), after effective quantifier elimination and a chosen Boolean circuit. No supplied time bound is required.*

These constants may be extremely large. The report neither instantiates a quantified-formula eliminator for every possible rule nor claims a small universal numerical bound, efficient uniform preprocessing, or an optimal arity. Total arity is the number of external natural inputs plus K_Φ , at most $9 + K_\Phi$ timed or $8 + K_\Phi$ untimed for fixed labels. Mass two does not mean two witnesses. Coefficients and their bits, formula length, and Boolean-circuit size remain paid parts of a concrete instantiation.

Uniqueness here comes from the *post-elimination arithmetic compiler*, not from CA determinism: an untimed target may be reached at many or infinitely many times. No corresponding real or rational witness characterization is claimed. No analogous semilinearity or finite-fold statement for arbitrary recursively enumerable sets follows from this decidable low-mass result.

[write] **A second route, at a higher degree.** Corollary 46.1 rests on the general fact that a semilinear set has a single-fold Diophantine representation, and for that fact the collection already has a sharper statement: Theorem `cdc:of:thm:singlefolds1` of Part XI of *canonical-diophantine-certificates* gives, for every semilinear set, an integer polynomial of degree at most *two*, nonnegative on the real nonnegative orthant, with exactly one natural witness for each member and none otherwise, built in the same way (Presburger quantifier elimination, then deterministic evaluation of the order and congruence atoms, with the comparison of `cdc:of:lem:unboundedcompare` and a Euclidean congruence gadget, and Boolean gates). The compiler of this section is a second route to that fact, of degree four (squared residuals) rather than an orthant-nonnegative quadratic, which carries the exact ledger (46.2). What is source 13's here is the semilinearity of the low-mass relation (Theorem 45.1), to which either compiler applies. Theorem 45.1 is related only loosely to Part I's rigidity theorem (Theorem 13.1): there a fixed globally nonnegative quadratic forces a semilinear zero set, here low-mass dynamics forces a semilinear reachability relation.

47 Reproduction and evidence

The accompanying ZIP contains editable LaTeX, the rendered PDF, standard-library Python implementations, coefficient-explicit fixtures, completed finite local tables, structured test receipts, and a SHA-256 manifest. It contains no copies of third-party papers. Source links and exact attribution limits are given in the bibliography and provenance document.

47.1 What each layer of verification establishes

The producer compiler constructs each sparse polynomial as a canonical list of integer-coefficient monomials. An export includes the public input/table descriptor, variable and residual ordering, canonical witness values, row aliases, degree and scalar ledgers, and every residual coefficient. Its bound verifier regenerates the declared circuit and compares it before evaluating the witness. This catches deletion of equations, changed input binding, changed row aliases, altered accounting, and malformed scalar types. Merely checking that a supplied witness satisfies some unbound list of equations would be weaker. A caller seeking one particular input must still compare the public descriptor with that intended input; no file can infer an unstated intention.

An independent implementation separately constructs the preferred row-selector polynomials. Its cross-check compares residual coefficient multisets under an explicit wire-name correspondence, not only decoded trajectories or final zero scores. Separate exact dynamics tests compare sparse and dense physical evolution. A third set of source checks verifies the corrected rule schemes, legal source macrosteps, mass-fiber completion, and the false-observer example. Finally, the low-mass

audits test encounter acceleration and query arithmetic; they are not exhaustive tests of all CA rules.

Finite tests support the implementations and expose edge cases; the general correctness results rest on the proofs in this report. None of the code constitutes a proof-assistant formalization, an exhaustive literature search, or a test of every natural witness.

47.2 Selected literal fixtures

Fixture	V	R	Multiplications	Additions
Binary three-way collision	288	243	1,614	1,424
Same, orthant-exact	288	261	1,830	1,559
Gap 10^{100}	108	88	540	495
Mixed ternary mass	610	370	3,694	3,287

All counts use the literal convention (40.1). The huge-gap example has maximum witness bit length 334 and residual-coefficient bit length 333, illustrating that sparse dimension can remain small while geometric values and coefficients grow. The mixed-mass fixture has largest witness bit length four. The orthant collision fixture adds eighteen residuals, exactly $2MT$.

The restricted-row doubling fixture is bound to the full completed local table but uses the declared 97-row simulation subset. It verifies an actual first pulse, rather than merely simulating a source trajectory outside the polynomial. The larger independently replayed $n = 2$, $T = 41$ case has 257,872 witnesses, 157,683 residuals, 1,263,585 nonzero residual terms, 1,694,649 literal multiplications, and 1,421,268 additions. Its largest witness is 45. Both coefficient-explicit instances are shipped. The smaller $n = 0$, $T = 7$ instance has 45,712 witnesses, 28,585 residuals, 222,845 residual terms, 299,323 multiplications, and 251,430 additions. Its largest witness is 25. The counts for each are generated and checked by the included script.

47.3 Audit coverage

The independent dense/sparse dynamics tests cover 13,200 randomized steps and 73,728 steps across all 36 binary conservative local permutations, for 86,928 comparisons. The independent preferred literal compiler checks 36 instances containing a total of 17,058 witnesses and 9,663 core residuals, and verifies the $2MT$ orthant increment. The earlier factorized independent compiler checks are retained separately with their own ledgers; those results are not relabeled as preferred-backend tests.

The preferred producer/independent coefficient cross-check matches 54 compilers, 21,096 witnesses under explicit renaming, and 11,844 residuals coefficientwise. It additionally checks 20 uniform-loader instances, 24 paid loader-plus-observer instances (12 valid first pulses and 12 correctly rejected earlier pulses), six restricted-row coefficient comparisons, and ten accounting or declaration mutations. Restricted-domain exits and a restricted table omitting the vacuum are rejected. These tests use the final preferred backend, not the older factorized compiler.

The source audit exhaustively enumerates 373 valid three-state source machines under its stated syntax and endpoint conventions, checking 8,672 legal macrosteps with both counters between zero and three. It also checks 3,000 seeded random source machines with 2–12 states for local properties, and 22,004 legal macrosteps on the first 300. Every rule family is exercised. There are 672 startup-signal predecessor tests. The doubling source is replayed for $n = 0, \dots, 10$, including

exact microtimes, final counters, and persistent result centers. The nonuniversal doubling and false-signal completions contain 17,576 and 9,261 rows respectively.

The independent mass-two arithmetic audit checks 93,663 cases: 13,656 exhaustive and 20,000 random encounter instances; 20,000 anchored-word, 20,000 unanchored-word, and 20,000 exact-configuration queries; and seven large-integer or unbounded cases. A separate encounter regression checks 6,750 far segments and a 100-digit encounter. These are tests of the affine arithmetic used in the proof, not claims about that many distinct CA rules.

The semilinearity component regression checks 1,697,593 cases: 484,788 exhaustive and 1,210,000 random free-prefix guards, four hundred-digit time cases, 81 inequality and 1,452 signed congruence bounded-fiber cases, twelve Boolean truth-table cases, 256 composed-polynomial inputs, and 1,000 single-variable mutations. An independent audit checks another 837,150 arithmetic and actual-CA cases, including 394,740 timed configurations with splitting, merging, recurrent near templates, and terminal free flights. Both audits include exact affine and negative-quotient edge cases. Neither implements arbitrary-rule quantifier elimination or constitutes formal verification.

The release runner regenerates outputs in a separate working directory and compares stable results with the shipped receipts. Its README lists commands, dependencies, and the optional larger reproduction. The manifest detects accidental modification; it is an integrity record, not a replacement for the mathematical or semantic checks.

[write] **The release in this report.** The programs are shipped as `code/13-sparse-lattice-*`, the receipts, generic fixtures and the two completed tables as `data/13-sparse-lattice-*`, and the provenance record as `13-sparse-lattice-SOURCE-PROVENANCE.md`; the delivered sub-directories `replay/`, `receipts/` and `tables/` are flattened, and the three programs named `audit.py` keep their directory in their shipped names (README of this report). The two completed tables (`data/13-sparse-lattice-doubling-complete.csv` and `data/13-sparse-lattice-false-signal-complete.csv`) are kept byte for byte with their delivered CRLF line ends. The two coefficient-explicit Morita fixtures named above (the $n = 0$, $T = 7$ and $n = 2$, $T = 41$ instances, 0.84 and 4.37 MB compressed) are *not* shipped: the shipped generator rebuilds them in about 14 s and 94 s with identical decompressed payloads, and the README gives the commands; their receipts are shipped. The delivered manuscript (six L^AT_EX files), PDF, README, checksum ledger, packager and one byte-identical copy of the Morita audit are not shipped. The programs need the delivered layout and POSIX (LF) newlines, and must be run on a copy (README). Since batch 80 the corrected code edition (next paragraph but one) adds the exactness suite `code/13-sparse-lattice-test_poly_exactness.py`, its receipt `data/13-sparse-lattice-poly_exactness.json` and the correction note `13-sparse-lattice-CORRECTION.md`.

[write] **The repository’s review.** The programme reviewed this archive in full [30]: it found the generic history theorem, the loader and observer, the corrected source interface, the diameter barrier, the mass-two decision proof and the semilinearity argument sound within their stated domains, regenerated both large Morita fixtures, and ran both complete eighteen-stage replays. Its one finding is a low-level exactness defect: the public Poly constructor of `sparse_mass.py` accepts floating coefficients and mutable terms, so a hand-built floating residual can round -1 to 0. The high-level compiler and the JSON decoder already require exact coefficients, so the finding refutes no theorem and no exported certificate. The review’s repair is the patch `sparse_mass_exact_polynomials.patch` in the programme’s directory [30]; the shipped `code/13-sparse-lattice-sparse_mass.py` was the delivered, unpatched file until batch 80 (see the paragraph after this one). Two later reductions of the programme bear on this Part’s ledgers without changing its theorems: a wire projection of the preferred compiler with a fixed initial configuration [32] removes $T[7M + M(M - 1)]$ natural

variables and as many rows, leaving $V = T[M(S + 7) + 4M(M - 1)]$ and $R = T[10M + 4M(M - 1)]$ with quadratic residuals and a bijection of natural zero sets; and a five-witness congruence atom [33] lowers the witness count (46.2) of the post-elimination compiler from $2I + 6C + G$ to $2I + 5C + G$ at the same residual count. Both are the programme's, are not re-proved here, and are not gate counts.

[write] **Corrected code edition (added 2 October 2026, batch 80).** `Sparse_Lattice_Diophantine_Certificates_corrected.zip` (arrival 4e270aa46, batch 80, manuscript 07; placed by 8a4e64732, batch 80, cluster K1; the archive dates its correction 3 October 2026) repairs the review's finding; its authors read the review's commit but did not execute its code. `Poly` now validates directly supplied terms: immutable tuples, exact integer indices and coefficients (not Booleans, floats or subclasses), nonzero coefficients, sorted indices, distinct sorted monomials; `Poly.make` remains the canonicalizing constructor, and negative (parameter) and repeated (power) indices and the zero and constant polynomials stay valid. The new suite `code/13-sparse-lattice-test_poly_exactness.py` rejects 24 malformed constructor categories and the floating false zero (the exact residual at $2^{60} + 1$ is -1 , with score 1), checks canonicalization and mutation isolation, and compares 2,700 exact expressions on 300 seeded pairs; with `-baseline-module` it reproduces the defect on the original (residual 0.0, accepted witness) and compares the same expressions with the original producer. The release runner runs it first (nineteen stages with `-regenerate-source`, eighteen before). The repair is written independently and is equivalent to the programme's patch: the patched original passes the suite and the unpatched original fails it. The patch is not to be applied to the shipped file, which already contains the repair (applied anyway, it would insert a second constructor check). The manuscript, the fixtures, both tables, every receipt value other than the producer hash and the new stage, and every theorem and ledger of this Part are unchanged; at the batch-79 write the shipped file was the original, and since batch 80 it is the corrected edition's. The correction note is `13-sparse-lattice-CORRECTION.md`, and the shipped provenance record `13-sparse-lattice-SOURCE-PROVENANCE.md` now carries the corrected edition's dated preface. The programme's correction audit [36] confirms the repair, that removing the new guard leaves the module's syntax tree identical to the original's, and the unchanged mathematics.

48 Further questions and boundaries

The strongest next steps are narrowly defined rather than general claims of universality improvement.

1. **Print one safe universal source.** Choose and export a concrete universal counter simulator, retain the history and prime-encoding invariants, and provide its actual m , complete local table, valid input encoding, and proof of reachable legality. This would replace the present existence interface with a fully numerical instance
2. **Compress sparse microtime when structure permits it.** The certificate pays one layer per physical lattice step. Large counter gaps create long round trips. An exact arithmetic macrostep compiler for the source could remove these journeys without changing the finite-mass semantics, but would have to pay for branch tests, timing, and valid intermediate routes
3. **Separate mass, state, radius, and degree costs.** The row-selector simplification reduces residual count dramatically while retaining q^3 witnesses and long sums. Other explicit lookup or collision organizations may trade these resources. None of the displayed counts is a lower bound or minimality record
4. **Resolve small-particle model comparisons.** Check the full particle-complexity construc-

tions and their precise observation, state-weight, radius, and reversibility conventions before claiming a sharper universal mass threshold. The mass-two theorem is a rigorous exclusion in its stated positive-weight model; it is not a proof of a matching upper bound ([write] added 2 October 2026, batch 80: answered in substance by Parts V and VI. The number of weight-one symbols is the decisive convention: with many, three units suffice for a fixed reversible automaton, Part V; with at most one, masses three and four are decidable, so integer-state automata need at least five particles, Part VI. Still open: the full Alhazov–Imai construction, a reversible numerical threshold, a binary upper bound, and the least number of weight-one symbols for universality at mass three)

5. **Keep exact targets separate from halt signals.** A reduction to a specified complete target needs its own output-normalization argument. The safe-source pulse interface alone cannot supply it ([write] added 2 October 2026, batch 80: answered for the three-mass weighted automaton by source 15, Theorem 67.1, whose output normalization is inverse execution of a fresh-entry source, with a target y_N that depends on the input; the pulse interface of this Part is unchanged)

The construction gives a family of finite-horizon Diophantine certificates with explicit variable growth. Treating T as an additional unknown in a single fixed-arity universal equation, or hiding all horizons behind an unspecified MRDP representation, would be a different result and would not inherit these scalar counts. No such transfer is made for the universal higher-mass sparse family. The horizon-free fixed-arity construction in Sections 45–46 is instead for the decidable mass-at-most-two subclass and pays its actual post-elimination circuit size.

[write] Two of these questions touch questions elsewhere in the collection. The first is the lattice counterpart of Part III’s literal source: Part III prints a complete universal conservative signal machine, while source 13 deliberately leaves its safe universal source as an existence construction. The certificate of Theorem 38.1 also bears on the question “Local certificates on adaptive support geometry” of Part XVI of *canonical-diophantine-certificates*, which asks for a sparse lattice support checked canonically at a cost proportional to support and boundary, paying for sorting, distinctness and nearest-neighbour tests: source 13 pays exactly those costs (pairwise equality and a bubble-sorting network) for one-dimensional conservative partitioned automata, at size $O(T(M^2 + Mq^3))$, which is not proportional to support plus boundary, and not for sandpiles. That question stays open.

Part V

Three conserved mass units suffice (sources 14 and 15)

[write] Part V prints two AI-assisted research manuscripts of ProveIt’s incoming reports, both dated 3 October 2026 and arrived in batch 80 (commit 4e270aa46). *Source 14* is batch-80 manuscript 10, *Three mass units in a reversible weighted cellular automaton*, author line “Mathematical construction and reproducibility report”, from the archive `Three_Mass_Reversible_Computation.zip`; its labels carry the prefix `smc:tm:`, and its sections are Sections 49–63 here. *Source 15* is batch-80 manuscript 02, *Exact targets with three conserved mass units*, author line “A clean return extension of the reversible weighted CA construction”, from the archive `Exact_Targets_Three_Mass_Units.zip`;

its labels carry the prefix `smc:ct:`, and its sections are Sections 64–73. They are called sources 14 and 15 after the prefixes of their shipped files. Source 15 is a declared extension of source 14 (“a separate, 15-page mathematical extension of the frozen three-mass construction”, in its delivery README): it vendors five of source 14’s programs byte for byte and pins source 14’s L^AT_EX source, PDF and checksum ledger by SHA-256 as its “frozen companion release 9”, and all three pins match source 14’s archive. Neither manuscript names a repository commit or cites Part IV or any repository file. Statements keep their numbering by section, and equations are numbered by section. The programme has reviewed both archives [45] (Section 1.14).

[write] **Relation to Part IV.** [Editorial note: source 14’s Section 59 proves Theorem 44.1 of Part IV (source 13) again, by the same argument: the same walker maps α, d , the same near/far threshold $2r$ and encounter inequality, the same count $|Q_2| + 2r|Q_1|^2$ of near shapes, the same translating tail and the same fixed-rule polynomial bound. Its two-mass arithmetic test is source 13’s audit (the shipped `code/14-three-mass-test_two_mass_arithmetic.py` differs from `code/13-sparse-lattice-two-mass-audit.py` only by omitting the receipt-writing line), and its receipt is byte-identical to `data/13-sparse-lattice-two-mass-arithmetic.json`, which is therefore not shipped twice. The section is printed as a pointer, keeping source 14’s statement and its use in Theorem 49.2; source 14 calls that proof “self-contained” and does not cite source 13.] The certificate gate $(E_t - e_{tj})(e_{tj} + u_{tj})$ of source 14 (and of source 15) is the inactive-product gate of *quadratic-orthant-certificates*, Part VI (`qoc:rn:lem:gates`); a note after Theorem 58.1 says so. Part IV and Part V both draw on Morita and Imai’s reversible two-counter normal form [19] (source 14 cites its Proposition 3.4, source 15 adds its Lemma 3.5); Part IV also interfaces with their partitioned automaton, whose mass is $m + 19$ for an m -state source, as source 14 recalls (Section 62); source 14’s automaton keeps three units for every source, on finite-support configurations with no background. Source 14 does not cite Kong’s thesis or Imai’s KAKEN report, which concern the numerical model of Part VI; a note in Section 62 points there.

Source 14 (batch-80 manuscript 10): Three mass units in a reversible weighted cellular automaton

Mathematical construction and reproducibility report, 3 October 2026. Its abstract follows.

A fixed one-dimensional reversible cellular automaton of radius at most four can simulate a universal reversible two-counter machine using configurations of total mass exactly three. The model has a finite alphabet of Boolean channel subsets, with subset cardinality as conserved weight and a unique zero-weight vacuum. Two counters are encoded by a single gap $12 \cdot 2^a 3^b$. A reversible residue query, rational gap changes, and finite instruction-tag cleanup give an explicit local permutation schema. Spatial blocking gives radius one at the same clock with four times as many types. A coordinate-preserving phase alternative has a fourfold clock cost. Both preserve the mass. A self-contained mass-at-most-two decidability proof makes three the sharp threshold for fixed-rule undecidable finite-pattern occurrence in this weighted model. We also give a quadratic natural-number certificate at a specified source-step horizon, with exact physical time and a fully stated resource ledger. The source compiler and literal finite examples are executable. Universality uses the classical existence of a universal reversible two-counter machine; this release does not contain an expanded universal source-machine table. The three-unit statement does not transfer to integer-state number-conserving CA, where the state itself is its numerical mass.

49 Results and scope

All configurations in the simulation live on the infinite line $\mathbb{Z}$, with finite support. A finite-support initial condition is an ordinary finite object even when its nonzero sites are very far apart. There is

no finite-ring assumption and no active zero-weight background.

Theorem 49.1 (Three-unit compiler). *For every finite deterministic reversible two-counter machine M in the separated-instruction form of Section 50, there is an effective finite-alphabet CA F_M with the following properties.*

1. F_M and its inverse have radius at most four. A weight function w takes positive integer values on all non-vacuum symbols, has a unique zero-weight vacuum, and is conserved on finite configurations.
2. The canonical encoding of (q, a, b) has total mass exactly three and gap $12N$, where $N = 2^a 3^b$. It is computable from the source input.
3. Each enabled source transition produces precisely the next canonical encoding after the exact finite time specified in Section 56. Nonhalting encoded trajectories never depend on unspecified collision rows.
4. A designated one-cell symbol $\text{Ready}(q_f)$ occurs somewhere if and only if the encoded source computation reaches its halt state. The observation is taken after all finite scratch and instruction data have been reset.

Theorem 49.2 (Sharp threshold in the weighted model). *There exists one fixed reversible CA satisfying Theorem 49.1 for which occurrence somewhere of a fixed finite pattern is undecidable on a computable family of initial configurations of mass exactly three. In contrast, for every fixed deterministic conservative CA with a finite alphabet, finite radius and a unique zero-weight vacuum, exact finite-configuration reachability and finite-pattern occurrence are decidable from every initial configuration of mass at most two. Reversibility is unnecessary for the lower-mass assertion. Therefore three is the least mass allowing fixed-rule undecidable finite-pattern occurrence in this model.*

Section 57 strengthens the upper bound to a fixed rule of radius one, with the same three mass units. Its inverse also has radius one. Radius zero cannot yield this undecidable observation problem, so both the weighted mass threshold and the radius threshold are attained.

The lower-mass statement includes anchored observations. The upper-mass result here uses occurrence *somewhere*; it does not assert anchored halt observation, or exact-target configuration undecidability. The latter would require an additional construction when output counters are unknown. Universality means computation-universality under an effective input encoder and designated observation, not intrinsic universality or an efficiency bound.

The local alphabet is $2^{\mathcal{T}}$ for a finite type set $\mathcal{T}$. Many distinct symbols have weight one. The finite type information is charged to alphabet size, while each occupied channel contributes one unit of mass. An ordinary integer-state number-conserving CA has just one symbol whose numerical state is one. Merely renumbering our symbols cannot preserve all their weights. Thus this is not a three-numerical-particle theorem for that more restrictive model.

[write] [The lower-mass half of Theorem 49.2 is Theorem 44.1 of Part IV (see Section 59). Part VI makes the model point of the last paragraph precise: with at most one weight-one symbol, masses three and four are decidable (Theorems 74.1 and 83.1).]

50 Reversible counter machines and the gap encoding

A source configuration is $(q, a, b) \in Q \times \mathbb{N}^2$, where $\mathbb{N} = \{0, 1, 2, \dots\}$. Instructions have a source state, target state, counter index in $\{0, 1\}$, and one of five operations: increment, decrement, no-op, zero test, or positive test. Tests leave both counters unchanged; decrement is enabled only for a positive counter. The final state q_f has no outgoing instructions.

The separated syntax has at most one outgoing motion instruction at a state, or zero/positive test alternatives on one counter. Dually, incoming instructions at a state consist of at most one motion instruction, or zero/positive tests on one counter. Distinct tests with the same endpoint must have disjoint guards. These are the syntactic conditions underlying the reversible normal form of Morita and Imai [19]. They are sufficient for forward determinism and backward uniqueness on counter configurations. Fused guarded arithmetic is not being substituted for this form.

Number the instructions $1, \dots, m-1$ and reserve 0 for “no enabled instruction.” Encode the two counters by

$$N = 2^a 3^b \geq 1, \quad n = 12N. \quad (50.1)$$

Increment and decrement multiply N by 2, $1/2$, 3, $1/3$; a zero test on a counter is nondivisibility by its associated prime. The six-valued residue $N \bmod 6$ determines both zero/positive statuses. Prime-exponent encoding in one distance and finite-state divisibility protocols already appear in the three-agent grid-exploration construction of Brandt, Portmann and Uitto [37]; neither device is claimed as new here.

Consequently there are total finite tables

$$I : Q \times \mathbb{Z}_6 \longrightarrow \mathbb{Z}_m, \quad H : Q \times \mathbb{Z}_6 \longrightarrow \mathbb{Z}_m.$$

Here $I(q, r)$ returns the enabled outgoing instruction or 0. For H , a unique incoming motion is selected without consulting the residue, while incoming test alternatives are distinguished by their guard on the unchanged output counters. If no incoming instruction applies, set $H = 0$. On the output of any actual source step, H returns that step’s unique instruction. No search through predecessor counter values is required.

The input cost is geometric as well as finite-state. A sparse binary-coordinate encoding of $-12 \cdot 2^a 3^b$ uses $\Theta(a + b + 1)$ bits, while its numerical gap is exponential in the counter values. If a, b themselves are supplied in binary, this conversion need not be polynomial in their input bit length. The reduction requires a computable encoder, and makes no stronger efficiency claim. A fixed source table has a fixed finite channel set; allowing a larger table changes that set and its alphabet cost.

The classical existence dependency is precise: Proposition 3.4 of Morita and Imai [19], using their Definitions 3.1–3.3, states that a deterministic reversible two-counter machine can simulate any Turing machine. Apply it to a fixed universal Turing machine and fix the resulting finite source machine. The compiler theorem is effective once that finite table is supplied. This release supplies small literal test tables, rather than that expanded universal table. The earlier universal-source result is Morita [20].

[write] [Part IV’s source interface (Section 41) uses the number-conserving partitioned automaton of the same paper, with a corrected sign; source 14 uses only its two-counter normal form and Proposition 3.4. The programme’s review checked Definitions 3.1–3.3, Proposition 3.4 and Lemma 3.5 in the primary paper [45].]

51 The globally reversible channel construction

For every $t \in \mathcal{T}$, fix an integer velocity $v(t) \in [-4, 4]$. At each lattice site the symbol is a subset $A \subseteq \mathcal{T}$. Each channel is Boolean: the same type cannot occur twice at one site. Set $w(A) = |A|$ and use the empty subset as vacuum. Different channels may occupy the same cell, so number of occupied cells and mass are different quantities.

One tick is

$$F = S \circ \Pi, \quad (51.1)$$

where Π applies a single permutation π of $2^{\mathcal{T}}$ independently at each site, and S shifts channel t by $v(t)$. In particular, the output bit for channel t at site x is the t -bit of π at site $x - v(t)$ before the tick.

Lemma 51.1 (Finite completion). *Suppose the prescribed singleton rules form a permutation, and the prescribed two-element subset rules form a finite partial injection. They extend effectively to a cardinality-preserving permutation of $2^{\mathcal{T}}$, fixing the empty subset and every subset of cardinality at least three.*

Proof. The directed graph of a partial injection on the finite set of unordered pairs has in-degree and out-degree at most one. Each component with edges is a cycle or a finite directed path. Close each noncyclic path by the edge from its terminal vertex back to its initial vertex. Keep every pair outside this support fixed. Distinct components are disjoint, so the result is a permutation. Combine this pair permutation with the singleton permutation and the identities in all other cardinality sectors. $\square$

Identity on *all* unspecified pairs before path closure would generally fail: a prescribed image could collide with such an identity image. The completion procedure is part of the construction.

Lemma 51.2 (Global reversibility and conservation). *The completed rule in Lemma 51.1 defines a CA F that is a bijection on the full configuration space $(2^{\mathcal{T}})^{\mathbb{Z}}$. Both F and F^{-1} have radius at most four, and F conserves total mass on every finite-support configuration.*

Proof. Both Π and S are full-space bijections. Their inverse composition is $\Pi^{-1}S^{-1}$; the latter first shifts each channel by the negative of its velocity. Each local output depends on positions within distance four. The local permutation preserves cardinality, and shifting individual Boolean channels permutes their sites without creation or loss. Summing channel occupancies proves conservation. Vacuum is fixed by both operations. $\square$

All nontrivial rows below are in the singleton or pair sector. Use fresh labels for distinct routines, waiting modes, shuttle directions and data. Labels that are intentionally shared are identified explicitly.

52 Canonical sections and finite gates

The right-marker type R is stationary at coordinate 0. A left-controller type $L_{j,x}$ and a stationary shuttle S_j are colocated at $-n$, where

$$x = (q, i, z) \in Q \times \mathbb{Z}_m \times \mathbb{Z}_6, \quad j \in \{0, \dots, 7\}.$$

Thus a canonical configuration occupies two cells but has three units of mass. At stage zero, $i = z = 0$ defines the clean symbol

$$\text{Ready}(q) = \{L_{0,(q,0,0)}, S_0\}.$$

During flights the same three units usually occupy three cells. Finite control is inside the weight-one controller type, never in a mass-free track.

A finite data permutation G between static stages j and k is implemented by the pair rules

$$\{L_{j,x}, S_j\} \mapsto \{L_{k,G(x)}, S_k\}. \quad (52.1)$$

Both output velocities are zero, so this consumes exactly one tick. Each output stage has exactly one preceding routine. Since G is a permutation, these rules have distinct outputs; fresh internal labels separate them from unrelated routine outputs.

53 Exact reversible residue queries

Each query label ℓ has fresh shuttles

$$P_{\ell,d,r}, M_{\ell,d,r}, \quad d \in \{+, -\}, \quad r \in \mathbb{Z}_{144},$$

with velocities $+1, -1$ according to d . On singleton inputs, π increments the P phase and decrements the M phase. All waiting controllers and R are fixed as singletons. Define the finite lookup $h(r) = \lfloor r/24 \rfloor$, for representatives $0 \leq r < 144$.

For $\epsilon \in \{+1, -1\}$, the query from stage j to k uses the following pair rows; phase arithmetic is modulo 144 and scratch arithmetic modulo 6:

$$\{L_{j,x}, S_j\} \mapsto \{L_{\ell,P,x}, P_{\ell,+1}\}, \quad (53.1)$$

$$\{R, P_{\ell,+r}\} \mapsto \{R, P_{\ell,-,r+1}\}, \quad (53.2)$$

$$\{L_{\ell,P,(q,i,z)}, P_{\ell,-,r}\} \mapsto \{L_{\ell,M,(q,i,z+\epsilon h(r))}, M_{\ell,+r-1}\}, \quad (53.3)$$

$$\{R, M_{\ell,+r}\} \mapsto \{R, M_{\ell,-,r-1}\}, \quad (53.4)$$

$$\{L_{\ell,M,x}, M_{\ell,-,0}\} \mapsto \{L_{k,x}, S_k\}. \quad (53.5)$$

The exit row is prescribed *only* at phase zero. It never erases an arbitrary phase. For the stage-zero outgoing query, launch rows with source state q_f are omitted.

Lemma 53.1 (Residue query). *At gap $n = 12N$, the routine preserves the gap and q, i , maps z to $z + \epsilon(N \bmod 6)$, and exits at exactly $4n + 1$ ticks with no clock garbage.*

Proof. Set the launch time to zero. After the first tick the shuttle has moved one cell right and has phase one. At time n it reaches R with phase n . The next collision tick also updates the phase, so at time $2n$ it is back at the left controller with phase

$$r = 2n \bmod 144 = 24(N \bmod 6).$$

Therefore $h(r) = N \bmod 6$. The turnaround row performs the required scratch translation and starts the negative clock. At time $3n$ the shuttle reaches R with phase $r - n$ modulo 144. At time $4n$ it is back at the left controller with phase zero. The exit row produces the stationary stage at time $4n + 1$.

Only the shuttle moves, and $n \geq 12$, so there are no simultaneous endpoint or triple collisions. The two reflection families are injective in phase. The turnaround's output phase $r - 1$ recovers r , after which subtraction recovers the old z . Other finite data are retained. Thus every pair family is injective. The signs use separate routine labels. The minus query independently traverses the same geometry and subtracts its measured residue; it is an inverse operation on finite scratch, rather than a claim that its physical trajectory is the reversed plus trajectory. $\square$

The phase updates on collision ticks are necessary. Omitting the $+1$ or -1 in a reflection or turnaround would alter both the residue and the boundary time. The modulus $144 = 2 \cdot 12 \cdot 6$ simultaneously accounts for the round trip, the lattice scaling and the six-valued residue.

54 Exact rational gap changes

The reflected-shuttle geometric mechanism has a direct three-agent predecessor in Brandt, Portmann and Uitto [37], Section 3, which credits the factor-two case to Dobrev and coauthors [38]. Its rational-speed agents give the same ratio after time rescaling. The construction here supplies integer-lattice alignment, reversible local channel rules and cleanup.

For a moving instruction tag i with factor $p/d \in \{2, 1/2, 3, 1/3\}$, set

$$c = p + d, \quad u = d - p.$$

Use a controller B_x of velocity u and shuttles $A_{i,+}, A_{i,-}$ of velocities $c, -c$. All these types are singleton-fixed. Retain the full data $x = (q, i, z)$. The rows are

$$\{L_{4,x}, S_4\} \mapsto \{B_x, A_{i,+}\}, \quad (54.1)$$

$$\{R, A_{i,+}\} \mapsto \{R, A_{i,-}\}, \quad (54.2)$$

$$\{B_x, A_{i,-}\} \mapsto \{L_{5,x}, S_5\}. \quad (54.3)$$

No-op and test instructions use the static row from stage four to stage five, with the full tag retained.

Lemma 54.1 (Rational gap primitive). *For a valid factor p/d applied to N , the new gap is $n' = pn/d = 12N'$. The moving routine exits after $n/d + 1$ ticks. A test or no-op exits in one tick. The maximum absolute velocity is four.*

Proof. Temporarily translate the initial left endpoint to 0 and R to n . During the outward flight the shuttle is at ct and the left controller at ut . The shuttle arrives at R at $t_R = n/c$. After reflection its position is $2n - ct$. It meets the controller at

$$T = \frac{2n}{c+u} = \frac{n}{d}, \quad n' = n - uT = \frac{p}{d}n.$$

The collision is present at time T ; the next tick produces the stationary exit pair without moving it, so the total duration is $T + 1$.

For prime two, $(c, u) = (3, -1)$ or $(3, 1)$; for prime three, $(c, u) = (4, -2)$ or $(4, 2)$. The factor 12 makes n/c integral. Valid division requires $d \mid N$, hence N' is integral and remains positive. Multiplication has $d = 1$. All encounters are therefore at integer positions and times.

Since $|u| < c$, the shuttle separates immediately, visits R first and then meets the controller. When $u \leq 0$, the marker gap increases. When $u > 0$, its minimum up to the closing encounter is $n' > 0$. The markers cannot meet, and no triple collision or crossed-but-unseen encounter occurs. Distinct tags have distinct shuttles and remain in x at exit, making the prescribed rows injective. $\square$

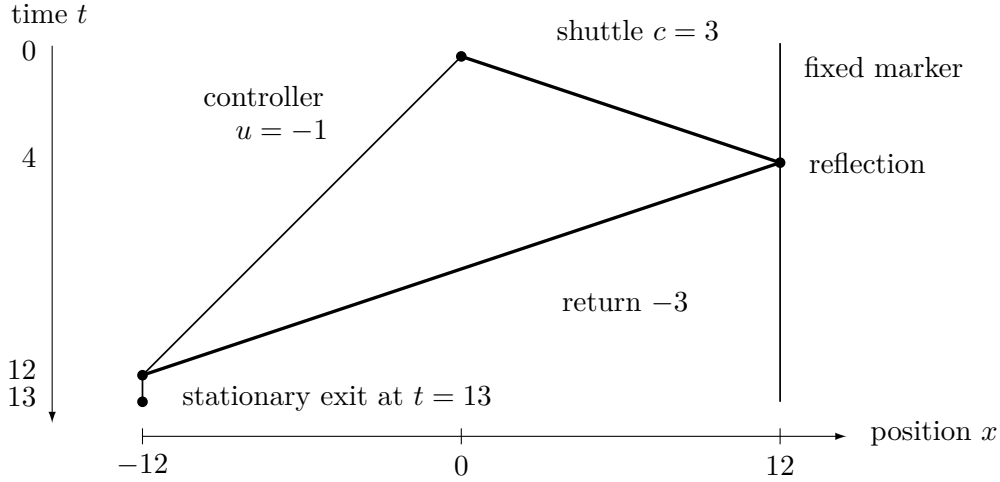


Figure 4: One exact multiplication by two at initial gap 12. The shuttle reflects at $t = 4$, meets the moving controller at $t = 12$, and the closing collision produces the stationary pair one tick later. The final gap is 24. Lines show velocities between integer-time configurations; this is the arithmetic primitive alone, not the surrounding source-step queries.

55 Instruction cleanup and the simulation proof

The fixed sequence of eight routines is

$$0 \xrightarrow{Q_+} 1 \xrightarrow{G_I} 2 \xrightarrow{Q_-} 3 \xrightarrow{G_q} 4 \xrightarrow{A} 5 \xrightarrow{Q_+} 6 \xrightarrow{G_H} 7 \xrightarrow{Q_-} 0.$$

Here the four queries have distinct labels, but each is shared by all instruction tags. The gates are

$$\begin{aligned} G_I(q, i, z) &= (q, i + I(q, z), z), \\ G_H(q, i, z) &= (q, i - H(q, z), z). \end{aligned}$$

For fixed $i \neq 0$, G_q transposes the source and target states of instruction i and fixes all other states, retaining i, z . If source equals target, that transposition is the identity; for $i = 0$ use the identity. Thus all three gates are permutations on *every* finite data value, not only valid inputs.

Start at $\text{Ready}(q)$ with $i = z = 0$. The first query loads $r = N \bmod 6$. Gate G_I writes the enabled instruction $i = I(q, r)$, and the first unquery resets $z = 0$. For $i \neq 0$, G_q changes q to the instruction's target. Arithmetic produces N' and still retains i . The output query loads $N' \bmod 6$; backward uniqueness gives

$$H(q', N' \bmod 6) = i.$$

Gate G_H therefore clears i , and the last unquery clears z . The exit is exactly $\text{Ready}(q')$ at gap $12N'$.

No hidden tag erasure. The arithmetic exit (54.3) retains i in $L_{5,x}$ while it removes the tag from the shuttle. All branches enter the same stage-five query. The instruction is cleared only by the common stage-six gate, while q, z determine its predecessor. The final unquery also has a common label. There is no later many-to-one merger of instruction-specific labels after the instruction register has become zero.

Disabled instructions. If $I = 0$, the stage-four row emits a fresh shuttle of velocity -1 and a fresh stationary trap controller carrying x . The shuttle leaves leftward, away from both stationary markers, and never collides again. Its type is singleton-fixed. The row is injective, and the resulting infinite escape creates no $\text{Ready}(q_f)$. This explicitly covers decrement at zero and any nonfinal source state with no applicable instruction.

Lemma 55.1 (Union of prescribed rows). *All prescribed singleton and pair rows satisfy Lemma 51.1 simultaneously.*

Proof. Singleton clocks are disjoint finite cycles; every other singleton is fixed. They lie in a different weight sector from the pair rows. Gate outputs are separated by static stage and their data permutations. Query reflections carry R and a fresh routine, mode and direction. Query turnarounds retain phase information; only the unique zero phase exits. Different waiting modes and routine labels separate these families. Arithmetic dispatch retains all finite data in its moving controller, and its return retains the instruction tag in the common static output. Static ratio-one branches and moving branches have disjoint instruction tags. Trap labels are fresh. Finally each static stage has one preceding routine, including the common clean exit at stage zero. These observations establish distinct domains and images for the union, not merely for each macro map in isolation. Outputs are allowed to be the next routine's inputs; this is intended and does not violate injectivity. $\square$

Proof of Theorem 49.1. Complete the pair map using Lemma 51.1. Global reversibility and conservation follow from Section 51. Lemmas 53.1 and 54.1, together with the cleanup argument, give the canonical-section simulation inductively. Each nonfinal enabled step uses only prescribed rules, with no triples; a disabled step follows the prescribed escape forever. Hence unspecified completion rows cannot manufacture a false halt on an intended nonhalting source computation.

Observe only the cell symbol $\text{Ready}(q_f)$, with both registers zero. An intermediate occurrence of controller state q_f at a different stage is not acceptance. The clean output symbol occurs exactly at a committed halt section, including time zero if the source is already final. Launch rows from the final section are deliberately unspecified. Their completed future is irrelevant after observation. A reversible dynamics cannot enter a periodic orbit from outside it; it may nevertheless make the entire orbit, including the initial state and its first observed halt, periodic. No global freezing or eventual fixed point is asserted. $\square$

55.1 Exact type and collision counts

For this ledger let $q = |Q| \geq 1$ include the halt state, let $m - 1$ be the number of source instructions and let d be the number of increment or decrement instructions. State labels are distinct and every instruction endpoint belongs to Q . These are syntactic counts for the stated compiler, including types and rows that may be unreachable from a particular encoded input.

Put $D = 6qm$, the number of finite data triples. The disjoint type families have the following sizes:

Type family	Number
Static controllers in eight stages	$8D$
Waiting controllers in four queries	$8D$
Query shuttles with direction and clock	$4 \cdot 2 \cdot 2 \cdot 144 = 2304$
Static shuttles	8
Moving arithmetic controllers	$6qd$
Arithmetic shuttles	$2d$
Trap controllers	$6q$
Escape shuttle and right marker	2

Thus the exact channel count s and full native cell-alphabet size are

$$s = 96qm + 6qd + 2d + 6q + 2314, \quad |2^{\mathcal{T}}| = 2^s.$$

The three gates contribute $3D$ prescribed pairs. Each query contributes $146D + 288$: one launch and one return per datum, 144 turnarounds per datum and 288 right-marker reflections. The first query omits $6m$ halt launches. Arithmetic contributes D dispatches (including static and trap dispatches), $6qd$ returns and d reflections. Therefore

$$P = 3528qm + 6qd + d - 6m + 1152$$

is the exact number of prescribed pair rows.

For completeness the number of rows after path closure is also explicit. Let A, B be the domain and image sets of the prescribed partial injection. Their intersection contains only static pairs. The domain omits the $6m$ stage-zero halt pairs, and the image omits the $6q$ stage-five pairs with instruction register zero. Those omissions are disjoint, so

$$X = |A \cap B| = 8D - 6m - 6q.$$

The nonstatic outputs are not domains: their shuttle direction or fresh trap label separates them from any prescribed collision input. There are $|A \setminus B| = P - X$ path starts; closing one path adds exactly one row, while existing cycles need none. Hence the stored completed pair-support count is

$$C = 2P - X = 7008qm + 12qd + 2d + 6q - 6m + 2304.$$

Unlisted pairs are implicitly fixed; C is not the number of all possible pairs or all cell states. In either radius-one realization below, s becomes $4s$ and the full alphabet has 2^{4s} states. No analogous claim about the lifted pair-row count is needed. These formulas yield a numerical universal alphabet only after a particular universal source and its q, m, d have been supplied.

56 Exact microtime

Write $n = 12N$ and $n' = 12N'$ at consecutive clean sections. There are two queries at each gap, three one-tick finite gates, and one arithmetic routine. Thus the whole source-step duration is

$$\tau(N, N', p, d) = \begin{cases} 8n + 8n' + n/d + 8, & \text{increment or decrement,} \\ 16n + 8, & \text{test or no-op.} \end{cases} \quad (56.1)$$

For a motion instruction the four query constants contribute four ticks, the gates three and the arithmetic exit one. For a static instruction the arithmetic itself is one tick. In particular

operation	$\times 2$	$\div 2$	$\times 3$	$\div 3$	static
$\tau(N)$	$300N + 8$	$150N + 8$	$396N + 8$	$132N + 8$	$192N + 8$

with divisions restricted to enabled inputs. These formulas agree with the primitive boundary convention $F = \text{SII}$, with the ready shuttle initially colocated with the left controller.

For a valid h -step source run, with N_t the gap integer at source step t , the exact committed physical time is

$$T_h = \sum_{t=0}^{h-1} \tau_t(N_t), \quad T_0 = 0. \quad (56.2)$$

A source horizon and a CA tick horizon are different parameters. The certificate below records source horizon h and can optionally constrain T_h without expanding every physical tick.

57 Radius one realizations

Both realizations below keep the mass exactly three and use four times as many channel types as the native rule. Spatial blocking preserves every tick and groups four old positions into one new cell. Internal phase dilation preserves the old spatial coordinates and multiplies physical time by four. Neither radius reduction is a novelty claim.

57.1 Spatial blocking with the native clock

There is also a time-preserving radius-one realization by the standard operation of grouping four consecutive old sites into one new site. The new labels record positions within a block, rather than clock phases.

Theorem 57.1 (Time-preserving spatial blocking). *Let F be the native CA, with s Boolean channel types and both forward and inverse radius at most four. There is a globally reversible mass-conserving radius-one CA H on $2^{\mathcal{T} \times \{0,1,2,3\}}$ and a bijective mass-preserving spatial recoding $\mathcal{B}$ such that $H\mathcal{B} = \mathcal{B}F$. No time dilation is required.*

Proof. For every old coordinate, use its unique Euclidean decomposition $x = 4j + a$, with $j \in \mathbb{Z}$ and $a \in \{0, 1, 2, 3\}$, including when $x < 0$. Define

$$(\mathcal{B}c)_j = \{(t, a) : t \in c_{4j+a}\}.$$

This is a bijection of the two full configuration spaces. Each old occupied channel becomes exactly one new occupied channel, so it preserves total mass on finite configurations and maps vacuum to vacuum. Set $H = \mathcal{B}F\mathcal{B}^{-1}$. The old sites influencing a new output block j lie between $4j - 4$ and $4j + 7$. They belong only to blocks $j - 1, j, j + 1$, proving radius at most one. The same argument for F^{-1} proves radius at most one for H^{-1} .

An explicit channel implementation first applies the completed old permutation π independently to each of the four offset projections. A post-collision lane (t, a) in block j is then transported to

$$\left(j + \left\lfloor \frac{a + v(t)}{4} \right\rfloor, \quad t, \quad (a + v(t)) \bmod 4 \right).$$

Its block displacement is $-1, 0$ or 1 . This transport is a permutation of all Boolean lanes; its inverse uses the opposite old displacement. The four on-site permutations and this transport preserve mass also on arbitrary mixed-offset configurations. Thus this implements the stated conjugacy on the full shift, not merely on the intended three-unit runs. $\square$

The canonical sites $-12N$ and 0 become blocks $-3N$ and 0 , with all three channel labels at offset zero. The blocks are distinct since $N \geq 1$. Observe the exact block symbol containing only the offset-zero versions of $L_{0,(q_f,0,0)}$ and S_0 . Every committed halt in the encoded native orbit has this block symbol, and its occurrence implies the old one-cell halt symbol. Hence the same first-halt observation occurs at the same tick; the whole reduction survives unchanged except for the spatial encoding. The new full alphabet has 2^{4s} states. The counts P, C above refer to the native pair dictionary, not to a flattened pair dictionary for either radius-one realization.

For example, the nine-step certificate fixture halts at tick 5340 in this spatially blocked radius-one CA. The coordinate-preserving phase realization instead uses tick 21360. Both use three mass units and $4s$ types. Native or spatially blocked time uses `clock_scale=1`; the four-phase realization uses `clock_scale=4`. The branch forms and certificate degree and slot ledger are otherwise the same.

57.2 Internal phase dilation with unchanged coordinates

Theorem 57.2 (Radius-one slowdown). *Let $F = S_v \circ \Pi$ be a Boolean-channel CA on 2^T , where the local permutation π preserves cardinality and $|v(t)| \leq r$ for an integer $r \geq 1$. There is a globally reversible, cardinality-conserving CA G of radius at most one on $2^{T \times \mathbb{Z}_r}$ and a mass-preserving phase-zero embedding E such that $G^r \circ E = E \circ F$.*

Proof. Identify a cell with $(a_0, \dots, a_{r-1}) \in (2^T)^r$ and put

$$\Pi_0(a_0, \dots, a_{r-1}) = (\pi(a_0), a_1, \dots, a_{r-1}).$$

Let P advance every occupied phase by one modulo r . For $k \in \{0, \dots, r-1\}$ define

$$\varepsilon(t, k) = \begin{cases} \text{sgn}(v(t)), & k < |v(t)|, \\ 0, & k \geq |v(t)|, \end{cases} \quad \delta(t, p) = \varepsilon(t, p-1 \bmod r).$$

Take $G = S_\delta \circ P \circ \Pi_0$. The first two factors acting in each cell are permutations, and S_δ is an independent shift of each Boolean lane. All three preserve cardinality, including on arbitrary mixed-phase configurations. Their inverses give $G^{-1} = \Pi_0^{-1} \circ P^{-1} \circ S_\delta^{-1}$. Since $\delta \in \{-1, 0, 1\}$, both global maps have radius at most one.

On a phase-zero input, only the first of the next r ticks applies π to a nonempty phase-zero subset. The next phases are $1, \dots, r-1, 0$; cardinality preservation gives $\pi(\emptyset) = \emptyset$. For $1 \leq j \leq r$, each post-collision type t has phase $j \bmod r$ and has moved

$$\sum_{k=0}^{j-1} \varepsilon(t, k) = \text{sgn}(v(t)) \min(j, |v(t)|).$$

At $j = r$ this displacement is $v(t)$, proving the identity. The velocity used is that of the type after π , so type-changing collisions cause no exception. Boolean capacity is respected: the three stages are permutations of cell states or of full-shift Boolean coordinates. $\square$

The operation Π_0 uses the cardinality of the phase-zero projection, not the total cell cardinality. Thus a phase-zero pair still undergoes its old collision when other-phase types share its site. This detail is essential to the stated full-alphabet implementation. The already completed original permutation supplies Π_0 directly; no new partial-rule completion is needed.

For the construction here take $r = 4$. The alphabet has $2^{4|T|}$ states, with weight equal to occupied channel count and with a unique zero-weight vacuum. All three original mass units remain charged; the phase is their finite internal label, rather than a background clock. Original durations are multiplied by four, including static and collision ticks. In particular, a query of duration $4n + 1$ takes $16n + 4$ new ticks.

The new observer requires the exact cell state consisting of the co-located phase-zero versions of $L_{0,(q_f,0,0)}$ and S_0 . On encoded orbits all mass units have phase $j \bmod 4$ at time $4k + j$, so no intermediate time $0 < j < 4$ can trigger this observer. At times $4k$ it is exactly the original committed-halt observation. Hence the undecidability reduction and the total mass three are preserved.

Finally, radius zero is impossible for this fixed-pattern occurrence undecidability. A reversible radius-zero CA is a sitewise finite permutation, so all configurations have a common finite period dividing the least common multiple of its state-cycle lengths. Finite-pattern occurrence from an explicitly presented finite-support input can therefore be checked over one period and finitely many relevant positions. Thus radius one is minimal in this model and for this observation notion; this asserts neither intrinsic universality nor a minimal alphabet.

For the certificates in Section 58, write Θ for native radius-four time, also the spatially blocked radius-one time. The phase-dilated radius-one observation time is 4Θ . The exporter and checker accept the request setting `clock_scale` equal to 1 (the default) or 4. They retain native ticks in each branch row and use `clock_scale` · Θ for the aggregate physical-time expression, extracted time coordinate and optional time constraint. Replacing $(\Theta - T)^2$ by $(4\Theta - T)^2$ changes no degree or slot count.

58 Natural number quadratic certificates

Fix a source machine, initial state q_0 , a positive raw gap integer N_0 and an externally specified source horizon $h \in \mathbb{N}$. Require that q_f have no outgoing instructions. The certificate below recognizes *first committed halt exactly at source step h* . Its variables are natural numbers. The horizon is a parameter of the compiler, not one of the polynomial's variables.

A raw integer $N \geq 1$ need not be of the form $2^a 3^b$. It uniquely factors as $C 2^a 3^b$ with $\gcd(C, 6) = 1$. The arithmetic and residue guards preserve C and correctly simulate the valuations a, b . Thus the raw-gap macro and the CA lemmas remain valid; the standard source input family is the special case $C = 1$. No smoothness test is silently asserted by accepting arbitrary positive raw inputs.

58.1 Residue-expanded branches

Expand each zero test at prime p into $p - 1$ branches, one for each nonzero residue $r \in \{1, \dots, p - 1\}$. Every other instruction gives one branch. Let B be the resulting number of branches, and let s_j, t_j be numerical codes for branch j 's source and target states, using distinct integers for distinct states.

For each step $0 \leq t < h$ and branch $1 \leq j \leq B$, introduce two natural variables e_{tj}, u_{tj} . The first selects the branch; the second is its nonnegative magnitude or quotient offset. The following

affine forms describe the old value A_j , new value C_j and exact tick duration D_j , with arguments (e, u) suppressed:

Branch	A_j	C_j	D_j
Increment at p	$e + u$	$p(e + u)$	$108A_j + 96C_j + 8e$
Decrement at p	$p(e + u)$	$e + u$	$96A_j + 108C_j + 8e$
Positive test at p	$p(e + u)$	$p(e + u)$	$192A_j + 8e$
Zero test at p , residue r	$pu + re$	$pu + re$	$192A_j + 8e$
No-op	$e + u$	$e + u$	$192A_j + 8e$

For $e = 1$, the increment/no-op forms cover every positive integer; decrement and positive-test forms cover precisely positive multiples of p ; and the zero-test forms cover precisely positive integers with residue r . All forms vanish at $e = u = 0$. Tick coefficients follow directly from (56.1), including the eight constant ticks only on the active branch.

Set, using the forms at step t ,

$$E_t = \sum_j e_{tj}, \quad U_t = \sum_j A_j(e_{tj}, u_{tj}), \quad V_t = \sum_j C_j(e_{tj}, u_{tj}),$$

$$S_t = \sum_j s_j e_{tj}, \quad Q_t = \sum_j t_j e_{tj}, \quad \Theta = \sum_{t,j} D_j(e_{tj}, u_{tj}).$$

The interfaces are affine substitutions, not additional variables:

$$\hat{N}_t = \begin{cases} N_0 & t = 0, \\ V_{t-1} & t > 0, \end{cases} \quad \hat{q}_t = \begin{cases} \text{code}(q_0) & t = 0, \\ Q_{t-1} & t > 0. \end{cases}$$

Write $V_{-1} = N_0$ and $Q_{-1} = \text{code}(q_0)$ for the horizon-zero endpoint convention.

58.2 The polynomial and its exactness

Define

$$P_h = \sum_{t=0}^{h-1} \left[(E_t - 1)^2 + (S_t - \hat{q}_t)^2 + (U_t - \hat{N}_t)^2 \right] + (Q_{h-1} - \text{code}(q_f))^2 + \sum_{t=0}^{h-1} \sum_{j=1}^B (E_t - e_{tj})(e_{tj} + u_{tj}). \quad (58.1)$$

This is one integer polynomial of degree at most two. Each square has an affine argument. Every product is nonnegative on the full natural orthant, since $E_t - e_{tj} = \sum_{k \neq j} e_{tk} \geq 0$ and $e_{tj} + u_{tj} \geq 0$.

Theorem 58.1 (Unique natural witness at a fixed horizon). *For fixed $N_0 \geq 1$, q_0 and h , the natural solutions of $P_h = 0$ are in bijection with first-halt source executions of length exactly h . Since the source is deterministic, there is exactly one full natural witness if such a run exists and no witness otherwise. The affine output V_{h-1} is its final raw gap integer, and Θ is its exact physical CA time.*

Proof. All summands of (58.1) are nonnegative, so a zero forces every summand to vanish. The square $(E_t - 1)^2$ gives $E_t = 1$. Natural selectors then have exactly one value one and all others zero; no separate Boolean equations are needed. For an inactive branch, the product is $1 \cdot u_{tj}$, which

forces $u_{tj} = 0$. The active branch's product vanishes automatically and permits its nonnegative magnitude.

Thus U_t, V_t, S_t, Q_t reduce to the selected branch's forms and endpoint codes. Control and input squares connect the branch to the actual preceding state and raw integer. The table enforces its guard and arithmetic exactly by natural quotients. A zero-test branch is uniquely selected by the nonzero residue. Induction from $N_0 \geq 1$ proves that every successive raw value is positive. The terminal square forces the last state to be q_f . Since q_f has no outgoing branch, it could not have occurred earlier in this h -step run. The affine tick expression is then the sum of the already proved physical durations.

Conversely, select the actual branch of a first-halt run at each step, set its e to one and all other branch variables to zero. For an increment or no-op put $u = N - 1$; for a decrement or positive test put $u = N/p - 1$; for a zero test with residue r put $u = (N - r)/p$. These are uniquely determined naturals, and every summand vanishes. Determinism determines the execution, the residue determines any expanded branch and the affine input form determines its u . Therefore every variable is unique. When $h = 0$, there are no core variables and the sole core square compares q_0 with q_f ; success has output N_0 and time zero. $\square$

The inactive-product factor $e + u$ is deliberate. A weaker gate that left an inactive branch's quotient unrestricted would destroy uniqueness and could contaminate the interface sums. Here every inactive variable is forced to its canonical zero value.

[write] [The inactive-product gate $(E_t - e_{tj})(e_{tj} + u_{tj})$ is the gate of Lemma `qoc:rn:lem:gates` of Part VI of *quadratic-orthant-certificates*, whose products $(\sum_{s \neq t} e_{js})(e_{jt} + \sum_p x_{jtp})$ force the inactive magnitudes to zero in the same way, and a form of Part III's selector/copy device (note after Theorem 36.1) and of the one-hot selectors of *canonical-diophantine-certificates*; the raw-integer branch forms, such as $pu + re$ for a residue, are source 14's own. The natural-domain counterexample below is consistent with that report's real-orthant one-hot lemma: one-hot selection survives over the nonnegative reals, divisibility does not.]

The natural domain is essential. With a prime-two zero-test branch and raw input $N_0 = 2$, the real or rational assignment $e = 1, u = 1/2$ satisfies $2u + e = 2$. For a one-step zero-test-to-halt machine it makes the entire polynomial zero although the integer guard is false. The certificate therefore does not claim real- or rational-domain exactness. Neither nonnegativity alone nor a squared input relation enforces divisibility.

58.3 Resource ledger and endpoint constraints

For the fixed-raw-input core (58.1), the exact *syntactic slot counts* are

$2Bh$ natural variables,	$3h + 1$ affine squares,	Bh nonnegative affine products.
--------------------------	--------------------------	-----------------------------------

The degree is at most two. A slot can simplify to zero for a degenerate instance; these are the counts emitted before such simplification, not lower bounds on irreducible polynomial complexity. Interface sums, state codes, raw values and clocks do not introduce auxiliary variables.

To require a fixed final raw value N_* , add $(V_{h-1} - N_*)^2$. To require a fixed physical time T_* , add $(\Theta - T_*)^2$. Each adds one affine-square slot and no variable. Instead, exporting a free natural coordinate for either endpoint adds one variable and that same square. In a successful fixed-input fiber these outputs are uniquely determined.

A free positive raw input is represented by $N_0 = x + 1$, with $x \in \mathbb{N}$. It adds one input coordinate, no square, and no hidden positive-domain convention. The theorem then states uniqueness *for each fixed input coordinate*; different accepted inputs may give different polynomial zeros.

The arity grows linearly with the externally chosen h and with the fixed source's expanded branch count B . Eventual halting is the union over all horizons. No claim is made that merely existentially quantifying an unknown horizon preserves this finite arity, this degree or this ledger.

58.4 An explicitly paid bounded counter loader

For fixed numerical counter inputs (a, b) , an external computable encoder may calculate the constant $N_0 = 2^a 3^b$ before constructing the polynomial. This is not an internal polynomial representation of exponentiation with free unbounded exponents.

A fully explicit optional loader supports free counter coordinates within supplied fixed bounds $0 \leq a \leq A$, $0 \leq b \leq C$. Introduce

$$K = (A + 1)(C + 1)$$

natural selectors λ_{ij} for $0 \leq i \leq A$, $0 \leq j \leq C$, and add the three affine squares

$$\left(\sum_{i,j} \lambda_{ij} - 1 \right)^2, \quad \left(a - \sum_{i,j} i \lambda_{ij} \right)^2, \quad \left(b - \sum_{i,j} j \lambda_{ij} \right)^2.$$

Substitute the affine expression

$$N_0 = \sum_{i,j} 2^i 3^j \lambda_{ij}$$

directly into the initial interface. The coefficients are precomputed integers. Natural one-hot selectors force the unique pair $(i, j) = (a, b)$, or no witness if the counters are outside the supplied bounds. This adds exactly K loader variables and three affine-square slots; each genuinely free a or b also adds its input coordinate. No separate N_0 variable is needed and the degree remains at most two. The cost is numerical in the fixed bounds, not a uniform logarithmic-size exponentiation gadget. This finite loader makes no assertion for free unbounded counter values.

59 Why masses at most two are decidable

This section gives the lower bound needed by Theorem 49.2. Let F be any deterministic translation-invariant CA of finite radius r over a finite alphabet Q . Assume a quiescent state 0, integer weights $w(0) = 0$, $w(q) \geq 1$ for $q \neq 0$, and conservation on finite configurations. Define $Q_j = \{q : w(q) = j\}$. Translation by a is denoted T_a . For nonempty x , let $\ell(x) = \min \text{supp}(x)$ and normalize by $\bar{x} = T_{-\ell(x)}x$.

Theorem 59.1 (Two-unit decidability). *From a finite configuration of mass at most two, one can decide exact finite-configuration reachability and occurrence of an explicitly supplied full finite word, either at a specified anchor or somewhere. For fixed F , these decision problems have polynomial complexity in the input bit length: configurations use sparse coordinate/state lists with binary coordinates, and the complete pattern length counts toward input size.*

[write] [Editorial note: this is Theorem 44.1 of Part IV (source 13), stated there for the same model with the same complexity claim, and proved there by the same argument. Source 14's

four subsections (“Free particles and the first encounter”, “Near configurations and recurrence”, “Observations at every intermediate time” and “Polynomial complexity for a fixed rule”) give that proof again step by step: the finite walker maps $\alpha : Q_1 \rightarrow Q_1$ and $d : Q_1 \rightarrow [-r, r] \cap \mathbb{Z}$; the near/far threshold $2r$, with the first encounter solved phase by phase ($k = \lceil (A - 2r)/(-D) \rceil$ for an intercept $A > 2r$ and a negative drift D); the count $K_{\text{near}} = |Q_2| + 2r|Q_1|^2$ of normalized near shapes; the translating tail $F^{i+s+kp}x = T_{k\delta}F^{i+s}x$; the tests at every intermediate time and at all later translates; and the fixed-rule polynomial bound, which is not uniform in the rule. Reversibility is not used. These subsections are not reprinted; they survive in the arrival archive.]

Proof of Theorem 49.2. The lower bound is Theorem 59.1. For the upper bound, choose a fixed universal source machine in the separated reversible form and apply Theorem 49.1. If fixed-pattern occurrence for its mass-three encodings were decidable, the computable input encoding followed by that decision procedure would decide the universal source’s halting problem. This is impossible. Combining the two assertions proves the threshold in precisely the common weighted finite-alphabet model. $\square$

60 Geometric and finite-ring boundaries

Fixed mass does not imply bounded support diameter. In the three-unit simulation the unbounded variable is the gap. A specified diameter bound would change the decision problem fundamentally.

Proposition 60.1 (Bounded diameter). *Given F , a finite initial configuration x and a bound D , it is decidable whether the orbit ever exceeds diameter D . Under the promise of staying within that diameter, exact reachability and anchored or unanchored finite-pattern occurrence are decidable.*

[write] *Pointer.* This is Proposition 43.1 of Part IV (source 13), proved there by the same finite-shape argument: simulate until the diameter escapes or a normalized shape repeats, a repeat giving the translating tail. Source 14’s proof is not reprinted. $\square$

For the undecidable mass-three family, no computable input-dependent bound can cover the whole pre-halt diameter on every successful computation. If such a bound existed, the preceding procedure could answer yes at a hit, no upon prior escape, or decide the translating tail upon a repeat, thereby deciding halting. This argument concerns a *complete* bound through every possible successful witness; it does not preclude bounded individual runs or unbounded sparse certificates. [write] [This is Proposition 43.2 of Part IV, for the mass-three family; its finite-search step is `cdc:bd:prop:nobound.`]

On a specified ring of L cells, a fixed-mass sector is finite, and orbit exploration decides reachability. Its size is the coefficient of z^M in

$$\left(\sum_{q \in Q} z^{w(q)} \right)^L.$$

For fixed M this is polynomially bounded in the numerical ring length, which is not necessarily polynomial in its binary encoding length. A finite ring can reproduce an infinite-line run through a specified tick horizon T by sufficient padding, for example $L > D_0 + 2rT$ for initial diameter D_0 . This is only a finite-horizon replacement. [write] [Same as Part IV’s paragraphs on finite rings and global halting (Section 43).]

61 Literal examples and reproducibility

The archive contains the exact tested channel generator, standalone arithmetic and inverse checks, the certificate compiler, literal finite rule exports, machine-readable receipts, and this LaTeX source. Only source generated for this construction is packaged; third-party papers are linked in the references rather than redistributed.

The principal literal example has states q_0, q_1, q_2, q_3, q_f and the four instructions

$$q_0 \xrightarrow{\text{inc}_0} q_1, \quad q_1 \xrightarrow{\text{inc}_1} q_2, \quad q_2 \xrightarrow{\text{dec}_0} q_3, \quad q_3 \xrightarrow{\text{dec}_1} q_0.$$

The final state is unused in this cyclic example. For $N = 1$, the committed sections are

tick	0	308	1108	2016	2420
state	q_0	q_1	q_2	q_3	q_0
N	1	2	6	3	1

The full configuration returns to its initial value at tick 2420. This is a test of all four gap factors and exact cleanup, not a universal machine.

This instance has 4872 channel types, 89446 prescribed pair rows and 177752 pair-support rows after completion. Its full cell alphabet has 2^{4872} subsets. The exported sparse permutation specifies every singleton action and every pair-support action; all other pairs and all sectors of cardinality zero or at least three are identities. This is an exact finite representation of the entire local rule, without enumerating all 2^{4872} symbols. [write] [The three literal rule exports of this section (`examples/cycle_rule.json`, `merge_rule.json` and `trap_rule.json`, 17.2 MB together) are not shipped in this report: the shipped `code/14-three-mass-export_examples.py` rebuilds them byte for byte in seconds, and the shipped `data/14-three-mass-rule_index.json` records the size and SHA-256 digest of each (README, “Reconstructing the excluded data”).]

Two additional source examples test merging inverse guards and a disabled decrement. The merging example has a zero-test instruction and positive-test instruction from different states into a common final state. The output residue distinguishes the predecessor and clears its tag. At $N = 1, 3$ the zero branch halts after 200,584 ticks, and at $N = 2, 6$ the positive branch halts after 392,1160 ticks. The disabled decrement at $N = 1$ takes the explicit escape route.

61.1 A literal certificate with nine source steps

The separate certificate fixture `mixed_request.json` has the successive operations

$$\text{inc}_0, \text{inc}_1, \text{positive}_0, \text{positive}_1, \text{dec}_0, \text{dec}_1, \text{zero}_0, \text{zero}_1, \text{nop},$$

with a fresh control state after every operation and halt after the ninth. For fixed initial $N_0 = 1$, its raw values and native times are

t	0	1	2	3	4	5	6	7	8	9
N_t	1	2	6	6	6	3	1	1	1	1
T_t	0	308	1108	2268	3428	4336	4740	4940	5140	5340

The prime-three zero test contributes two residue branches, so $B = 10$. The core has 180 auxiliary variables, 28 affine-square slots and 90 product slots. The literal exported version also records two natural output coordinates, `output_N` and `physical_time`: its total arity is 182, with 30

square slots and the same 90 products. These are output coordinates, not extra free inputs. Its unique witness has `output_N = 1` and native time 5340; phase-dilated radius-one time is 21360, while spatially blocked radius-one time remains 5340. The native expanded polynomial contains 16593 nonzero monomials and maximum absolute coefficient 682112 (20 bits). These last counts refer to this one fully specified fixture, with both output coordinates. [write] [The expanded certificates of this fixture and of its phase-dilated version (`examples/mixed_certificate.json` and `mixed_radius_one_certificate.json`, about 1.8 MB each) are not shipped either; `code/14-three-mass-certificate.pyexport` rebuilds them, and their requests, witnesses and checker receipts are shipped as `data/14-three-mass-mixed*`.]

61.2 What the verification establishes

The reproducibility script reruns the following independent checks against the packaged sources and records receipts with the executed source hashes.

- The local CA audit checks all 177752 completed pair-support rows as a permutation, 3000 random global inverse cases, 144 query cases and 136 arithmetic cases with exact boundaries and stepwise inverses.
- The certificate audit checks 1125 schema and ledger instances, 2160 independently expanded polynomial evaluations and 43586 fully enumerated natural assignments in finite boxes. It constructs 246 unique witnesses and rejects 834 incompatible guards or horizons.
- A separate simulator replays 555394 original-CA microsteps, including 191 enabled macro transitions, three complete mixed runs and six disabled-decrement escape cases. It reads the serialized local maps rather than calling the producer’s step or ready predicates.
- The phase-radius-one audit checks every supported pair under the four-phase embedding, arbitrary-cardinality and mixed-phase inputs, both inverse orders, all synchronized collision boundaries and halt phases. The four cycle runs use 115136 slowed ticks in total.
- The spatial-blocking audit checks all 177752 supported native pairs at all four offsets, 19488 singleton/offset cases, 5000 random configurations and 37460 complete-trajectory ticks. The blocked and native trajectories agree at every tick, including the 5340-tick mixed halt. Its 1368268 checks also pass with optimization enabled.
- The two-unit arithmetic tests cover 93663 encounter and observation cases, including large binary coordinates and later translating phases. They validate the arithmetic components; they are not a complete implementation of the general lower-mass decision procedure. [write] [This is source 13’s mass-two audit (Section 47): the program is `code/13-sparse-lattice-two-mass-audit.py` without its receipt-writing line, and its receipt is source 13’s.]
- The size-ledger audit checks all type families, overlap and path-closure counts on 33 finite sources. The independent clock audit performs 28820 explicit checks over 1623 semantic cases and 54 CLI runs, both normally and with optimization enabled.
- The serialized-checker and strict-input tests exercise coefficient, branch, ledger, endpoint, scalar-type, ownership and witness corruptions. The final receipts give exact mutation counts for the packaged revision.

The producer also runs its own simulation, certificate and literal-export suites. None of these finite tests substitutes for the all-input proofs, and none supplies a universal source-machine table.

61.3 Replaying the release

Python 3.10 or later and its standard library suffice for every executable check. From the extracted archive root, run

```
python3 verify_manifest.py
sh replay.sh
```

The replay writes fresh evidence to `.replay/` without replacing the published examples or receipts. It checks the stored literal rules, exports new copies, compares their hashes and rebuilds the native and phase-radius-one certificate examples. The source and fixture hashes are listed in `SHA256SUMS`. Assertions are required: do not run Python with `-O` or `PYTHONOPTIMIZE` set.

To export a certificate, construct its unique witness when one exists, and run the independent checker directly:

```
mkdir -p .replay
python3 code/certificate.py export examples/mixed_request.json \
    .replay/mixed.json --expanded
python3 code/certificate.py witness .replay/mixed.json \
    .replay/mixed-witness.json
python3 code/checker.py .replay/mixed.json \
    .replay/mixed-witness.json
```

The JSON request declares the source table, initial state, source horizon, input representation, optional endpoints and physical clock convention. The checker reconstructs the expected schema independently before checking its natural witness; it does not trust a bare assertion that the submitted polynomial encodes a source run.

The PDF is rebuilt with `sh build.sh` using a normal TeX Live installation with the packages named in `tex/report.tex`. The archive includes the modular LaTeX sources and a standalone combined `three-mass-report.tex`. PDF byte identity additionally depends on the installed TeX and font versions; the script fixes the document date.

[write] **The release in this report.** These are the delivered names. Source 14's programs are shipped as `code/14-three-mass-*` (the two programs of `code/legacy/` as `code/14-three-mass-legacy-*`), its examples and receipts as `data/14-three-mass-*`, and its `SOURCE_PROVENANCE.md` as `14-three-mass-SOURCE_PROVENANCE.md`. Not shipped: the manuscript (printed here) in its standalone and modular forms, its PDF and README, the checksum ledger `SHA256SUMS` with `verify_manifest.py`, the five literal exports named above, and the two-mass receipt, which is a byte copy of source 13's. `replay.sh` and `build.sh` need the delivered layout, and `build.sh` the unshipped `tex/report.tex`; the README of this report gives a rerun recipe in a scratch copy of the archive.

62 Provenance and limits of the claims

The mathematical statements in this report separate a compiler theorem, a classical source-existence dependency and finite implementation checks. The generator is parameterized by a source-machine table. Fixing an existing universal source establishes a fixed universal CA mathematically, but the literal example export in this archive is not a frozen universal transition table. No concrete universal alphabet cardinality is being claimed from the small examples.

The three-unit compiler was checked independently for local injectivity, full-space inverse behavior, exact collision times, finite cleanup and the committed observation. The mass-at-most-two argument was independently checked, including anchored observations in later translates and binary-coordinate complexity. The tests exercise finite instances and proof components; the proofs provide the all-input and all-source-machine conclusions.

There is substantial close prior art. Brandt, Portmann and Uitto [37] already use three finite-state agents, reflected-shuttle multiplication and division, walking-clock divisibility, and prime exponents encoded in one gap. Their algorithm is a grid-exploration protocol; the full-shift reversible CA extension and reversible information cleanup are separate obligations proved here. The factor-two geometric predecessor is credited to Dobrev and coauthors [38].

The publisher’s abstract for Dobrev, Kráľovič and Pardubská, *Busy agents on a line* [39], reports Turing-machine simulation by three synchronous distinguishable finite-state agents on $\mathbb{Z}$. That abstract’s universality statement does not assert global CA reversibility. The full journal proof was not obtained for this report. Consequently even a nonreversible three-agent universality claim would have a close published predecessor.

Durand-Lose [5] establishes reversible conservative rational signal-machine universality. Its collision geometry uses rational space and time and additional counter/history components; it is not a mass-three integer-lattice CA theorem. Morita and Imai [19] already give a reversible number-conserving partitioned CA with mass $m + 19$ for an m -state reversible source machine. A three-component numerical partitioned alphabet itself permits multiple states of the same total weight. The equal-weight-type distinction therefore separates the present claim from literal single-integer-state NCCA, not from every partitioned predecessor.

The 2016 paper by Alhazov and Imai [24] reports a universal finite number-conserving CA with five numerical particles in its primary abstract. That abstract does not claim reversibility. Its complete paper was not verified for this report, so no hidden lower bound or reversible strengthening is attributed to it. Its numerical-state mass model differs from the typed weighted model here. This report makes no literature-wide novelty, priority or smallest-alphabet claim.

[write] [For the numerical-state model, Part VI shows that three and four particles do not suffice for undecidable observations, and credits two 2021 precursors that source 14 does not cite: Kong’s thesis [41] (binary three-particle nonuniversality) and Imai’s KAKEN report [42].]

The certificate is over natural numbers, not real solutions with a silently inferred divisibility condition. The source horizon is externally specified; the arity grows with it. An unbounded existential time cannot be inserted into a fixed-arity polynomial merely by writing a variable in place of the horizon. Free exponent inputs require their own stated loader and resource payment. No unrelated numerical record or previously used variable ledger is imported into this construction.

The unique-zero and positive-weight assumptions are indispensable for the lower bound. Active zero-weight states, signed cancellation or an infinite nonzero background would invalidate the mass-to-support argument. The construction and threshold are for the infinite line, not a finite ring, and acceptance is a finite observed event rather than cessation of a reversible global evolution.

63 Questions left open by this construction

The proofs resolve the mass and radius thresholds within the stated weighted model. Several sharper or more concrete tasks remain distinct.

1. **Expand a universal source.** Supply a particular universal reversible two-counter table in

the separated syntax, then freeze its generated channel names and local permutation. This would turn the existence-based instance into one explicit universal rule with an exact alphabet count. ([write] Added 2 October 2026, batch 80: open. Source 15’s Question 1 and Part IV’s first question ask the same.)

2. **Reduce the finite-state overhead.** Can fewer clock phases, channel types or collision rows implement the same cleanup and residue query? The phase implementation pays a factor four in types and time; spatial blocking pays a factor four in types while preserving time. These are constructive bounds for different spatial conventions, not optimal overhead theorems. ([write] Compare source 15’s Question 3.)
3. **Develop quantitative observation extensions.** Construct and cost an anchored halt observer, for example by carrying a halt message to the fixed marker, while minimizing extra channel types and rules. Exact finite-target encodings also require explicit treatment of the final counters. Neither extension is part of the validated generator here. ([write] Added 2 October 2026, batch 80: the exact finite-target part is answered by source 15 (Theorem 67.1): its target y_N is computed from the input, and the final counters are restored by inverse execution rather than treated explicitly. The anchored halt observer is still open.)
4. **Change the mass convention.** What is the sharp reversible threshold when cell states are nonnegative integers equal to their conserved mass? The abundance of equal-weight types used here cannot be carried over by relabeling. ([write] Added 2 October 2026, batch 80: answered in part by Part VI. With at most one weight-one symbol, exact reachability and pattern occurrence are decidable at masses three and four, so undecidable observations need at least five units (Corollaries 79.1 and 83.2); no reversibility is assumed there. The sharp reversible threshold is still open. Compare source 15’s Question 4.)
5. **Cost an unbounded-time encoding for this frontend.** Instantiate known fixed-arity Diophantine coding for this particular simulation with an explicit degree and variable ledger, and determine which witness properties survive the composition. Uniqueness at each fixed horizon does not by itself yield a finite-fold or single-fold representation after introducing such coding. This is a concrete construction and accounting task, not an open question about existence of ordinary Diophantine representations. ([write] Compare source 15’s Question 5.)
6. **Complete the prior-art comparison.** A literature-wide priority assessment would require the uninspected full texts and a precise comparison of the update, reversibility, particle and observation conventions. No such assessment is asserted here.

Source 15 (batch-80 manuscript 02): Exact targets with three conserved mass units

A clean return extension of the reversible weighted CA construction, 3 October 2026. [write] Its “companion construction” and “companion report” are source 14, printed above; its delivered bibliography entry for it (“Companion construction report, frozen release 9”) is replaced by references to source 14’s statements. Its abstract follows.

A fresh-entry reversible two-counter program can be wrapped so that it first runs forward, then executes its logical inverse, and finally changes to a designated terminal state with its entire input restored. Applying the existing three-mass cellular-automaton compiler to this wrapped program gives undecidable exact finite-configuration reachability for one fixed reversible weighted CA. Both configurations in the reduction have mass three and the same two-site support. The target

is computable from the input, without knowing whether or when the computation halts. Spatial blocking yields radius one with no clock penalty; a coordinate-preserving phase construction costs a factor four in time. Decidability at mass at most two makes three the sharp mass threshold for exact-pair reachability in this model. We prove the wrapper, the exact first-target clock, and a compact quadratic natural-number certificate, including an affine bijection of its zero fibers with those of the full wrapped computation. The universal-source existence dependency, finite implementation checks, and limits of the result are stated separately.

64 Result and scope

The previous construction established undecidable occurrence of a designated clean halt symbol somewhere on the infinite line. Its final counter values were generally unknown. That observation did not by itself provide an exact target configuration that could be computed before running the source. The extension here removes this obstacle by restoring the input before announcing completion.

The change is at the source-program level. The same local-rule generator is applied to a different finite source table. Thus the resulting CA generally has different control types and a different fixed rule from the one compiled from the original program. The earlier release is unchanged.

Theorem 64.1 (Exact-pair reachability at mass three). *There is a fixed one-dimensional reversible CA A over a finite alphabet, with a unique zero-weight vacuum and positive integer weights on all other symbols, for which the following hold.*

1. *Total weight is conserved on every finite configuration. Both A and A^{-1} have radius at most four.*
2. *There is a computable family of finite configurations (x_N, y_N) , $N \geq 1$, of total mass exactly three, such that deciding whether $A^t(x_N) = y_N$ for some $t \geq 0$ is undecidable. [write] [The undecidable family is the image of the source's input encoding ($c = 1$, or any N for a many-one complete source), as Corollary 67.2 states.]*
3. *The input and target have the same support, at coordinates $-12N$ and 0 . Their only difference is the finite control state in the left pair. The target depends on N and is computed without knowing the halt time or the final counters.*
4. *There are also fixed radius-one rules with radius-one inverses satisfying the same exact-pair undecidability conclusion and conserving the same three mass units.*

For every fixed deterministic conservative finite-alphabet CA in the same one-dimensional weighted model, exact finite-configuration reachability from mass at most two is decidable. Thus three is the least mass permitting the stated fixed-rule undecidability.

Here and below $\mathbb{N} = \{0, 1, 2, \dots\}$. Configuration equality is equality at every absolute lattice coordinate, including vacuum everywhere else. The problem is not equality up to translation, a pattern query, a limit configuration, or an assertion that evolution stops. In fact a bijection cannot carry a distinct initial configuration into a fixed point: a fixed point already has itself as its unique predecessor.

The model matters. The alphabet is a power set of finitely many Boolean channel types, and a symbol's weight is its cardinality. There are many different weight-one symbols. This is not the ordinary single-integer-state number-conserving CA convention in which a state's numerical label

is itself its mass. The result is for finite-support configurations on $\mathbb{Z}$, with unbounded possible separation, not for a prescribed finite ring or an active mass-free background.

What is proved anew. The new ingredients are the clean source wrapper, exact whole-configuration reduction, first-hit clock identity, and certificate composition and lift. The local residue and gap routines, global reversible completion, radius reductions, and low-mass decision theorem are dependencies from the companion construction [source 14, Part V of this report]. We state their precise interface and summarize the lower-bound argument so the reduction can be checked without reconstructing the earlier report. The archive contains the exact executable dependencies, not an expanded universal source table.

[write] [In this report the restated interface is printed with pointers to source 14’s statements (Section 65), and the summary of the lower-bound argument is replaced by a pointer to Part IV’s Theorem 44.1 (Section 67).]

65 The source model and the inherited compiler

65.1 Separated reversible instructions

A source configuration is $(q, a, b) \in Q \times \mathbb{N}^2$. Each instruction has a source state, target state, counter index in $\{0, 1\}$, and an operation among INC, positive-only DEC, NOP, ZERO, and POS. Tests preserve both counters. A zero test is enabled exactly when its counter is zero; a positive test and decrement require a positive counter.

The separated syntax imposes the following conditions on both the incoming and outgoing instructions at every state: a group is empty, a single instruction, or disjoint zero/positive tests on one and the same counter. In particular arithmetic cannot be mixed with another instruction on that side. These conditions imply forward determinism and uniqueness of a predecessor, including the guards. The designated halt state q_h has no outgoing instruction. Getting stuck elsewhere is possible and does not count as halting.

We require the additional structural condition

$$\text{no instruction has target } q_0. \quad (65.1)$$

It must hold for the whole finite table, not merely for transitions enabled on one chosen input. The necessity of this condition and its classical availability are addressed in Section 66.1.

65.2 Raw integers and exact canonical sections

Represent the two counters by a positive integer

$$N = c 2^a 3^b, \quad \gcd(c, 6) = 1. \quad (65.2)$$

The usual encoded source inputs use $c = 1$. Allowing arbitrary raw $N \geq 1$ is also sound: prime-two and prime-three valuations are the counters, and the cofactor c is invariant. An increment or decrement at prime $p \in \{2, 3\}$ changes N by p or $1/p$; zero and positive tests inspect nondivisibility or divisibility by p .

For a source table M , the inherited compiler constructs a finite set $\mathcal{T}_M$ of typed channels, velocities $v : \mathcal{T}_M \rightarrow [-4, 4] \cap \mathbb{Z}$, and a cardinality-preserving local permutation π on $2^{\mathcal{T}_M}$. One native

CA tick is

$$A_M = S\Pi, \quad (65.3)$$

where Π applies π independently at all sites and S shifts each channel by its velocity. The empty subset is vacuum. Its weight is zero; all other weights are positive integers.

Let R be the stationary right-marker channel. A canonical section $C_M(q, N)$ is exactly

$$C_M(q, N)(x) = \begin{cases} \{L_{0,(q,0,0)}, S_0\}, & x = -12N, \\ \{R\}, & x = 0, \\ \emptyset, & \text{otherwise.} \end{cases} \quad (65.4)$$

The controller holds the state and finite instruction and residue registers, both zero here. The stationary shuttle S_0 is a separate unit. Consequently the section has two occupied cells and mass three, not mass two.

Proposition 65.1 (Compiler interface used in this extension). *For every separated deterministic reversible source M with terminal state q_h , the inherited compiler has the following properties.*

1. A_M is a bijection of the full configuration space. Both directions have radius at most four, and total mass is conserved on finite configurations.
2. Each enabled source instruction from (q, N) to (q', N') carries $C_M(q, N)$ to $C_M(q', N')$ after the exact duration in Table 7. The right marker stays at zero throughout.
3. Intended macro executions use only the prescribed local collision rows. Intermediate stages cannot contain an unrelated clean stage-zero final pair.
4. A stuck nonfinal source enters a prescribed escape: its shuttle moves permanently left away from the stationary markers. This continuation cannot later produce a clean terminal section and uses no arbitrary completion row.

Source operation	Raw update	Exact native ticks
INC at p	$N' = pN$	$108N + 96N' + 8$
DEC at p	$N' = N/p, p \mid N$	$96N + 108N' + 8$
ZERO, POS, NOP	$N' = N$	$192N + 8$

Table 7: The complete source-step clock, including residue queries, finite gates, and cleanup. These are not arithmetic-flight times alone.

[write] [Interface with source 14. Item 1 of Proposition 65.1 is Theorem 49.1(1), with the path-closure completion of Lemma 51.1 and Lemma 51.2; items 2 and 3 are Theorem 49.1(3), by Lemmas 53.1, 54.1 and 55.1 and the cleanup argument of Section 55; item 4 is source 14's disabled-instruction escape there. The section $C_M(q, N)$ is source 14's canonical configuration with the clean symbol $\text{Ready}(q)$ (Section 52), and Table 7 is (56.1) at $n = 12N$: $8n + 8n' + n/d + 8$ equals the displayed entry for each of the five operation types. Source 15's two paragraphs recalling the path-closure completion and deriving the clock from the residue queries ($4n + 1$ ticks) and gap routines ($n/d + 1$ ticks) are not reprinted.]

66 A clean return source program

Let R be a source satisfying Section 65. Form two disjoint copies of its states and one new state,

$$Q_{\text{cl}} = \{F: q : q \in Q\} \dot{\cup} \{B: q : q \in Q\} \dot{\cup} \{H\}.$$

For each instruction $q \xrightarrow{\text{op}} q'$, introduce

$$F: q \xrightarrow{\text{op}} F: q', \quad B: q' \xrightarrow{\text{op}^{-1}} B: q. \quad (66.1)$$

The inverse exchanges increments with decrements, retains the counter index, and leaves both kinds of test and no-op unchanged. Add just two bridges:

$$F: q_h \xrightarrow{\text{NOP}} B: q_h, \quad B: q_0 \xrightarrow{\text{NOP}} H. \quad (66.2)$$

The new initial state is $F: q_0$ and the sole designated terminal state is H . In particular the two copies of q_h are ordinary, nonterminal states for the CA compiler.

Lemma 66.1 (Preservation of reversible syntax). *The program R_{cl} defined by (66.1)–(66.2) is separated, deterministic, and reversible. Its initial state has no incoming instruction and H has no outgoing instruction.*

Proof. The forward copy retains each original incoming and outgoing group. In the backward copy, outgoing groups are the inverses of original incoming groups, and incoming groups are inverses of original outgoing groups. A singleton arithmetic instruction remains singleton. A test group remains the same pair of disjoint tests on the same unchanged counter. Thus reversal preserves the syntactic conditions.

The first bridge uses an empty outgoing side at $F: q_h$ and an empty incoming side at $B: q_h$; both are empty because q_h had no outgoing instruction. The second uses the empty outgoing side at $B: q_0$, by freshness (65.1), and the fresh incoming side at H . Each added group is a legal singleton no-op. No other side is changed. The claimed entry and exit properties follow immediately. $\square$

Theorem 66.2 (Restoration and first source halt). *Suppose the original run from (q_0, N_0) first reaches q_h after h instructions. Write its sections as (q_t, N_t) , $0 \leq t \leq h$, with q_h also denoting its final state. Then the wrapped run is exactly*

$$(F: q_0, N_0), \dots, (F: q_h, N_h), \\ (B: q_h, N_h), (B: q_{h-1}, N_{h-1}), \dots, (B: q_0, N_0), (H, N_0).$$

It first reaches H after $2h + 2$ instructions and restores the whole raw integer, including its cofactor. If the original run does not reach q_h , the wrapped run does not reach H .

Proof. Before the first bridge, the forward copy executes exactly the original instructions. The bridge is available precisely at $F: q_h$ and preserves N_h . Every reversed arithmetic instruction is enabled on the actual preceding output: the inverse of an increment is a decrement on a number just multiplied by the relevant prime, and the inverse of a decrement is an always-enabled increment. Each inverse test sees the same unchanged raw number. These operations restore the previous raw value exactly. Backward uniqueness gives no alternative predecessor branch.

Inducting in reverse order reconstructs $(q_{h-1}, N_{h-1}), \dots, (q_0, N_0)$. The original initial state cannot occur at a positive source time, because no instruction targets it. Therefore the second

bridge cannot be encountered prematurely. It preserves N_0 and changes only the state to H . The count is $h + 1 + h + 1 = 2h + 2$. The invariant cofactor in (65.2) is unchanged by every operation and is restored along with both valuations.

If the original run is infinite, its forward copy is infinite and never takes the first bridge. If it is stuck nonfinally, the forward copy is stuck at that same section. Neither case reaches a backward-copy state or H . When $h = 0$, the hypotheses imply $q_0 = q_h$ is isolated; the run consists of the two bridges and the same proof applies. $\square$

66.1 Why the fresh entry condition matters

Consider the separated reversible source

$$q_0 \xrightarrow{\text{ZERO}_0} q_1, \quad q_1 \xrightarrow{\text{INC}_0} q_0, \quad q_0 \xrightarrow{\text{POS}_0} q_h. \quad (66.3)$$

From $N_0 = 1$ its trace is $(q_0, 1), (q_1, 1), (q_0, 2), (q_h, 2)$. A backward copy first arrives at $B: q_0$ with $N = 2$, before undoing the increment. An unconditional exit there would restore the wrong input. It also conflicts syntactically with the inverse decrement already leaving $B: q_0$. The implementation rejects this source. Rejecting only an incoming instruction enabled on the current input would be insufficient.

Nor can an arbitrary reversible source be freshened merely by prepending a no-op into its old initial state: that new incoming motion may conflict with existing incoming instructions. The source-existence dependency must genuinely supply the needed normal form. Morita and Imai [19], Proposition 3.4, give universal reversible two-counter simulation, and Lemma 3.5 supplies a simulator with no instruction entering its initial state while preserving reversibility. Apply these results after choosing one universal source problem. The finite counts below start after this normalization; no bound on its overhead is claimed. A terminal halt may be enforced by deleting outgoing instructions from the designated halt state, preserving first-halt behavior and the separated conditions.

66.2 Relation to classical uncomputation

Forward computation followed by inverse execution to clear temporary information is a classical idea. Bennett's 1973 construction [40] restores the input after retaining the desired output and clearing intermediate information. Here the source is already reversible, so its inverse execution needs no new history tape. A terminal control label records success while both counter values are restored. This report claims neither novelty nor priority for the forward-and-reverse idea; it proves the particular source syntax, finite geometry, exact target, and resource consequences needed for this CA.

67 Exact configurations and the sharp threshold

Apply the inherited compiler to R_{cl} with H as its designated final state, and abbreviate its CA to A . Set

$$x_N = C_{R_{\text{cl}}}(\mathbf{F}: q_0, N), \quad y_N = C_{R_{\text{cl}}}(H, N). \quad (67.1)$$

Both descriptions are computed directly from the supplied N . They have identical absolute support and differ only in the controller state inside the weight-two left symbol.

Theorem 67.1 (Whole-configuration equivalence). *For every positive raw input N ,*

$$(\exists t \in \mathbb{N} : A^t(x_N) = y_N) \iff R \text{ reaches } q_h \text{ from } (q_0, N).$$

On a first source halt at h , the target is first reached at the end of cleaned source instruction $2h + 2$.

Proof. On a halt, Theorem 66.2 gives the canonical section (H, N) after the indicated number of source instructions. Proposition 65.1 resets the finite registers at every section. The right marker has stayed at zero and the left pair returns to $-12N$. There are exactly the three prescribed channels and vacuum elsewhere, so the resulting object is the entire target in (67.1), not just a matching halt symbol.

Before this section, Theorem 66.2 lists all canonical source states, none equal to H . Intermediate macro stages have different stage or shuttle types and cannot equal the clean stage-zero target. The prescribed-row simulation therefore rules out an earlier hit.

For a nonhalting source, an infinite forward execution continues through its intended macros without an H section. A finite stuck execution takes the explicit escape continuation: the escaping channel moves forever away and never reassembles a clean pair. These are complete descriptions of both possible nonhalting continuations, not finite-prefix checks. No arbitrary completion row supplies a later false target. The equivalence follows. $\square$

The theorem concerns the first exact hit. It makes no assertion that the target is stationary, that it occurs only once, or that the subsequent orbit remains a simulated source run. Completion may determine post-hit behavior. None of those later facts is needed for reachability.

Corollary 67.2 (One fixed rule). *Fix a universal reversible fresh-entry source as above. Then exact-pair reachability for its fixed cleaned CA is undecidable on the computable family (67.1). Exact-pair reachability is recursively enumerable: direct simulation semidecides a hit.*

Proof. Encode an instance of the fixed source's halting problem into its initial counter values, then into N and the pair (x_N, y_N) . Both endpoints are computed without the halt time or output values. A decision procedure for this restricted CA reachability family would decide source halting by Theorem 67.1. Direct CA simulation enumerates finite times and checks finite configuration equality, proving the semidecision statement. With a standard many-one complete halting source, the same reduction gives recursively enumerable completeness. $\square$

This supplies a fixed rule, not a variable rule supplied alongside each input. It does not supply one fixed target for every N . The dependence of y_N on N is explicit and is part of the reduction.

67.1 Why mass two cannot do the same

[write] [Source 15 summarizes here, “from the companion proof”, the mass-two decision argument: finite walker maps, the near/far threshold $2r$, the $|Q_2| + 2r|Q_1|^2$ near shapes, the translating recurrent tail, exact target tests at every phase and at all later translates, and polynomial bit complexity for a fixed rule. That is Theorem 44.1 of Part IV, proved there, and Section 59 of source 14; the summary is not reprinted.]

Combining this decision theorem with Theorem 67.1 proves the mass threshold in Theorem 64.1. The assumptions exclude zero-weight active computation and signed mass cancellation. A prescribed finite ring has a finite mass sector and is decidable by orbit search, so it is a different problem.

68 Radius one and the exact clock

68.1 The first target time

Let Θ be the native microtime of the original forward source trace and N_h its final raw value. Reversal exchanges an increment $N \rightarrow N'$ with a decrement $N' \rightarrow N$. Table 7 gives

$$108N + 96N' + 8 = 96N' + 108N + 8.$$

The decrement/increment pair has the same symmetry. Inverse tests and no-ops leave the raw value unchanged and have the same duration. Hence the reversed half takes exactly Θ , in reverse order of macro durations. The two bridges cost $192N_h + 8$ and $192N_0 + 8$.

Theorem 68.1 (Exact first-target clock). *If the original first halt is after h source instructions, the native CA first reaches its exact target at*

$$T_{\text{cl}} = 2\Theta + 192(N_h + N_0) + 16. \quad (68.1)$$

For $h = 0$ this is $384N_0 + 16$, with both no-op bridges retained.

This is logical inverse execution using forward CA time and different finite controller types. It is not the microscopic inverse orbit under A^{-1} , and need not reverse particle worldlines. The time identity follows from the explicit macro formulas, not from global reversibility by itself.

68.2 Four-site spatial blocking

Group old sites $4k, 4k+1, 4k+2, 4k+3$ into blocked site k . Every old channel gets an offset 0, 1, 2, 3. This is a conjugacy of full shifts, with weight equal to the sum of the four old weights. A native radius-four update and its radius-four inverse become radius-one updates because the four-site block needs only itself and its two neighboring blocks.

The native right marker stays at blocked coordinate zero, offset zero. Since $-12N = -4(3N)$, the left pair is at blocked coordinate $-3N$, also offset zero. Conjugacy preserves exact equality at every tick, and therefore preserves the first-hit time T_{cl} . No extra particle or background is added.

68.3 Four-phase internal dilation

The alternative radius-one construction keeps native coordinates and gives every channel four phase copies. A native velocity in $[-4, 4]$ is spread over four single-cell submoves; the local collision permutation acts once per four-phase cycle. On synchronized phase-zero configurations, four new ticks implement one native tick. Each individual step remains a full-space bijection, including on configurations outside the synchronized encoded sector, as established by the inherited phase construction.

The target uses phase-zero channels. At every time all three encoded tokens have phase $t \bmod 4$, so an off-section time cannot equal that nonempty target. At time $4t$, its state equals the native state after t ticks under the phase-zero encoding. The first exact target is consequently at $4T_{\text{cl}}$, with the same coordinates $-12N$ and 0. Both radius-one alternatives have four times the native number of channel types. Spatial blocking preserves the clock; internal dilation preserves the native coordinate convention. [write] [The two constructions are source 14's Theorems 57.1 and 57.2, applied to the wrapped source.]

Radius zero cannot support the stated undecidable exact-pair problem: sites neither move nor communicate, and a finite-support configuration has a finite orbit of independent finite-state updates. Thus radius one also attains the smallest possible radius for this result.

69 Finite program and local rule costs

Let the normalized original source have $q = |Q|$ states, J instructions, instruction-register modulus $m = J + 1$, and d arithmetic instructions. Let B be its residue-expanded branch count: a zero test at prime three contributes two branches, and each other instruction contributes one. The wrapper does not simplify or remove unreachable states.

Quantity	Original	Cleaned
States	q	$q' = 2q + 1$
Instructions	J	$J' = 2J + 2$
Instruction modulus	m	$m' = 2m + 1$
Arithmetic instructions	d	$d' = 2d$
Expanded branches	B	$B' = 2B + 2$
First-halt source horizon	h	$h' = 2h + 2$

Table 8: Exact unsimplified wrapper counts. The first five rows count a finite program; the last counts one successful execution.

The inherited verified local-rule formulas can be applied directly to these primed quantities:

$$K_{\text{types}} = 96q'm' + 6q'd' + 2d' + 6q' + 2314, \quad (69.1)$$

$$K_{\text{prescribed}} = 3528q'm' + 6q'd' + d' - 6m' + 1152, \quad (69.2)$$

$$K_{\text{completed}} = 7008q'm' + 12q'd' + 2d' + 6q' - 6m' + 2304. \quad (69.3)$$

The latter two count native pair rows before completion and pair-support rows after completion. The full native cell alphabet has $2^{K_{\text{types}}}$ symbols. It is specified by sparse singleton and pair permutations with identities elsewhere, without enumerating all subsets. [write] [These are the counts s , P and C of source 14's Section 55.1 at the primed values.]

Both radius-one alternatives use $4K_{\text{types}}$ channels. Equation (69.3) is not a flattened count of blocked local rows: spatial blocking uses a composed finite description. The counts are constructive upper descriptions for this generator, not smallest-alphabet or optimal-overhead theorems. They do not assign a numerical universal alphabet size, because no expanded universal normalized source is supplied here.

70 Compact quadratic certificates

The finite source horizon h is an external parameter. The certificate is a polynomial over natural numbers whose zeros encode first halt of the original source at that horizon. Reusing this certificate avoids independently guessing the inverse computation. We give the original encoding and the precise sense in which its witness determines a full cleaned witness.

70.1 Forward branch forms and exactness

Expand zero tests at prime p into residues $r \in \{1, \dots, p-1\}$. For every step $0 \leq t < h$ and branch $1 \leq j \leq B$, use natural variables e_{tj} and u_{tj} . The affine forms in Table 9 describe an old raw value A_j , a new raw value C_j , and duration D_j . Fix an injective integer coding of states; let s_j and t_j be branch j 's source and target codes. They may coincide for a self-loop.

Branch	$A_j(e, u)$	$C_j(e, u)$	$D_j(e, u)$
INC at p	$e + u$	$p(e + u)$	$108A_j + 96C_j + 8e$
DEC at p	$p(e + u)$	$e + u$	$96A_j + 108C_j + 8e$
POS at p	$p(e + u)$	$p(e + u)$	$192A_j + 8e$
ZERO at p , residue r	$pu + re$	$pu + re$	$192A_j + 8e$
NOP	$e + u$	$e + u$	$192A_j + 8e$

Table 9: All forms vanish when $e = u = 0$. At $e = 1$, natural u enforces the exact enabled arithmetic or residue class.

Define the affine sums, separately at every t ,

$$E_t = \sum_j e_{tj}, \quad U_t = \sum_j A_j(e_{tj}, u_{tj}), \quad V_t = \sum_j C_j(e_{tj}, u_{tj}),$$

$$S_t = \sum_j s_j e_{tj}, \quad Q_t = \sum_j t_j e_{tj}, \quad \Theta = \sum_{t,j} D_j(e_{tj}, u_{tj}).$$

Set $V_{-1} = N_0$ and $Q_{-1} = \text{code}(q_0)$. These are affine interfaces, not new variables. The core polynomial is

$$P_h = \sum_{t=0}^{h-1} [(E_t - 1)^2 + (S_t - Q_{t-1})^2 + (U_t - V_{t-1})^2]$$

$$+ (Q_{h-1} - \text{code}(q_h))^2 + \sum_{t=0}^{h-1} \sum_{j=1}^B (E_t - e_{tj})(e_{tj} + u_{tj}). \quad (70.1)$$

It has degree at most two. Every summand is nonnegative on the full natural orthant, since $E_t - e_{tj} = \sum_{k \neq j} e_{tk} \geq 0$. [write] [Table 9 and (70.1) are source 14's branch table and certificate (58.1), with q_h for q_f ; the gate is discussed in the note after Theorem 58.1.]

Lemma 70.1 (Forward zero fibers). *For fixed input coordinates, P_h has a natural zero exactly when the original run first halts at horizon h . If a zero exists, its full forward witness is unique. Its affine endpoints are N_0 , $N_h = V_{h-1}$, and Θ .*

[write] *Pointer*. This is source 14's Theorem 58.1, for the same polynomial. Source 15's proof repeats source 14's (the square $(E_t - 1)^2$ and natural selectors force one-hot selection, every inactive product forces its offset to zero, the active offset is $N - 1$, $N/p - 1$ or $(N - r)/p$, the terminal square forces a first halt, and $h = 0$ leaves only the constant halt square); it is not reprinted. $\square$

70.2 The compact clean certificate and roles of coordinates

Use the forward polynomial unchanged and replace its exposed physical-time form by (68.1). By Theorems 66.2 and 67.1, its zero already proves the exact cleaned target and determines every source

section in both halves. The fixed compiler then determines every intervening CA microconfiguration. This is a certificate for the specified encoded source computation, not a general certificate for arbitrary CA trajectories.

To impose a supplied exact microtime T , add just

$$(\kappa[2\Theta + 192(N_h + N_0) + 16] - T)^2, \quad \kappa = \begin{cases} 1 & \text{native or spatial,} \\ 4 & \text{phase.} \end{cases} \quad (70.2)$$

For fixed T this adds no variable. To expose time as a free output coordinate, introduce one natural T and the same square; it is uniquely determined in each accepted input fiber. The target raw value is the already existing input expression N_0 . Its restoration needs neither a separately guessed output value nor another endpoint square.

The exact core ledger is

$$2Bh \text{ natural variables,} \quad 3h + 1 \text{ affine-square slots,} \quad Bh \text{ affine-product slots.} \quad (70.3)$$

These are unsimplified emitted slots; a degenerate term may vanish identically. By comparison, applying the original exporter directly to the cleaned program at horizon $2h + 2$ would give

$$8(B + 1)(h + 1) \text{ core variables,} \quad 6h + 7 \text{ squares,} \quad 4(B + 1)(h + 1) \text{ products.} \quad (70.4)$$

The smaller ledger is justified by source composition and the zero-fiber lift below, not by dropping constraints from an arbitrary reverse run.

70.3 Input modes and their paid costs

A fixed positive raw input is a constant. A free raw input uses $N_0 = x + 1$ with one genuine natural input coordinate x and no loader square. Fixed counters (a, b) can be encoded externally by the computable constant $2^a 3^b$.

For an explicitly bounded counter interface, fix bounds $0 \leq a \leq A$ and $0 \leq b \leq C$. Introduce $K = (A + 1)(C + 1)$ natural selectors λ_{ij} and three squares

$$\left(\sum \lambda_{ij} - 1\right)^2, \quad \left(a - \sum i \lambda_{ij}\right)^2, \quad \left(b - \sum j \lambda_{ij}\right)^2.$$

Use $N_0 = \sum 2^i 3^j \lambda_{ij}$ by affine substitution. The selectors are uniquely one-hot on accepted inputs. Every genuinely free a or b is an additional input coordinate; fixed values are constants. This finite table is not an unbounded exponentiation gadget.

Let r count genuine input coordinates, k loader selectors, $\ell \in \{0, 3\}$ loader squares, $s \in \{0, 1\}$ a free time output, and $\epsilon \in \{0, 1\}$ whether a time equality is included. The full compact ledger is

$$\begin{aligned} \text{natural variables} &= 2Bh + r + k + s, \\ \text{affine squares} &= 3h + 1 + \ell + \epsilon, \\ \text{affine products} &= Bh, \quad \text{degree} \leq 2. \end{aligned} \quad (70.5)$$

For $h = 0$, the time form is $384N_0 + 16$ before any phase scaling, and all optional input and time costs are still charged.

70.4 An affine lift to the full cleaned witness

Index original expanded branches by $j = 0, \dots, B-1$ for this subsection. In the full cleaned exporter, branches j are the forward copies, $B+j$ are their inverse copies, and $2B, 2B+1$ are the bridges. Its steps are indexed $0, \dots, 2h+1$. Denote its branch variables by $\tilde{e}_{s,k}, \tilde{u}_{s,k}$.

The following affine substitution L maps compact witnesses to full ones:

$$\tilde{e}_{t,j} = e_{t,j}, \quad \tilde{u}_{t,j} = u_{t,j}, \quad 0 \leq t < h, \quad (70.6)$$

$$\tilde{e}_{2h-t,B+j} = e_{t,j}, \quad \tilde{u}_{2h-t,B+j} = u_{t,j}, \quad 0 \leq t < h, \quad (70.7)$$

$$\tilde{e}_{h,2B} = 1, \quad \tilde{u}_{h,2B} = N_h - 1, \quad (70.8)$$

$$\tilde{e}_{2h+1,2B+1} = 1, \quad \tilde{u}_{2h+1,2B+1} = N_0 - 1. \quad (70.9)$$

Every unlisted full-core coordinate is zero. Preserve all genuine input coordinates, loader selectors, and any time output. Coordinate names are merely labels: rename them consistently if a user-specified public name collides with a larger full-core variable name.

Theorem 70.2 (Affine bijection of natural zero fibers). *With the same fixed source, horizon, input interface, and optional time condition, L restricts to a bijection from compact natural zeros to natural zeros of the full cleaned-source certificate at horizon $2h+2$. Projection onto its original forward variables and public coordinates is its inverse, after any consistent renaming.*

Proof. Start with a compact zero. Lemma 70.1 gives a genuine forward trace and positive endpoints N_0, N_h . Thus the two bridge magnitudes in (70.8)–(70.9) are natural, and every coordinate of L is natural.

For an original increment, $N = e + u$ and $N' = p(e + u)$. Its inverse decrement on N' uses exactly the same e, u , so exchanges the old and new affine raw values. An original decrement has the same property with increment as its inverse. Tests and no-ops have unchanged raw values and retain the same parameters, including the zero-test residue. Consequently (70.7) realizes the unique inverse trace step by step. The bridge rows are no-ops at N_h and N_0 . All unselected coordinates are zero, so their inactive products vanish. The source and raw interface squares match the concatenated trace, and the full terminal state is H .

The unchanged loader satisfies its original three squares when present. Summing the full duration forms gives (68.1); hence any time endpoint square also agrees. The lifted assignment is therefore a full natural zero.

Conversely, the forward-certificate exactness theorem applied to the full cleaned exporter turns any of its natural zeros into a first-halt cleaned execution of length $2h+2$. The only route to H traverses a forward first halt, both bridges, and the unique inverse trace. If that forward length is k , Theorem 66.2 gives total length $2k+2$, so $k = h$. Projecting to the first h rows and original public coordinates gives a compact zero; the clock identity supplies its time condition. Full witness uniqueness at fixed input forces all remaining rows to be precisely (70.7)–(70.9). Thus the two maps are inverses on zero fibers. $\square$

The restriction to zero fibers is essential. The bridge expression $N_h - 1$ can be negative at a general natural assignment that is not a compact zero. The claim is not a map of the entire natural orthant, nor an identity of the two polynomial values off their zero sets. The natural domain is also essential for the original encoding: fractional branch magnitudes can bypass discrete divisibility or residue guards. No assertion for nonnegative rational or real solutions is made. [write] [The programme's review [45] exhibits both limitations exactly: for a one-step increment at $N_0 = 1$ and

the off-zero assignment $e = u = 0$, the compact polynomial has value 3, its lift has value 18 and a bridge coordinate -1 ; and a one-step prime-two zero test at $N_0 = 2$ has the nonnegative rational zero $e = 1$, $u = 1/2$ of source 14's Section 58. It records them as documented limitations, not defects, and checks 45 compact-to-full natural zero-fibre bijections.]

Finally, this is a family indexed by externally fixed h . Eventual halting is a union over all finite horizons. Merely replacing h by a variable does not give a fixed-arity unbounded-time quadratic polynomial, and fixed-horizon uniqueness does not imply a finite-fold unbounded-time representation.

71 A complete exact-target example

Take six states $q_0, \dots, q_5$, halt q_5 , and the five-instruction chain

$$q_0 \xrightarrow{\text{INC}_0} q_1 \xrightarrow{\text{INC}_1} q_2 \xrightarrow{\text{DEC}_0} q_3 \xrightarrow{\text{DEC}_1} q_4 \xrightarrow{\text{INC}_0} q_5.$$

It is fresh-entry and separated reversible. Use $N_0 = 5$, whose two counter valuations are initially zero and whose preserved cofactor is five. The original raw values are

$$5, 10, 30, 15, 5, 10.$$

The source halts after $h = 5$ with $N_h = 10 \neq N_0$ and native time $\Theta = 13\,480$. The different endpoint value makes this a genuine restoration test rather than a run whose original output happens to equal its input.

Cleaned source step	State	Raw N	Native or spatial tick
0	F: q_0	5	0
1	F: q_1	10	1 508
2	F: q_2	30	5 476
3	F: q_3	15	9 984
4	F: q_4	5	11 972
5	F: q_5	10	13 480
6	B: q_5	10	15 408
7	B: q_4	5	16 916
8	B: q_3	15	18 904
9	B: q_2	30	23 412
10	B: q_1	10	27 380
11	B: q_0	5	28 888
12	H	5	29 856

Table 10: Every clean section of the example. The inverse source half restores all intermediate raw values in reverse order. Its microscopic paths need not be reversed.

The initial configuration is the clean F: q_0 controller and shuttle at -60 , with R at 0. The exact target uses the clean H controller and shuttle at -60 , the same R at 0, and vacuum everywhere else. Its first time is

$$2(13\,480) + 192(10 + 5) + 16 = 29\,856.$$

Spatial blocking moves the left pair to -15 , offset zero, and keeps that time. The phase construction keeps coordinate -60 and first reaches its phase-zero target at 119 424 ticks.

Here $q = 6$, $J = 5$, $m = 6$, $d = B = 5$, while the cleaned values are $q' = 13$, $J' = 12$, $m' = 13$, $d' = 10$, $B' = 12$. The native generator gives 19 416 channels, 598 096 prescribed pair rows, and 1 188 236 completed pair-support rows. These are example counts, not universal-machine counts.

The compact certificate has 50 core variables, 16 core squares, and 25 products. The packaged free-time version has 51 total variables and 17 squares, with time output 29 856. By comparison, the full cleaned exporter has 288 core variables, 37 core squares, and 144 products before the same time-output charge. The affine lift reuses the forward witness, with bridge magnitudes $N_h - 1 = 9$ and $N_0 - 1 = 4$.

The archive’s request, literal expanded certificate, unique witness, and independent checker receipt reproduce Table 10. They describe a small finite chain rather than a universal program.

72 Validation and reproducibility

72.1 What the implementations check

The cleanup exporter validates structural freshness, builds the forward/inverse source table, composes the original polynomial, and exposes the cleaned clock. A separate checker reconstructs the wrapped program and affine outputs instead of trusting these claims in the exported JSON. It invokes the unchanged independent forward checker and validates the natural witness, optional literal polynomial expansion, and full resource ledger. Its decoded cleaned trace is compared with actual source execution and literal lattice evolution.

The producer tests exercise all primitive operations, both nonzero prime-three residues, guarded branch/merge behavior, a mixed chain with unequal original endpoint values, $h = 0$, immediate and delayed stalls, a growing nonhalting prefix, raw cofactors, loader interfaces, invalid freshness, clock symmetry, and maliciously altered certificates. Successful lattice checks compare the entire target and every canonical section, test conservation and inverses, and require prescribed pair rows through the first hit. Selected cases execute all native, spatial, and phase ticks.

The independent lattice harness separately builds the wrapper, executes source instructions, recognizes the full canonical configurations, and steps the generated collision and stream maps without calling the compiler’s ordinary step or ready helpers. Its certificate audit exercises valid interfaces and rejects deliberate mutations. The separate lift audit compares explicit forward/reverse variable reuse with the full cleaned exporter and its frozen independent checker.

Finite evidence and theorem evidence. Finite tests verify concrete implementations and boundary cases. The all-input restoration and nonhalting exclusion follow from the proofs, the full-space reversible behavior from the compiler, and undecidability from the classical source dependency. No finite collection of successful traces proves universality, and a growing finite prefix does not alone prove nonhalting. The archive records each test’s actual scope rather than conflating these forms of evidence.

Replay summary. The completed release receipt records the exact test counts and subject hashes. The independently checked lattice sample comprises 14 small sources and 35 runs, including 26 successful cleanups and nine stuck inputs. It covers 111 712 native ticks, 9 456 spatial ticks, and 37 824 phase ticks. The independent certificate pass checks 61 valid model/interface cases, rejects 174 mutations and 19 malformed syntax or API cases, and enumerates 1 536 natural assignments for prime-three zero-test branches. The producer checks 77 affine-lift cases; the independent lift audit

checks 24 cases and 96 public-coordinate naming-collision regressions. The producer lattice suite covers 45 halting runs, 21 stalls, 197 374 native ticks, 21 120 spatial ticks, and 84 480 phase ticks; it also checks 140 clock-symmetry cases. Normal and optimized Python runs are recorded separately.

72.2 Portable replay and source provenance

The portable archive includes this standalone editable LaTeX source and PDF, the cleanup code and independent checker, literal examples, the five exact frozen Python dependencies, independent audit harnesses, receipts, and build/replay scripts. Python replay uses only the standard library. PDF rebuild uses a conventional TeX installation and the packages declared at the top of the source. No third-party paper PDFs are redistributed.

All package files are covered by a SHA-256 manifest except the manifest itself. A verifier checks both hashes and the expected file set. The archive safety check rejects absolute paths, parent traversal, duplicate members, and symbolic links. Fresh replay is performed from an extracted copy rather than from the source working directory. Temporary generated outputs remain outside the sealed file set. The final PDF is rendered and every page is visually checked.

The provenance ledger records hashes for the five frozen dependencies, the exact extension sources, source-audit scripts, and the companion report that establishes inherited results. Independent harnesses are made portable only by changing dependency/output path selection; that adaptation is recorded explicitly and the adapted copies are rerun. Receipts identify which subject code was actually tested. References link to primary publications; packaging does not silently substitute a different local-rule generator.

[write] **The release in this report.** These are the delivered names. Source 15’s top-level programs and its `audit/` programs are shipped as `code/15-clean-targets-*`; its `examples/`, `receipts/` and `provenance/` records as `data/15-clean-targets-*`; and `CLEAN-TARGET-THE OREM.md`, `SOURCE_PROVENANCE.md` and `audit/INDEPENDENT-AUDIT.md` as `15-clean-targets-*` at the report root. “The five exact frozen Python dependencies” (`vendor/*.py`) are byte copies of source 14’s programs and are shipped once, as `code/14-three-mass-*`. Not shipped: the manuscript, PDF and README, the SHA-256 manifest and `verify_manifest.py`. The programs import `vendor/` by its delivered name, so they must be run in a restored layout; on Windows `verify_manifest.py` rejects every nested member (it compares native paths with `/`-separated names), and the replay’s byte-exact example comparison fails on CRLF output although the content agrees, so the replay is run without the manifest there (README).

73 Boundaries and precise next questions

The construction settles the mass threshold for exact-pair reachability in the stated weighted model. It also settles radius one as a sufficient radius with three conserved mass units. Several stronger or more concrete claims remain distinct.

1. **A fully expanded universal instance.** Instantiate a particular universal reversible two-counter table with fresh entry and terminal halt, provide its input encoder, and freeze the resulting cleaned local rule. This would yield an exact universal alphabet and row count. The small examples in this archive do not perform that instantiation. ([write] Compare source 14’s Question 1 and Part IV’s first question; all three are open.)
2. **One target independent of the input.** Determine whether a suitable fixed reversible

weighted CA can have undecidable reachability to one chosen finite configuration from a computable input family. The present restoration construction gives y_N , not a single y . Injectivity and the backward orbit of a fixed target must be handled explicitly; no claim of impossibility follows from this report.

3. **Smaller finite-state overhead.** Reduce the duplicate state space, residue clock, or two bridge costs while retaining global reversibility and exact scratch cleanup. Compare improvements using the same target convention and all resource ledgers. A faster logical inverse cannot be assumed to reverse microscopic paths. ([write] Compare source 14’s Question 2.)
4. **The integer-state mass convention.** Find the corresponding sharp reversible threshold when each cell state’s integer value is its conserved mass. Equal-weight channel types used here cannot be transferred by relabeling. This is a different model-specific question. ([write] Compare source 14’s Question 4, answered in part by Part VI, as the note there says.)
5. **A costed unbounded-time certificate.** Compose this frontend with an explicit fixed-arity Diophantine encoding of unbounded computation, and state degree, variable costs, and the resulting fiber properties. The fixed-horizon affine lift alone supplies no finite-fold or single-fold unbounded-time conclusion. ([write] Compare source 14’s Question 5.)
6. **Mechanized compiler composition.** Formalize the separated syntax, inverse source transform, prescribed collision routes, and zero-fiber correspondence in one proof system. The key interfaces to preserve are exact absolute configurations, natural domains, and the distinction between source steps and CA ticks.

There is close geometric prior art. Brandt, Portmann and Uitto [37] use three finite-state agents, a prime-exponent gap, and reflected-shuttle arithmetic. The companion report [source 14, Section 62] compares those mechanisms with the global reversible CA and its cleanup obligations. The extension here does not claim a literature-wide priority result or a new general principle of uncomputation. Its conclusion is the explicit exact-target strengthening, under the model and dependencies stated above.

Part VI

A single unit symbol: decidability at masses three and four (sources 16, 17 and 18)

[write] Part VI prints two further AI-assisted research manuscripts of batch 80 (arrival 4e270aa46), both dated 3 October 2026. *Source 16* is batch-80 manuscript 05, *Three mass units with a single unit symbol*, author line “Decidability theorem and executable checks”, from the archive `Single_Unit_Three_Mass_Decidability (1).zip` (with a space before the parenthesis); its labels carry the prefix `smc:su:`, and its sections are Sections 74–82 here. That archive is the “attribution revision” (its `REVISION.md`, shipped as `16-single-unit-REVISION.md`) of the batch’s manuscript 06, `Single_Unit_Three_Mass_Decidability.zip`: the revision credits Kong’s 2021 thesis and Imai’s 2021 KAKEN report in its abstract, its source comparison and two bibliography entries, and leaves the mathematics, programs, receipts and audit unchanged (seven files byte-identical). Manuscript 06 is therefore superseded and not printed. *Source 17* is batch-80 manuscript 03, *Four mass units and exact reachability*, author line “Decidability proof and a binary timing certificate”, from the archive

`Four_Mass_Decidability_Package.zip`; its labels carry the prefix `smc:fm:`, and its sections are Sections 83–92. It builds on source 16, which it calls “the separate 3 October 2026 companion report”, and restates the ingredients it needs (its Section 84, printed with pointers). Neither manuscript names a repository commit; source 16 cites Part IV (source 13) by its title, “Section 9”, for the post-elimination compiler. Statements keep their numbering by section; equations, which the sources tagged (1), (2), $\dots$, are numbered by section. The programme has reviewed all three archives [46] (Section 1.14). ([write] Added 3 October 2026: a third source, batch-82 manuscript 19, follows source 17 as *source 18*; it gives canonical timed quartics for a fixed input and answers source 17’s last question. It is printed after source 17 with its own opening, and Section 1.17 describes it.)

[write] **Relation to Part IV and Part V.** [Editorial note: source 16’s mass-two case and its architecture are Part IV’s. The mass-two case is Theorems 44.1 and 45.1 of Part IV, proved there for any number of weight-one symbols; source 16’s Lemma 75.2 and Sections 76–79 follow the architecture of Sections 44–45 (first entry with the absolute position, finite templates, no product k_0n), and its quartic is Part IV’s compiler of Section 46. Bracketed notes mark each place.] The decisive new ingredient is source 16’s no-split lemma (Lemma 75.1), which uses the single unit symbol: in the comoving frame an isolated unit is fixed, so a bounded mass-two packet never separates into two distant pieces. Part VI is consistent with Part V, which uses many weight-one types (source 14, Section 49; source 16, Section 79).

Source 16 (batch-80 manuscript 05): Three mass units with a single unit symbol
Decidability theorem and executable checks, 3 October 2026. Its abstract follows; the two bracketed notes in it are [write].

For every fixed one-dimensional deterministic conservative cellular automaton with a finite alphabet, positive integer non-vacuum weights, and at most one weight-one symbol, the complete timed evolution relation at total mass at most three is effectively Presburger-definable in all input and output positions. Exact reachability and finite-pattern occurrence, at a fixed position or anywhere, are decidable. In the stationary-unit frame, every bounded two-unit cluster remains bounded and has finite-phase motion; three-unit encounters then reduce to a finite core. Composition with a canonical arithmetic compiler gives horizon-free quartic certificates with zero or one natural witness tuple. Consequently, standard nonnegative integer-state number-conserving CA cannot realize undecidable finite halt-pattern occurrence with three numerical particles. This separates them from typed weighted three-mass constructions [such as that of Part V (source 14), which uses many weight-one types]. Kong’s June 2021 thesis already proves binary three-particle nonuniversality using a periodic-orbit decomposition. The present weighted, uniform arithmetic strengthening is stated explicitly; its novelty and the current mass-four boundary remain unresolved here. [Added 2 October 2026: the mass-four boundary is settled within the same class by source 17 (Theorem 83.1).]

74 Model and main result

Let A be a finite alphabet with a distinguished vacuum 0 and a weight $w : A \rightarrow \mathbb{N}$, where $w(0) = 0$ and $w(a) > 0$ for every $a \neq 0$. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, time-independent, translation-equivariant CA of finite radius R . The vacuum is quiescent. For every finite-support configuration c , assume

$$M(F(c)) = M(c), \quad M(c) = \sum_{i \in \mathbb{Z}} w(c_i).$$

There is at most one symbol u with $w(u) = 1$. Other symbols may share weights, including weights two and three. No reversibility assumption is made. The vacuum is the only *zero-weight* symbol;

other constant configurations are allowed to be fixed.

Write $\tau_a c$ for translation of a configuration by a sites to the right. A sparse configuration is specified by a strictly increasing position tuple $x_1 < \dots < x_n$ and its nonzero label tuple $\alpha = (\alpha_1, \dots, \alpha_n)$. At mass at most three, $n \leq 3$, and only finitely many label tuples are possible.

Theorem 74.1 (Uniform timed evolution). *For each fixed F satisfying these assumptions and each pair of nonzero label tuples α, β of total weight at most three, an algorithm constructs a Presburger formula*

$$\text{Rel}_{\alpha, \beta}^F(x_1, \dots, x_n, t, y_1, \dots, y_m)$$

which holds exactly when $t \in \mathbb{N}$, both position tuples are strictly increasing, and $F^t(c_{\alpha, x}) = c_{\beta, y}$. The vacuum is handled as a separate case. The construction is effective from the finite local rule, radius, and weights, under the stated hypotheses.

“Uniform” means that the formula includes all sparse input positions, output positions, and times for one fixed rule. It does not assert a single Presburger formula in which arbitrary CA rule tables are themselves encoded as an additional numerical parameter. The effective equivalence with semilinear sets is classical [7]. Presburger arithmetic here is first-order arithmetic over $\mathbb{Z}$ with addition, order, and integer constants. Quantification over $\mathbb{N}$ is expressed by nonnegativity constraints.

Corollary 74.2. *For any such fixed F , the following problems are decidable on finite inputs of mass at most three:*

1. *Exact finite-configuration reachability, including specified-time reachability and reachability up to translation.*
2. *Occurrence of a given finite pattern at a specified position, or at some position, at some nonnegative time.*
3. *Any fixed finite collection of such observations at specified relative time offsets.*

For fixed input and fixed observation, its set of occurrence times is effectively semilinear and hence eventually periodic.

[write] [The mass-two case is Theorems 44.1 and 45.1 of Part IV, proved there for any number of weight-one symbols. Lemma 75.2 and Sections 76–79 follow the architecture of Part IV’s Sections 44–45 (first entry with the absolute position preserved, finite templates after entry, no product $k_0 n$), one mass higher. Source 16 cites Part IV only for its compiler (Section 80).]

75 The stationary frame and two-unit clusters

If no unit symbol exists, every nonempty configuration of mass at most three has exactly one occupied site. Conservation forces its successor to have exactly one site of the same weight. There are finitely many labels and bounded one-step displacements, so this is a finite-state walker. Its motion is effectively eventually periodic up to translation, and Theorem 74.1 follows immediately. Henceforth assume u exists.

Conservation forces $F(u \text{ at } 0)$ to be exactly one copy of u , at some site $\delta \in [-R, R]$. Define

$$G = \tau_{-\delta} \circ F, \quad S = \max\{1, R + |\delta|\}.$$

Then G has radius at most S , is conservative, and fixes each isolated copy of u . Moreover $F^t = \tau_{\delta t} \circ G^t$. It suffices to prove the theorem for G ; replacing each output position y_j by $y_j - \delta t$ gives the formula for F . If the comoving rule has radius zero, each occupied coordinate stays fixed and its label follows a finite weight-preserving map; finite transient and periodic label conditions prove the result directly. The padding $S \geq 1$ also includes this case in the argument below without dividing by a zero radius.

Connect two occupied sites if their distance is at most $2S$. Components always refer to this graph. If distinct components are separated by more than $2S$, no radius- S input neighborhood sees both. Quiescence therefore gives exact independence for one step: the global successor is the disjoint union of the successors of those components evolved alone. Their radius- S output neighborhoods are disjoint as well.

Lemma 75.1 (Two-unit diameter invariance). *If a mass-two configuration has diameter at most $2S$, then every iterate under G has diameter at most $2S$.*

Proof. There are only two possibilities. A single weight-two symbol has its successor inside an interval of width $2S$ by finite propagation.

Otherwise the configuration is u at sites 0 and d , where $1 \leq d \leq 2S$. If $d > S$, each original site has exactly its isolated-unit radius- S neighborhood. Both original sites therefore output u . These two sites already exhaust the conserved mass, so the entire configuration is fixed.

If $d \leq S$, consider an output site other than 0 and d . If its neighborhood sees no unit, it outputs zero by quiescence. If it sees just one unit, its local input agrees with an isolated-unit configuration at that output site; since isolated units are fixed, it again outputs zero. A new nonzero site must see both units, and thus lies in

$$[d - S, S].$$

This interval also contains 0 and d , and has width $2S - d \leq 2S$. All output support is therefore contained in it. Induction proves the claim. $\square$

This lemma excludes a potentially misleading geometric scenario. Under a coarse propagation bound alone, a pair could appear to split, let one fragment join a third unit, and leave the other behind without a full encounter. Here that cannot happen across distinct $2S$ -components: a mass-two component never splits into two such components. The unique stationary unit label and mass conservation are essential to this conclusion. [write] [This no-split lemma is the decisive new ingredient of source 16. It has no counterpart in Part IV, where several weight-one types let a near pair separate into two walkers (Section 44).]

Lemma 75.2 (Finite phase motion). *For each normalized mass-two configuration q of diameter at most $2S$, its isolated orbit has a computable finite transient and a translated periodic tail. There exist $\mu \geq 0$, $p \geq 1$, $D \in \mathbb{Z}$, and explicitly computed finite configurations $q_0, \dots, q_{\mu+p-1}$ such that*

$$G^{\mu+r+np}(q) = \tau_{nD} q_{\mu+r} \quad (0 \leq r < p, \ n \geq 0).$$

Each configuration on this orbit has diameter at most $2S$.

Proof. Normalize every nonempty configuration by moving its leftmost occupied site to zero. Lemma 75.1 leaves finitely many normalized mass-two shapes: one for each weight-two symbol, and one pair u, u at each distance $1, \dots, 2S$. The successor shape and change in the leftmost position depend only on the current shape. Iterating this finite deterministic transition system reaches a repeated shape. The repeat supplies μ, p, D and the stated identity by translation equivariance. $\square$

[write] [Same argument as Proposition 43.1 and (43.2) of Part IV, applied to the finitely many shapes left by Lemma 75.1.]

If N_2 denotes the number of weight-two labels, there are exactly $N_2 + 2S$ possible normalized shapes. No speed bound on the movement of the leftmost occupied site is needed: its displacement on a shape transition is simply computed from the rule. A mass-two input of diameter greater than $2S$ is two independent unit symbols and is fixed by G .

76 The bounded encounter core

For mass exactly three, define the encounter core by the diameter condition

$$\text{Core} = \{c : M(c) = 3, \text{diam}(\text{supp } c) \leq 4S\}.$$

The core includes every connected mass-three configuration, and may include some disconnected ones. Its use requires no claim that a configuration entering it has already undergone a local interaction. Modulo translation, the core is finite. If N_3 is the number of weight-three labels, its number of normalized shapes is

$$N_3 + 8SN_2 + \binom{4S}{2}.$$

These are respectively one weight-three site, a weight-two site together with a unit site, and three unit sites. One step from the core has diameter at most $6S$, by finite propagation.

Every mass-three configuration outside the core has one of two forms:

1. Three mass-one components, each an isolated u . They are mutually separated by more than $2S$ and the entire configuration is fixed by G .
2. One mass-two component of diameter at most $2S$, and one isolated u .

A weight-three singleton is always in the core. Thus these cases exhaust all possibilities, including an initial weight-two singleton far from a unit.

Lemma 76.1 (Exact evolution before the core). *Suppose an outside-core configuration consists of a mass-two component $\tau_a q$ and a unit at b . Let*

$$z_k = \tau_a G^k(q)$$

be the isolated mass-two evolution. Consider the counterfactual union of its support with the fixed site b , and let h be the first $k \geq 0$ at which the span of that union is at most $4S$, if there is such a k . Before time h , and also at time h when finite, the actual orbit is exactly z_k together with the unit at b . The union at those times has disjoint occupied sites. If h does not exist, this describes the entire orbit.

Proof. By Lemma 75.1, $\text{diam } z_k \leq 2S$. If b were within $2S$ of any occupied site of z_k , the whole union would have span at most $4S$. Therefore, whenever the counterfactual span is greater than $4S$, the mass-two component and the unit are separated by more than $2S$. Their next step is independent and their outputs are disjoint. Induction proves equality up to and including the first core time. Although a later counterfactual superposition could have overlapping sites, no such overlap occurs in the part used here. $\square$

Lemma 76.2 (Effective first entry). *The first entry in Lemma 76.1, or the fact that no entry occurs, is computable without stepping through the initial gap.*

Proof. The finite transient from Lemma 75.2 is checked directly. On each periodic phase, write $k = \mu + r + np$ with $n \geq 0$; the leftmost site is $a + A_r + nD$, and the diameter is a constant $d_r \leq 2S$. Core membership is exactly the pair of inequalities

$$b - 4S \leq a + A_r + nD \leq b + 4S - d_r. \quad (76.1)$$

For a fixed phase this is an integer interval condition on n . If $D = 0$ it is always true or always false. If $D > 0$, compute the first nonnegative integer above the lower bound and check the upper bound; if $D < 0$, negate the inequalities. Take the earliest time among the finitely many valid phase and transient candidates. $\square$

[write] [Lemmas 76.1 and 76.2 are the exact acceleration to the next encounter of Part IV ((44.2) and the paragraph after it), with a bounded mass-two packet in place of a walker. The core and its shape count are new.]

77 Finite encounter dynamics and complete orbits

Starting from any normalized core shape, take one exact CA step. If the output is again in the core, this gives a transition with elapsed time one and a computed translation. If it is outside, apply the classification above and Lemma 76.2. Either there is no return, or the first return gives another core shape with a positive integer elapsed time and an integer translation. All these data are effective, and depend only on the starting normalized shape.

This produces a finite deterministic graph of core shapes. Every outgoing edge either reaches a core shape with a fixed time/translation increment, or enters a completely described nonreturning orbit. There are two terminal forms: a fixed configuration, or the independent combination of a stationary unit and the finite-phase mass-two walker.

Proposition 77.1 (Eventual orbit structure). *From each fixed finite configuration of mass at most three, the complete orbit under G is effectively eventually of one of the following forms:*

1. *The entire configuration is periodic up to a single translation: there are $T \geq 0$, $P \geq 1$, $d \in \mathbb{Z}$ with $G^{t+P}(c) = \tau_d G^t(c)$ for all $t \geq T$.*
2. *A stationary unit and a mass-two component evolve independently, with the latter periodic up to translation after a finite transient.*

The fixed configuration case is included in the first form with $P = 1, d = 0$.

Proof. An initially unbounded separation is resolved by Lemma 76.2, or gives an immediate nonreturning description. After a core entry, there are finitely many normalized core shapes. If a nonreturning edge is taken, its description gives the second form or a fixed configuration. Otherwise a normalized core shape repeats: at two times $t_0 < t_1$ the actual configurations satisfy $G^{t_1}(c) = \tau_d G^{t_0}(c)$. Determinism and translation equivariance imply

$$G^{t_1+n}(c) = \tau_d G^{t_0+n}(c) \quad (n \geq 0).$$

This identity covers every intermediate configuration, including every part of every subsequent excursion, not just the core-return times. $\square$

[write] [Same argument as Part IV's finite near-shape recurrence ((44.3) and the paragraphs after it), with the mass-three core in place of the near shapes.]

For a fixed starting configuration, these descriptions immediately yield a Presburger relation between time and all occupied output positions. A phase $t = T + r + nP$ gives constant labels and positions of the form $c_j + nd_j$. The drifts d_j are either all equal, or are the walker drift together with zero for the stationary unit. Any finite initial segment can be included by a finite disjunction. Positions, labels, and finite-pattern observations at intermediate times are retained by this construction.

This proves a per-input semilinearity statement. The next section supplies the additional step needed for one formula covering arbitrarily large initial gaps.

78 Uniform Presburger construction

For every normalized core shape c , the preceding finite computation gives an effective formula $K_c(a, t, y)$ describing the whole orbit from $\tau_a c$. There are finitely many such formulas. This notation suppresses the finitely many output-label cases; all displayed formulas may be expanded separately for each label tuple.

The mass-zero, mass-one, and mass-two cases were already handled by finite phase motion and separated stationary units. Initial core configurations are covered by the formulas K_c . Initial configurations with three mass-one components are fixed in the stationary frame.

It remains to treat the outside-core pair-plus-unit case, with bounded normalized pair shape q , pair translation a , and unit position b . There are only finitely many choices of q . Its isolated profile from Lemma 75.2 is uniformly Presburger in (a, k) : each periodic phase has time $k = \mu + r + np$ and positions obtained by adding $a + nD$ to fixed constants. Its finite transient contributes finitely many cases.

Let $H_q(a, b, k)$ express that the span of the counterfactual support together with b is at most $4S$. Formula (76.1) and its finite transient cases show that H_q is Presburger. Define its first-hit predicate by

$$E_q(a, b, k) \equiv k \geq 0 \wedge H_q(a, b, k) \wedge \neg \exists j (0 \leq j < k \wedge H_q(a, b, j)).$$

Presburger arithmetic is closed under these quantifiers and Boolean operations.

For output time t , use the union of two cases:

1. *No entry through time t .* Require $t \geq 0$ and $\neg \exists j (0 \leq j \leq t \wedge H_q(a, b, j))$. Describe the output as the isolated pair profile at time t together with the unit at b .
2. *Entry by time t .* Existentially choose $0 \leq k \leq t$ satisfying $E_q(a, b, k)$. For each normalized core shape c , express that the entry configuration is $\tau_e c$, for some integer e . Then require $K_c(e, t - k, y)$.

The entry-shape condition is Presburger: its labels and relative positions belong to a finite list, while the pair-profile positions, b , and the translation e are related by linear equalities. The minimum-position choice for e , disjoint support, and ordering can all be expressed by finite conjunctions or disjunctions. The post-entry period and drift depend only on the finitely chosen core shape, not on the unbounded initial gap or first-entry time; this introduces no product of two quantified variables. Lemma 76.1 guarantees that these are valid configurations at the first entry.

Every initial label/position case reduces to one of the listed cases through finitely many comparisons and bounded-offset equalities. The result is a Presburger formula for G . Substituting $y_j - \delta t$ gives the formula for F , completing Theorem 74.1.

[write] [Same architecture as Part IV’s linear guard, first entry and templates ((45.2) and (45.3): “There is no product $k_0 n$ ”). New here is the composition of a bounded mass-two packet with a distant stationary unit.]

79 Observation and computational consequences

Exact reachability is obtained by existentially quantifying t . Reachability up to translation also quantifies one spatial shift. To test a finite pattern p at position z , quantify the at-most-three output positions and labels, and require the specified state at every pattern site. In particular, a zero in the pattern means that no occupied output position is present at that site; zeros are not ignored. The finite union over possible output label tuples is Presburger. Quantify t and, for occurrence anywhere, z . A finite collection of relative-time observations uses a conjunction of the same timed relation at $t + d_1, \dots, t + d_k$, with independent output position tuples. All these constructions are effective. [write] [Same tests as Part IV’s at every intermediate time (Section 44), written in Presburger arithmetic.]

Corollary 79.1 (Numerical particle lower bound). *Let F be a fixed one-dimensional finite-alphabet number-conserving CA with nonnegative integer states and weight $w(q) = q$. From configurations whose numerical state sum is at most three, finite-pattern occurrence and exact finite-configuration reachability are decidable. Consequently such a mass-three family cannot encode arbitrary Turing-machine halting through a computable finite-configuration encoder and a designated finite halt-pattern occurrence event.*

Proof. An integer-state alphabet has at most one state of numerical value one, and all its nonzero states have positive integer weight. The theorem applies. An effective reduction of undecidable halting to the stated decidable observation problem would be a contradiction. $\square$

The conclusion concerns ordinary effective computation with finite input encodings and finite halt observations. It is not a claim about intrinsically universal simulation, noncomputable output interpretation, signed conserved quantities, infinite alphabets, spatially nonuniform rules, time-dependent rules, or active zero-weight backgrounds. Those are different models.

Multiple distinct weight-one symbols invalidate the stationary-unit argument: isolated particles can carry finite internal states or have different motions. A three-mass construction in such a typed weighted alphabet therefore does not contradict Corollary 79.1. [write] [Part V (source 14) is such a construction.] Conversely, simply renumbering those unit symbols as distinct positive integer states cannot preserve their unit weights. The precise distinction is the number of available labels of mass one, not the number of geometrically occupied sites.

80 Horizon free canonical quartic certificates

The uniform relation has a direct Diophantine consequence. Encode a signed coordinate x by natural inputs (x^+, x^-) with

$$x = x^+ - x^-, \quad x^+ = 0 \text{ or } x^- = 0.$$

This is a unique representation of each integer. Include its domain condition, the ordering constraints, and $t \geq 0$ in the Presburger predicate. Apply effective Presburger quantifier elimination, for example Cooper’s procedure [25], to obtain a quantifier-free predicate on the chosen natural inputs. For untimed reachability, eliminate time as well. Leaving a reaching time as an auxiliary witness would generally destroy uniqueness, because a target may be visited at several or infinitely many times.

Use the canonical quantifier-free Presburger-to-quartic compiler in Section 9 of the companion report [Part IV of this report (source 13), Section 46]. Its precise interface is as follows. Normalize atoms to affine inequalities $L \geq 0$ and fixed-modulus congruences $L \equiv 0 \pmod{d}$, and choose an AND/OR/NOT circuit. Let I, C, B be its actual counts of inequality atoms, congruence atoms, and Boolean gates. The compiler gives an integer polynomial, a sum of squares of quadratic residuals, with degree at most four and

$$K = 2I + 6C + B \quad \text{natural witnesses,} \quad E = 2I + 5C + B + 1 \quad \text{residuals.} \quad (80.1)$$

(The companion report calls the gate count G ; B avoids confusion with the stationary-frame CA in this article.) Its natural witness fiber is a singleton when the predicate is true and empty when false. The atom encodings assign canonical truth bits and slack or Euclidean-division data, and the acyclic Boolean circuit has uniquely determined outputs; the final residual requires the output bit to equal one. The established compiler is reused here rather than replaced by a new implementation. [write] [The signed encoding and the elimination of time are those of Part IV’s Section 45, last subsection, and (80.1) is (46.2).]

Corollary 80.1 (Canonical quartics at mass three). *For every fixed CA and fixed input/output label case in Theorem 74.1, timed and untimed exact reachability admit effectively constructible integer polynomials of degree at most four with zero or one natural auxiliary witness tuple. Their witness arity is the formula-dependent constant K in (80.1), after quantifier elimination and a chosen Boolean circuit. No supplied time horizon is required. The same conclusion applies to each fixed finite-pattern observation predicate from Corollary 74.2.*

Proof. Apply Theorem 74.1, the stated natural input encoding, quantifier elimination, and the canonical compiler in that order. The first three steps preserve the predicate; the last step has the asserted degree, counts, and natural-fiber property. $\square$

[write] [Corollary 80.1 is Part IV’s Corollary 46.1 one mass higher, with up to twelve rather than eight external natural inputs.]

For exact reachability there are at most twelve external natural coordinate inputs, and thirteen when time is retained. Thus total arity is at most $12 + K$ untimed or $13 + K$ timed for fixed labels. The formula-dependent quantity K is not a universal numerical arity bound. Quantifier-elimination cost, formula size, circuit size, integer coefficients and their bit lengths remain paid parts of any concrete instantiation; no efficient preprocessing or optimal arity is claimed. The executable package validates the orbit accelerator, not a general-purpose rule-to-formula or quantifier-elimination implementation.

Uniqueness comes from post-elimination arithmetic, not merely from deterministic CA evolution. No corresponding exactness of the nonnegative-real or rational witness fiber is claimed. This corollary concerns the decidable single-unit-label class and gives no finite-fold conclusion for arbitrary recursively enumerable sets or for a universal typed three-mass construction.

[write] **Added 2 October 2026 (batch 80).** Two sharper compilers apply. The programme’s five-witness congruence atom [33], already recorded for Part IV (Section 47), applies verbatim after

elimination and gives $K = 2I + 5C + B$ natural witnesses at the same E , keeping the single-fold property and degree four; source 16’s provenance record says that it retains the verified six-witness bound, and the programme’s review of this source [46] identifies the substitution as a formula-dependent saving, with no total-gate claim. Theorem `cdc:of:thm:singlefolds1` of Part XI of *canonical-diophantine-certificates* gives the same semilinear relation an orthant-nonnegative single-fold representation of degree two, as the note after Part IV’s Corollary 46.1 says for mass two. No novelty is claimed for Corollary 80.1.

81 Primary sources and what remains unresolved

There is a direct primary-source precursor. Gil-Tak Kong’s June 2021 thesis [41], Section 2.4.1, Theorem 2, printed page 14 (PDF page 15), proves non-strong-Turing-universality with three particles. Its definitions are explicitly one-dimensional and binary: Definition 1, printed page 8 (PDF page 9), uses alphabet $\{0, 1\}$. The proof distinguishes eventual periodicity up to translation of the whole configuration from eventual decomposition into independent periodic components. Thus binary three-particle nonuniversality and this orbit-decomposition approach have an explicit 2021 precursor.

The present article states and proves a weighted extension with at most one unit label, effective uniform timed Presburger formulas in all sparse positions, exact and finite-pattern observation consequences, and the canonical quartic composition. These are the precise claims to compare with earlier work; we do not claim that the brief sources establish or exclude these stronger statements, or that their absence from a short source would establish novelty.

Imai’s KAKEN final report dated 25 June 2021 [42], PDF page 4, Section 4, item (2), also reports that motion representation was used to establish decidability when the particle number is at most three. The Japanese sentence leaves the decision problem implicit; it does not identify exact reachability, pattern occurrence, an input encoding, or a complexity bound. This establishes a numerical-NCCA decidability precursor, while leaving its relation to the fully specified weighted timed claim here undetermined. PDF page 3 distinguishes the need to encode NCCA internal state geometrically from the capabilities of pebble automata and discusses the earlier five-particle construction.

The publisher abstract of Alhazov and Imai [24] describes finite-support, nonnegative-integer-state NCCAs. It reports a radius-one universal construction via two-register machines and a universal construction with five particles when a sufficiently large finite radius is allowed. The accessible abstract does not claim minimality. The full paper was not accessible in this check, so its detailed discussion of smaller particle counts could not be verified.

A Kyoto University RIMS report [43], printed page 93, records Kong and Imai’s presentation on particle complexity on 2 February 2021; the KAKEN project record [44] places that title under academic year 2020. The thesis above supplies a directly inspectable mathematical source.

Kong’s thesis conclusion, printed page 15 (PDF page 16), describes the four-particle case as open in 2021. That is a historical statement, not evidence that the question remains open today. The current article proves insufficiency of three for the stated observation task in its weighted single-unit-label class, while the 2016 abstract supplies a five-particle numerical upper bound. It does not settle the current four-versus-five boundary or the novelty of its explicit weighted, uniform Presburger, or Diophantine strengthenings.

[write] **Added 2 October 2026 (batch 80).** The four-versus-five boundary is settled within

this weighted single-unit class by source 17 (Part VI): exact reachability and pattern occurrence are decidable at mass four (Theorem 83.1), so the numerical lower bound is five (Corollary 83.2); a five-particle upper bound rests on the Alhazov–Imai abstract only. The programme’s review [46] read the cited pages of Kong’s thesis (the binary Definition 1, Theorem 2, and the four-particle case open in 2021) and retrieved Imai’s KAKEN report, whose Japanese text it could not extract and does not certify; it obtained no full text of the Alhazov–Imai paper.

Two further directions remain outside this proof: classifying the effect of two or more unit labels, including the least unit-label count permitting a universal three-mass construction; and deriving quantitative bounds for encounter acceleration, Presburger elimination, and the resulting quartic ledger. [write] [Both are open; they are also source 17’s Questions 5 and 3.]

82 Executable checks and limits of validation

The accompanying Python script implements the finite mass-two phase profile, exact first-core acceleration for positive, zero, and negative drift, and whole-orbit acceleration using the 4S core. It compares accelerated configurations against direct simulation, including all intermediate sampled times. The tests use:

- Sixteen binary conservative radius-two rules obtained from disjoint parallel swaps of selected 10 pairs, in both directions. The selection is false for an isolated unit, so its drift is zero.
- All five conservative elementary binary rules, 170, 184, 204, 226, and 240, each shifted into its unique-unit stationary frame. Their conservation is certified exactly by a potential on the four-vertex de Bruijn graph; all 256 elementary rules are tested against that finite identity.
- An ordinary four-state NCCA with numerical states 0, 1, 2, 3, using locally guarded translation, a 2 + 1 merge into 3, and a reverse split. Its label values equal its weights.
- A locally guarded weighted rewrite CA with a unique unit label, several weight-two labels, and two weight-three labels. Its cases include positive and negative walkers, zero-drift phase motion, merging, splitting inside a bounded cluster, recurrent encounters, and terminal separation.

Each sample rule has a direct conservation argument: binary moves swap disjoint occupied/empty pairs; weighted rewrites have disjoint guarded supports and preserve the mass on each rewritten support. The code also tests dense, multi-center configurations as a consistency check.

The recorded run passed 7,127 orbit cases and 1,140,320 direct-versus-accelerated time comparisons across 23 rules. It checked 102 mass-two profiles, 5,103 signed linear interval cases, and 2,760 dense conservation cases. A binary pair with period two and drift one was tested against an initial unit at $10^{30} + 17$; the accelerator handled the enormous separation using 13 core visits, without iterating through the gap.

These counts describe actual CA orbit cases and direct-versus-accelerated configuration comparisons, not counts of symbolic proof templates. These finite tests exercise the implementation and the proposed normal forms. They are neither an exhaustive enumeration of all conservative CA rules nor a machine-checked proof of the theorem. A separate executable audit passed 1,186 boundary orbits with 118,600 direct time comparisons, 4,109 independent first-core checks, and 216,160 finite-window conservation cases. It also tested reflected negative-drift gaps, zero drift, sorting and translations, with closed-form checks through time $10^{80} + 123$; its full ledger is included in the package. The mathematical argument, including the no-split lemma, uniform first-entry composition, and all-intermediate-time cycle identity, was reviewed independently as well.

[write] **The release in this report.** These are the delivered names. The script and the independent audit are shipped as `code/16-single-unit-test_single_unit_acceleration.py` and `code/16-single-unit-audit_accelerator.py`, with `code/16-single-unit-run-tests.sh` and `code/16-single-unit-build.sh`; their receipts as `data/16-single-unit-test-results.json` and `data/16-single-unit-audit-results.json`; and `REVISION.md`, `SOURCE-PROVENANCE.md` and `independent/AUDIT.md` as `16-single-unit-*` at the report root. The manuscript, PDF, README and checksum ledger are not shipped, and nothing of manuscript 06 is (its programs, receipts and audit are the same files). `run-tests.sh` copies the two programs to a temporary directory and compares the receipts; it needs the delivered layout (README). The programme’s review reran both programs and reproduced both receipts [46].

Source 17 (batch-80 manuscript 03): Four mass units and exact reachability

Decidability proof and a binary timing certificate, 3 October 2026 (its PDF metadata name the author as “Mathematical research report”). [write] Its “companion” report is source 16, printed above. Its abstract follows.

For every fixed one-dimensional finite-alphabet conservative cellular automaton with a unique zero-weight vacuum, positive integer non-vacuum weights, and at most one weight-one symbol, exact finite-configuration reachability and finite-pattern occurrence are decidable at total mass at most four. In the stationary-unit frame, recurrent separated interactions reduce to a finite-control additive gap counter. A translating three-mass gadget either escapes or brings all four masses into a bounded encounter; it cannot retain a second independent gap. The remaining expanding shuttles have spatial phases affine in their excursion number and quadratic accumulated time. This also handles anchored observations in the original moving frame.

Timed semilinearity does fail at mass four: an explicit binary radius-six NCCA has anchored return times $t_k = k^2 + (2d - 11)k$, $d \geq 7$. Its full local table has an exact de Bruijn conservation certificate. A single squared quadratic gives a horizon-free quartic timing certificate with exactly one natural witness at a hit. Primary 2021 lower-particle precursors are credited; no priority claim or present-day literature-status conclusion is inferred from an unsuccessful search.

83 Model and results

Let A be finite, with a distinguished vacuum 0, and let $w : A \rightarrow \mathbb{N}$ satisfy $w(0) = 0$ and $w(a) > 0$ for $a \neq 0$. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, time-independent, translation-equivariant cellular automaton of finite radius R . Assume that $0^{\mathbb{Z}}$ is fixed and that, for every finite-support configuration c ,

$$M(F(c)) = M(c), \quad M(c) = \sum_{i \in \mathbb{Z}} w(c_i).$$

There is at most one symbol of weight one. Labels of higher weights may be repeated. Write $\tau_a c$ for translation to the right by a sites.

Theorem 83.1 (Four-mass decision theorem). *For every fixed F satisfying these assumptions, the following problems are decidable for finite initial configurations of mass at most four:*

1. *Exact reachability of a specified finite configuration, including reachability up to translation.*
2. *Occurrence of a specified finite pattern at a specified position, or anywhere, at some nonnegative time.*

The procedures are effective from the finite rule, radius and weights, under the stated hypotheses. No bound on their computational efficiency is asserted.

A pattern may contain zeros, and these are enforced. The theorem concerns finite inputs and finite observations. It does not cover infinite initial backgrounds, signed weights, an infinite alphabet, an active zero-weight background, multiple unit labels, or time-dependent or spatially nonuniform rules.

Corollary 83.2 (Ordinary numerical particles). *A fixed one-dimensional finite-alphabet NCCA whose states are nonnegative integers and whose weights equal those integers cannot encode arbitrary Turing-machine halting by a computable finite-configuration encoder of total numerical mass at most four and a designated finite halt-pattern observation. The same conclusion holds for exact-target halting observations.*

Proof. There is at most one numerical state equal to one. Such an encoding would reduce an undecidable problem to a decision problem in Theorem 83.1. $\square$

The 2016 publisher abstract of Alhazov and Imai reports a universal numerical NCCA with five particles [24]; the full paper was not available for this check. The present lower bound matches that reported numerical upper bound for the usual effective finite-input halting-observation interpretation. [write] [It would match, if the 2016 construction uses an effective finite input encoding and a finite halt-pattern observation; only its abstract was checked, and source 17's delivery README says "compatible with".] It does not establish a five-particle upper bound for the *binary* subclass.

84 Stationary units and small packets

Assume first that the unit symbol u exists. Its isolated successor must be exactly one u , at a fixed displacement $\delta \in [-R, R]$. Put

$$G = \tau_{-\delta}F, \quad S = \max(1, R + |\delta|).$$

Then G has radius at most S , fixes each isolated u , and $F^t = \tau_{\delta t}G^t$. Components below join occupied sites at distance at most $2S$. If two components are separated by more than $2S$, their next steps evolve independently and have disjoint output support.

[write] [This section restates the ingredients of source 16 that source 17 needs (Section 75 and the next two); the mass-two case for any number of weight-one symbols is Part IV's Theorem 44.1. The proofs of the two lemmas are pointers.]

Lemma 84.1 (Bounded mass-two packets). *A mass-two configuration of diameter at most $2S$ stays within that diameter under G . Its orbit is effectively eventually periodic up to translation.*

[write] *Pointer.* This is source 16's Lemmas 75.1 and 75.2, proved there by the same argument (the cases $d > S$ and $d \leq S$, with all new support in $[d - S, S]$, then finitely many normalized shapes); not reprinted. $\square$

An isolated packet profile can therefore be written, after a finite transient, as

$$G^{\mu+r+kp}(q) = \tau_{kD}q_r, \quad 0 \leq r < p, \quad k \geq 0. \quad (84.1)$$

Its labels, phase shapes, period, drift and transient are explicit finite data. Two units farther than $2S$ apart are fixed.

Lemma 84.2 (Effective three-mass profiles). *Every fixed mass-three input has an effectively computable finite prefix and one of these tails:*

1. *the whole configuration is periodic up to a single translation;*
2. *a stationary unit and an independent mass-two packet with finite-phase motion.*

Every intermediate configuration is included in the resulting description.

[write] *Pointer*. This is source 16's Proposition 77.1, with Lemmas 76.1 and 76.2 for the first entry into the core of diameter $4S$ and the finite graph of core returns; not reprinted. $\square$

For total mass at most three, these profiles already give a fixed-input timed Presburger description, and the frame translation δt preserves it. Thus all conclusions of Theorem 83.1 at mass at most three are settled here; the remaining component analysis assumes mass exactly four.

This is the small-packet ingredient of the companion mass-three report [source 16, Part VI]. The present argument does not require its stronger uniform timed-Presburger conclusion.

85 A finite library of local encounters

Let $\mathcal{E}$ be the finite set of normalized connected mass-three configurations. Every seed has diameter at most $4S$. When a bounded mass-two packet first comes within $2S$ of a unit, their union is such a seed. At the first-contact time their supports are still disjoint: they evolved from components separated by more than $2S$ at the preceding step.

Tag each entry seed by the target unit's coordinate within it. This finite bookkeeping tag matters when the incoming packet itself consists of two unit symbols. It adds no physical label; it records which incoming coordinate was the stationary target.

For each $e \in \mathcal{E}$, compute Lemma 84.2's orbit profile and choose a time T_e giving one of two normalized outcomes:

Compact outcome. The complete mass-three configuration has a translated-periodic tail.

Emission outcome. A stationary unit at coordinate a_e remains, and a mass-two packet has nonzero periodic drift and recedes from that unit forever. Its launch phase and coordinates are fixed relative to the seed.

If an independent packet has zero drift, its support is bounded and periodic, so reclassify the whole tail as compact. With nonzero drift, prolong the finite prefix by whole periods until every future packet phase is farther than $2S$ from the stationary unit on the drift side. Earlier detachments and returns are part of the finite prefix. Infinitely recurrent detachments within a translated periodic three-mass orbit belong to its compact phase library.

Choose an integer $B \geq 4S$ bounding the absolute value of every occupied coordinate, relative to its normalized seed, in every finite prefix through T_e and every initial-period phase representative of either outcome. Include the residual marker coordinate a_e . Then compact phase diameters are at most $2B$, and emitted head-to-marker offsets have magnitude at most $2B$. Set

$$N = 4B + 12S, \quad W = N + 2B. \quad (85.1)$$

These constants are noncircular: B is computed from the original span- $4S$ seed family, not from all mass-three configurations of diameter W .

Lemma 85.1 (Safe local resolution). *Suppose a configuration consists of $\tau_a e$, $e \in \mathcal{E}$, and a fourth unit at b , with $|b - a| > B + 2S$. The complete isolated seed prefix through T_e evolves independently of that unit. Subsequently:*

1. *a compact outcome either stays independent forever, or first contacts the fourth unit in a complete configuration of diameter less than W ;*
2. *an emission changes the nearby residual marker by a fixed seed-dependent offset and leaves the fourth unit fixed.*

All alternatives and the first-contact times are effectively computable.

Proof. During the prefix the seed support lies in $[a - B, a + B]$, at distance greater than $2S$ from b . Exact independence applies step by step, including its terminal time. A compact phase has diameter at most $2B$. Its zero drift or drift away from b gives independence forever. If its drift points toward b , compute first proximity at most $2S$ from its affine phase profiles. Such a contact exists: finite propagation prevents a packet from crossing the stationary unit without proximity at most $2S$. At contact the total diameter is at most $2B + 2S < W$. The emission statement is the computed isolated profile. $\square$

An outside- W seed-plus-unit configuration automatically satisfies the safety condition: the seed diameter is at most $4S$, so $|b - a| > W - 4S > B + 2S$. Thus a translating mass-three gadget is not collapsed into a constant-time local reaction. Its arbitrarily long trip is either terminal or ends in a bounded full encounter.

86 The global scattering section

Call every complete mass-four configuration of diameter at most W an *ordinary state*; there are finitely many up to translation. A *live emission section* consists of two stationary units at $L < R$ and one bounded mass-two packet at a prescribed launch offset from one endpoint, with nonzero drift toward the other endpoint. Write $D = R - L > N$. The finite mode records the source side, emitted packet type and phase, and tagged seed information. The geometric ordinary and live sets can overlap. We use this deterministic representation convention: at a designated emission checkpoint, select the live tag whenever $D > N$, even if the span is at most W ; at other dispatch states apply the ordinary span test first. These tags belong to the analysis and do not add memory to the CA. A live step retains its tag through the next designated emission or an explicit ordinary reset.

Lemma 86.1 (Additive gap transitions). *Every continuing live-to-live transition, for sufficiently large gap as already ensured by (85.1), has the form*

$$q' = f(q, D \bmod M), \quad D' = D + c, \quad L' = L + e, \quad h = \alpha D + \beta, \quad (86.1)$$

where the constants are selected from a finite effective list, $c, e \in \mathbb{Z}$, $\alpha > 0$ is rational, and h is integer on the permitted residue class. M is one fixed positive integer. All intermediate configurations have effective finite-phase descriptions.

Proof. The emitted packet is permanently independent of the source unit. Follow it until first proximity at most $2S$ to the target unit at b . The resulting connected mass-three seed has left edge

a' with $|a' - b| \leq 4S$. The old source marker p therefore satisfies

$$|p - a'| \geq D - 4S > N - 4S > B + 2S.$$

Lemma 85.1 applies to its complete local prefix, however long that fixed prefix is. For an emission from the new seed, the new stationary marker is $p' = a' + a_{e'}$, so

$$|p' - b| \leq 4S + B.$$

Only the contacted marker moves. Since $D > N > B + 4S$, marker order cannot reverse. The new gap and left marker thus change by bounded constants. Unit identities can switch within the encounter; the new coordinates, not identities, are what the section records.

If the outgoing packet points outward, the tail is terminal independent motion. If it points inward with new gap at most N , its complete launch configuration has diameter at most $N + 2B = W$ and is ordinary. A compact outcome is accelerated separately by Lemma 85.1; it either escapes or resets to an ordinary state. Carrying a marker inside a translating three-mass packet cannot preserve an extra unbounded gap through that reset.

For the arithmetic, reflect coordinates if necessary so the packet drift is $d > 0$, with period p . In phase r , let its left edge be $p_0 + h_r + kd$ and its width be $d_r \leq 2S$, where p_0 is the source marker. Contact with target $p_0 + D$ is exactly

$$D - 2S - d_r \leq h_r + kd \leq D + 2S. \quad (86.2)$$

The hull criterion is exact: because packet width is at most $2S$, the union of the radius- $2S$ neighborhoods of its occupied sites is the single expanded hull interval. There is no hole that falsely counts as a contact.

On each residue $\rho \bmod d$, every candidate period count is $(D - \rho)/d$ plus an integer constant. The choice of earliest valid phase therefore depends only on ρ ; all candidate times have leading coefficient p/d . The bound $D > N$ ensures nonnegative candidate counts in the periodic regime. Finite propagation ensures that one phase is valid. The entry seed and its tag are fixed relative to the target. Add the fixed local-prefix time to obtain (86.1). Take M to be a common multiple of the nonzero emission drifts. Every precontact phase and every local-prefix configuration is retained, not merely the transition endpoint. $\square$

Lemma 86.2 (Complete effective dispatch). *From an arbitrary mass-four configuration, one can effectively obtain a terminal independent profile, an ordinary state, or a live emission section, with an exact description of the intervening orbit. From an ordinary state, the next such section can be chosen at positive elapsed time.*

Proof. The possible $2S$ -component mass partitions are

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

A connected component of mass m has diameter at most $2S(m - 1)$. The first partition is ordinary since $6S < W$; four unit components are fixed. For $3 + 1$, an outside- W state has a connected seed and a safely distant unit, so Lemma 85.1 applies.

For $2 + 2$, both packets have finite-phase independent motion until first proximity at most $2S$. Use a common phase period; each signed relative gap is affine in the period count. Finite transient checks and integer interval calculations find first contact or prove none occurs. First contact has total span at most $6S < W$.

For $2 + 1 + 1$, compute the packet's first proximity to either stationary unit by the same finite-phase calculation. If absent, the entire tail is independent. At contact, a connected three-mass seed is formed. A simultaneous contact connects all four masses and has span at most $6S$. Otherwise either the complete configuration is ordinary, or its remaining unit is safely distant and Lemma 85.1 applies. Before every first contact, independence is exact; it also holds at the first-contact time itself.

From an ordinary state take one exact step. If still ordinary, retain that step. Otherwise its span is at most $W + 2S$, and the exhaustive dispatch above applies. Its result is terminal, ordinary, or live. This supplies a positive-time outgoing section edge for each of the finitely many normalized ordinary states. Live edges include a nonempty flight. Zero-time section loops do not arise. $\square$

87 Termination and the expanding shuttle form

Enlarge the finite live mode by $D \bmod M$. A repeated finite control then has gap difference divisible by M , and consequently the same residue itinerary on each repeated cycle. Translation L is recorded as output information; it is not used as another control counter.

Proposition 87.1 (Effective orbit alternatives). *For every fixed mass-four input, its G -orbit is effectively eventually one of:*

1. *independent finite-phase objects;*
2. *a whole configuration periodic up to translation;*
3. *an expanding shuttle cycle, whose corresponding-section gap and left marker are affine in a cycle number n , whose complete spatial phases are Presburger in n and one within-flight period count, and whose accumulated section time is quadratic in n with positive leading coefficient.*

Proof. Follow the effective section machine. If an ordinary normalized state repeats, the complete configurations at those two times differ by a translation. Determinism gives alternative 2 at every subsequent intermediate time.

During an uninterrupted live excursion, store the finite controls. A control repeats after finitely many transitions. Let Δ be the change in gap between those repeats. If $\Delta = 0$, the complete launch configuration repeats up to translation, again giving alternative 2. If $\Delta > 0$, all intermediate large-gap guards remain valid forever and the same residue itinerary repeats. This gives alternative 3.

If $\Delta < 0$, compute the maximal number of cycle repetitions for which *every intermediate gap* remains greater than N . Fast-forward those repetitions, summing their constant anchor updates and affine durations, then execute the finite remaining cycle prefix until the ordinary region is reached. This is exact finite arithmetic. There are only finitely many ordinary normalized states, so one cannot visit them indefinitely without repetition. The section analysis therefore terminates with one of the stated alternatives.

For the expanding case, write the corresponding-section coordinates as

$$D_n = D_0 + n\Delta, \quad L_n = L_0 + nE, \quad \Delta > 0. \quad (87.1)$$

There are finitely many edges per cycle. A bounded scattering-prefix phase has positions affine in n . A flight phase has marker positions affine in n and packet positions affine in (n, j) , where $j \geq 0$ is

its packet-period count. Its allowable range is given by linear inequalities. The finite union of these phase formulas describes every visited configuration and is Presburger.

Each live edge duration is affine in its incoming gap, with positive leading coefficient. Consequently the complete cycle duration is affine in n and strictly increases, and its start time has the form

$$T_n = T_0 + A \frac{n(n-1)}{2} + Cn, \quad A > 0. \quad (87.2)$$

Coefficients are computable rationals yielding integer times. There is a computable C_1 for which every configuration throughout cycle n has support inside

$$[L_0 + nE - C_1, \quad L_0 + nE + D_0 + n\Delta + C_1]. \quad (87.3)$$

Local prefixes have uniformly bounded offsets; each free flight stays between its two markers up to bounded phase deviations. Thus (87.3) includes every phase, including all short local transients, rather than just sections. $\square$

Corollary 87.2 (Untimed spatial description in the stationary frame). *For each fixed input of mass at most four, the set of configurations visited by G has an effectively constructible Presburger description in their occupied coordinates, separately for each possible output-label tuple.*

Proof. Use Proposition 87.1. Independent finite-phase and translated periodic tails have affine descriptions. Expanding spatial phases were described in its proof. All earlier parts are finite, or finitely many explicitly given finite-phase segments, so can be included effectively. An enormous finite prefix may be enumerated; no efficiency claim is involved. $\square$

88 Anchored observations in the original frame

Translation-invariant questions, including exact reachability up to translation and occurrence anywhere, have the same answers for F and G . Existentially quantify the spatial shift in the preceding corollary.

For anchored questions, if the eventual orbit is independent finite-phase or translated periodic, its fixed-input timed relation is Presburger. Adding δt to all coordinates preserves that property. If the orbit is an expanding shuttle and $\delta = 0$, its untimed spatial description suffices.

Suppose now $\delta \neq 0$. For $\delta > 0$, throughout cycle n every occupied coordinate in the F -orbit is at least

$$\delta T_n + L_0 + nE - C_1,$$

which tends to $+\infty$ by (87.2). For $\delta < 0$, every occupied coordinate is at most

$$\delta T_n + L_0 + nE + D_0 + n\Delta + C_1,$$

which tends to $-\infty$. These estimates are uniform over the whole cycle because $t \geq T_n$ and (87.3) holds throughout.

Given a finite target window, compute an integer cycle index n_* beyond which the appropriate quadratic bound lies permanently outside it. Simulate or accelerate the finite prefix up to section T_{n_*} . A pattern with a nonzero symbol can occur only in this prefix; an all-zero pattern occurs afterward. A nonempty exact target has the same cutoff. The vacuum case is immediate from conservation. This proves anchored decidability without a general quadratic Diophantine solver and without claiming timed Presburger definability.

For a pattern, every zero site is enforced by absence of an occupied coordinate there. Since mass is at most four, only finitely many nonzero label tuples are possible. Exact configuration equality and these finite pattern tests are thus effective in all the described cases.

If there is no unit symbol, mass at most four occupies at most two sites. An isolated site of weight two or three is a finite-state walker; a weight-four local configuration may split into two weight-two walkers. A bounded encounter core has finitely many normalized states. Outside it there are two independent finite-phase walkers, whose first encounter is computed by common-period linear inequalities. The same finite-core return argument gives an eventually translated-periodic or independent-two-walker orbit. This handles the missing case and completes Theorem 83.1.

89 An explicit binary expanding shuttle

The loss of timed semilinearity already occurs for an ordinary binary NCCA. Join consecutive occupied sites whose distance is at most two. On every such component, apply these replacements up to translation:

$$\begin{array}{llll}
 \{0, 1\} & \mapsto & \{1, 2\} & \text{right-moving pair,} \\
 \{0, 2\} & \mapsto & \{-1, 1\} & \text{left-moving pair,} \\
 \{0, 1, 3\} & \mapsto & \{-1, 1, 4\} & \text{right bounce,} \\
 \{0, 2, 4\} & \mapsto & \{0, 3, 4\} & \text{left bounce.}
 \end{array} \tag{89.1}$$

Leave every other component unchanged.

Proposition 89.1 (A genuine conservative local rule). *Rule (89.1) defines a binary conservative CA of radius at most six.*

Proof. Each replacement preserves the number of occupied sites, and its output hull extends the input hull by at most one site. Distinct input components have separation at least three, so these expanded output hulls still have disjoint lattice sites. This proves global conservation for every finite configuration.

A recognized component is an exact pattern of span at most four, with no occupied site within two beyond either endpoint. For any output site, every recognized pattern that can change it, together with both endpoint guards, is contained in its radius-six neighborhood. All other sites retain their old value. Thus the component definition is a local rule, including on arbitrarily large unrecognized components. The vacuum and isolated units are fixed. $\square$

The rule is not injective: the distinct configurations $\{0, 1, 4, 6\}$ and $\{1, 2, 3, 5\}$ both map to $\{1, 2, 3, 5\}$. No reversibility is asserted.

The package exports all $2^{13} = 8192$ local outputs. Independently, a potential P on the $2^{12} = 4096$ de Bruijn vertices satisfies every identity

$$f(w_0, \dots, w_{12}) - w_6 = P(w_1, \dots, w_{12}) - P(w_0, \dots, w_{11}). \tag{89.2}$$

The complete table and potential are included as a compact JSON certificate. Summation over the lattice telescopes, yielding another exact conservation proof, not merely a dense-window test.

Proposition 89.2 (Exact quadratic hit times). *From occupied sites $\{0, 3, 4, d\}$, $d \geq 7$, the anchored pattern 10011 at sites $0, \dots, 4$ occurs exactly at*

$$t_k = k^2 + (2d - 11)k, \quad k \in \mathbb{N}. \tag{89.3}$$

Its occurrence-time set is not eventually periodic and is not Presburger.

Proof. The adjacent pair travels from $\{3, 4\}$ to $\{d - 3, d - 2\}$ in $d - 6$ steps. The component $\{d - 3, d - 2, d\}$ then becomes $\{d - 4, d - 2, d + 1\}$ in one step. The left-moving pair returns to $\{2, 4\}$ in $d - 6$ steps, after which the component $\{0, 2, 4\}$ becomes $\{0, 3, 4\}$. One cycle takes $2d - 10$ steps and increments d by one. Summing cycle durations proves (89.3). Between returns the adjacent right-moving pair is strictly farther right, or the left-moving pair has distance two and cannot occupy 3, 4 together. There are no extra pattern hits.

Successive hit gaps are $2k + 2d - 10$, which are unbounded. Every infinite ultimately periodic subset of $\mathbb{N}$ has bounded gaps after a finite prefix. The claimed failure follows; the equivalence of Presburger and semilinear sets is classical [7]. $\square$

[write] [The hypothesis $d \geq 7$ is safe but not sharp: a direct simulation of rule (89.1) at this write, like the placement check, gives exactly the times (89.3) also for $d = 6$, through $t \leq 2500$, while the formula fails for $d = 5$. Section 90 writes $d = 7 + x$ with natural x .]

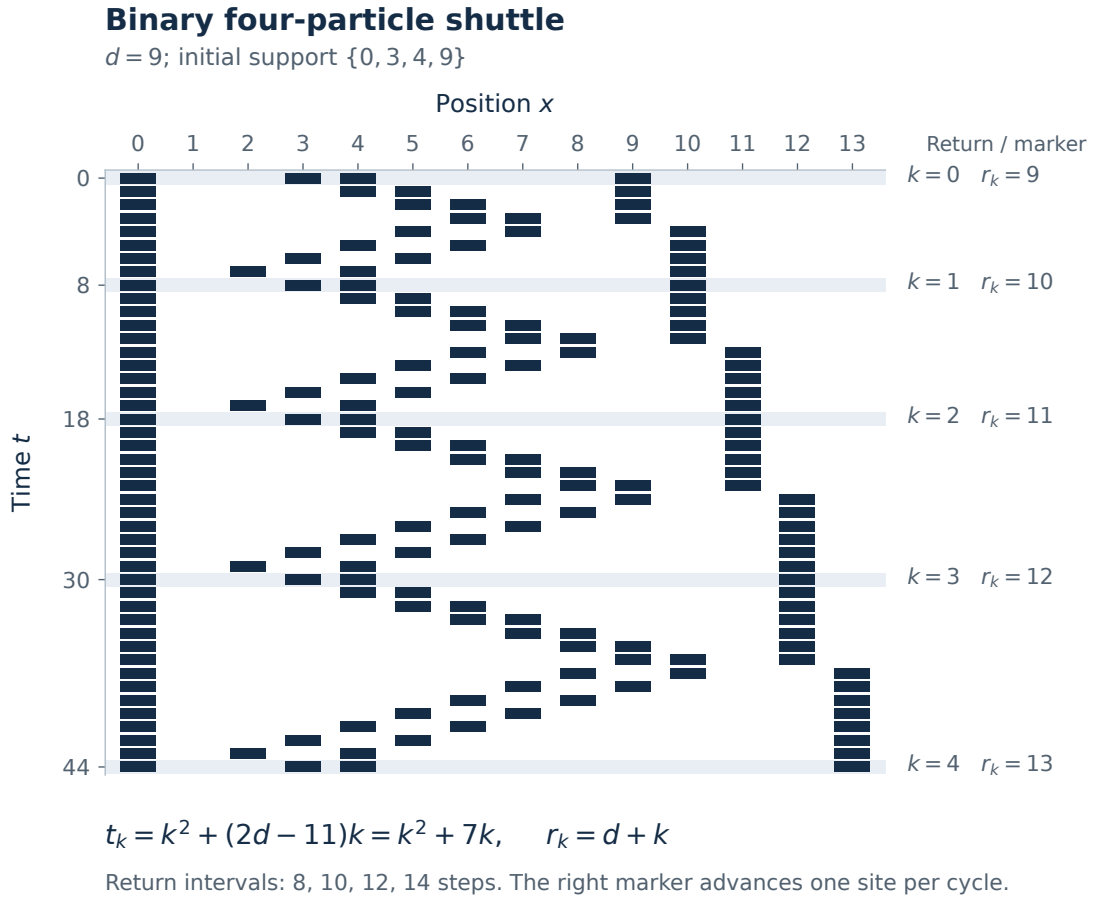


Figure 5: An actual trace of the exported binary local rule. The left marker stays fixed; the right marker moves outward by one after each cycle. The two remaining units alternate between an adjacent right-moving pair and a separated left-moving pair. The exact trace data and plotting source accompany the package.

90 A one-witness quartic timing certificate

Write $d = 7 + x$, where $x \in \mathbb{N}$ is an input. Define

$$\mathcal{P}(x, t; k) = [k^2 + (2x + 3)k - t]^2. \quad (90.1)$$

For natural inputs x, t , this degree-four polynomial has one natural auxiliary witness k , one squared quadratic residual, and three total variables. Its natural witness fiber is a singleton exactly when pattern 10011 occurs at time t , and empty otherwise. Indeed the residual must vanish, and

$$[(k + 1)^2 + (2x + 3)(k + 1)] - [k^2 + (2x + 3)k] = 2k + 2x + 4 > 0.$$

No time horizon or orbit tableau is supplied.

For natural x, t , a rational root k of the monic integer polynomial $k^2 + (2x + 3)k - t$ is integral, so the same fiber statement holds over nonnegative rationals. Over nonnegative reals the quadratic increases from zero to infinity; every natural t , including non-hit times, has a root.

Evaluating $k(k + 2x + 3) - t$ and squaring uses four additions or subtractions and two multiplications; at fixed gap the counts are two and two. These are straight-line evaluator counts, excluding input construction and bit complexity. The certificate is specific to this timing formula: mere decidability does not imply a general fixed-arity single-fold quartic compiler.

[write] [By Theorem `cdc:of:thm:classification` of Part XI of *canonical-diophantine-certificates* ($D_2^+ = D_3^+ = \text{SL}$), the natural zero set of a polynomial of degree at most three that is nonnegative on the real nonnegative orthant has semilinear projections. The hit set $\{(x, t)\}$ of (90.1) is not semilinear (its sections are not eventually periodic, Proposition 89.2), so it has no such representation of degree at most three, single-fold or not, and the degree four of (90.1) is the least possible in that class. The programme’s review [46] confirms the uniqueness of the witness over the naturals and the nonnegative rationals, the real counterexample at $x = 0, t = 1$ (witness $(\sqrt{13} - 3)/2$), the loss of uniqueness over signed integers, and the schedule of two multiplications and four additions.]

91 Primary attribution and scope of the literature check

Kong’s June 2021 thesis [41] studies one-dimensional *two-state* NCCAs explicitly. Its Definition 1 is on printed p. 8 (PDF p. 9). Section 2.4 proves translated eventual periodicity for two particles and three-particle non-strong-universality; Theorem 2 appears on printed p. 14 (PDF p. 15). The following conclusion, printed p. 15 (PDF p. 16), describes the four-particle case as open *at that time*. These are direct primary precursors. Their binary scope must not be silently enlarged to arbitrary weighted alphabets. [write] [Theorem 83.1 covers two-state NCCAs, whose unique unit state is 1, for its decision problems, and so bears on the four-particle case that Kong’s thesis called open in 2021, subject to the thesis’s notion of strong universality, which is not compared here.]

Imai’s final KAKEN report dated 25 June 2021 [42], PDF p. 4, reports a motion-representation decidability result at particle count at most three. Its brief statement does not identify the exact decision problem. Its methods discussion on PDF p. 3 distinguishes pebble-machine state from numerical NCCA state that must be encoded geometrically, and discusses the five-particle construction and smaller-particle analysis. The unspecified word “decidable” should not be read as a sourced statement of uniform timed Presburger definability or of the full weighted theorem here.

The Alhazov–Imai 2016 publisher abstract [24] describes finite-support nonnegative-integer-state NCCAs and reports universality with five particles at sufficiently large finite radius. This is an upper bound, not an abstract claim of minimality. Its full paper was unavailable in this check. The thesis’s nearby upper-bound reference appears mismatched, so the 2016 result is cited independently rather than inherited through that reference.

No later primary four-particle resolution was located in the targeted search. This is not evidence that the 2021 question remains open today and is not evidence of novelty of the present proof. The mathematical claim in this report rests on its argument, not on a literature-status assertion.

92 Checks and remaining questions

The general decision theorem is a mathematical proof, independently inspected; there is no implemented rule-to-section compiler and no machine-checked formalization. Finite examples cannot verify a theorem over all rules. The explicit binary rule, by contrast, has a complete exported local table and an exact finite conservation certificate.

The producer’s reproducible standard-library Python checks include all 8192 identities in (89.2), 262,144 exhaustive width-eighteen conservation cases, 8,192 component-rule/local-table comparisons, 10,000 larger random locality and translation comparisons, 302,400 direct trajectory steps, 2,424 predicted hits, and 20,000 finite unique-witness checks. A separately written replay, importing no producer evaluator, checked all 8192 local outputs, all 8192 potential identities, 49,640 direct steps over 40 initial gaps, and 840 exact pattern hits. A second independent guarded-pattern detector reconstructed every local output, checked 1,396 complete closed-form transitions and hit-exclusivity cases through section index 10^{80} , and made 25 large-integer quartic/discriminant checks. The section arithmetic audit also checked 951 first-contact instances and huge residue shifts, 272 weighted-shuttle transitions, and 27,360 original-frame support bounds. These ledgers supplement the exact rule and timing proofs; they do not establish the general theorem by testing.

The portable package contains this editable L^AT_EX source and PDF, the full rule table and potential, reproduction scripts, recorded test ledgers, independent audit materials, and exact figure data. The earlier mass-three release is not changed by this package.

[write] **The release in this report.** These are the delivered names. The programs are shipped as `code/17-four-mass-*` (those of `independent/` and `figures/` flattened), the receipts, the conservation certificate and the figure data as `data/17-four-mass-*`, the article’s figure as `figures/17-four-mass-binary-four-particle-shuttle.pdf`, and the proof-detail and audit notes (`BINARY-EXPANDING-SHUTTLE.md`, `EXPANDING-SHUTTLE.md`, `SOURCE-PROVENANCE.md`, `finite-seed-section-lemma.md`, and `independent/BINARY-INDEPENDENT-AUDIT.md`, `FINAL-REVIEW.md` and `INDEPENDENT-AUDIT.md`) as `17-four-mass-*` at the report root. The typed weighted shuttle whose 272 transitions are counted above is described in `17-four-mass-EXPANDING-SHUTTLE.md` and tested by `code/17-four-mass-test_expanding_shuttle.py`. The figure table `data/17-four-mass-binary-four-particle-shuttle.csv` keeps its delivered CRLF line ends. Not shipped: the manuscript, PDF, README, checksum ledger and the raster copy of the figure. `test_binary_expanding_shuttle.py` rewrites the sealed conservation certificate in place, with CRLF on Windows, after which `test_exact_boundaries.py` fails its hash test; run the suite on a copy and restore the certificate first (README). The programme’s review reran all nine invocations and reproduced every receipt [46].

Several questions remain useful without being asserted as new open problems:

1. Determine the current published status and priority of the four-particle boundary by obtaining the missing full papers and later primary results. ([write] Open; the programme’s review [46] did not obtain the full Alhazov–Imai paper either.) ([write] Added 3 October 2026: touched by Part VII, whose sources 19 and 20 add pebble-automaton and plane-walking literature (Sections 115 and 127); source 19 did not locate a primary theorem for its class and draws no priority conclusion.)
2. Investigate uniform *untimed* spatial Presburger definability in the stationary frame; the theorem here only needs a fixed-input construction. ([write] Added 3 October 2026: touched for a fixed input by source 20, whose visited sets are semilinear with quasipolynomial box counts in every dimension (Theorem 118.2); the uniform question stays open.)
3. Give explicit complexity bounds or an implemented general section compiler, accounting for the potentially enormous finite seed library. ([write] Added 3 October 2026: open; source 19’s nearly parallel example (Section 107) shows that naive geometric bounds fail, and its Question 3 asks the same in every dimension.)
4. Determine what stronger arithmetic normal form describes original-frame uniform timed relations. The binary shuttle rules out Presburger arithmetic. ([write] Added 3 October 2026: touched. For a fixed input the complete timed orbit is a disjoint union of charts with at most two parameters and quadratic outputs (source 18, Lemma 94.1), and with drift the original-frame site set can be nonsemilinear with counts $3N - 3\sqrt{N} + O(1)$ (source 20, Theorem 125.1); the uniform question stays open.)
5. Classify the effect of several weight-one labels. The stationary-unit and no-split arguments used here no longer apply. ([write] Part V shows that with many weight-one types three units suffice; the least number of weight-one symbols that permits a universal construction at mass three or four is open, as source 16 also says.) ([write] Added 3 October 2026: still open in every dimension; source 19’s Question 4.)
6. Study minimal radius or other resource bounds for the binary timing example separately from universal-computation particle bounds. ([write] Added 3 October 2026: touched by source 21, whose planar binary rule has exact radius six and exactly 78 essential cells (Theorem 137.1), with no minimality claimed; the one-dimensional question stays open.)
7. Audit a fixed-input timed quartic compiler built from canonical disjoint phases of the orbit normal form. Quadratic time expressions and unique phase parameters suggest a route, but no such general compiler is proved here. ([write] Added 3 October 2026: answered for a fixed input by source 18 below (Theorem 93.1 and Proposition 96.1), conditional on Proposition 87.1, with natural witnesses; source 19’s Appendix I lifts it to $\mathbb{Z}^d$. The rule-to-chart preprocessing is not implemented.)

Source 18 (batch-82 manuscript 19): Canonical quartic certificates for timed four mass orbits

Research addendum 13. Complete fixed input orbits and exact witness accounting, 3 October 2026 (its PDF metadata name the author as “Mathematical research report”). [write] Source 18 is an AI-assisted research manuscript of ProveIt’s incoming reports, batch 82, manuscript 19 (arrival db37d18c8), from the archive `Timed_Four_Mass_Quartic_Certificates.zip`; it was added to Part VI on 3 October 2026. Its labels carry the prefix `smc:tq:`, and its sections are Sections 93–100 here. It is the research pipeline’s “addendum 13” and a declared addendum to source 17, which

it calls “companion report 12” (source 17 is the pipeline’s Report 12); source number, manuscript number and pipeline number all differ, and bracketed notes translate them. It answers source 17’s last question (Question 7) for a fixed input, conditional on source 17’s normal form (Proposition 87.1), with natural witnesses. Its binary rule and its return quartic are source 17’s and are printed as pointers (Section 98); its chart compiler is a second route to Part I’s Theorem 10.2 (note in Section 96). It names no repository commit, and the Hilbert’s-tenth programme had not reviewed it when this was written (Section 1.19). Its abstract follows; the bracketed note in it is [write].

Fix a conservative one-dimensional cellular automaton with at most one weight-one symbol and an initial configuration of mass at most four. We construct a canonical natural-witness quartic for its complete timed orbit from the effective normal form in companion report 12 [source 17]. Disjoint half-open charts have affine domains and quadratic outputs. The baseline compiler uses $B + K + M$ witnesses for B charts, K parameters, and M inequalities. An explicit binary shuttle admits a four-witness, seven-square specialization, uniform in the natural gap parameter $x = d - 7$; at integer external inputs it is also exact for nonnegative real witnesses. The general theorem remains fixed-input and conditional on the imported normal form, with no generic real-witness, uniform-input, or Diophantine finite-fold claim.

93 The relation and its precise scope

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Let A be a finite alphabet with a unique zero-weight symbol 0, with positive integer weights on all other symbols. Let $F : A^{\mathbb{Z}} \rightarrow A^{\mathbb{Z}}$ be a deterministic, translation-equivariant cellular automaton of finite radius. Assume $0^{\mathbb{Z}}$ is quiescent and total mass is conserved on every finite-support configuration. There is at most one symbol of weight one. Fix F and a finite-support configuration c with total mass at most four.

For a fixed ordered tuple $a = (a_1, \dots, a_m)$ of nonzero symbols, define

$$R_a(t, x_1, \dots, x_m) \iff \begin{cases} t \in \mathbb{N}, & x_1 < \dots < x_m, & x_i \in \mathbb{Z}, \\ F^t(c)_{x_i} = a_i & (1 \leq i \leq m), \\ F^t(c)_x = 0 & \text{for every } x \notin \{x_1, \dots, x_m\}. \end{cases}$$

The empty tuple is permitted. It describes exactly the vacuum at time t . If the requested symbols have incompatible total mass, the relation is empty.

Theorem 93.1 (Canonical timed quartic). *Assume the effective fixed-input orbit normal form in companion report 12 [source 17, Part VI]. One can effectively construct an integer-coefficient polynomial*

$$C_a(t, x_1, \dots, x_m; w_1, \dots, w_N)$$

of total degree at most four such that, for $t \in \mathbb{N}$ and $x_i \in \mathbb{Z}$,

$$\#\{w \in \mathbb{N}^N : C_a(t, x; w) = 0\} = \begin{cases} 1, & R_a(t, x), \\ 0, & \text{otherwise.} \end{cases}$$

The witness count N is fixed independently of t and x . A product-gated version is nonnegative for all real external coordinates and natural witnesses. A sum-of-squares version is nonnegative on the whole real space.

The coefficients, number of charts, witness count, and finite preprocessing may depend on both F and c . A computable but enormous finite history may be fully materialized. There is no complexity

bound here, nor a bound uniform over arbitrary initial configurations. The explicit one-parameter binary family in Section 99 has a separate, stronger conclusion.

The theorem concerns natural witness fibers at integer external inputs. Global nonnegativity alone does not make rational or real witness zeros exact. Section 98.4 proves the stronger nonnegative-real conclusion only for the explicit binary specialization. Time stays external: forgetting time can destroy uniqueness, even for a stationary orbit.

94 The imported normal form and the chart interface

The only dynamical theorem imported from report 12 is its effective fixed-input eventual orbit classification, including all finite phase data. Its proof reduces separated recurrent interactions to finite control and a single additive gap counter. In a frame where the unique unit symbol is stationary, the eventual orbit is one of the following:

1. A whole configuration periodic up to translation
2. Finitely many mutually independent finite-phase objects, with a common clock after their finite transients
3. An expanding shuttle, with a finite control cycle, an affine growing marker gap, affine marker anchors, and finite-phase flights separated by fixed finite local scattering prefixes

The no-unit-symbol case and mass-at-most-three cases use the first two possibilities. Coherent translating three-mass tails are separately accelerated to a bounded full encounter or to an independent terminal tail. They are not treated as fixed-duration continuing shuttle scatterings. The classification itself is not re-proved in this addendum. [write] [The imported classification is Proposition 87.1 of source 17, with its section times (87.2) and the acceleration of Section 87; “report 12” is source 17 throughout this source.]

Lemma 94.1 (Sufficient chart interface). *It suffices to construct an effective finite chart family with the following properties. Chart i has $k_i \leq 2$ natural parameters z_i , a domain*

$$D_i = \{z_i \in \mathbb{N}^{k_i} : A_{ir}(z_i) \geq 0 \ \forall r, \quad H_{is}(z_i) = 0 \ \forall s\},$$

where A_{ir} and H_{is} are integer-affine, and a fixed output-symbol tuple. Its output $P_i(z_i)$ consists of time and sorted positions, is rational-polynomial of total degree at most two, and is integer-valued on D_i . Before filtering by output labels, the disjoint union of all chart domains maps bijectively to all timed configurations of the orbit.

The required bijection includes parameter uniqueness, not just uniqueness of a list of phase names. In particular, a finite union of overlapping parametrizations is insufficient. Once this interface is proved, retain only charts with symbol tuple a . Their outputs are exactly R_a , still without duplication. The algebraic compiler of Section 96 then proves Theorem 93.1. The supplied compiler checks exact chart syntax and degree; it does not decide this semantic bijectivity hypothesis.

95 A half open partition of every time

95.1 Finite histories and periodic tails

Let T_0 be the computable start of the eventual profile. Materialize the configuration at every integer $0 \leq t < T_0$, making each a parameter-free chart. This is allowed even if an accelerated negative-gap

excursion makes T_0 impractically large. The boundary T_0 belongs only to the tail.

For a translated-periodic tail with period $p > 0$, each residue $r \in \{0, \dots, p-1\}$ has one natural parameter n and

$$t = T_0 + r + pn.$$

Its occupied coordinates are affine in n and its symbols are fixed. Euclidean division of $t - T_0$ gives exactly one r, n .

For an independent terminal tail, advance the common start beyond every object's transient and take a common multiple p of their periods. At each phase r , all positions again have affine displacements in the same n . The absorbed finite prefix remains disjoint. Giving different objects independent free clocks would be incorrect: their phases are synchronized by actual time. Stationary objects have zero displacement. The vacuum is an ordinary one-clock stationary tail, with no spatial coordinates.

95.2 Quadratic section times in an expanding shuttle

Number repetitions of one expanding control cycle by $n \geq 0$. The refined control includes the necessary gap residues. At corresponding sections,

$$D_n = D_0 + n\Delta, \quad L_n = L_0 + nE, \quad \Delta > 0.$$

Each of finitely many continuing transitions ℓ in the cycle has a nonempty remote flight and a fixed local scattering prefix. Its gap and anchor are affine in n , with the same Δ and E slopes. Its total duration is affine,

$$\tau_\ell(n) = A_\ell n + B_\ell.$$

Although the original flight calculation may use rational coefficients, $\tau_\ell(n)$ is integral for every $n \geq 0$ along this fixed itinerary. Thus $B_\ell = \tau_\ell(0)$ and $A_\ell = \tau_\ell(1) - \tau_\ell(0)$ are integers. The flight coefficient is positive. Summing the durations gives $An + B$ with $A > 0$, $B > 0$, so cycle n starts at

$$T(n) = T_0 + \frac{An(n-1)}{2} + Bn. \quad (95.1)$$

This integer-valued quadratic satisfies $T(n+1) - T(n) = An + B > 0$ and tends to infinity. Consequently the half-open intervals $[T(n), T(n+1))$ partition the entire tail. The start $S_\ell(n)$ of transition ℓ is $T(n)$ plus the sum of its preceding transition durations, hence is also quadratic and integer-valued.

95.3 Flight phases and local prefixes

Write the flight duration as $h_\ell(n) = \alpha_\ell n + \beta_\ell$ and the local-prefix length as the fixed integer $b_\ell \geq 0$. Transition ℓ owns

$$[S_\ell(n), S_\ell(n) + h_\ell(n) + b_\ell).$$

Its flight owns the first $h_\ell(n)$ times; the local prefix owns the remaining b_ℓ times. The next launch belongs to the next transition or cycle. A zero-length prefix supplies no chart.

If the flight packet has period p , then for every fixed $r \in \{0, \dots, p-1\}$ use natural parameters n, j with

$$t = S_\ell(n) + r + pj, \quad h_\ell(n) - r - pj - 1 \geq 0. \quad (95.2)$$

The domain inequality is affine. It says exactly that the integer elapsed time $r + pj$ is strictly less than $h_\ell(n)$. Euclidean division selects one r, j at every flight time. Marker positions are affine in n ; packet positions are affine in (n, j) , with symbols fixed by the phase. Empty phase domains may be retained without introducing any zero.

For each fixed $0 \leq s < b_\ell$, the local-prefix chart has just n and

$$t = S_\ell(n) + h_\ell(n) + s.$$

The finite control fixes the gap residue, contact phase, and tagged seed. Thus the contacted seed anchor is a fixed offset from an affine marker coordinate. Every occupied coordinate during its fixed local prefix is affine in n , including the distant old marker.

Uniqueness is now proved in order: first the cycle n , then its transition ℓ , then flight versus local prefix, and finally r, j or the fixed prefix index s . Every integer time is included, and no boundary is included twice. Time, phase, quotient, and prefix length have all been accounted for by the chart choice and its at most two parameters; no extra clock witness is needed.

95.4 Original coordinates and strict affine sorting

If the unique unit moves by δ sites per step under F , write $F^t = \tau_{\delta t} G^t$, where the unit is stationary for G . For any indexed occupied site v , let g_v be its stationary-frame coordinate in a chart. Its original-frame coordinate is

$$x_v = g_v(n, j) + \delta t(n, j). \quad (95.3)$$

This has degree at most two. Importantly, $x_v - x_w = g_v - g_w$ is still affine.

There is no need to assume a global ordering of moving objects. Refine each chart by every permutation π of its finitely many indexed occupied sites, adding the domain inequalities

$$g_{\pi(k+1)} - g_{\pi(k)} - 1 \geq 0 \quad (1 \leq k < m). \quad (95.4)$$

If rational coefficients arise in an affine description, clear positive denominators first. On integer sites these inequalities are exactly strict spatial order. Actual sites are distinct, so precisely one permutation passes at each original chart point. Symbols follow their indexed sites; repeated symbols do not multiply the accepted orders. Now filter by a . When a phase's order is already proved fixed, the redundant sorting refinement can be omitted. This completes Lemma 94.1.

96 An exact quartic compiler

The construction in this section applies to any family satisfying Lemma 94.1, independently of its dynamical origin. Suppose there are B charts, $K = \sum_i k_i$ private parameters, M domain inequalities, H domain equalities, and Q external output coordinates $y_1, \dots, y_Q$. For timed m -site configurations, $Q = m + 1$. First multiply each rational-affine domain constraint by its own positive common denominator. All domain forms below therefore have integer coefficients. A natural slack must be introduced *after* this step: for example, $(n - 1)/2 \geq 0$ does not mean that $(n - 1)/2$ itself is a natural integer.

96.1 Private parameters and constant term lifting

For each chart i introduce a natural selector e_i , its private natural parameters z_i , and a private natural slack u_{ir} for each inequality. Set

$$E = \sum_{i=1}^B e_i, \quad Z_i = \sum_{j=1}^{k_i} z_{ij} + \sum_r u_{ir}.$$

For a polynomial f belonging to chart i , define

$$\mathcal{L}_i f = f(z_i) - f(0) + f(0)e_i. \quad (96.1)$$

This changes only the constant term. Multiplying the quadratic monomials by e_i would instead raise an output residual to degree three and its square to degree six. Such multiplication is unnecessary because the gates below force every inactive chart's private variables to zero.

Use residuals

$$R_0 = E - 1, \quad R_{ir}^{\geq} = \mathcal{L}_i A_{ir} - u_{ir}, \quad R_{is}^{\leq} = \mathcal{L}_i H_{is}, \quad (96.2)$$

$$R_q^{\text{out}} = L_q \left(y_q - \sum_{i=1}^B \mathcal{L}_i P_{iq} \right), \quad 1 \leq q \leq Q. \quad (96.3)$$

Here $L_q > 0$ is a common integer denominator for all rational coefficients in the *pooled* output expression. Clearing branch outputs separately and then adding the independently rescaled results would alter the equation. For example, outputs $z/2$ and $z/3$ require a single output residual such as $6y - 3z_1 - 2z_2$, not $y - z_1 - z_2$. Every residual has integer coefficients and degree at most two.

96.2 Two polynomial formats

Let the common squared part be

$$S = R_0^2 + \sum_{i,r} (R_{ir}^{\geq})^2 + \sum_{i,s} (R_{is}^{\leq})^2 + \sum_{q=1}^Q (R_q^{\text{out}})^2.$$

The product-gated and literal sum-of-squares forms are respectively

$$C_{\text{prod}} = S + \sum_{i=1}^B (E - e_i) Z_i, \quad (96.4)$$

$$C_{\text{SOS}} = S + \sum_{i=1}^B ((1 - e_i) Z_i)^2. \quad (96.5)$$

All product gates in (96.4) are nonnegative on the natural witness orthant: $E - e_i$ is the sum of the other natural selectors. They have degree at most two. The SOS gates are squares of quadratic residuals. Both forms have degree at most four; the former is nonnegative for real external coordinates and natural witnesses, the latter for all real inputs. One could use $E - e_i$ inside the squared gate as well, since it equals $1 - e_i$ at selector zeros.

Proposition 96.1 (Exact zero fibers). *The natural zeros of either polynomial over a fixed external output y are in bijection with the admissible chart points mapping to y .*

Proof. Every summand is nonnegative on the intended domain. At a zero, $R_0 = 0$ gives $E = 1$. Natural selectors therefore have exactly one value equal to one and all others equal to zero. Separate Boolean-selector equations are not needed.

Let i be the selected chart. For every inactive chart h , its gate has coefficient one, so $Z_h = 0$. Every private parameter and slack is natural; therefore every individual summand in Z_h is zero. Equation (96.1) then gives $\mathcal{L}_h f = 0$ for each inactive domain or output form. On the selected chart it gives $\mathcal{L}_i f = f(z_i)$. All surviving residuals say exactly

$$A_{ir}(z_i) = u_{ir} \geq 0, \quad H_{is}(z_i) = 0, \quad y = P_i(z_i).$$

Conversely, an admissible chart point determines a zero: select that chart, use its parameters, set its slacks to the uniquely determined integers $A_{ir}(z_i)$, and set every other private coordinate to zero. These two constructions are inverses. The chart bijection then proves the empty-or-singleton assertion of Theorem 93.1. $\square$

Naturality is essential to the inactive-zero step. Signed private variables could cancel in Z_h . Free difference pairs for private witnesses would also admit many representations and destroy the stated uniqueness. The compiler uses no such encoding.

[write] [Second route. The device of this compiler is Part I's: Theorem 10.2 multiplies the rows of disjoint chambers by natural selectors that sum to one, and Proposition 96.1 is a second route to it for chart families with private parameters. The inactive gate $(E - e_i)Z_i$ is the gate of Part V's certificate (source 14, Theorem 58.1) and the inactive-product gate of Lemma `qoc:rn:lem:gates` in Part VI of *quadratic-orthant-certificates*. Source 18's own refinements are the constant-term lifting (96.1), which keeps quadratic outputs at degree two, the pooled output denominators L_q , and the selector elimination below. Source 18 calls its proof elementary algebra and cites none of these; no novelty is claimed for the device.]

96.3 Exact witness and residual ledgers

For $B \geq 1$, both formats use precisely

$$N = B + K + M \leq 3B + M \tag{96.6}$$

natural witnesses when $k_i \leq 2$. The slots before deletion of zero terms are as follows.

Quantity	Product-gated	Sum of squares
Selectors	B	B
Private parameters	K	K
Private inequality slacks	M	M
Squared residuals	$1 + M + H + Q$	$1 + M + H + Q + B$
Nonsquared nonnegative products	B	0
Total degree upper bound	4	4

These counts do not bound coefficient sizes or the preprocessing needed to find the chart family. The polynomial may have degree less than four. If $B = 0$, use the constant polynomial 1 with zero witnesses. If $B = 1$, the product gate is identically zero because there is no inactive chart; the SOS gate can be retained. Parameter-free charts and syntactically retained empty domains cause no difficulty.

96.4 Removing one selector in the SOS form

For $B \geq 1$, the SOS construction can omit its last selector. Retain natural $e_1, \dots, e_{B-1}$, set $E_* = \sum_{i < B} e_i$, and substitute $e_B = 1 - E_*$ everywhere. Replace R_0 by $E_*(E_* - 1)$, keeping every other squared residual and every squared gate. At a zero, the natural integer E_* is either zero or one, so the implicit selector is also zero or one and exactly one of all B selectors is one. The inactive-zero and uniqueness proof is unchanged. Every residual remains quadratic; the result has $B - 1 + K + M$ natural witnesses and the same $1 + M + H + Q + B$ square slots. This refinement is for the SOS form. Its whole-space nonnegativity does not require the implicit selector to be nonnegative away from zeros. For $B = 1$ the selector residual is identically zero and may be deleted.

97 Signed external positions without extra witnesses

Signed integer positions are already permitted as external inputs in the theorem. For an entirely natural external coordinate relation, replace each x_i by the canonical pair $(x_i^+, x_i^-) \in \mathbb{N}^2$, imposing

$$x_i = x_i^+ - x_i^-, \quad x_i^+ x_i^- = 0.$$

Substitute $x_i^+ - x_i^-$ in the output residuals. Add the nonnegative product $x_i^+ x_i^-$ to C_{prod} , or its square to C_{SOS} . There are m extra product slots or m extra squared residuals, respectively, and *no additional witnesses*: these pairs are external inputs. The number of external coordinates becomes $1 + 2m$; the original $Q = m + 1$ output equations remain. Total degree is at most four.

For any signed integer x , the unique accepted pair is $(\max(x, 0), \max(-x, 0))$. Without the product condition, the value -1 would be encoded by both $(0, 1)$ and $(1, 2)$, among infinitely many pairs. Those unrestricted difference pairs describe a different external relation. This external coding must not be confused with an encoding of the private parameters or slacks, whose literal natural values were needed in the proof.

98 A complete binary shuttle orbit

98.1 The conservative radius six rule

[write] [Pointer. This subsection restates source 17's binary rule: its four component replacements up to translation, with the roles right-moving pair, left-moving pair, right bounce and marker shift, and left bounce, are rule (89.1); the proof that the component prescription conserves mass and is a genuine local binary rule of radius six is that of Proposition 89.1; and the 8192-entry rule table with its exact conservation potential on all 4096 length-twelve binary words is the certificate (89.2). Source 18 reuses source 17's three files for it byte for byte (they are shipped once, as source 17's). The omitted text survives in the arrival archive. Its closing sentence is kept:] These finite algebraic identities certify global conservation; bounded orbit simulations serve a different purpose.

98.2 Two disjoint sorted charts

Fix $d \geq 7$ and start at $\{0, 3, 4, d\}$. Set

$$T(n) = n^2 + (2d - 11)n, \quad r = d + n.$$

Cycle n begins with $\{0, 3, 4, r\}$. The right-moving pair takes $r - 6$ steps to reach $\{r - 3, r - 2\}$. Its right bounce gives $\{0, r - 4, r - 2, r + 1\}$. The left-moving pair takes another $r - 6$ steps to reach $\{2, 4\}$, and its next bounce starts cycle $n + 1$. One cycle therefore takes $2r - 10$ steps, which agrees with $T(n + 1) - T(n)$.

There are exactly two charts, both with natural n, j and the inequality

$$d + n - 6 - j \geq 0. \quad (98.1)$$

Their outputs are

$$\text{Right: } t = T(n) + j, \quad (x_0, x_1, x_2, x_3) = (0, 3 + j, 4 + j, d + n), \quad (98.2)$$

$$\text{Left: } t = T(n) + d + n - 5 + j, \quad (x_0, x_1, x_2, x_3) = (0, d + n - 4 - j, d + n - 2 - j, d + n + 1). \quad (98.3)$$

Both use $0 \leq j \leq r - 6$. The right chart owns $[T(n), T(n) + r - 5]$, including its right contact configuration. The next step is the bounce, owned by the left chart at $j = 0$. The left chart owns $[T(n) + r - 5, T(n + 1))$, including its left contact. The next step belongs to cycle $n + 1$. Since $r \geq 7$, both displayed coordinate lists are strictly increasing. Thus no sorting inequalities need to be added. These charts describe every state, not only returns. [write] [The hypothesis $d \geq 7$ is safe but not sharp, as for Proposition 89.2: in the placement check the two charts also reproduce every state through $t < 2500$ for $d = 6$.]

98.3 An explicit eight witness baseline certificate

Use the ordered witness tuple

$$w = (e_R, e_L, n_R, j_R, u_R, n_L, j_L, u_L).$$

There are three selector/domain residuals,

$$\begin{aligned} r_0 &= e_R + e_L - 1, \\ r_R &= (d - 6)e_R + n_R - j_R - u_R, \\ r_L &= (d - 6)e_L + n_L - j_L - u_L, \end{aligned}$$

one time residual

$$\begin{aligned} r_t &= t - (n_R^2 + (2d - 11)n_R + j_R \\ &\quad + n_L^2 + (2d - 10)n_L + (d - 5)e_L + j_L), \end{aligned}$$

and four position residuals,

$$\begin{aligned} r_{x_0} &= x_0, \\ r_{x_1} &= x_1 - (3e_R + j_R + (d - 4)e_L + n_L - j_L), \\ r_{x_2} &= x_2 - (4e_R + j_R + (d - 2)e_L + n_L - j_L), \\ r_{x_3} &= x_3 - (de_R + n_R + (d + 1)e_L + n_L). \end{aligned}$$

Let $S_d = r_0^2 + r_R^2 + r_L^2 + r_t^2 + \sum_{v=0}^3 r_{x_v}^2$. The explicit product-gated certificate is

$$C_d = S_d + e_L(n_R + j_R + u_R) + e_R(n_L + j_L + u_L). \quad (98.4)$$

The SOS variant replaces these two product terms by

$$((1 - e_R)(n_R + j_R + u_R))^2 + ((1 - e_L)(n_L + j_L + u_L))^2.$$

Its ledger is $B = 2$, $K = 4$, $M = 2$, $H = 0$, $Q = 5$: eight natural witnesses, eight squared residuals plus two product slots, or ten squared residuals and no product slots. Both polynomials have degree exactly four.

For example, the minimal gap $d = 7$ gives the following boundary checks. The selected branch parameters and its slack determine all eight witnesses; the inactive branch's coordinates are zero.

t	Occupied sites	Branch	n	j	u
0	$\{0, 3, 4, 7\}$	Right	0	0	1
1	$\{0, 4, 5, 7\}$	Right	0	1	0
2	$\{0, 3, 5, 8\}$	Left	0	0	1
3	$\{0, 2, 4, 8\}$	Left	0	1	0
4	$\{0, 3, 4, 8\}$	Right	1	0	2

At $t = 2$ the witness is $(0, 1, 0, 0, 0, 0, 1)$; at $t = 4$ it is $(1, 0, 1, 0, 2, 0, 0)$. The two changes of branch exhibit the half-open ownership convention explicitly.

98.4 A four witness binary specialization

The two binary charts share their domain and their quadratic cycle time. This permits a stronger specialized SOS certificate with shared natural witnesses (e, n, j, u) . Let $e = 0$ select the right chart and $e = 1$ the left. For $d \geq 7$ define seven residuals

$$\begin{aligned} q_e &= e(e - 1), \\ q_u &= d + n - 6 - j - u, \\ q_t &= t - (n^2 + (2d - 11)n + j + e(d + n - 5)), \\ q_0 &= x_0, \\ q_1 &= x_1 - (3 + j + e(d + n - 7 - 2j)), \\ q_2 &= x_2 - (4 + j + e(d + n - 6 - 2j)), \\ q_3 &= x_3 - (d + n + e). \end{aligned}$$

Then

$$C_d^{(4)} = q_e^2 + q_u^2 + q_t^2 + q_0^2 + q_1^2 + q_2^2 + q_3^2$$

has degree four, four natural witnesses, seven squared residuals, and no nonsquared product slots. At a natural zero, $q_e = 0$ forces $e \in \{0, 1\}$, and $q_u = 0$ forces the shared domain $0 \leq j \leq d + n - 6$ with its unique slack. For $e = 0$ the output is exactly (98.2); for $e = 1$ it is exactly (98.3). Their proved timed partition therefore gives exactly one witness tuple at every valid complete timed configuration and none otherwise. In particular, the quadratic time term is shared rather than multiplied by a selector, so no cubic residual appears.

At $d = 7$, the five states in the preceding table have witnesses $(0, 0, 0, 1)$, $(0, 0, 1, 0)$, $(1, 0, 0, 1)$, $(1, 0, 1, 0)$, and $(0, 1, 0, 2)$. This is an upper bound for this binary family, not a witness-minimality claim or a generic shared-parameter optimization for arbitrary charts.

Exactness on the nonnegative real witness orthant. For this specialized certificate, integer external inputs in the stated domain force every real zero's witnesses to be integers. Indeed $q_e = 0$ gives $e \in \{0, 1\}$, and $q_3 = 0$ gives $n = x_3 - d - e \in \mathbb{Z}$. Then $q_1 = 0$ gives $j = x_1 - 3$ when $e = 0$, or $j = d + n - 4 - x_1$ when $e = 1$. Finally $q_u = 0$ gives $u = d + n - 6 - j \in \mathbb{Z}$. If the four real witnesses are nonnegative, they are therefore natural, and the proved singleton-fiber conclusion applies. This also covers nonnegative rational witnesses. It uses the spatial output equations and does not extend to the one-witness return-time certificate or arbitrary chart compilers.

Unrestricted real witnesses still fail. At $d = 7$, $t = 0$, the sorted external output $(0, 5, 7, 8)$ makes all seven residuals vanish at $(e, n, j, u) = (1, 0, -2, 3)$, although the true initial state is $\{0, 3, 4, 7\}$. In the uniform version this is $x = 0$; the negative j is essential.

99 A uniform gap specialization with the same cost

For this particular rule and initial family, set $d = 7 + x$ and make $x \in \mathbb{N}$ an additional external input. Substitute $7 + x$ for d in every residual of Section 98.3. The new monomials include xe_R , xe_L and xn ; all residuals still have total degree at most two in all external and witness variables. Consequently both baseline polynomials still have degree four and eight natural witnesses, with unchanged slot counts. The same substitution in Section 98.4 gives degree four, four natural witnesses, and seven squares. The selector-eliminated baseline SOS likewise keeps its seven witnesses and ten squares.

Corollary 99.1 (One-parameter complete orbit family). *The substituted four-witness certificate has a singleton natural witness fiber exactly at the complete timed orbit of the initial configuration $\{0, 3, 4, 7 + x\}$, for each natural external x . The same statement holds for nonnegative real witnesses. All other integer output configurations at a natural time have no such witness.*

Proof. For each fixed $x \in \mathbb{N}$, the substituted polynomial is precisely the four-witness fixed-gap polynomial with $d = 7 + x \geq 7$, already proved correct. The simultaneous degree bound follows from the explicit residuals, rather than from an unproved parameterized version of the generic compiler. $\square$

The implementations represent x symbolically in the baseline `uniform_binary_shuttle` and the compact `compact_binary_uniform_sos`. Its tests include $x = 0$, compare the expanded polynomial with independently compiled fixed-gap polynomials, and compare the certificate with direct orbit charts. This is uniformity over one explicit one-parameter input family. It is not a generic extension to arbitrary externally parameterized chart domains or arbitrary initial configurations.

The smaller return-hit certificate in report 12 [source 17's (90.1), written there with the witness k and $d = 7 + x$] is

$$[n^2 + (2d - 11)n - t]^2,$$

with just one natural witness. It describes the times when the anchored pattern 10011 appears at sites $0, \dots, 4$, equivalently these cycle returns. It does not output every intermediate complete configuration. That narrower relation explains its smaller cost. The return gaps grow without bound; their nonperiodicity also separates this quartic representation from a claim of timed Presburger definability.

100 Verification and mathematical provenance

100.1 Exact implementation and reproducibility

The release contains the standalone source of this report, its PDF, exact Python code, machine-readable expanded polynomials, and deterministic test receipts. The code uses only standard-library integer and rational arithmetic. Charts and sparse polynomials snapshot mutable constructor inputs. Boolean and floating-point coefficients, malformed degrees and arities, and negative natural witnesses are rejected before arithmetic. There are no approximate-zero tests or pre-validation value-keyed caches.

The chart tests perform the following finite exact checks.

Check	Count
Direct binary CA states	5,208
Complete eight-witness cube evaluations across both forms	13,122
Wrong output rejections	1,920
Rational-coefficient single-chart evaluations	250
Malformed input rejections	40
Immutable snapshot checks	5
Strict affine sorting refinements	10
Original-frame quadratic shifts	160
Large-integer nonnegativity checks	200
Canonical signed external pairs	50
Uniform-gap orbit certificate comparisons	1,456
Uniform versus fixed expanded-polynomial comparisons	2,560

The separately frozen four-witness suite checks 5,208 direct CA states, 26,040 wrong-output rejections, 3,125 full four-witness cube evaluations, 122 valid-cube phase comparisons, and 32 malformed-input rejections. The selector-elimination suite checks 2,187 seven-witness cube points, 872 uniform-gap orbit comparisons, 10,935 uniform/fixed cube comparisons, 18 one-chart no-selector cases, two parameter-free cases, and 14 malformed inputs. The refinements were mathematically reviewed and added in separate modules; the original compiler and fixtures remain byte-identical.

A missing-gate counterexample is also exercised: deleting inactive-chart gates admits a false witness, whereas the actual compiler rejects it. The finite cube includes invalid selectors and nonzero inactive branches. The separately included binary-rule suite verifies all conservation-potential identities, compares its component and radius-table rules, and checks a larger collection of direct trajectories. These computations supplement the exact arguments; they do not prove the normal form or decide injectivity of an arbitrary chart description.

Run `./run-tests.sh` from the extracted bundle. The script checks its manifest, reruns all four suites, and requires byte-identical regenerated JSON fixtures. Run `./build.sh` to compile the standalone LaTeX source with pdfLaTeX. Detailed replay and visual-review scope is recorded in the verification files. No network access or external service is needed.

[write] **The release in this report.** These are the delivered names. The programs of `code/phase/` and the top-level `build.sh` and `run-tests.sh` are shipped as `code/18-timed-quartics-*`, the JSON fixtures, receipts and verification records as `data/18-timed-quartics-*`, and `verification/independent-audit.md` as `18-timed-quartics-independent-audit.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README, checksum ledger and

manifest checker `verify_manifest.py`, and the three files of `code/` that are byte copies of source 17's (`test_binary_expanding_shuttle.py`, `binary-radius6-conservation-certificate.json` and `binary-shuttle-test-results.json`, shipped once as `code/17-four-mass-test_binary_expanding_shuttle.py` and `data/17-four-mass-*`). `run-tests.sh` needs the delivered layout; the README gives the rerun recipe. At placement on Windows the binary suite and the selector-elimination suite passed and their regenerated fixtures were identical to the delivered ones apart from line ends, the two exhaustive cube suites exceeded the 170-second cap and were not run to completion, and the binary test rewrote the sealed conservation certificate with CRLF line ends, after which the manifest check failed, as for source 17 (Section 92).

100.2 Audit correction and boundaries of the result

Independent mathematical review covered the chart partition, compiler, ledgers, binary formulas, and SOS refinements, conditional on the companion normal form. Test reruns establish reproducibility, not new algebraic proofs. A preliminary small-cube expectation was corrected from twelve roots to fourteen; the compiler and theorem were unchanged.

The following boundaries remain:

- The effective bijective low-degree charts, not decidability alone, supply the quartic construction
- Forgetting time can lose uniqueness; a stationary orbit repeats forever
- No general uniform-input arity bound or general single-fold or finite-fold Diophantine conclusion, including for MRDP, is asserted
- Only the four-witness binary specialization is proved exact for nonnegative real witnesses at integer external inputs. The generic compiler makes no such claim; unrestricted real witnesses can give spurious zeros

Further directions. Implement the effective rule-to-chart preprocessing, characterize when untimed projection retains canonical witnesses, and optimize arithmetic evaluation costs. These are research directions, not literature-level open-problem assertions or witness-minimality claims.

100.3 Attribution and classical context

The normal form and binary rule come from unchanged companion report 12 [source 17, Part VI]. Its lower-particle precursors include Kong's explicitly binary thesis [41] and Imai's project report [42]; neither is attributed this weighted-alphabet four-mass normal form or compiler. Ginsburg and Spanier [7] supply the classical Presburger/semilinear context. The displayed compiler proof is elementary algebra. No novelty, priority, or current open-problem status is inferred from these comparisons or a literature search failure.

Part VII

Mass at most four in every dimension: orbit geometry, counting and a binary planar shuttle (sources 19, 20 and 21)

[write] Part VII prints three AI-assisted research manuscripts of ProveIt’s incoming reports, batch 82 (arrival db37d18c8), all dated 3 October 2026 and added on that day. Each is a numbered report of one research pipeline: *source 19* is batch-82 manuscript 06, the pipeline’s Report 29; *source 20* is manuscript 17, Report 30; *source 21* is manuscript 01, Report 31. Their author lines are “Research report 29”, “Research report 30” and “Research report 31”; they name no assistant and no repository commit. The three numbers of each manuscript differ; brackets translate the pipeline’s report numbers wherever they occur (Report 12 is source 17, addendum 13 is source 18, Reports 29–31 are sources 19–21; Reports 16 and 26 are batch-82 manuscripts placed in the collection’s report `five-particle-binary-automata`, where Report 16 is printed as Part II).

The three form one chain on top of Part VI. Source 19 lifts source 17’s four-mass decision theorem from $\mathbb{Z}$ to every $\mathbb{Z}^d$ (Theorem 101.1), proving the case $d = 1$ again by the same architecture (notes in its Sections 101–111), and its Appendix I lifts source 18’s fixed-input timed quartics to $\mathbb{Z}^d$. Source 20 draws quantitative orbit geometry out of source 19’s normal form: semilinear visited sets, quasipolynomial box counts of sharp degree, quadratic clocks, first arrivals, and the nonsemilinear effect of drift, with explicit weighted planar automata and canonical first-visit quartics. Source 21 realizes source 20’s planar wedge and drift regime by an ordinary binary rule, built by lifting source 17’s binary shuttle into the plane. Each pins its predecessors by SHA-256 (source 19 pins source 17’s L^AT_EX source, source 18’s and those of Reports 16 and 26; source 20 pins source 19’s; source 21 pins those of sources 17, 18 and 20), and the pins were verified at placement. Source 19’s sharp four-versus-five boundary is conditional on Reports 16 and 26, as it says itself (note after Corollary 101.2).

[write] **Literature.** Sources 19 and 20 compare the encounter normal form with pebble automata (Delorme and Mazoyer [49], Lacalle and Gajardo [50], Blum and Sakoda [51]), plane-walking automata (Salo and Törmä [52]), two-dimensional reversible number-conserving universality (Morita, Tojima, Imai and Ogiro [53]) and Schaeffer’s physically universal automaton [54], besides Kong’s thesis [41] and the Alhazov–Imai abstract [24] already credited in Part VI; source 20 uses Woods’ Presburger counting theorems [55]. These pebble-automaton, plane-walking and counting references are new to the collection. Several were checked only through publisher abstracts, as the sources say, and none of the three claims novelty or priority.

[write] **Review status.** The Hilbert’s-tenth programme has reviewed source 19 (review `_dimension_independent_four_mass.md`, commit 7b547f686; Section 116) and source 21 (review `_binary_planar_four_particle31.md`, commit 16f50dcc6; Section 139), and made a scoped intake of source 20’s original-frame certificate, with a follow-up on its timed dependency in source 19 (review `_original_frame_first_hit30_intake.md` and `_original_frame_first_hit30_fixture.md`, commits c875bad40 and 5b4c091e6; Appendix K); source 18 has not been reviewed. Section 1.19 gives the relation to the rest of the report, and Section 1.20 collects what the four batch-82 sources do not claim.

Source 19 (batch-82 manuscript 06): Dimension independent particle thresholds

Effective reachability at mass four and the inherited mass five boundary, “Research report 29”, 3 October 2026. [write] From the archive `Dimension_Independent_Particle_Thresholds_Package.zip`; its labels carry the prefix `smc:zd:`, its sections are Sections 101–117 here, and its two appendices are Appendices I and J, after Part III’s. The Hilbert’s-tenth programme has reviewed its proof (note at the end of Section 116). Its abstract follows; the bracketed notes in it are [write].

For a translation invariant finite range cellular automaton on $\mathbb{Z}^d$, with a finite positively weighted alphabet, a unique weight zero vacuum, and at most one weight one symbol, exact finite configuration reachability and finite pattern occurrence are decidable from every input of mass at most four. The result is uniform in the finite dimension and does not require reversibility. A mass two packet has finitely many normalized shapes. Its successful flights toward a stationary unit follow finitely many exact lattice rays. Two consecutive successful flights in nonparallel directions force the marker separation into an effectively computable finite set. Outside that set, the dynamics reduces to finite control and one additive scalar gap, with the vector anchor retained as output. Expanding cycles have quadratic section times; a drift cutoff restores anchored observations in the original frame. We give the full effective proof, all weighted and no unit cases, a decision algorithm, and reproducible finite arithmetic checks. An appendix extends Report 13’s [source 18’s] fixed input timed quartic certificate compiler using canonical higher dimensional phase charts. The previously established one dimensional five particle universal construction embeds linewise into every finite dimension, giving the inherited sharp four versus five boundary [conditional on batch-82 Reports 16 and 26, which this report does not print; note after Corollary 101.2]. This is an ordinary mathematical proof, not a formalization, an implemented general rule compiler, or a claim of novelty or priority.

101 Statement and scope

The mass of a finite configuration $c: \mathbb{Z}^d \rightarrow A$ is

$$\mathbf{m}(c) = \sum_{x \in \mathbb{Z}^d} w(c(x)).$$

Here A is finite, $0 \in A$ is the unique weight zero symbol, and every other symbol has a positive integer weight. A finite configuration equals 0 outside a finite support. A cellular automaton (CA) is deterministic, translation equivariant, and has a finite neighborhood. Conservation below means conservation of this finite total mass for every finite configuration. Write $\tau_v c(x) = c(x - v)$.

Theorem 101.1 (Uniform finite input decision theorem). *Let $d \geq 1$ be finite. Let F be a CA on $\mathbb{Z}^d$ with a quiescent vacuum and the weights just specified. Assume mass conservation and at most one weight one symbol. Given its finite local rule, a finite neighborhood bound, the weights, and a finite input c of mass at most four, there is an effective procedure deciding each of the following:*

1. *whether $F^t(c) = b$ for a specified finite configuration b and some $t \in \mathbb{N}$;*
2. *whether $F^t(c) = \tau_v b$ for some $t \in \mathbb{N}$ and $v \in \mathbb{Z}^d$;*
3. *whether a specified finite pattern occurs in its prescribed window at some time;*
4. *whether that pattern occurs at some translated window at some time.*

Pattern zeros are enforced. Reversibility is not required. The rule hypotheses are promises; the procedure need not certify conservation for an arbitrary submitted rule.

[write] [For $d = 1$ this is Theorem 83.1 of Part VI (source 17, the pipeline’s “Report 12”), whose proof the present one generalizes step by step: the stationary-unit frame, bounded packets, the

tagged seed library with B fixed first, the thresholds N before W , one additive gap, the dispatch over the five mass partitions, termination in three forms, and the drift cutoff. Brackets below mark each place; at $d = 1$ the steps there are a re-proof by the same architecture, and no novelty is claimed for them. The higher-dimensional ingredients are new: contact through actual occupied sites rather than a bounding box (Figure 6), exact contact rays with a Bezout functional and residues (Section 106), the finite nonparallel switch set (Lemma 107.1), compact triples that miss a remote unit (Section 105), and a transverse offset folded into the finite mode (Section 108). Source 19 itself cites source 17 only in Section 114 and in its lineage appendix.]

The proof constructs a fixed input orbit description. In the frame fixing isolated mass one particles, its untimed part is effectively Presburger definable. Its eventual form is independent finite phase objects, a whole configuration periodic up to translation, or an expanding one counter shuttle. In the last case exact elapsed time is retained, but is quadratic in the shuttle cycle index. No assertion of uniform timed Presburger definability is made.

The input dimension may be part of the finite rule description. Every construction below is a finite enumeration or effective integer calculation for that dimension. No useful complexity bound is asserted. In particular, “effective” does not mean that the normalized state libraries or resulting formula are small.

Corollary 101.2 (Inherited sharp boundary). *In the effective finite input and finite observation simulation setting of Reports 16 and 26 [batch-82 reports of the same pipeline; note below], the sharp conserved mass boundary is four versus five in every finite dimension: mass at most four has the decision property of Theorem 101.1, and the previously established binary five particle universal construction embeds in every $\mathbb{Z}^d$.*

The upper part inherits the source construction and compiler proofs in those reports; this report proves the higher dimensional lower part and the linewise inheritance argument. Section 114 states the distinction precisely. It does not transfer undecidability to an arbitrary observation convention lacking the inherited effective encoding and halt observation.

[write] [The upper half of Corollary 101.2 is not proved in this report. It is the linewise embedding (Section 114) of the binary five-particle universal construction of batch-82 Reports 16 and 26 of the same research pipeline, which source 19 pins by SHA-256 (Appendix J). Both were placed in the collection’s report `five-particle-binary-automata` (file prefixes `16-literal-source-` and `26-parallel-involutions-`; placement commits `bca6383e9` and `7d2b1b245`). Report 16 is printed there as Part II (labels `fpa:ls:`); Part I of that report, Reports 14 and 15, proves with source 17 that five particles are the exact one-dimensional binary threshold (its Corollary `fpa:bc:cor:sharp`). Report 26 is placed there but not yet written. The four-versus-five boundary in every dimension is therefore sharp only conditionally on those releases having their stated properties. The Hilbert’s-tenth programme has reviewed both. Its review of Report 16 (`review_literal_universal_reversible_source16.md`, commit `69730d3e8`) [61] found no source-interface or arithmetic contradiction; it records that Report 16 closes the literal-source and clean-nonblocking obligations of the earlier five-particle compiler, keeps the published Neary–Woods universality theorem and the earlier compiler theorems as explicit dependencies, and notes that Report 16’s own threshold claim inherits source 17’s mass-four theorem, which it does not recertify. Its review of Reports 26–28 (`review_parallel_particle_reports.md`, commit `fb7e3cb47`) [57] found the two-involution mechanism of Report 26 sound but states that it did not reconstruct the complete source compiler and that Report 26’s full source binding remains outside its replay. The lower half, decidability at mass at most four in every finite dimension, is proved in full here. Source 19’s own adversarial review says the same: “the sharpness statement remains conditional on the cited five-particle source

releases having their stated properties” (19-mass-four-zd-independent-review.md).]

102 Stationary unit frame and independent components

Use the maximum norm on $\mathbb{Z}^d$. A radius bound $R \geq 0$ means that the output at x depends only on $x + [-R, R]^d$. For a nonempty finite configuration,

$$\text{diam}(c) = \max\{\|x - y\|_\infty : x, y \in \text{supp}(c)\}.$$

Normalize it by subtracting its lexicographically first occupied site. Translation preserves lexicographic order, so this normalization is translation compatible. Bounded diameter and bounded mass leave only finitely many normalized configurations, and their successors can be computed by evaluating the rule on a finite neighborhood of their support.

Suppose the unique possible weight one symbol u exists. An isolated u must evolve to exactly one u : positivity and conservation leave no other output of mass one. Quiescence and finite range confine its displacement δ to $\|\delta\|_\infty \leq R$. Put

$$G = \tau_{-\delta}F, \quad S = \max(1, R + \|\delta\|_\infty).$$

Then G has radius at most S , fixes each isolated u , and

$$F^t = \tau_{t\delta}G^t. \tag{102.1}$$

The case with no unit symbol is treated separately in Section 112.

Join occupied sites at distance at most $2S$. A connected component of mass m has at most m occupied sites and therefore diameter at most $2S(m - 1)$. Components mutually farther than $2S$ have disjoint radius S influence neighborhoods. Their one step evolutions are independent and have disjoint output supports. By induction, the independent evolutions agree with the full evolution through the first configuration at which two such supports have distance at most $2S$. The next step is computed using the full rule. A geometric contact need not imply a nontrivial collision, but processing it exactly is harmless.

This observation is used with *actual occupied sites*. In several dimensions an expanded bounding box is generally larger than the union of the contact boxes of those sites. Replacing the latter by the former is incorrect.

103 Bounded mass two packets

[write] [At $d = 1$ the lemma below is source 17’s Lemma 84.1, which restates source 16’s no-split Lemma 75.1; the coordinatewise intersection of two radius- S cubes is the $\mathbb{Z}^d$ form of that argument. With one weight-one symbol and $d = 1$, the mass-two case is also a special case of Part IV’s Theorem 44.1, proved there for any number of weight-one symbols.]

Lemma 103.1. *A mass two configuration of diameter at most $2S$ remains within that diameter under G . It has an effectively computable finite transient and a periodic profile up to a vector translation.*

Proof. A weight two singleton produces support in its radius S cube, of diameter at most $2S$. Otherwise the configuration is two units, translated to 0 and v , with $\|v\|_\infty \leq 2S$.

If $\|v\|_\infty > S$, each original occupied site sees only its own unit and stays occupied by u . These two sites already exhaust the total mass, so the configuration is fixed. If $\|v\|_\infty \leq S$, any newly occupied site other than those two must see both units: a neighborhood seeing zero units gives vacuum, and a neighborhood seeing just one gives the stationary singleton output. The intersection of the two radius S cubes contains both original sites, and its coordinate widths are $2S - |v_i| \leq 2S$. The entire successor support lies in that intersection.

There are finitely many normalized mass two shapes of this diameter. Each has one next normalized shape and a translation increment. A repeated shape gives the finite transient followed, after resetting the time origin, by

$$P(r + kp) = kD + P_r, \quad 0 \leq r < p, \quad k \in \mathbb{N}, \quad (103.1)$$

where $D \in \mathbb{Z}^d$, the period $p \geq 1$, and all labeled phase shapes P_r are explicit. This finite graph can be enumerated. $\square$

A pair of units at distance greater than $2S$ is fixed. The lemma includes all weight two singleton labels and two unit shapes, so it is not restricted to binary alphabets. The word “packet” will denote such a bounded mass two object, and “dimer” will be used synonymously even when its support is a singleton.

104 Mass three profiles and a finite encounter library

Lemma 104.1. *Every fixed mass three input has an effective finite prefix followed by either a whole configuration periodic up to a vector translation, or an independent stationary unit and a finite phase mass two packet. Three stationary units are included as a constant case. Mass at most three consequently has a fixed input timed Presburger description.*

Proof. Take as the core every normalized mass three configuration of diameter at most $4S$. Outside this finite core, the $2S$ component partition is $2 + 1$ or $1 + 1 + 1$. The latter is fixed. In the former case Lemma 103.1 gives the dimer’s finite transient and phases. In an eventual phase its occupied coordinates are $p + kD$ for a finite list of offsets p , while the other unit is stationary.

Entry of the complete configuration into the diameter $4S$ core is a finite conjunction of coordinatewise linear inequalities in k . Their common integer solution set is an interval, possibly empty or unbounded. Compute its first nonnegative integer, compare the corresponding actual times over all phases, and also check the finite transient. If no core entry exists, there is no later dimer unit contact: such a contact would be connected and have diameter at most $4S$.

From each normalized core state, take one exact step and then its just computed first return when needed. The result is a finite deterministic graph of core returns, labeled by finite elapsed times and translation vectors, with possible terminal independent tails. Repetition of a normalized core state gives equality of two complete configurations up to translation. Determinism and equivariance propagate this equality to all future times, including every intermediate excursion. The corresponding finite period can be enumerated even if enormous.

In the independent case, take a common period for the finite phase objects. In either case coordinates and time become affine in one nonnegative integer after finitely many phases. Restoring δt keeps them affine. $\square$

[write] [In $d = 1$, source 16 proves the stronger uniform form, Theorem 74.1 (uniform in all positions and the time), and source 17’s Lemma 84.2 is the fixed-input form used here. Lemma 104.1 is weaker in $d = 1$ but holds in every $\mathbb{Z}^d$, and source 19 needs only the fixed-input form.]

[write] [At $d = 1$ the tagged seed library and its bound B are source 17’s (Section 85).] Enumerate every normalized *connected* mass three seed. Its diameter is at most $4S$, so this is a finite effective list. A dimer’s first contact with a unit is such a seed. Include a finite bookkeeping tag for the incoming target unit’s coordinate. The tag records the geometric role of that coordinate; it introduces neither a new CA symbol nor a persistent physical identity.

For each tagged seed, compute the preceding lemma’s profile and extend its finite prefix to one of two checkpoints:

Compact

The whole triple is periodic up to translation, with finitely many bounded phase representatives.

Emission

A stationary unit remains and the independent dimer has nonzero drift D . From the checkpoint onward it is purely periodic up to D and never comes within $2S$ of the residual unit.

If an independent dimer has $D = 0$, the complete triple is eventually periodic and belongs to Compact. If $D \neq 0$, choose a coordinate with nonzero drift. Its linear escape allows the finite prefix to be extended effectively until every later phase remains beyond the residual unit’s contact interval. This produces Emission.

Use a common seed anchor frame: normalize each triple seed with occupied anchor 0, and measure every occupied coordinate through its checkpoint in that frame. Choose B to bound all these coordinates, every compact phase representative, and every emission first period representative, including residual unit offsets. On reanchoring an emission at its residual unit, include all resulting finite difference offsets as well, enlarging B if necessary. The list is finite. This B is computed solely from the original connected mass three seeds, *before* any mass four core is chosen. Compact phase diameters are at most $2B$.

105 Compact triples reset or escape

Place a tagged library seed at an anchor a and a fourth unit at b , with $\|b - a\|_\infty > B + 2S$. Its entire precomputed prefix through the checkpoint is independent of the fourth unit. At an emission, the remote unit remains fixed; the new residual unit is at a seed dependent fixed offset from the target anchor.

For a compact checkpoint, compute the first contact with the fourth unit by enumerating the compact phases and the actual occupied sites in each. For each candidate site the distance at most $2S$ conditions are coordinatewise linear inequalities in the period count. Check the finite prefix separately. A translating triple can miss the remote unit in higher dimensions. If it does, this is a terminal independent profile. If it hits, the full configuration has diameter at most $2B + 2S$.

Thus an arbitrarily long trip of a compact triple either ends in a bounded full encounter or remains independent forever. It cannot carry two independently changing unbounded gaps through a continuing interaction. This includes triples that detach and recombine during their periodic orbit: their entire period, not just a section shape, is retained in the compact profile.

106 Exact contact rays

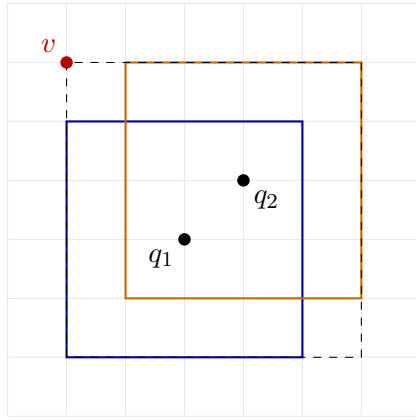
At an emission checkpoint put the source residual unit at 0 and write the dimer profile as

$$Q(r + kp) = kD + Q_r, \quad D \neq 0, \quad 0 \leq r < p, \quad k \in \mathbb{N}.$$

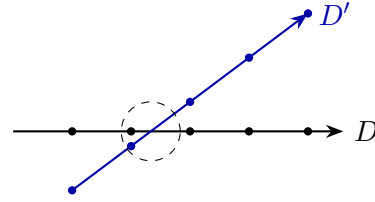
It never meets its source again in this isolated profile. For a target unit at v , contact at time $r + pk$ occurs exactly when

$$v = a + kD, \quad a \in Q_r + ([-2S, 2S]^d \cap \mathbb{Z}^d). \quad (106.1)$$

The notation denotes the union over the actual occupied sites of Q_r , with the phase r retained. All offsets a form a finite effective set. Repeated witnesses are harmless. Minimize the *actual integer time* $r + pk$ over all valid witnesses. If there is no witness with $k \geq 0$, the packet and both markers remain independent forever.



Actual contact boxes
leave holes in their bounding hull



Nonparallel flight equations
have at most one pair of counts

Figure 6: Left: for $S = 1$, sites $(0, 0)$, $(1, 1)$ do not contact $(-2, 3)$, although their expanded bounding hull contains it. Right: schematic arithmetic rays; the finite exceptional region is computed from exact determinants, not from the drawing's angle or scale.

To express the successful rays in finite control, write $D = m\nu$, where ν is the primitive integer vector on its unoriented line, with first nonzero coordinate positive, and $m \in \mathbb{Z} \setminus \{0\}$. By Bezout's identity compute an integer linear functional λ with $\lambda(\nu) = 1$. Every vector has the unique decomposition

$$v = n\nu + z, \quad n = \lambda(v), \quad z = v - \lambda(v)\nu, \quad \lambda(z) = 0.$$

For each contact offset define $\ell = \lambda(a)$ and $z_a = a - \ell\nu$. Equation (106.1) becomes

$$z = z_a, \quad n \equiv \ell \pmod{|m|}, \quad k = (n - \ell)/m \geq 0, \quad (106.2)$$

$$t = (p/m)n + r - (p/m)\ell. \quad (106.3)$$

For $|n| > \max_a |\lambda(a)|$ on the drift facing sign, all eligible counts are nonnegative. On the opposite sign there is no candidate. On the eligible sign, the transverse offset, residue, and phase identify the possible witnesses and the earliest constant term in (106.3).

Relative to the target, the entire contact dimer shape is $Q_r - a$. Hence its tagged seed is also determined by these finite data. If several witnesses attain the same actual time they describe

the same complete configuration; a fixed witness ordering removes bookkeeping ambiguity. All intermediate flight phases remain available: their coordinates are affine in the marker vector and one period variable, with a linear first contact cutoff. Every local encounter prefix phase is retained too.

107 The effective nonparallel exception

Let a dimer fly from marker A to marker B . At the next emission the contacted marker is $B + c$, with c in a finite library of marker update vectors, while the old source A remains stationary. Thus the incoming marker vector $v = B - A$ and outgoing target vector v' satisfy

$$v' = A - (B + c) = -v - c.$$

If both flights succeed, with nonzero drifts D, D' , their contact ray equations give

$$v = a + kD, \quad v' = a' + k'D', \quad kD + k'D' = -(a + a' + c), \quad k, k' \in \mathbb{N}. \quad (107.1)$$

The offsets a, a', c range over finite computable lists.

Lemma 107.1 (Finite nonparallel switch set). *For all nonparallel pairs of emission drifts, the marker vectors in (107.1) form an effectively computable finite set.*

Proof. Choose coordinates i, j with

$$\Delta_{ij} = D_i D'_j - D_j D'_i \neq 0, \quad H = -(a + a' + c).$$

The only possible rational solution of the two selected equations is

$$k = \frac{H_i D'_j - H_j D'_i}{\Delta_{ij}}, \quad k' = \frac{D_i H_j - D_j H_i}{\Delta_{ij}}. \quad (107.2)$$

Retain it only when both entries are nonnegative integers and every remaining coordinate equation in (107.1) holds. Compute $v = a + kD$. There are finitely many drift and offset inputs, so this enumerates the full finite exceptional set. $\square$

This is an exact calculation, rather than a qualitative appeal to divergent directions. In particular, a bound depending only on the small offsets is insufficient. Let $M = 10^6$ and set

$$D = (M, 1), \quad D' = (-M - 1, -1), \quad a = a' = 0, \quad c = (0, -1).$$

The determinant is 1, while (107.1) has the solution $k = M + 1, k' = M$. The exceptional vector $v = (M + 1)D$ has

$$\|v\|_\infty = M^2 + M = 1\,000\,001\,000\,000.$$

Higher dimensional examples follow by adjoining zero coordinates. This shows why nearly parallel drift vectors can create a very large but finite exception.

Outside a computable bound every continuing sequence of successful dimer exchanges uses one fixed *unoriented* rational line. Opposite orientations on that line are permitted. A nonparallel outgoing packet that misses the old marker is terminal, not another counter mode. A compact triple is already covered by Section 105.

108 Finite control and one scalar gap

For every primitive line ν in the emission library, choose one Bezout functional λ . Order the two markers by their λ coordinates, calling them L and R , and write

$$R - L = x\nu + z, \quad x = \lambda(R - L) > 0, \quad \lambda(z) = 0. \quad (108.1)$$

This is a bookkeeping order for the chosen line, not a global linear order of $\mathbb{Z}^d$. For a successful long flight the transverse offset belongs to the finite set of $\pm(a - \lambda(a)\nu)$ from contact offsets on this line. Its sign depends on the source side.

The finite mode records ν , the source side, the complete emission profile and launch shape, the transverse offset, and any needed tagged seed information. The absolute vector anchor L is retained as output; it does not affect the next rule application.

108.1 Noncircular threshold selection

[write] [At $d = 1$ these are source 17's thresholds (85.1), $N = 4B + 12S$ and $W = N + 2B$, also chosen in that order; here N must in addition exceed the Bezout and exceptional-switch quantities.] Choose an integer N larger than all required finite quantities:

1. every $|\lambda(a)|$ in the contact lists, to validate the large gap residue formulas;
2. every $|\lambda(c)|$ for marker shifts, so the marker order cannot reverse;
3. every $|\lambda(v)|$ for exceptional switch vectors, for the relevant incoming functionals;
4. a safety bound separating the remote marker from every local seed prefix.

For the fourth item it suffices, with further finite launch offsets included if needed, to take

$$N > \max_{\lambda} \|\lambda\|_1 (B + 10S + 1),$$

since $|\lambda(v)| \leq \|\lambda\|_1 \|v\|_{\infty}$. In particular $x > N$ makes the remote marker safely farther than $B + 6S$ from the incoming seed anchor. All data used here were already computed from the original seed and phase libraries.

Now choose W larger than $6S$, $B + 6S$, $2B + 2S$, and the diameter of every successful emission launch with $0 < x \leq N$. This last list is finite: the lines and transverse offsets are finite, the scalar ranges over a finite interval, and all launch shapes have fixed finite offsets. More explicitly, a successful vector satisfies $n = \lambda(a) + km$ with $k \geq 0$. If $|n| \leq N$, then

$$0 \leq k \leq \frac{N + |\lambda(a)|}{|m|}.$$

Enumerate those finitely many integers for every offset and profile, and include their full launch diameters in W . This also includes $n = 0$ and every degenerate marker order. Thus W is computed *after* N , without enlarging and recomputing the seed library.

Call a configuration of diameter at most W ordinary. At designated emission checkpoints use the live representation when the next cross marker flight succeeds and $x > N$. Otherwise use an ordinary state if its diameter is at most W , or an explicit terminal independent profile if the flight misses. At other dispatch states test ordinary first. Ordinary and tagged live states may overlap; the dispatch convention is part of the algorithm, not an extra state of the CA.

108.2 The transition formula

On a live to live edge Lemma 107.1 prevents a nonparallel line switch. Only the contacted marker changes, by one vector c . Since the order remains fixed,

$$x' = x + c_0, \quad L' = L + e, \quad z' = z + c_\perp,$$

with constants from finite effective lists. Fold z into the mode. Let M_0 be the least common multiple of the positive numbers $|m|$ in all emission drift decompositions. Equations (106.2)–(106.3), plus the fixed local prefix duration, give

$$q' = f(q, x \bmod M_0), \quad x' = x + c_0, \quad L' = L + e, \quad h = \alpha x + \beta, \quad (108.2)$$

where $\alpha > 0$ is rational and h is a positive integer on the permitted residue. Here $\alpha = p/|m|$ for the flight. Positivity follows because live targets are safely distant. [write] [At $d = 1$ this is source 17's Lemma 86.1, equation (86.1), with the marker gap D in the role of the scalar x .]

An updated successful gap at most N has bounded full diameter and resets to ordinary. An emission that misses is terminal. A compact outcome is analyzed by Section 105; it then resets or is terminal. The *choice to continue live or exit through a compact outcome or emission miss*, apart from bounded lower guard effects, depends only on mode and residue. Whether a compact outcome subsequently hits the remote marker may depend on the actual gap, and is calculated separately. It is not a live edge and must not be treated as residue determined.

There is no further control clock: the CA is time independent, the units are stationary in G , and packet phase is already recorded. Neither absolute anchor nor accumulated time is queried by the control. The vector separation is consequently not a pair of freely testable scalar counters.

109 Complete dispatch from arbitrary inputs

For total mass four the possible $2S$ component mass partitions are

$$4, \quad 3 + 1, \quad 2 + 2, \quad 2 + 1 + 1, \quad 1 + 1 + 1 + 1.$$

Lower masses were covered above. Each mass partition is handled as follows. [write] [At $d = 1$ the same five partitions are source 17's Lemma 86.2.]

Connected mass four

Its diameter is at most $6S$, so it is ordinary. This includes a weight four singleton and all weighted connected shapes.

Four unit components

All are stationary.

Mass three plus one

Outside ordinary, the connected triple has diameter at most $4S$. Since $W > B + 6S$, the remote unit is far enough from the chosen triple seed anchor for the library prefix to be independent. Apply the tagged seed library and Section 105, or enter an emission checkpoint.

Two dimers

Combine their finite transients and then take a common phase period. At each common phase the relative occupied coordinates are affine in one nonnegative integer period count. Their first distance

at most $2S$ event is the earliest solution among finitely many coordinatewise interval tests, one for each pair of occupied sites. If absent, the tail is independent. At contact, each packet has diameter at most $2S$, so the full configuration has diameter at most $6S$ and is ordinary.

A dimer and two units

Use the exact contact calculation against each unit, including any finite dimer transient. If the initial dimer has eventual drift $D = 0$, inspect its transient and one entire positioned period against both units; the nonzero drift ray formula is unnecessary. If neither is hit, the tail is independent. A first simultaneous contact with both yields a connected mass four state of diameter at most $6S$. Otherwise the contact forms a tagged triple seed. If the full state is ordinary, reset. If not, the remaining unit satisfies the safety separation and the library applies.

Actual times are compared across all sites, phases, and possible target units. Ties are retained at the level of complete configurations. No argument assumes a packet must hit a marker merely because one coordinate moves toward it.

From an ordinary state take at least one exact CA step before invoking dispatch. This yields a positive time edge from one of finitely many normalized ordinary states to another ordinary state, a live checkpoint, or an explicit terminal profile. All intervening configurations are kept as finite phase descriptions. A live edge has a positive length flight. Thus the section graph has no artificial zero time loops.

110 Termination and the complete orbit form

Enlarge the finite mode by $x \bmod M_0$. Follow the section machine from the given input. A repeat of a normalized ordinary state means two complete configurations are translates, so the whole future is periodic up to translation, at every intermediate time.

Within an uninterrupted live excursion, the unoriented line is fixed and there are finitely many enlarged modes. At a repeated mode let Δ be the net scalar change around the recorded cycle. The residue has repeated, hence Δ is divisible by M_0 and the same residue itinerary is available on later repetitions.

$$\Delta = 0$$

The whole launch configuration repeats up to the anchor translation. Determinism gives a translated periodic future.

$$\Delta > 0$$

Every intermediate lower guard valid in the recorded cycle remains valid in all later repetitions. The same itinerary persists forever, giving an expanding shuttle.

$$\Delta < 0$$

Compute the greatest number of complete repetitions for which every intermediate scalar stays above N . This is a finite collection of linear inequalities. Execute or accelerate those repetitions. The remaining finite cycle prefix exits the live region to ordinary or terminal. An above threshold mode dependent exit would already have appeared during the recorded cycle, because the enlarged control itinerary is the same.

The decreasing case can be simulated without acceleration: there are finitely many repetitions, however large that number is. Acceleration merely gives a compact exact description of the finite prefix.

There are only finitely many normalized ordinary states. Therefore the analysis terminates with a finite prefix and one of three eventual forms:

- A. independent finite phase objects;
- B. the entire configuration periodic up to translation;
- C. an expanding shuttle.

All described prefixes and phases are effective. This is an unbounded termination proof, not an inference from finite experiments. [write] [At $d = 1$ this is source 17's Proposition 87.1, with the same three cases for Δ .]

110.1 Expanding shuttle coordinates and time

Suppose the cycle in case C has $\Delta > 0$ and total anchor update $E \in \mathbb{Z}^d$. At its corresponding sections,

$$x_n = x_0 + n\Delta, \quad L_n = L_0 + nE, \quad n \in \mathbb{N}. \quad (110.1)$$

There are finitely many edges and local phases per cycle. During a local encounter prefix every occupied coordinate is affine in n . During a flight it is affine in (n, j) , where $j \in \mathbb{N}$ is the packet period count subject to linear first contact bounds. Labels are fixed within a phase. These descriptions include the whole orbit between sections. Their finite union is Presburger definable when elapsed time is forgotten.

The duration of every live edge is $\alpha x + \beta$. Its duration at corresponding cycles grows by $\alpha\Delta > 0$. Summing finitely many edges gives a positive affine cycle duration, so section times have the form

$$T_n = T_0 + A \frac{n(n-1)}{2} + Cn, \quad A > 0. \quad (110.2)$$

The effective rational coefficients take integral values on all natural cycle indices. The duration of cycle n is $An + C > 0$.

There are effective constants $K_0, K_1 \geq 0$ such that every stationary frame occupied point y in cycle n satisfies

$$\|y\|_\infty \leq K_0 + K_1 n. \quad (110.3)$$

Both markers are affine in n ; every local offset is bounded; the flight period count is bounded above by an affine function of n ; and every packet drift and phase offset is fixed finite data. Taking maxima over finitely many phases yields the bound. This reasoning requires neither coordinatewise monotonicity nor a line through the spatial origin.

111 Deciding observations and restoring translation

In cases A and B, a common period expresses all occupied coordinates and time affinely in a nonnegative integer, with finitely many phases and a finite transient. Exact targets and finite patterns are finite Presburger predicates on these coordinates. For example, a prescribed nonzero site must equal some occupied site with the prescribed label; a prescribed zero site must differ from every occupied site. There are at most four occupied sites and finitely many label possibilities. Exact configuration equality also requires that there be no additional occupied sites. Quantify an integer vector for translated observations. Restoring (102.1) preserves affine formulas here.

In case C, the finite union of the affine in (n, j) stationary frame phase formulas decides every *untimed* exact target and pattern query for G , anchored or translated. Translation invariant queries have the same answer for F and G by (102.1). When $\delta = 0$, this finishes all queries.

Suppose $\delta \neq 0$ and consider an anchored query for F . Choose a coordinate i with $\delta_i \neq 0$. Since $t \geq T_n$ throughout cycle n , equations (102.1) and (110.3) imply

$$y_i(F^t(c)) \geq \delta_i T_n - K_0 - K_1 n \quad \text{if } \delta_i > 0, \quad (111.1)$$

$$y_i(F^t(c)) \leq \delta_i T_n + K_0 + K_1 n \quad \text{if } \delta_i < 0. \quad (111.2)$$

The reversed inequality in the second line is essential. The nonzero quadratic leading term eventually dominates the linear term uniformly over every phase and every time in the cycle. [write] [At $d = 1$ this cutoff is source 17's Section 88.]

For any specified finite target window, compute n_* beyond which the appropriate bound lies permanently beyond that window. One may solve a rational quadratic inequality, or increase n until the bound is beyond the window and its forward difference points outward; discrete convexity then proves permanence. Consequently a nonempty exact target, or an anchored pattern prescribing any nonzero site, can occur only before T_{n_*} . That finite time interval can simply be simulated. Its possible enormous size does not affect decidability. A wholly zero anchored pattern occurs after the cutoff. An empty exact configuration is settled by mass conservation; mass zero input is handled immediately.

This completes the proof of Theorem 101.1 when a unit symbol exists. In particular, the proof restores absolute position without invoking unrestricted quadratic Diophantine solvability and without introducing an absolute position control counter.

112 Alphabets with no unit symbol

When no symbol has weight one, mass at most four occupies at most two sites. A singleton of weight two or three cannot split into two nonvacuum sites, so it is a finite state walker. Its label evolution and vector increments become periodic. A weight four singleton may split into two weight two walkers; at the split their separation is at most $2R$.

Take a finite normalized core of singleton and two site configurations of diameter at most $2\max(1, R)$. Outside it the only nontrivial case is two independent weight two walkers. Their finite transients and common eventual phases give coordinatewise affine inequalities for the first core encounter. Solve them exactly, or certify no encounter. A core return has a computable finite elapsed time and translation. Its finite core return graph gives an eventually translated periodic whole configuration or a terminal pair of independent walkers, exactly as in Lemma 104.1. These profiles have timed Presburger descriptions and decide all four observations.

Symbols of weight greater than four never appear at these total masses, by conservation and positivity. This also covers every missing small mass and weighted singleton case, completing the theorem.

113 The effective decision algorithm

The construction may be implemented eagerly or by computing only visited normalized states. It uses the promised rule hypotheses as follows.

1. Check mass zero and immediate label or mass impossibilities. If there is no unit symbol, use Section 112.
2. Compute the isolated unit displacement δ , the stationary unit rule G , and S .
3. Enumerate normalized mass two states of diameter at most $2S$ and compute their finite profiles.
4. Enumerate normalized mass three states of diameter at most $4S$, build their return graph, and compute the tagged connected seed library, its prefixes, compact and emission profiles, and B .
5. Enumerate actual site contact offsets and phases. Compute primitive drift directions and Bezout functionals.
6. Enumerate marker shifts and solve (107.1) for every nonparallel profile and offset combination, retaining only valid nonnegative integral solutions in all coordinates.
7. Select N and then W as in Section 108.1. Construct ordinary dispatch and finite live mode residue transitions as needed.
8. Follow Section 110 until case A, B, or C is established, retaining exact durations, vector anchors, and all intermediate phases.
9. Test the finite prefix and eventual profiles using Section 111, with the effective original frame cutoff when necessary.

Every enumeration is finite. Unbounded event searches reduce to one variable integer linear equations or inequalities, or to the proved one counter termination argument. Presburger decision is applied only after the explicit finite formulas have been constructed. No step replaces an unbounded question by an unjustified finite search.

114 The inherited mass five upper bound

Let H be the previously established one dimensional binary five particle universal CA. For $x = (x_1, \dots, x_d)$, define $\hat{H}$ by applying H independently on each line parallel to the first coordinate axis. It has the same finite alphabet and radius bound when the one dimensional neighborhood is regarded as a subset of $\mathbb{Z}^d$. A finite configuration meets finitely many such lines, and conservation on each line gives conservation of the finite total particle count.

If H is reversible, apply H^{-1} independently on the same lines. This is the inverse of $\hat{H}$. If $H = I_2 \circ I_1$ with full shift involutions I_1, I_2 , their linewise lifts remain involutions on the entire higher dimensional full shift, and preserve finite particle number when the original involutions do. An input supported on a single line has exactly its original one dimensional dynamics on that line; all other lines remain vacuum.

Therefore every effective input encoding, clean target or halt observation, exact clock, and source simulation proved for the chosen one dimensional construction is inherited on a fixed line. Reports 16 and 26 [batch 82; Report 16 is Part II of `five-particle-binary-automata`, Report 26 is placed there; note after Corollary 101.2] supply the relevant universal source and compilation statements. This paragraph does not independently reconstruct that source or materialize a universal rule array. Together with Theorem 101.1, it gives Corollary 101.2 in that inherited simulation setting.

Report 12's [source 17's, Proposition 89.2] binary four particle quadratic shuttle also embeds linewise. Hence decidability at four does *not* imply eventual timed semilinearity. Report 13

[source 18, Part VI] already treats fixed input timed quartic certificates in the one dimensional setting; it is a precursor, not a result reclassified here as a discovery.

The hypotheses delimit the conclusion. Multiple unit labels can give independent finite state walkers with different drifts, invalidating the stationary dimer argument. Negative weights, weight zero nonvacuum symbols, infinite alphabets, active vacuum, infinite input scenery, and time or position dependent rules are outside the theorem. Infinite marked axes or a free external tape cannot be silently supplied without changing the model.

115 Relationship to earlier literature

The literature check included close pebble and walker models as well as conservative cellular automata. It did not locate a primary theorem deciding reachability for the exact class in Theorem 101.1. That bounded search result supplies no novelty or priority conclusion. The accompanying literature record separates inspected full texts from publisher abstracts.

Delorme and Mazoyer study pebble automata on the unbounded plane and report a three pebble intrinsic universality threshold, with no intrinsically universal two pebble automaton [49]. Their publisher abstract was checked, but the full 2002 article was not retrieved. Their finite state head and passive movable pebbles are a close predecessor model. Intrinsic universality is distinct from the exact reachability question here. A naive unique unit encoding charges at least mass two for a stateful head and one for each pebble; this bookkeeping observation is an inference, not their theorem or an established complete encoding.

Lacalle and Gajardo analyze two distinguishable pebbles on a one symbol grid [50]. Their full preprint provides one pebble periodicity, geometric ray escape arguments, and real time two counter recognition of two pebble traces. Its conclusion explicitly extends its geometric argument to $\mathbb{Z}^d$. Trace recognition by a counter machine is not itself a reachability decision theorem. The present proof must additionally account for CA cluster splitting, fusion, control transfer, weighted labels, and synchronous intermediate phases. The transient bound in their local one pebble argument should not be applied to an arbitrary distant head and pebble start.

Blum and Sakoda report, with attribution to Sipser, that two planar finite automata and one pebble simulate a universal two counter machine [51]. This cautions against equating agent count with conserved mass: two independently stateful agents and a pebble have naive minimum unique unit mass $2 + 2 + 1 = 5$. Their discussion of marked axes likewise cannot provide a free finite mass background. Their two pebble nonuniversality statement is not substituted for our decision algorithm.

Salo and Törmä's plane walking model has independently controlled heads reading a separate supplied configuration [52]. Its sparse input and hierarchy results are relevant, but static marks carry information and must be counted in a conservative encoding. Independently controlled heads cannot automatically be represented by interchangeable weight one symbols.

Morita, Tojima, Imai, and Ogiro give genuinely two dimensional reversible number conserving universal constructions [53]. The accessible publisher abstract does not establish the present mass four or unique unit restriction. Schaeffer's number conserving physically universal CA has a diffusion theorem bounding the duration of finite configuration computation [54]; it is not a small finite seed Turing universality counterexample. Block phases and velocity tracks also need explicit encoding when comparing its model to an ordinary translation invariant CA.

Kong's thesis discusses the one dimensional binary small particle hierarchy and leaves four

particles open in its concluding discussion [41]. It neither proves the weighted higher dimensional theorem nor explicitly gives this uniform reachability procedure. The separately identifiable particle complexity paper is Alhazov and Imai [24]; its full construction was not independently reverified in this literature pass. For the inherited upper bound, the supplied Reports 16 and 26 [batch 82; Part II of, and placed in, `five-particle-binary-automata`] remain the actual proof dependencies.

116 Executable evidence and reproducibility

The package includes the unchanged arithmetic audit source and its deterministic expected receipt. It tests the formulas used in the proof, not the class of all CA rules. Normal and optimized Python runs have the same exactly typed receipt. The audit uses explicit failure checks, not optimization removable assertions.

Finite check	Count
First contact profiles in dimensions 1 to 3	600
Exact contact ray shifts exceeding 10^{60}	600
Tied earliest contact witness cases	30
Common clock two packet comparisons in dimensions 1 to 5	500
Nonparallel switch equations in dimensions 2 to 6	500
Two unit intersection box checks in dimensions 1 to 4	892
Original frame escape cutoffs	1,000
Expanded hull false positive regression	1
Nearly parallel exceptional separation	1 000 001 000 000

The final row is a norm, not a number of trials. Randomized finite cases use the fixed seed 20261003. Bounded direct traces are compared only inside their tested windows; they are not used to infer nonoccurrence beyond those windows. Huge ray shifts are tested by exact integer arithmetic.

The replay entrypoint verifies an independently fixed required inventory and the pinned scientific source digest before running any packaged Python helper. The full manifest checks every payload byte and rejects extra files and symlinks. Tests execute in fresh external temporary copies, in both normal and optimized modes. Their receipts are checked recursively with exact value *and type* equality against the expected schema, so booleans cannot masquerade as integer counts. The original scientific payload is checked before and after for identical bytes, permission modes, and modification times. Outputs are written outside the release directory.

A tamper regression exercises changed and resealed scientific code, missing or extra payloads, and symlink substitution. This is accidental corruption and dependency substitution protection around a trusted entrypoint, not cryptographic authentication against an adversary who replaces the entire package and all trust anchors. The ZIP hash must be obtained from a trusted distribution if authenticity matters.

Run `./reproduce.sh -output-dir ../report29-replay` from the extracted release, after inspecting the script if desired. Run `./build.sh ../report29-build` to rebuild the PDF into an external directory using pdfLaTeX. The README gives dependencies, scope, and exact output semantics. Copyrighted literature reading copies are excluded.

A fresh independent arithmetic audit additionally checks 4,335 exact contact comparisons (including zero drift), 2,172 finite control seed and time invariances, 7,040 nonparallel solution recoveries, 58,800 small launch bounds, and 28,800 indices beyond quadratic cutoffs. It also rejects

a remaining coordinate inconsistency and a bounding hull false positive, and exercises a distinct nearly parallel exception of norm 1 000 002 000 000. Its delivered copy changes only the proof binding hash to that of the sanitized proof copy; the arithmetic source and expected values are otherwise unchanged.

The proof was independently inspected as ordinary mathematics. The arithmetic evidence strengthens confidence in the contact and cutoff calculations; it does not constitute a formal proof, a general rule to section compiler, or a complete end to end simulation test for every promised input.

[write] **The release in this report.** These are the delivered names. `build.sh`, `reproduce.sh`, `replay.py`, `tamper_regression.py`, `scientific/audit_arithmetic.py`, `scientific/check_independent_arithmetic.py` and `scientific/timed/check_corollary.py` are shipped as `code/19-mass-four-zd-*` (the last as `code/19-mass-four-zd-timed-check_corollary.py`); the four `verification/*.json` records as `data/19-mass-four-zd-*`; and `scientific/PROOF.md`, `audit.md`, `independent-review.md`, `literature.md`, `timed/PROOF.md` and `timed/REVIEW.md` as `19-mass-four-zd-*.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README and checksum ledger. The replay entry point checks an exact inventory and pinned hashes under the delivered names, so it runs only in the delivered layout (README). At placement on Windows, `replay.py` passed in normal and optimized mode (receipts typed-equal to the expected ones) and `tamper_regression.py` rejected every tamper case.

[write] **The programme’s review.** The Hilbert’s-tenth programme reviewed this archive’s placed proof in `review_dimension_independent_four_mass.md`, with its checker and receipt, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit 7b547f686, 3 October 2026) [58]: the proof “passes this mathematical review”. It read the complete proof, including the no-unit case and the absolute-frame observation cutoff, and a separate reviewer of the proof and the new checker found no gap. Its new checks are arithmetic fixtures, not a cellular-automaton analyzer: support contact against 216 targets of a three-dimensional packet profile, 125 ray translations, the residue and transverse decomposition, a marker missed despite movement toward it, six nonparallel pairs on 1,470 right sides against a bounded search (with a parallel opposing pair rejected), and three counter cycles. It executed none of the archive’s Python. It states that the hypotheses are necessary to the argument, that the five-particle upper bound is inherited from separate source and compiler proofs which it does not reprove, and that no arithmetic-circuit lower bound or smaller universal polynomial follows.

117 Further questions

Several next steps would add information without changing the present theorem.

1. Give a precise comparison theorem between the CA encounter normal form and classical two pebble machines. The reduction must preserve splitting, fusion, geometric tags, and observation semantics, rather than merely count objects.
2. Implement and verify the finite rule to encounter library compiler for bounded small rules. The present arithmetic audit does not implement that construction. A small verified compiler would test the integration points between exact local evolution, first contacts, and section dispatch.
3. Quantify state library sizes, determinant exceptions, and Presburger formula growth as functions of dimension, radius, alphabet, and weights. The proof makes no complexity claim; the nearly parallel example excludes especially naive small geometric bounds.

4. Classify which relaxations of unit uniqueness already allow undecidable fixed mass dynamics. Multiple labels can provide independent walker control, but that observation alone is not a complete conservative CA construction at a specified mass. ([write] Added 3 October 2026: open. In $d = 1$ Part V shows that many weight-one types already allow undecidable pattern occurrence at mass three (Theorem 49.2); the least relaxation is open, as source 17’s Question 5 also says.)
5. Optimize the input dependent canonical timed certificates of Appendix I, particularly the spatial sorting and fixed pattern refinement costs, without confusing those bounds with a uniform arity theorem over arbitrary inputs. ([write] Added 3 October 2026: open. Source 18’s selector elimination (Section 96) saves one witness per chart family, and its four-witness binary certificate (Section 98.4) shows what shared parameters can save in one explicit family; neither is a general optimization.)
6. Formalize the dimer invariance, first contact arithmetic, nonparallel exception, and section termination lemmas in a proof assistant. The currently supplied evidence is mathematical exposition and finite executable checking.

Source 20 (batch-82 manuscript 17): Sparse orbit geometry and exact counting

Semilinear spatial sets, quadratic clocks, and the effects of drift, “Research report 30”, 3 October 2026. [write] From the archive `Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip`; its labels carry the prefix `smc:og:`, its sections are Sections 118–128 here, and its appendix is Appendix K, after source 19’s. It builds on source 19 (“Report 29”), which it pins by SHA-256. The Hilbert’s-tenth programme made a scoped intake of its original-frame certificate (note in Appendix K). Its abstract follows; the bracketed note in it is [write].

For a fixed finite input of conserved mass at most four, the orbit classification in Report 29 [source 19] has quantitative geometric consequences. In the frame fixing the unique possible unit symbol, the set of distinct occupied sites is effectively semilinear and lies in finitely many translates of one rational subspace of rank at most two. Centered box counts are eventually quasipolynomial of degree zero, one, or two. An expanding shuttle has sharp degree $r = \text{rank}_{\mathbb{Q}}\{E, \nu\}$, discovery count $\alpha c^{-r/2} T^{r/2} + O(T^{(r-1)/2})$, and first arrival times piecewise rational quadratic in the site coordinates. We retain the exact quadratic clock and its floor oscillations, prove inverse estimates, and give a stronger exact rank one inverse theorem. A fully specified radius six weighted cellular automaton realizes a planar wedge. Restoring a nonzero unit drift can destroy both semilinearity and eventual box quasipolynomiality: the same example has an exact floor root count $3N - 3\sqrt{N} + O(1)$. A third direction is genuinely needed for another drifted realization, even against finite unions of bounded width planes. A constructive appendix gives ordinary degree at most four Diophantine certificates with exactly one natural witness tuple for each stationary frame visited site and each reachable canonical complete configuration in either frame. The stationary first time uses one shared cycle variable; full original frame targets use whole orbit injectivity. The mathematical proofs are separated from finite computational checks. The report uses existing Presburger counting theorems and states its close relationship to classical two pebble work; no novelty, binary realization, or reversibility claim is made.

118 Scope and the inherited classification

Let F be a deterministic, translation equivariant, finite radius cellular automaton on $\mathbb{Z}^d$, for a finite $d \geq 1$. Its finite alphabet A has a unique weight zero vacuum 0, positive integer weights on every other symbol, and at most one symbol u of weight one. Assume that F preserves the total weight

of every finite configuration. Fix a finite input c_0 of mass at most four. The vacuum is quiescent. These are exactly the promise hypotheses of the lower mass classification in Report 29 [source 19, Part VII].

If u exists, its isolated evolution is a single unit translated by a fixed vector $\delta \in \mathbb{Z}^d$. With the convention $\tau_v c(x) = c(x - v)$, put

$$G = \tau_{-\delta} F, \quad F^t = \tau_{t\delta} G^t. \quad (118.1)$$

Then G fixes isolated units. If no unit symbol exists, use $G = F$ and $\delta = 0$. All unlabeled counts below count the union of occupied sites, regardless of the state occupying a site on a particular visit.

Definition 118.1. The stationary frame spatial set, centered box count, and time discovery count are

$$\begin{aligned} S &= \bigcup_{t \geq 0} \text{supp}(G^t(c_0)), \\ C(N) &= \#(S \cap [-N, N]^d), \quad N \in \mathbb{N}, \\ V(T) &= \# \bigcup_{0 \leq t \leq T} \text{supp}(G^t(c_0)), \quad T \in \mathbb{N}. \end{aligned}$$

Each site contributes once. These are not counts of visits, occupied site and time pairs, or phase witnesses. Write S_F and C_F for the corresponding original frame quantities.

118.1 Exactly what is used from Report 29

The present report does not repeat the decision theorem or the full proof that a sparse CA has the following classification. It uses the effective orbit construction of Report 29, Sections 2–12 [source 19, Sections 102–112], particularly the fixed line reduction, additive scalar gap, and complete intermediate phase description. After a computable finite prefix, precisely one of the following forms is supplied:

- A. Independent finite phase objects. Taking a common period, each occupied coordinate is one of finitely many $p + kD_i$, $k \in \mathbb{N}$. In the stationary unit frame there are at most two nonzero drift directions: a genuinely mobile object costs mass at least two.
- B. A whole configuration periodic up to translation. Its occupied coordinates are finitely many rays $p + kD$ with one common drift.
- C. An expanding shuttle. A repeated finite control cycle has scalar gap, anchor, and exact section times

$$x_n = x_0 + n\Delta, \quad L_n = L_0 + nE, \quad T_n = T_0 + cn^2 + bn, \quad (118.2)$$

where $\Delta > 0$, $c > 0$, and all data are effective. Section times are integral and strictly increasing; b, c may be rational. A fixed primitive integer vector $\nu \neq 0$ gives the common unoriented flight line. Every edge, encounter prefix, packet period phase, label, and within cycle time offset is retained.

The source writes $T_n = T_0 + An(n-1)/2 + Cn$, so here $c = A/2$ and $b = C - A/2$. This is a change of notation, not an extra timing assumption. The exact affine within cycle offsets used later also come from those retained phases. Knowledge only of an unordered spatial union and $T_n \sim cn^2$ would not suffice for the first arrival assertions.

The entire dependency is fixed input. “Effective” means that a finite description can be constructed from the promised rule and this input. It does not mean a small description, a polynomial time algorithm, an implemented general classifier, or a single uniform timed Presburger description covering arbitrary inputs.

118.2 Main conclusions

A rational affine plane is a translate of a real subspace spanned by integer vectors. Its rank is the dimension of that subspace. A set has finite width in such a plane if it is contained in the plane plus a fixed bounded set.

Theorem 118.2 (Stationary frame geometry and counting). *Under the stated hypotheses, the following descriptions are effective.*

1. *S is semilinear, equivalently Presburger definable. Its characteristic generating function is rational after partitioning $\mathbb{Z}^d$ into disjoint sign orthants and reflecting each orthant into $\mathbb{N}^d$.*
2. *S lies in finitely many exact affine translates of one rational subspace of rank at most two, hence in a finite width rational affine plane of rank at most two.*
3. *For a nonempty orbit, $C(N)$ is eventually quasipolynomial of degree zero, one, or two. Degree zero is equivalent to S being finite. If S is infinite, then*

$$C(N) = \kappa N^r + O(N^{r-1}), \quad \kappa \in \mathbb{Q}_{>0}, \quad r \in \{1, 2\}. \quad (118.3)$$

4. *Counts in boxes with independently variable integer lower and upper endpoints are piecewise quasipolynomial. Any fixed affine one parameter family of finite integer boxes consequently has an eventual quasipolynomial count of degree at most two, possibly zero or of smaller degree.*

For independent or translated periodic tails the degree is one if any occupied ray has nonzero drift, and zero otherwise. For an expanding shuttle it is exactly $r = \text{rank}_{\mathbb{Q}}\{E, \nu\}$. The empty orbit has identically zero count.

A rank two containing plane does not force degree two: two nonparallel independent rays already need such a plane but have only linearly many sites in a box.

Theorem 118.3 (Expanding shuttle timing). *For an expanding shuttle, let $D(K)$ count the distinct sites in the fixed prefix and the first K complete eventual cycles. Use the half open cycle intervals $[T_n, T_{n+1})$. Then D is effectively eventually quasipolynomial and*

$$D(K) = \alpha K^r + O(K^{r-1}), \quad \alpha \in \mathbb{Q}_{>0}, \quad r = \text{rank}_{\mathbb{Q}}\{E, \nu\}. \quad (118.4)$$

Consequently

$$V(T) = \alpha c^{-r/2} T^{r/2} + O(T^{(r-1)/2}). \quad (118.5)$$

The first visit time $\tau(y)$ on S is effectively piecewise rational polynomial of degree at most two on a finite semilinear partition. For visited sites escaping every bounded set,

$$\tau(y) = \Theta(\|y\|_{\infty}^2). \quad (118.6)$$

The largest first visit time of a site in $S \cap [-N, N]^d$ is $\Theta(N^2)$.

The last assertion is coverage of the sites this particular orbit eventually visits, not coverage of the entire lattice box. The count $V(T)$ need not be eventually quasipolynomial in ordinary time.

Theorem 118.4 (Drift restoration). *For an expanding shuttle, S_F is contained in finitely many affine translates of $\text{span}_{\mathbb{R}}\{E, \nu, \delta\}$, of rank at most three. If $\delta \neq 0$, then $C_F(N) = \Theta(N)$. Nevertheless S_F need not be semilinear and C_F need not be eventually quasipolynomial. These failures and rank three sharpness occur for explicit finite range, globally weight conserving CAs with one unit symbol and a mass four input. Independent and whole translated periodic tails remain semilinear in the original frame.*

119 Complete phase coordinates and the common plane

At edge i of cycle n , the two geometrically designated markers have coordinates

$$L_0 + nE + e_i, \quad L_0 + nE + e_i + (x_0 + n\Delta + d_i)\nu + z_i.$$

Here e_i, d_i, z_i are fixed data in the finite mode. The symbols need not distinguish the markers: their names are bookkeeping roles. A packet emitted from either marker begins at a bounded offset from it and has a fixed period drift $m_i\nu$, with nonzero integer m_i . Its retained phase sites have the form

$$y = p_i + n(E + \varepsilon_i\Delta\nu) + jm_i\nu, \quad \varepsilon_i \in \{0, 1\}. \quad (119.1)$$

The integer domain is a finite conjunction or finite union of linear inequalities and congruences in n, j . After splitting finitely many residues and boundary phases, it has $0 \leq j \leq a_in + b_i$. Each genuine growing full flight phase has $a_i > 0$. Constant offsets and initial gap contributions are incorporated in p_i . Encounter prefixes and marker positions have the same affine form without j .

Thus all these points belong to $p_i + \text{span}_{\mathbb{R}}\{E, \nu\}$. There are finitely many p_i . Any finite prefix is included by adding one translate per exceptional point. This proves the common subspace assertion, including exact finite translates rather than only a tube estimate.

Proposition 119.1 (Segment confinement and pointed cone). *Every support in cycle n is within a fixed distance of the segment joining*

$$L_0 + nE \quad \text{and} \quad L_0 + n(E + \Delta\nu).$$

If E and ν are independent, the eventual site set lies in a fixed width neighborhood of the pointed rational cone generated by E and $E + \Delta\nu$. Its opening in their plane is strictly less than 180 degrees.

Proof. A successful long flight begins within bounded distance of one marker and terminates within bounded distance of the other. Its period samples progress on the common line by a fixed nonzero multiple of ν . The retained within period offsets are bounded. These facts confine the whole flight to bounded distance from the joining segment, even if individual microscopic steps do not move monotonically. Every local encounter prefix has bounded duration and bounded relative support. Marker positions at the finitely many edges differ from the section markers by bounded vectors. These bounds are uniform in n .

If E, ν are independent, so are $E, E + \Delta\nu$. A real linear functional can take value one on both generators. Their positive cone is therefore pointed and lies in an open half plane of their span. If E is parallel to ν , the cone can degenerate to a ray or a line; only rank one is asserted there. $\square$

For an independent tail, stationary units add offsets but no drift. Every genuinely mobile independent component costs at least mass two, so at most two drift vectors occur. Taking their span proves the common rank two bound. If there is no unit symbol, at most two occupied sites exist and the classification supplies the same independent alternative. For a whole translated periodic tail, all phases use one drift vector. These observations cover terminal missed flights, independent dimers, zero drift periodic objects, and stationary configurations.

120 Semilinearity and exact distinct site counts

A subset of $\mathbb{Z}^d$ is semilinear if it is a finite union of sets

$$a + \mathbb{N}v_1 + \cdots + \mathbb{N}v_s, \quad a, v_i \in \mathbb{Z}^d.$$

Presburger arithmetic permits integer addition, order, congruences, Boolean operations, and quantifiers, but not products of variables. Its definable sets are precisely the semilinear sets.

120.1 Membership before enumeration

For each phase, existentially quantify n, j in its integer domain and impose equation (119.1). Disjoin all phases, marker and encounter lists, and finite prefix points. This gives an effective Presburger formula for $y \in S$. Independent rays have the same property.

The counted free variable must be y . Different cycles, periods, states, or phases can describe the same site. Summing the sizes of their parameter domains would count multiplicities and can change both coefficients and growth degree. Existential site membership removes them before counting.

Woods' Theorem 1.5 [55] supplies the standard equivalence with rational characteristic generating functions. For signed coordinates, use a disjoint orthant partition and reflection into $\mathbb{N}^d$, or the injective encoding

$$y \mapsto (y_1^+, y_1^-, \dots, y_d^+, y_d^-), \quad y_i^+ = \max(y_i, 0), \quad y_i^- = \max(-y_i, 0).$$

An unrestricted two sided Laurent sum may have incompatible expansions or no common domain of convergence. No such unspecified formal series is needed.

120.2 Existing Presburger counting theorem

The formula

$$y \in S \quad \wedge \quad -N \leq y_i \leq N \quad (1 \leq i \leq d)$$

has finite fibers in the spatial variable y . Woods' Theorem 1.10 gives an effective piecewise quasipolynomial count in N . In one nonnegative parameter, finitely many rational polyhedral pieces leave finitely many exceptional bounded intervals and eventual residue polynomials. This is eventual quasipolynomiality. With independently variable endpoints, the same theorem gives a multivariate piecewise quasipolynomial.

These are applications of existing counting results, not new general theorems about Presburger arithmetic. Their use is effective without a complexity promise. The special work here is extracting the correct finite fiber family and proving its sharp degree.

120.3 Sharp degree and common leading coefficient

Proof of the counting assertions in Theorem 118.2. A rational rank r affine plane has $O(N^r)$ lattice points in $[-N, N]^d$. Choose r coordinate projections independent on its direction space; on an exact affine plane these determine at most one point. Finitely many parallel translates still give $O(N^r)$. For finite width, bounded integer transverse offsets give the same estimate. Thus an expanding tail has $C(N) = O(N^r)$ for $r = \text{rank}_{\mathbb{Q}}\{E, \nu\}$.

Fix one genuine growing flight phase, present on all sufficiently large cycles or on one fixed positive density residue class. Its period parameter has at least $an - b$ values, with $a > 0$. For a sufficiently small constant $\eta > 0$, the sites with $n \leq \eta N$ all lie in the radius N box once finite exceptions are removed.

If $r = 2$, the vectors $E + \varepsilon \Delta \nu$ and $m\nu$ are independent. Hence $(n, j) \mapsto y$ is injective within this single phase. There are $\Theta(N)$ admissible cycle values and $\Theta(n)$ positions in cycle n , yielding $\Omega(N^2)$ distinct sites.

If $r = 1$, that two parameter map need not be injective. Instead choose just one admissible cycle n within a bounded distance of ηN . Its $\Theta(N)$ period values give distinct sites because $m\nu \neq 0$. This proves $\Omega(N)$ without intercycle injectivity.

Every eventual residue polynomial therefore has exact degree r . Its leading coefficient is common across residues: monotonicity $C(N) \leq C(N+1)$ orders consecutive coefficients, and following the whole residue cycle forces equality. The common coefficient κ is positive and rational. Lower order terms are uniformly $O(N^{r-1})$.

For independent and translated periodic tails, each nonconstant ray contributes $\Theta(N)$ sites, whereas constant rays contribute finitely many. This proves their degree claims and the finite set characterization of degree zero. $\square$

In particular, the ordinary generating function $\sum_{N \geq 0} C(N)z^N$ is rational. For some effective period M , after accounting for finitely many initial terms its denominator divides $(1 - z^M)^{r+1}$.

120.4 Why multibox chambers matter

For the basic wedge $\{(n, j) : n \geq 0, 0 \leq j \leq n\}$, the centered rectangle count equals

$$\begin{cases} (N_1 + 1)(N_1 + 2)/2, & N_1 \leq N_2, \\ (N_2 + 1)(N_2 + 2)/2 + (N_1 - N_2)(N_2 + 1), & N_1 \geq N_2. \end{cases}$$

The chamber boundary is real; one global multivariate quasipolynomial is not generally available. Along $N_2 = 0$ the degree drops to one. Translated or degenerate boxes can miss every unbounded component. A positive centered rational dilation in every coordinate preserves the sharp degree, after resolving floors by residues. A general affine family guarantees only the upper degree bound.

121 Quadratic clocks and discovery inverses

Constrain the phase witnesses by $0 \leq n < K$ to count all sites through complete cycles. The free variable remains y , so $D(K)$ is a finite fiber Presburger count. All its positions have norm $O(K)$ in the common rank r subspace. The same growing phase lower bounds prove $D(K) = \Theta(K^r)$. Monotonicity again forces one positive rational leading coefficient α across residues, proving (118.4).

In particular,

$$D(K+1) - D(K) = O(K^{r-1}). \quad (121.1)$$

For rank one, an entire late cycle discovers only boundedly many new sites even though its shuttle makes $\Theta(K)$ visits along a long flight.

121.1 Retaining the clock exactly

For sufficiently large T , let $n(T)$ be the unique cycle with $T_{n(T)} \leq T < T_{n(T)+1}$. Then

$$n(T) = \left\lfloor \frac{-b + \sqrt{b^2 + 4c(T - T_0)}}{2c} \right\rfloor = \sqrt{T/c} + O(1). \quad (121.2)$$

The discriminant is nonnegative in the eventual range. The increasing quadratic on nonnegative cycle indices makes the indicated root the appropriate one.

Since $D(n(T)) \leq V(T) \leq D(n(T) + 1)$, substitution in (118.4) gives (118.5). Thus rank one has square root discovery with bounded error, and rank two has linear discovery with error $O(\sqrt{T})$. Merely knowing $T_n \sim cn^2$ would prove the leading asymptotic equivalence, but the displayed error requires $T_n = cn^2 + O(n)$.

There is an exact description stronger than this asymptotic. For current cycle n and time offset s , count the distinct sites from the prefix, earlier cycles, and current phase occurrences with their affine elapsed time at most s . This is a Presburger finite fiber count $Q(n, s)$, hence effectively piecewise quasipolynomial. Consequently

$$V(T) = Q(n(T), T - T_{n(T)}). \quad (121.3)$$

The assertion of piecewise quasipolynomiality belongs to Q , not to its composition with the quadratic floor root. Section boundaries and within cycle oscillations have not been discarded.

121.2 Spatial and cycle inverses

For an infinite spatial set define

$$R(M) = \min\{N : C(N) \geq M\}, \quad J(M) = \min\{K : D(K) \geq M\}.$$

Comparing $\kappa N^r + O(N^{r-1})$ with the level M gives

$$R(M) = (M/\kappa)^{1/r} + O(1), \quad J(M) = (M/\alpha)^{1/r} + O(1). \quad (121.4)$$

Indeed a fixed sufficiently large shift of $(M/\kappa)^{1/r}$ changes the leading polynomial by enough to dominate the uniform $O(N^{r-1})$ error.

These inverses are also exactly effective. On each eventual residue $N = u + Pq$ for a fixed period P , substitute into the corresponding degree one or two polynomial for $C(N)$, solve the eventual increasing linear or quadratic inequality, and minimize over the finitely many residues and finite exceptional values. This produces finitely many rational affine ceiling formulas in rank one or quadratic root ceiling formulas in rank two. In rank one R is eventually quasi affine. In rank two its square root growth rules out eventual quasipolynomiality.

For discovery of M distinct sites, let

$$H(M) = \min\{T : V(T) \geq M\}.$$

Monotonicity and (118.5) give

$$H(M) = \begin{cases} (c/\alpha^2)M^2 + O(M), & r = 1, \\ (c/\alpha)M + O(\sqrt{M}), & r = 2. \end{cases} \quad (121.5)$$

An exact rank one improvement is available because of the bounded number of new sites per cycle.

Proposition 121.1 (Quadratic rank one discovery inverse). *If $r = 1$, then $H(M)$ is effectively eventually quasipolynomial of degree exactly two, with common leading coefficient c/α^2 . Accordingly $V(T)$ has an eventual exact description using a finite maximum of quadratic floor root formulas, without becoming a quasipolynomial in T .*

Proof. By (121.1), some fixed integer B bounds the number of genuinely new sites in every sufficiently late cycle. The eventual graph of $D(n)$ is Presburger because D is quasi affine. For each fixed $1 \leq \ell \leq B$, “at least ℓ new sites in cycle n have appeared by offset s ” is Presburger: use ℓ pairwise distinct site variables, occurrence by s , and absence from the fixed prefix and every earlier cycle. A finite disjunction over ℓ defines the earliest cycle $n(M)$ and earliest offset $s(M)$ at which the count reaches M . This also handles cycles that introduce no new site.

These are one parameter Presburger functions, hence eventually quasi affine. An elementary proof of the underlying piecewise affine fact appears in Lemma 122.1. Substitution in

$$H(M) = T_0 + cn(M)^2 + bn(M) + s(M)$$

gives an eventual quadratic quasipolynomial. Equation (121.5) supplies its common leading coefficient. Solving each eventual residue polynomial inequality $H(M) \leq T$ and taking the largest admissible M gives the stated floor root description of V . $\square$

122 First arrival times and coverage

The spatial union alone does not tell us when a site first appears. The complete phase form supplies the missing timing information. Let $R(y, n, h)$ be the Presburger relation asserting that site y is occupied at offset h in the half open cycle n . The offset is affine in the phase parameters: preceding edge durations are affine in n , and a packet period contributes its fixed period length times j plus a retained phase offset.

Exclude the finite set of sites first seen before T_0 , treating those separately. Define $n_*(y)$ as the least cycle containing y , then $h_*(y)$ as the least offset in that cycle. Their graphs are Presburger: impose occurrence and negate any occurrence at an earlier cycle, or at an earlier offset in the same least cycle. Minimization is legitimate here because the cycle and time domains are nonnegative integers and every y in the domain is eventually visited.

[write] [The next lemma is classical: Presburger-definable functions have semilinear graphs [7] and are therefore piecewise rational affine. Source 20 proves it directly; no novelty is claimed for it.]

Lemma 122.1 (Presburger functions are piecewise rational affine). *A Presburger definable function with integer input and output is rational affine on each part of an effective finite semilinear partition of its domain.*

Proof. Decompose its graph into finitely many linear sets. On one such component write the generators as (v_i, w_i) , with v_i in the input coordinates and w_i in the output coordinates. Every

integer relation $\sum_i a_i v_i = 0$ also satisfies $\sum_i a_i w_i = 0$: separate positive and negative coefficients to obtain two points of this graph component with the same input, then use functionality. Therefore $v_i \mapsto w_i$ extends to a rational linear map on the rational span of the v_i . Adding the base point makes the function rational affine on the component's domain. A finite Boolean refinement of the semilinear domains yields a partition. All steps are effective. $\square$

This argument allows lower dimensional input pieces and several output coordinates. It asserts no uniqueness of the semilinear parameter witnesses.

By the lemma, both $n_*(y)$ and $h_*(y)$ are piecewise rational affine. Refining their partitions jointly gives

$$\tau(y) = T_0 + cn_*(y)^2 + bn_*(y) + h_*(y) \quad (122.1)$$

as a rational polynomial of degree at most two on every cell. Earlier prefix sites contribute singleton cells. This proves the exact effective statement in Theorem 118.3.

The source support bound $\|y\|_\infty \leq K_0 + K_1 n$ implies that the earliest cycle of a distant site is at least a positive constant times $\|y\|_\infty$ minus a constant. Since section times are quadratic, $\tau(y) = \Omega(\|y\|_\infty^2)$. Conversely, finitely many affine formulas for $n_*(y)$ give $n_*(y) = O(1 + \|y\|_\infty)$; the within cycle duration is $O(1 + n)$. Hence $\tau(y) = O(1 + \|y\|_\infty^2)$. This proves (118.6).

To obtain box coverage time, the upper bound applies uniformly to sites of norm at most N . For the lower bound, choose fixed $0 < \varepsilon < 1$. Equation (118.3) gives $C(N) - C(\lfloor \varepsilon N \rfloor) > 0$ for all sufficiently large N , so some visited site in the box has norm at least εN . Its first arrival takes $\Omega(N^2)$ time.

Remark 122.2. Piecewise quadratic first arrival does not imply that the full space and time orbit is Presburger. The relation $t = cn^2 + bn + T_0$ contains a genuine quadratic clock. A piecewise quadratic formula by itself is not yet a canonical Diophantine certificate. Appendix K supplies the additional disjoint clause construction, signed quotient encoding, and uniqueness proof.

123 A globally conservative planar wedge automaton

We now give actual CAs realizing the geometric phenomena. Their unrestricted conservation is proved directly, rather than inferred from an abstract shuttle schedule or from finite simulations.

Work on $\mathbb{Z}^2$ with alphabet $\{0, u, E, W\}$ and weights

$$w(0) = 0, \quad w(u) = 1, \quad w(E) = w(W) = 2.$$

The letters E, W denote internal head states. There is only one unit symbol. Let $e_1 = (1, 0)$, $e_2 = (0, 1)$. A head at a is eligible if there is no other head within maximum norm distance four. Its action reads the old configuration. Any failed guard gives no rewrite. Every site not explicitly written stays unchanged.

Proposition 123.1 (Full rule validity). *Simultaneously applying Table 11 at eligible heads defines a translation equivariant radius six CA. It preserves total weight on every finite configuration, including configurations with many heads, obstructed destinations, and extra unit symbols. Its vacuum is quiescent and its isolated unit is stationary.*

Proof. Rules 1 and 2 have mutually exclusive forward guards; so do rules 3 and 4. All writes lie in $a + [-2, 2]^2$. Distinct eligible heads are more than four apart, so their write cubes are disjoint. No

Rule	Required old symbols	New symbols at written sites
1	E at a ; 0 at $a + e_1$	0 at a ; E at $a + e_1$
2	E at a ; u at $a + e_1$; 0 at $a + 2e_1 + e_2$	W at a ; 0 at $a + e_1$; u at $a + 2e_1 + e_2$
3	W at a ; 0 at $a - e_1$	0 at a ; W at $a - e_1$
4	W at a ; u at $a - e_1$; 0 at $a + e_2$ and $a - e_1 + e_2$	0 at a and $a - e_1$; E at $a + e_2$; u at $a - e_1 + e_2$

Table 11: The complete planar rewrite rule, in addition to the head isolation condition. All vacancy tests are part of the definition.

eligible rewrite can overwrite an ineligible head: a head at a written site would be within distance two and would itself disqualify the candidate. Each listed rewrite preserves the exact weight sum on its written sites. Every newly filled site is explicitly required to be vacant unless it is the head's own rewritten site. In particular no additional unit is silently erased.

For locality, the output at x only needs to consider heads within distance two of x and their radius four isolation tests. All this lies within distance six of x . The listed destination guards are within the same radius six neighborhood. If no action writes x , retain its old value. This is a well defined finite radius local rule. Disjoint, weight preserving rewrites prove conservation for all finite inputs. Vacuum and an isolated unit have no eligible head action and remain unchanged. $\square$

123.1 Sections and the exact spatial wedge

Fix $k \geq 3$. Start with units at $(0, 0)$ and $(k, 0)$ and an E head at $(1, 0)$. There is exactly one head forever, so its eligibility test always passes. Inductively the section at cycle n is

$$L_n = (0, n), \quad R_n = (k + n, n), \quad E \text{ at } (1, n). \quad (123.1)$$

Put $K = k + n$. There are $K - 2$ eastward free steps, one right bounce by rule 2, $K - 2$ westward free steps, and one left bounce by rule 4. The duration is $2(K - 1)$, so

$$T_n = n^2 + (2k - 3)n. \quad (123.2)$$

The right bounce moves the right unit to $(k + n + 1, n + 1)$. The left bounce moves the left unit and head up by one, yielding the next section. All guards hold along this intended orbit.

Every row n eventually has exactly the visited horizontal coordinates $0, 1, \dots, k + n$. The early arrival of the right unit on row $n + 1$ adds only that next row's endpoint. Therefore the entire occupied site union is exactly

$$S = \{(x, y) \in \mathbb{Z}^2 : y \geq 0, 0 \leq x \leq k + y\}. \quad (123.3)$$

This realizes $E = e_2$, $\nu = e_1$, $\Delta = 1$, $c = 1$, and rank two. No special rare label or section subsequence is being counted.

For $N \geq k$, direct summation gives

$$\begin{aligned} C(N) &= \sum_{y=0}^N (\min(N, k + y) + 1) \\ &= \frac{N^2 + (2k + 3)N + 2 + k - k^2}{2}. \end{aligned} \quad (123.4)$$

Thus $\kappa = 1/2$ and the density in centered square boxes is $1/8$. A finite union of bounded width lines has only $O(N)$ lattice sites there, so rank two is sharp even against finitely many lines. The wedge does not explore a half plane.

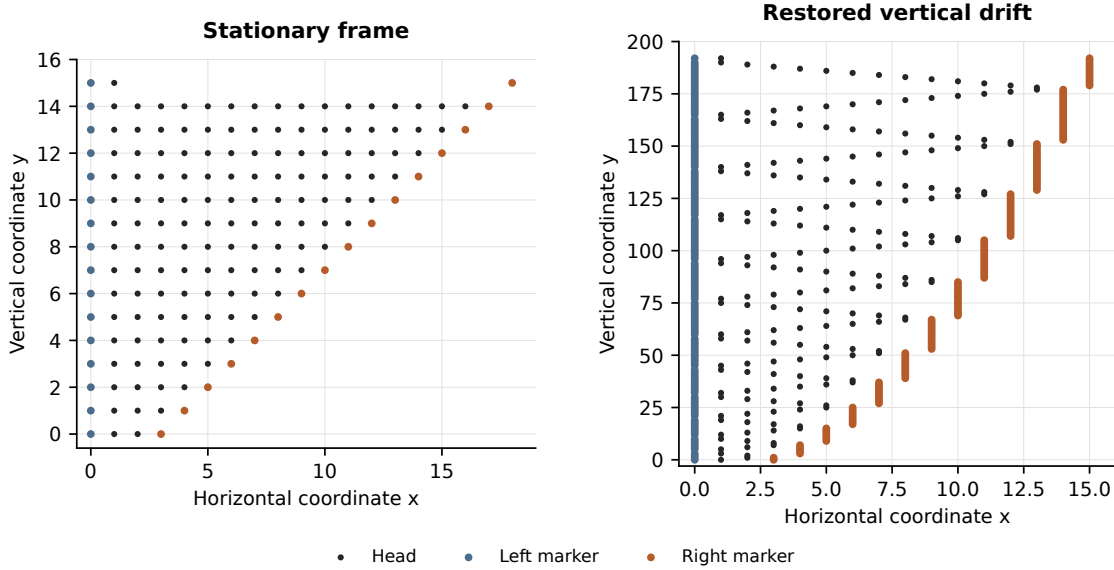


Figure 7: Actual occupied site unions from the authored local rule simulator, with $k = 3$. Left: the stationary frame wedge through section $n = 15$. Right: the same planar rule with a vertical shift after every step, through time 180. Colors distinguish the persistent left marker, head, and right marker in this intended one head orbit; the displayed sets are not an abstract flight schedule.

123.2 Exact discovery and first arrival for this example

At the section time T_n , the number of sites already visited is

$$B_n = \frac{n(n-1)}{2} + (k+1)n + 3. \quad (123.5)$$

Rows below n are complete and row n contributes its two left positions and right endpoint. For $0 \leq h < 2(k+n-1)$,

$$V(T_n + h) = B_n + \min(h, k+n-2) + \mathbf{1}_{\{h \geq k+n-1\}}. \quad (123.6)$$

The eastward flight discovers its interior one site per step. The right bounce discovers one next row endpoint. The westward flight revisits old sites, and the final left bounce creates two sites counted at the next section. Formula (123.6) exposes the within cycle oscillation hidden in $V(T) = T/2 + O(\sqrt{T})$.

For row $n \geq 1$, first arrival is

$$\tau(x, n) = \begin{cases} T_n, & x = 0 \text{ or } 1, \\ T_n + x - 1, & 2 \leq x \leq k+n-1, \\ T_n - (k+n-2), & x = k+n. \end{cases} \quad (123.7)$$

The last line is the preceding cycle's right bounce. On row zero, the three initial sites have time zero and each other $2 \leq x \leq k-1$ has time $x-1$. These formulas show explicitly the finite semilinear partition and rational quadratic first arrival description.

124 Rank one behavior and nonsemilinear drift

Degree zero occurs for one stationary unit. Degree one occurs for an isolated E head moving east under the preceding rule. These inputs have mass one and two, respectively.

For an expanding rank one example, change the planar rule in exactly two places:

1. In rule 2, change both the vacant destination guard and the new unit destination from $a + 2e_1 + e_2$ to $a + 2e_1$.
2. Replace rule 4 by: if W is at a and u is at $a - e_1$, change the head in place to E and leave that neighboring unit unchanged. Remove the former upward destination vacancy tests, since no new site is filled in this action.

The isolation convention and rules 1 and 3 are unchanged. This is a complete guard and write specification. The same disjoint rewrite proof establishes a radius six globally conservative CA.

From the same input, sections become $L_n = (0, 0)$, $R_n = (k + n, 0)$, with E at $(1, 0)$, and retain clock (123.2). Its full visited set is the nonnegative horizontal ray. The exact time count, after the initial interior has been visited, is

$$V(T) = \left\lfloor \sqrt{T + (k-1)(k-2)} \right\rfloor + 3, \quad T \geq k-2. \quad (124.1)$$

Indeed site $x \geq k+1$ is first reached by the right marker at time

$$(x-2)^2 - (k-1)(k-2).$$

Consequently

$$H(M) = (M-3)^2 - (k-1)(k-2), \quad M \geq k+2. \quad (124.2)$$

This directly realizes the rank one inverse theorem, with $\alpha = c = 1$. It also proves that an actual permitted CA can have nonquasipolynomial $V(T)$: an unbounded eventual quasipolynomial cannot grow like $\sqrt{T}$.

Now call this horizontal rule G and set $F = \tau_{e_2} G$. Translation commutes with G , so the isolated unit drift is e_2 and normalization recovers G . The full original frame occupied site set is

$$S_F = \{(x, t) : x \in \text{supp}(G^t(c_0)), t \geq 0\},$$

where the horizontal support is read as a set of x coordinates. Every row contains exactly three sites; all satisfy $x \geq 0$ and $x = O(\sqrt{t})$, yet x is unbounded.

Proposition 124.1. *The full set S_F in this example is not semilinear.*

Proof. Suppose it is a finite union of linear sets, and let (a, b) be a period vector of a nonempty component. Negative b is impossible because arbitrarily large period multiples would enter negative rows. If $b = 0$, finite rows force $a = 0$. If $b > 0$, the uniform sublinear bound on x forces $a = 0$: otherwise along its multiples $|x|/t$ tends to the nonzero ratio $|a|/b$. Hence every period is vertical, and finitely many component base points bound x globally, a contradiction. $\square$

This proof concerns every occupied site, not just special labels or sparse section times. Its nonsemilinearity alone does not rule out an eventual polynomial box count. The next example supplies that separate conclusion.

125 An exact nonquasipolynomial box count

Return to the planar wedge rule of Section 123 and again set $F = \tau_{e_2}G$, now shifting inside the wedge's plane. At time t in cycle n , let $\sigma(t) = 0$ before that cycle's right bounce finishes, and 1 afterward. The heights of the three objects in the original frame are

$$\text{left: } t + n, \quad \text{head: } t + n, \quad \text{right: } t + n + \sigma(t).$$

Every trajectory has strictly increasing height. Usually height increases by one; at its one upward move in each cycle it increases by two. The left marker and head have the same height sequence and different horizontal coordinates.

There are no cross time overlaps with the right trajectory. Before the right bounce it has the head's height at the same time but strictly larger x . After the bounce its height is either the head's next height or, at the left bounce, a missing head height. In the former case the right coordinate is $k + n + 1$, while the head's next coordinate is at most $k + n - 1$. The three entire trajectories are therefore pairwise disjoint, not merely distinct at each instant.

The left marker and head skip the heights

$$a_n = T_n + n - 1 = n^2 + (2k - 2)n - 1, \quad n \geq 1. \quad (125.1)$$

The right bounce finishes at $t = T_n + k + n - 1$, so the right trajectory skips

$$b_n = T_n + k + 2n - 1 = n^2 + (2k - 1)n + k - 1, \quad n \geq 0. \quad (125.2)$$

Each height sequence starts at zero and contains exactly the nonnegative integers outside its displayed skipped sequence. The number of its holes through height N is thus a quadratic inequality count.

For $N \geq 2k + 4$, the horizontal box constraint is inactive at all heights at most N . One way to see this is that the first right marker occurrence at coordinate $k + j$, $j \geq 1$, has height

$$(j - 1)^2 + (2k - 1)(j - 1) + k.$$

If its horizontal coordinate exceeds $N \geq k + 1$, this height exceeds N as well. The head is never farther right than the marker. We retain the convenient conservative threshold $2k + 4$ in the formula and audits.

Define

$$A(N) = \max \left(0, \left\lfloor \sqrt{(k - 1)^2 + N + 1} \right\rfloor - (k - 1) \right), \quad (125.3)$$

$$B(N) = \max \left(0, 1 + \left\lfloor \frac{-(2k - 1) + \sqrt{(2k - 1)^2 + 4(N - k + 1)}}{2} \right\rfloor \right). \quad (125.4)$$

These count (125.1) and (125.2), respectively, with their different starting indices.

Theorem 125.1 (Exact drifted wedge count). *For every integer $k \geq 3$ and $N \geq 2k + 4$, the original frame full occupied site count is*

$$C_F(N) = 3(N + 1) - 2A(N) - B(N). \quad (125.5)$$

In particular,

$$C_F(N) = 3N - 3\sqrt{N} + O(1), \quad (125.6)$$

and C_F is not eventually quasipolynomial. Thus S_F is not semilinear either.

Proof. The three trajectories are disjoint, and their heights are precisely the nonnegative integers minus the displayed holes. For N in the stated range, counting their points in the box is exactly counting these height sequences through N . This gives (125.5). Both $A(N)$ and $B(N)$ equal $\sqrt{N} + O(1)$, proving (125.6).

If C_F were eventually quasipolynomial, its linear growth and $C_F(N)/N \rightarrow 3$ would force every residue polynomial to be $3N$ plus a constant. Then $C_F(N) - 3N$ would be bounded, contradicting the negative square root term. A semilinear S_F would have an eventual quasipolynomial box count by the already cited Presburger theorem, so it too is excluded. $\square$

Formula (125.5) is the exact statement. Its bounded floor oscillations are retained; equation (125.6) is not a floor free expansion to all orders. The inverse box radius satisfies the useful two term estimate

$$\min\{N : C_F(N) \geq M\} = M/3 + \sqrt{M/3} + O(1). \quad (125.7)$$

Substitute $N = M/3 + \sqrt{M/3} + L$ into (125.6); a fixed sufficiently large positive or negative L brackets the level M . Exact inversion remains governed by the floor formula and monotonicity.

126 General drift bounds and rank three sharpness

For any expanding shuttle, add δ times its actual time to (119.1). Actual time is T_n plus an affine function of n, j . Every resulting point therefore lies in a fixed affine translate of

$$\text{span}_{\mathbb{R}}\{E, \nu, \delta\}.$$

There are only finitely many phase offsets. The quadratic degree in n is the source of the extra geometry. Drift restoration is not an affine change of the untimed spatial parameters alone.

In independent and translated periodic tails, actual time is affine in the common period index. Drift restoration simply changes each ray vector, preserving semilinearity.

126.1 Every nonzero drifting shuttle has linear box order

The stationary support norm bound and quadratic clock imply, uniformly over the at most four occupied sites at time t ,

$$y_F(t) = t\delta + O(\sqrt{t}). \quad (126.1)$$

If $\delta \neq 0$, any original frame site in the radius N box must occur at $t = O(N)$. There are at most four occupied sites at each time, so $C_F(N) = O(N)$.

For the matching lower bound, choose fixed small $\eta > 0$ and $T = \lfloor \eta N \rfloor$, so all support through T lies in the radius N box once N is large. Pick an occupied site at time T . Quiescence and finite radius imply that it has an occupied predecessor within a fixed radius at time $T - 1$; otherwise its entire input neighborhood would be vacuum. Repeat to time zero. This produces a bounded jump path of occupied sites contained in the box. A coordinate with $\delta_i \neq 0$ has endpoint displacement $\Theta(N)$ by (126.1). A bounded jump path spanning that displacement contains $\Omega(N)$ distinct sites. Thus $C_F(N) = \Theta(N)$ without assuming labeled particle identities or mistaking occurrences for distinct sites.

126.2 A three dimensional realization

Embed the planar wedge rule in $\mathbb{Z}^3$ using its e_1, e_2 directions and the same radius four head isolation test in the three dimensional maximum norm. The disjoint write and radius six proof is unchanged. Compose with the shift $\delta = e_3$.

At section times the original frame support contains

$$(0, n, T_n), \quad (k + n, n, T_n).$$

If all visited sites stayed within bounded distance of one affine plane of rank at most two, some nonzero real linear functional

$$\ell(x, y, z) = ax + by + cz$$

would stay bounded on them. Its value on the first sequence is $bn + c[n^2 + (2k - 3)n]$, so boundedness forces $b = c = 0$. Its value on the second then forces $a = 0$, a contradiction.

The stronger finite union assertion requires the flights, rather than only those two section sequences. The eastward flight contains

$$(1 + j, n, T_n + j), \quad 0 \leq j \leq k + n - 2. \quad (126.2)$$

On these points,

$$\ell = cn^2 + (b + c(2k - 3))n + (a + c)j + a.$$

If $a + c \neq 0$, a fixed bounded strip for ℓ contains only $O(1)$ of the integer j values in each cycle. If $a + c = 0$, this is a nonconstant unbounded polynomial in n unless $a = b = c = 0$, so the strip contains none of the flight for every sufficiently large n . A finite collection of strips therefore covers only $O(1)$ of its $\Theta(n)$ sites. Every bounded width affine plane of rank at most two lies inside one such strip.

Corollary 126.1. *The rank three upper bound is sharp even if finitely many bounded width affine planes of rank at most two are allowed, with arbitrary orientations.*

The original frame example still has only linear centered box order. Containing rank, growth degree, and semilinearity are distinct properties.

127 Prior work and mathematical interpretation

The geometric picture is close to classical two pebble automata. Lacalle and Gajardo [50], printed pages 2 and 5, describe conical confinement, zigzag coverage, and expanding oscillations versus periodic behavior, attributing core predecessors to Delorme and Mazoyer. Their conclusion on printed page 17 extends their geometric arguments to $\mathbb{Z}^d$. Their principal theorem concerns real time trace recognition, a different statement from the present fixed input site enumeration and weighted CA classification.

Blum and Sakoda [51], printed page 148, discuss sector exploration below a half plane and the inability of one finite automaton with two pebbles to explore an entire half plane. The pointed cone statement here should be read alongside that close predecessor geometry. Delorme and Mazoyer [49] remains a central reference. Its publisher abstract and the explicit account in Lacalle and Gajardo were checked; the complete 2002 text was not retrieved for this report. That limitation supplies no basis for a priority inference.

The explicit CAs use a mass two stateful head and two identical mass one markers. Report 29's [source 19's] classification also covers splitting, fusion, weighted symbols, and independent packet tails. Its proof is the required bridge; a general sparse CA should not silently be identified with a pebble machine. Conversely, the semilinearity to counting implication is existing Presburger mathematics [55].

For Diophantine computational substrates, the useful distinction is between untimed spatial definability and the extra quadratic clock. The spatial set admits effective linear arithmetic descriptions and rational generating functions, while first arrival and inverse discovery can be genuinely quadratic. Drift couples that clock into spatial coordinates, creating nonsemilinear sets and square root corrections despite a very small finite conserved mass. This is a concrete separation of descriptive resources, not an undecidability or universality claim.

127.1 What remains open here

1. Determine which of these sharp geometric and drift phenomena admit binary or reversible realizations under the same small mass restrictions. The weighted examples do not answer this. ([write] Added 3 October 2026: answered for binary rules by source 21, whose ordinary binary radius-six rule realizes a planar wedge, with two missing lines, and the drifted square-root regime (Theorems 133.1 and 136.2). Reversible realizations stay open: source 21's rules are not asserted reversible.)
2. Extract practical algorithms and size bounds for the effective semilinear partitions, counting chambers, and first arrival formulas. General effectivity alone need not be computationally useful.
3. Optimize the canonical first arrival Diophantine encoding of Appendix K, especially the cell count, quotient variables, and shared quadratic lifts, without mistaking input dependent arity for a uniform bound. ([write] Open; source 21's 614-auxiliary quartic certifies one local transition of its binary rule, not a first-arrival relation.)
4. Classify possible subleading terms and exact inverse formulas in the original frame. The present example exhibits a square root correction; the general $\Theta(N)$ theorem does not assert an exhaustive asymptotic taxonomy.
5. Clarify the exact correspondence with each classical pebble model, including indistinguishable markers, collision semantics, and its chosen observation convention, before making any novelty comparison.

128 Reproducibility and proof boundaries

The portable release contains this editable source and PDF, authored figures with their simulator, the proof and independent counting review, finite scientific checkers, expected JSON receipts, and a sealed replay entry point. No external paper cache is included. Bibliographic links are sufficient to locate the primary literature.

The main checker implements both explicit rule variants, the section clock, stationary box formula, and exact drifted count. It also samples malformed configurations and compares global rewrites with cellwise radius six evaluation. That cellwise strategy reuses the same action predicate: it is an independently evaluated local output route, not an independently reimplemented transition

rule. A separate included example checker independently transcribes the transition rules and checks first arrival and boundary arithmetic.

Every scientific correctness check uses explicit exceptions rather than removable Python assertions. A deliberately failing guard self test confirms enforcement under optimization. The replay runs fresh isolated normal and optimized copies, compares receipts recursively by exact type and value, and confirms that packaged file bytes, modes, and modification times remain unchanged. Booleans cannot masquerade as integer counts. An independently fixed inventory and pins are checked before invoking any packaged helper; a manifest by itself is insufficient when edited together with its payload.

The bundled tamper regressions test ordinary payload changes, removal paired with manifest editing, coordinated checker and manifest alteration, a type only receipt mutation, an extra file, and a symlink payload. Their aim is narrowly to exercise the stated verification boundary. This is not an adversarial security system against an attacker who replaces the trusted bootstrap and its reference pins together.

A PDF build writes only to an external directory. The release is not modified by replay or rebuild. Rebuilding requires a conventional L^AT_EX distribution; the geometry and example checkers use the Python standard library, while the optional first visit algebra fixtures use SymPy. Figure regeneration additionally uses Matplotlib. All pages of the final PDF are rendered and visually inspected for clipping, overlap, missing symbols, and reference errors.

Finite checks audit transcription, formula arithmetic, and selected boundary cases. They do not establish conservation on all configurations, classification termination, semilinearity, or asymptotic claims. Those conclusions rest on the unrestricted mathematical arguments in this report and the precisely stated Report 29 [source 19] dependency. No binary CA construction, reversibility proof, formal proof assistant certification, efficient general compiler, or new universality threshold is claimed.

[write] **The release in this report.** These are the delivered names. `build.sh`, `reproduce.sh`, `replay.py`, `tamper_regression.py`, `figures/make_figures.py` and the checkers `scientific/geometry/audit.py`, `scientific/examples/independent-audit.py` and `scientific/first-visit/audit.py` are shipped as `code/20-orbit-geometry-*` (sub-directories joined to the name by hyphens, for example `code/20-orbit-geometry-geometry-audit.py`); the normal and optimized receipts of the geometry and first-visit checkers and the four `verification/*.json` records as `data/20-orbit-geometry-*`; the figure as `figures/20-orbit-geometry-visited-geometry.pdf`; and the nine proof, review and supplement notes of `scientific/` as `20-orbit-geometry-*.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README and checksum ledger, and the two supplement checksum manifests `scientific/first-visit/MANIFEST.sha256` and `scientific/geometry/MANIFEST.sha256`. The replay entry point needs the delivered layout (README). At placement on Windows, `replay.py` passed in normal and optimized mode and `tamper_regression.py` rejected every tamper case.

Source 21 (batch-82 manuscript 01): A binary planar four particle shuttle

Exact orbit geometry and a one step quartic certificate, “Research report 31”, 3 October 2026. [write] From the archive `Binary_Planar_Four_Particle_Shuttle_Package.zip`; its labels carry the prefix `smc:bp:`, and its sections are Sections 129–140 here. It builds on sources 17, 18 and 20 (“Reports 12, 13 and 30”), which it pins by SHA-256, and modifies source 17’s binary shuttle; no code is shared. The Hilbert’s-tenth programme has reviewed it (Section 139). Its abstract follows; the bracketed notes in it are [write].

We give an ordinary binary, quiescent, conservative cellular automaton on $\mathbb{Z}^2$ with an explicit four-particle expanding shuttle. Its stationary-frame visited set is a wedge with exactly two missing lines, its centered-box count has leading coefficient $1/2$, and its first-arrival times and generalized inverse are explicit. Composing the rule with unit vertical translation gives four disjoint traces and the exact floor-root count $4N - 4\sqrt{N} + O_k(1)$, with inverse $M/4 + \sqrt{M/4} + O_k(1)$. Both specified rules have exact centered Chebyshev radius six and precisely 78 essential input cells. Their unique multilinear Boolean polynomial has degree 45. A separately constructed one-step sum-of-squares quartic has 614 canonical natural auxiliaries, 693 residuals, 79 external variables, and exactly 3403 collected terms. Its auxiliary tuple is unique even over the reals. The construction modifies the two bounce outputs of the one-dimensional binary shuttle in Report 12 [source 17] and retains its moving pairs and quadratic clock. It answers the binary-realization question left by Report 30’s weighted planar example [more precisely: it gives a binary realization of the planar wedge and of the drifted square-root regime of source 20’s weighted example, answering source 20’s first question for binary rules but not for reversible ones]. Full-shift correctness is proved, rather than inferred from finite tests. No reversibility, universality, optimal-radius, or general finite-fold Diophantine claim is made.

129 Scope and relation to the earlier reports

Report 30 established quantitative geometry for promised conservative cellular automata of fixed mass at most four and supplied a weighted planar realization [source 20, Part VII]. The present report supplies a binary realization directly. Every occupied site has weight one, and the selected orbit contains exactly four occupied sites at every time. The proof of this example is self-contained and does not require a general orbit-classification theorem. [write] [Its counts are exact instances of source 20’s Theorems 118.2 and 118.3 (an expanding shuttle of rank two, with leading coefficient $1/2$) and, after the vertical shift, of Theorem 118.4 (linear box order with a square-root deficit, as in source 20’s Theorem 125.1).]

The starting mechanism is the one-dimensional binary shuttle in Report 12, Section 7 [Section 89 here, rule (89.1)], “An explicit binary expanding shuttle” [source 17, Part VI]. Its replacements, up to translation, are

$$\{0, 1\} \mapsto \{1, 2\}, \quad \{0, 2\} \mapsto \{-1, 1\}, \quad \{0, 1, 3\} \mapsto \{-1, 1, 4\}, \quad \{0, 2, 4\} \mapsto \{0, 3, 4\}.$$

We keep both moving-pair replacements and all four input shapes. At the right bounce, only the outgoing right marker moves up one row. At the left bounce, the entire output triple moves up one row. These two transverse changes make successive sweeps occupy successive rows. Thus the four horizontal patterns are not presented as new patterns. The quadratic section clock is inherited from the same flight lengths; the planar visited set and vertical-drift geometry are proved below.

Report 13 distinguishes local or fixed-formula certificates from more general timed constructions [source 18, Part VI]. We preserve that distinction. The quartic in Section 138 describes *one local transition*, with a literal expanded polynomial and exact resource ledger. It is not an unbounded orbit tableau. Report 30’s [source 20’s Corollary K.4] separate complete-target theorem is discussed only in Section 140; it does not follow from the local quartic.

Throughout, $k \geq 7$ is an integer, $\mathbb{N} = \{0, 1, 2, \dots\}$, and a “natural witness” means a tuple of nonnegative integers. The notation $O_k(1)$ permits a bound depending on the fixed initial gap k . Spatial counts count distinct lattice sites ever occupied, not visits, particles with multiplicity, or space-time events.

130 The exact component rule

Identify a binary configuration with its occupied set $S \subseteq \mathbb{Z}^2$, which may be infinite. Join distinct occupied sites whenever their Chebyshev distance is at most two. A connected component that is a translate of an input A in Table 12 is replaced by the corresponding translate of B . All other components remain unchanged. Only translates are recognized; neither reflections nor rotations are implicitly included. Call the resulting map G .

Rule	Input A	Output B
E	$\{(0, 0), (1, 0)\}$	$\{(1, 0), (2, 0)\}$
W	$\{(0, 0), (2, 0)\}$	$\{(-1, 0), (1, 0)\}$
R	$\{(0, 0), (1, 0), (3, 0)\}$	$\{(-1, 0), (1, 0), (4, 1)\}$
L	$\{(0, 0), (2, 0), (4, 0)\}$	$\{(0, 1), (3, 1), (4, 1)\}$

Table 12: The four exact component replacements. A surrounding empty halo is part of recognition, not an optional condition.

Lemma 130.1 (Recognition and bounded matching). *The component prescription is unambiguous. On every finite or infinite configuration it gives a bijection from old occupied sites to new occupied sites whose displacements have max norm at most one. In particular, outputs from different components do not overlap.*

Proof. Each input shape is connected in the distance-two graph. An exact translate has a unique anchor, namely its minimum x coordinate and common y coordinate. The pair shapes have different gaps; the triple shapes have different gap lists. Therefore distinct rules cannot recognize the same component. Isolated sites, nonhorizontal components, larger components, and infinite components are unrecognized and stay fixed.

Use the following source-to-target bijections within recognized components:

$$\begin{aligned}
 E : & \quad (0, 0) \mapsto (1, 0), \quad (1, 0) \mapsto (2, 0), \\
 W : & \quad (0, 0) \mapsto (-1, 0), \quad (2, 0) \mapsto (1, 0), \\
 R : & \quad (0, 0) \mapsto (-1, 0), \quad (1, 0) \mapsto (1, 0), \quad (3, 0) \mapsto (4, 1), \\
 L : & \quad (0, 0) \mapsto (0, 1), \quad (2, 0) \mapsto (3, 1), \quad (4, 0) \mapsto (4, 1).
 \end{aligned}$$

Use the identity on every frozen component. Every displacement has norm at most one. Distinct components have all mutual distances at least three. If matched outputs from two components coincided, their source sites would have distance at most two by the triangle inequality, a contradiction. This includes a recognized component next to a frozen finite or infinite component. Outputs within each component are distinct by the displayed bijection. The combined matching is therefore a bijection onto the new occupied set. $\square$

For finite S , the lemma proves exact particle conservation. For infinite S , it proves a well-defined bounded-displacement matching without subtracting infinite cardinalities. Two output components may join after a step; the next step simply recognizes or freezes the newly formed component. Such joining does not invalidate the one-step argument.

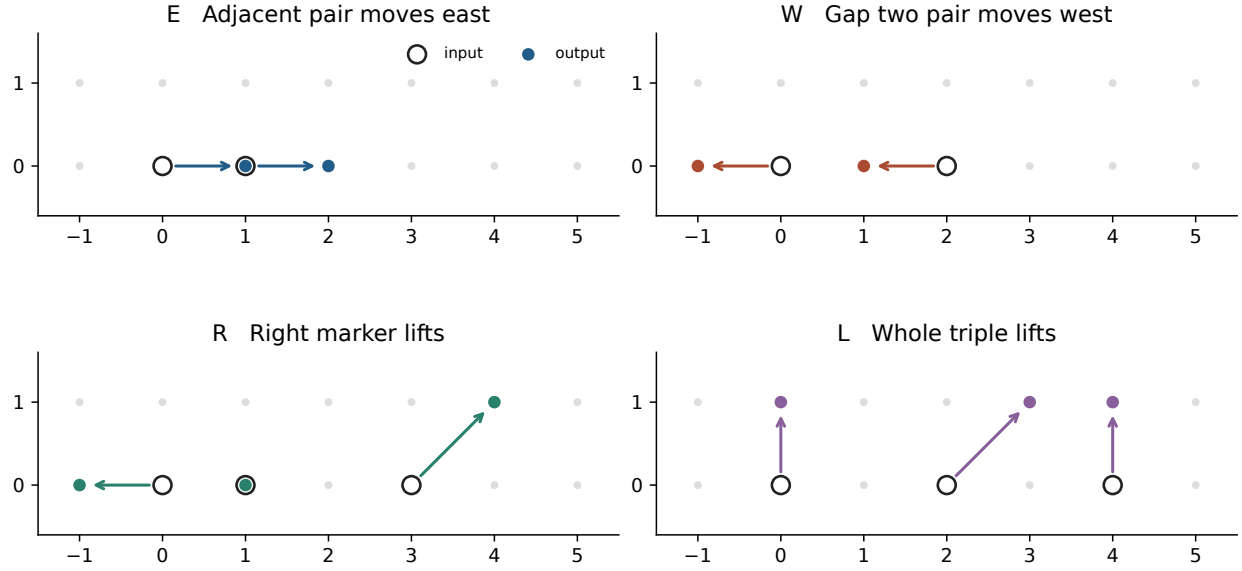


Figure 8: Exact primitive replacements and the source-to-target matching. Gray grid points are unoccupied reference sites. The two transverse bounce changes are visible in R and L . The required empty radius-two halo around each input is not drawn.

131 Locality on the full shift

For a translated input $A + a$, define its halo and recognition predicate by

$$H(A + a) = (A + a) + [-2, 2]^2, \quad P_{A,a}(S) \iff S \cap H(A + a) = A + a,$$

where intervals denote integer lattice intervals. This finite predicate is equivalent to saying that $A + a$ is exactly an occupied connected component. Indeed, the shape is connected, and the first site of any occupied path leaving it must lie in its halo. Conversely, an empty surrounding halo excludes every such path. This works on infinite configurations as well: a fragment of a larger or infinite component cannot be falsely rewritten because a window truncates the rest.

To evaluate the new bit at z , consider only anchors for which z is in a primitive input or output. If the old bit is zero, a successful predicate turns it on exactly when the corresponding replacement creates z . If the old bit is one, a successful predicate turns it off exactly when the replacement removes z . Otherwise the bit is preserved. Lemma 130.1 ensures that these local decisions agree with the component prescription on the entire binary full shift.

For every candidate and each input site p , one has $\|p - z\|_\infty \leq 4$. Its halo is therefore within radius six of z . More precisely, the changed-site cylinders read only

$$D_G = [-6, 6] \times [-3, 2], \quad |D_G| = 78. \quad (131.1)$$

The central unchanged bit lies in the same rectangle. All formulas commute with translations. No input shape is recognized in the vacuum, so G is quiescent. This proves the upper radius bound on all 2^{169} square radius-six windows without enumerating them. Section 137 proves that the smaller 78-cell dependency rectangle is exact.

Let τ be vertical translation by $(0, 1)$, with $\tau S = S + (0, 1)$, and set

$$F = \tau \circ G. \quad (131.2)$$

Translation equivariance gives $F^t(S) = \tau^t G^t(S)$. Evaluating F at z evaluates G at $z - (0, 1)$, so its read rectangle is

$$D_F = [-6, 6] \times [-4, 1]. \quad (131.3)$$

Thus F also has radius at most six, fixes the vacuum, and preserves every finite mass. Although the component matching for G moves sites by at most one, no such displacement bound is needed to reprove conservation after the global translation.

132 Four particles and the exact section clock

Start from

$$S_0 = \{(0, 0), (3, 0), (4, 0), (k, 0)\}, \quad k \geq 7.$$

Put

$$T_n = n^2 + (2k - 11)n \quad (n \in \mathbb{N}). \quad (132.1)$$

Theorem 132.1 (Complete phase description). *For every $n \geq 0$,*

$$G^{T_n}(S_0) = \{(0, n), (3, n), (4, n), (k + n, n)\}.$$

Writing $d = k + n$, the cycle $[T_n, T_{n+1})$ consists of these two inclusive phase lists:

$$G^{T_n+j}(S_0) = \{(0, n), (3 + j, n), (4 + j, n), (d, n)\}, \quad 0 \leq j \leq d - 6, \quad (132.2)$$

$$G^{T_n+d-5+q}(S_0) = \{(0, n), (d - 4 - q, n), (d - 2 - q, n), (d + 1, n + 1)\}, \quad 0 \leq q \leq d - 6. \quad (132.3)$$

In particular, each displayed support contains four distinct particles.

Proof. At the section, the adjacent pair is isolated from both markers. For $j < d - 6$, its separation from each marker is at least three, so E advances it by one. At $j = d - 6$, the right component is exactly the R triple $\{d - 3, d - 2, d\}$ in row n ; the left marker is separated because $d - 3 \geq 4$. The next state is the $q = 0$ state of (132.3).

The returning gap-two pair is separated from the lifted marker by three columns: $(d+1) - (d-2) = 3$. A row difference of one does not reduce this max-norm distance. The separation only grows during the return. For $q < d - 6$, the pair's leftmost site is at least three, so W applies and decreases both horizontal coordinates by one. At $q = d - 6$, the lower component is the exact L triple $\{0, 2, 4\}$, separated from the upper marker by at least $d - 3 \geq 4$. One L step gives the next section.

The duration is $(d - 6) + 1 + (d - 6) + 1 = 2d - 10$. Summing these durations for $d = k, k + 1, \dots, k + n - 1$ gives (132.1). This proves induction from $n = 0$ [read: this completes the induction on n]. The endpoints in the two lists cover exactly $2d - 10$ states and exclude the next section; they remain valid for $k = 7$. $\square$

[write] [Theorem 132.1 is the planar lift of source 17's Proposition 89.2: the four input shapes, the two moving pairs, the cycle of $2d - 10$ steps and the clock are source 17's, and so are the steps of the proof; only the transverse lifts of the right marker and of the left-bounce output are new, as source 21 says above.]

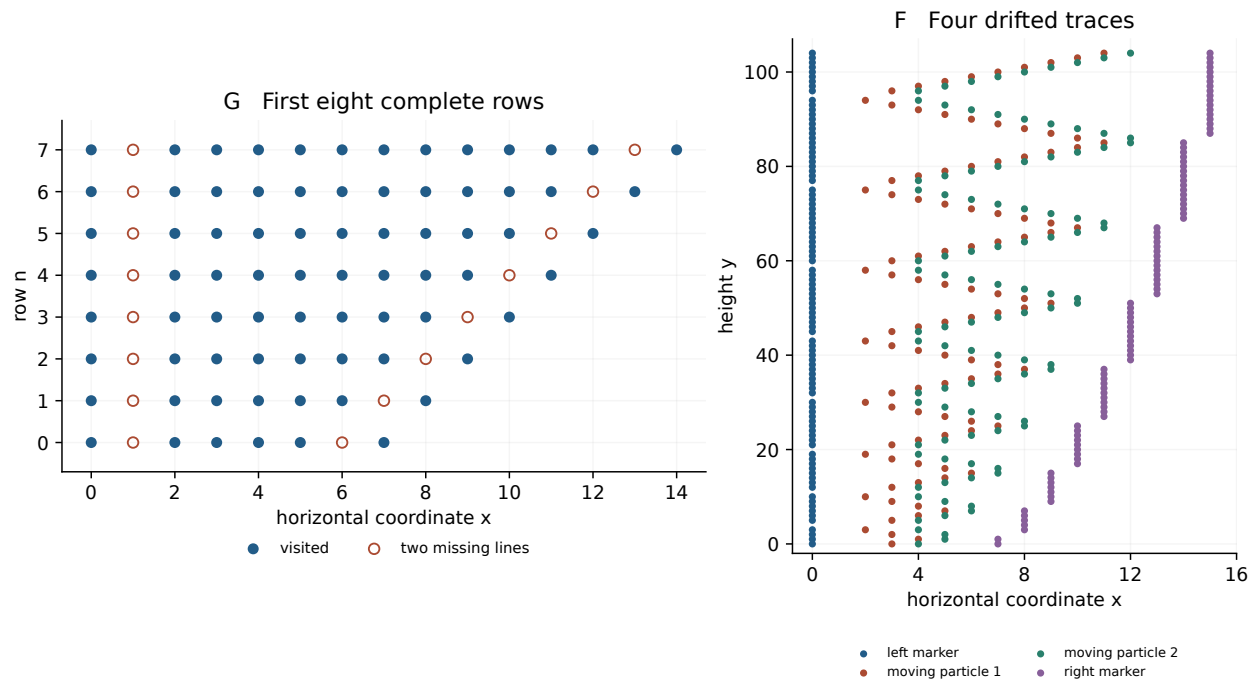


Figure 9: Actual simulation of the component rule from $k = 7$. Left: the first eight completed rows visited by G , simulated through T_8 , with the two missing families marked by open circles. Right: every occupied site of F through time 96, colored by the four particles using the bounded matching. The drifted traces have no repeated site. Both panels are generated from the rule, not drawn from the count formulas.

133 Visited sites and first arrivals

Theorem 133.1 (The wedge and its two missing lines). *The complete stationary-frame visited set is*

$$V_k = \{(0, n) : n \geq 0\} \cup \{(x, n) : n \geq 0, 2 \leq x \leq k + n - 2\} \cup \{(k + n, n) : n \geq 0\}. \quad (133.1)$$

It is the integer wedge $n \geq 0, 0 \leq x \leq k + n$ with precisely the two families $x = 1$ and $x = k + n - 1$ removed.

Proof. On row n , the east phase visits $[3, k + n - 2]$ and the west phase additionally reaches 2. It never reaches 1. The markers occupy 0 and $k + n$. The right marker may first occupy its row during the preceding cycle, but no extra sites result. Future cycles use higher rows. Hence the union is exactly (133.1). $\square$

Let $\theta_k(x, n)$ be the first time $G^t(S_0)$ occupies (x, n) , and leave it undefined at an unvisited site. For $n \geq 1$, put $d = k + n$. Then the exact disjoint branches are

$$\theta_k(x, n) = \begin{cases} T_n, & x \in \{0, 3, 4\}, \\ T_n + 2d - 11, & x = 2, \\ T_n + x - 4, & 5 \leq x \leq d - 2, \\ T_n - (d - 6), & x = d. \end{cases} \quad (133.2)$$

The same formula holds on row zero except that $\theta_k(k, 0) = 0$. For the right-marker branch, its creation time is $T_{n-1} + d - 6$, and $T_n - T_{n-1} = 2d - 12$, giving the displayed value. The interior branch is the leading east-moving member's first visit. The site 2 first appears in the last pre- L return state, whose relative time is $d - 5 + d - 6 = 2d - 11$. These branches partition every visited row even at the smallest gap. They establish piecewise-quadratic first-arrival data directly, without using a certificate compiler.

134 Exact centered counts and stationary inverse

For integer $N \geq 0$, define

$$C_{G,k}(N) = \#(V_k \cap [-N, N]^2), \quad R_{G,k}(M) = \min\{N \in \mathbb{N} : C_{G,k}(N) \geq M\}, \quad M \in \mathbb{N}.$$

Theorem 134.1 (Stationary count and complete inverse). *One has*

$$C_{G,k}(N) = \begin{cases} 1, & N = 0, \\ N(N + 1), & 1 \leq N \leq k - 2, \\ \frac{N^2 + (2k - 1)N - k^2 + 3k - 4}{2}, & N \geq k - 1. \end{cases} \quad (134.1)$$

The exact generalized inverse is

$$R_{G,k}(M) = \begin{cases} 0, & 0 \leq M \leq 1, \\ \left\lceil \frac{\sqrt{1 + 4M} - 1}{2} \right\rceil, & 2 \leq M \leq (k - 2)(k - 1), \\ \left\lceil \frac{\sqrt{8M + 8k^2 - 16k + 17} - (2k - 1)}{2} \right\rceil, & M > (k - 2)(k - 1). \end{cases} \quad (134.2)$$

Thus $C_{G,k}(N) = \frac{1}{2}N^2 + (k - \frac{1}{2})N + O_k(1)$ and $R_{G,k}(M) = \sqrt{2M} - k + \frac{1}{2} + O_k(1)$. [write] [The terms $-k + \frac{1}{2}$ are absorbed by $O_k(1)$; more precisely $R_{G,k}(M) = \sqrt{2M} - k + \frac{1}{2} + \vartheta_M + o(1)$ with $0 \leq \vartheta_M < 1$, the ceiling's rounding.]

Proof. All visited coordinates are nonnegative. For $1 \leq N \leq k - 2$, every included row contributes exactly N sites: 0 and $2, \dots, N$. At $N = 0$ only the origin contributes.

For $N \geq k$, the full wedge contributes

$$(N + 1)^2 - \frac{(N - k)(N - k + 1)}{2}.$$

Subtract the $N + 1$ holes on $x = 1$ and the $N - k + 2$ holes on $x = k + n - 1$. Simplification gives the final branch. At $N = k - 1$, the complete square loses $N + 1$ sites on $x = 1$ and one upper-edge hole; $N(N + 1) - 1$ agrees with that branch. Omitting either family of holes would produce the wrong polynomial.

The count is strictly increasing, and solving the corresponding quadratic inequality gives the two ceiling formulas. The threshold in (134.2) places the last inverse in the last count branch. The asymptotics follow by expanding its positive root; integer rounding contributes a bounded error. $\square$

135 Vertical drift and four disjoint traces

For each time t , let $n(t)$ be the unique integer satisfying $T_{n(t)} \leq t < T_{n(t)+1}$. Call the left marker and two matched moving particles the three bulk trajectories. Their F height is

$$h(t) = t + n(t).$$

The right marker has G row $r(t) \in \{n(t), n(t) + 1\}$, so its F position is $(k + r(t), v(t))$, where $v(t) = t + r(t)$. The functions $n(t)$ and $r(t)$ are nondecreasing, so $h(t)$ and $v(t)$ are strictly increasing.

Lemma 135.1 (Disjointness of all four traces). *Each particle's F trace has no self-intersection, and the four traces are pairwise disjoint. Every time contributes four new sites to the all-time visited set.*

Proof. The strictly increasing heights exclude self-repetition. All three bulk sites at a given time have the same height and distinct horizontal coordinates. Their heights agree across two times only when the times agree, so the three bulk traces are pairwise disjoint.

Suppose a marker site at time t equals a bulk site at time u . Write $r = r(t)$ and $n = n(u)$. Every bulk horizontal coordinate is at most $k + n - 2$, while the marker coordinate is $k + r$. Equality forces $n \geq r + 2$. Height equality gives $u = t + r - n \leq t - 2$. Monotonicity then implies $n(u) \leq n(t) \leq r(t) = r$, contradicting $n \geq r + 2$. Thus no marker site lies on a bulk trace. $\square$

The three bulk trajectories have one site at every nonnegative height except the skipped levels

$$a_n = T_n + n - 1 = n^2 + (2k - 10)n - 1, \quad n \geq 1. \quad (135.1)$$

Indeed $n(t)$ rises by one at T_n , so h jumps by two there. The right marker rises to row $n + 1$ at

$$\rho_n = T_n + k + n - 5.$$

It therefore skips precisely

$$b_n = \rho_n + n = n^2 + (2k - 9)n + k - 5, \quad n \geq 0. \quad (135.2)$$

Both sequences are strictly increasing. They interlace: $b_0 < a_1$, and for $n \geq 1$,

$$a_n < b_n < a_{n+1}, \quad b_n - a_n = n + k - 4, \quad a_{n+1} - b_n = n + k - 5.$$

No height is skipped by both the bulk and marker families. Notice that “four new sites per time” alone does not count sites in a spatial box: the four sites may enter that box at different heights and horizontal positions.

136 The exact drift count and its inverse

Write $V_{F,k} = \bigcup_{t \geq 0} F^t(S_0)$ and define $C_{F,k}(N)$ and $R_{F,k}(M)$ by the same centered-box and generalized-inverse conventions.

Lemma 136.1 (The horizontal cutoff). *Every F -visited site (x, y) satisfies $0 \leq x \leq \max(k + 1, y)$ and $y \geq 0$. Consequently, for $N \geq k + 1$, horizontal truncation removes no visited site of height at most N .*

Proof. For bulk sites in rounds $n = 0, 1$, $x \leq k + n - 2 \leq k - 1$. For $n \geq 2$, $T_n \geq T_2 = 4k - 18 \geq k - 2$, and hence $x \leq k + n - 2 \leq T_n + n \leq y$.

For the right marker, rows $r \leq 1$ give $x = k + r \leq k + 1$. When $r \geq 2$, its first time in that row is $\rho_{r-1} = T_{r-1} + k + r - 6$, so

$$y - x \geq T_{r-1} + r - 6 \geq T_1 - 4 = 2k - 14 \geq 0.$$

All coordinates are nonnegative by the phase formulas and the drift. The first lifted marker has $x = k + 1$, so the claimed threshold cannot simply be replaced by $N \geq k$. $\square$

Theorem 136.2 (Floor-root count and inverse). *For every integer $N \geq k + 1$, let $\alpha = 2k - 10$ and $\beta = 2k - 9$. Then*

$$A_k(N) = \#\{n \geq 1 : a_n \leq N\} = \left\lfloor \frac{\sqrt{\alpha^2 + 4(N + 1)} - \alpha}{2} \right\rfloor, \quad (136.1)$$

$$B_k(N) = \#\{n \geq 0 : b_n \leq N\} = 1 + \left\lfloor \frac{\sqrt{\beta^2 + 4(N - k + 5)} - \beta}{2} \right\rfloor, \quad (136.2)$$

$$\begin{aligned} C_{F,k}(N) &= 4(N + 1) - 3A_k(N) - B_k(N) \\ &= 4N - 4\sqrt{N} + O_k(1). \end{aligned} \quad (136.3)$$

As $M \rightarrow \infty$,

$$R_{F,k}(M) = \frac{M}{4} + \sqrt{\frac{M}{4}} + O_k(1). \quad (136.4)$$

Proof. Lemma 136.1 reduces the box count to a height count. The three disjoint bulk traces each omit $A_k(N)$ of the $N + 1$ heights, and the marker trace omits $B_k(N)$. Lemma 135.1 allows these counts to be added without overlap. Solving the two quadratic inequalities gives (136.1)–(136.2); the stated cutoff ensures that their expressions are nonnegative. Each is $\sqrt{N} + O_k(1)$.

For the inverse, set $m = M/4$ and $u = m + \sqrt{m}$. Then $u - \sqrt{u} = m + O(1)$. Rounding u to an integer changes the count by $O_k(1)$. Beyond the transient, increasing N by one adds four sites at a nonskipped height, one at a bulk skip, or three at a marker skip. The interlacing excludes a simultaneous skip. Thus an $O_k(1)$ count discrepancy can be corrected by $O_k(1)$ further radius, proving (136.4). $\square$

For completeness, the finite transient also has an exact formula requiring no limiting convention:

$$C_{F,k}(N) = \sum_{t=0}^N \#\{(x, y) \in G^t(S_0) : x \leq N, t + y \leq N\} \quad (N \geq 0). \quad (136.5)$$

All coordinates are nonnegative and the traces are disjoint; times $t > N$ have height greater than N . Thus (136.5) follows directly and can use the complete phase formulas. Together with (136.3), it gives an exact algorithm for every inverse value: check the finite transient, then find the least integer satisfying the monotone floor inequality. The implementation evaluates roots with integer square roots, including exact-square boundaries, rather than floating-point approximations.

137 Exact Boolean degree and all essential inputs

For input shape A at anchor a , define the exact-component indicator

$$M_{A,a}(c) = \prod_{p \in A+a} c_p \prod_{q \in H(A+a) \setminus (A+a)} (1 - c_q). \quad (137.1)$$

On binary inputs it equals one precisely for an isolated recognized component. The local rule is the following real multilinear polynomial on binary inputs:

$$g_z(c) = c_z + \sum_{(A,B)} \sum_{b \in B \setminus A} M_{A,z-b}(c) - \sum_{(A,B)} \sum_{a \in A \setminus B} M_{A,z-a}(c). \quad (137.2)$$

Its validity on the whole Boolean cube follows from exact recognition and disjoint outputs, not merely from checking the selected orbit. There are respectively 2, 4, 4, 6 signed translated indicators of types E, W, R, L . Each halo is a rectangle of height five and width 6, 7, 8, 9, respectively. The indicator degrees are consequently 30, 35, 40, 45.

Summing (137.2) over sites gives a second finite-input conservation proof: each translated-indicator sum is unchanged by the offset, and $|B \setminus A| = |A \setminus B|$. Only finitely many indicators can be nonzero on a finite support, so these sum rearrangements are legitimate.

Theorem 137.1 (Exact degree and exact dependency set). *The unique real multilinear polynomial for the local Boolean rule has degree exactly 45. Every cell of D_G is essential and no other cell is; F has exactly the translated dependency set D_F . Both specified rules therefore have exact centered Chebyshev radius six.*

Proof. The polynomial (137.2) is multilinear because each indicator uses every input bit at most once. The six L indicators have distinct rectangular halos of size 45. Each has three occupied factors and 42 complementary factors, so its degree-45 top coefficient is $(-1)^{42} = 1$, multiplied by the birth or removal sign. Their top monomials are distinct and cannot cancel one another; every other indicator has degree at most 40. This proves exact degree 45 by uniqueness of multilinear interpolation on the Boolean cube.

For the three births, the L halos have horizontal intervals $[-2, 6]$, $[-5, 3]$, $[-6, 2]$ and vertical interval $[-3, 1]$. For the three removals, the horizontal intervals are $[-2, 6]$, $[-4, 4]$, $[-6, 2]$ and the vertical interval is $[-2, 2]$. Their union is exactly D_G . Every one of its 78 variables therefore occurs in a nonzero degree-45 monomial.

If a Boolean function were independent of one such variable, the difference between the polynomial's one and zero restrictions would vanish on the Boolean cube. Uniqueness of multilinear interpolation would force this derivative polynomial to vanish identically, contradicting the nonzero monomial containing that variable. Thus all 78 variables are essential. Section 131 excludes any other dependency. Translating the evaluation point gives D_F and its essentiality. Both rectangles contain a horizontal coordinate of absolute value six, which proves the exact centered radius. $\square$

The release also supplies an explicit finite witness for each cell: two configurations differing at that cell only, with different central outputs. The file `essential-input-witnesses.json` in `scientific/audit/` records all 78 pairs. They are checked with both the direct component simulator and the independently encoded cylinder evaluator. Translating both inputs down one row gives the corresponding F witness. This is a concrete flip-witness confirmation independent of relying on the top-monomial argument. Exact radius here is a property of these two rules, not a least-radius claim over all possible binary constructions.

138 A literal one step quartic with unique auxiliaries

Enumerate the 78 cells of D_G in row-major order, increasing y and then x , and add one external output variable y_{out} . There are 79 external variables. For each of the sixteen indicators, use its own unshared multiplication chain, listing occupied factors first and complementary factors afterward. Within each group use row-major order. Start the chain with its first external occupied factor. At every subsequent factor introduce one new auxiliary z and one residual

$$z - uv \quad \text{or} \quad z - u(1 - v), \quad (138.1)$$

where u is the preceding chain value and v is the next external input. Indicator order and all variable indices are recorded canonically in the certificate.

There are

$$2 \cdot 29 + 4 \cdot 34 + 4 \cdot 39 + 6 \cdot 44 = 614 \quad (138.2)$$

auxiliaries and the same number of gate residuals. Add the 78 bit residuals $c_i(c_i - 1)$ and the output residual

$$y_{\text{out}} - c_{(0,0)} - \sum_{\text{births}} z_{\text{last}} + \sum_{\text{removals}} z_{\text{last}}. \quad (138.3)$$

Let P be the sum of squares of these $614 + 78 + 1 = 693$ residuals. The circuit gives an ordinary integer polynomial, not a compressed truth predicate or a symbolic placeholder for an unspecified computation. [write] [Arithmetizing a Boolean circuit by bit equations and product gates is the standard device of Part IV's gate residuals (Section 46); the counts 614, 693 and 3403 are source 21's own.]

Theorem 138.1 (The local zero set and canonical witnesses). *The total degree of P is exactly four. For any fixed external input-output tuple, its natural auxiliary fiber has exactly one element if that tuple is a valid binary local transition, and has no element otherwise. The same statement holds when the auxiliaries range over all real numbers.*

Proof. Every residual has degree at most two, so P has degree at most four. The squares of bit residuals supply positive fourth-power monomials of the external input variables, which cannot be canceled by the other squared residuals here; hence the degree is exactly four.

Since P is a sum of real squares, $P = 0$ is equivalent to simultaneous vanishing of every residual. The bit residuals force each external input into $\{0, 1\}$. Each acyclic chain equation then forces its next auxiliary uniquely, starting from an external input. All forced values are themselves zero or one. Finally (138.3) forces the output to equal (137.2), hence the actual binary local output. Conversely, for a valid transition these forced chain values make every residual zero.

Therefore the unique real auxiliary tuple is natural, and invalid external tuples have no real or natural auxiliary solution. This argument proves uniqueness by the specific deterministic chain structure. It is not an inference from finite perturbation tests or from decidability. $\square$

138.1 The exact algebra ledger

There are $2 \cdot 1 + 4 \cdot 1 + 4 \cdot 2 + 6 \cdot 2 = 26$ plain-product gates, each with two residual monomials, and $2 \cdot 28 + 4 \cdot 33 + 4 \cdot 37 + 6 \cdot 42 = 588$ complementary-factor gates, each with three. The bit residuals each have two monomials and the output residual has 18. Thus the uncollected ledgers are

$$\begin{aligned} \text{residual-monomial occurrences} &= 26 \cdot 2 + 588 \cdot 3 + 78 \cdot 2 + 18 = 1990, \\ \text{ordered square-expansion occurrences} &= 26 \cdot 4 + 588 \cdot 9 + 78 \cdot 4 + 18^2 = 6032. \end{aligned}$$

Collecting equal monomials in the literal exported polynomial gives Table 13. The last three values were independently reconstructed and compared exactly, not assumed from the circuit description.

Resource	Exact value
External input bits	78
External variables including output	79
Canonical natural auxiliaries	614
Total variables	693
Squared residuals	693
Residual-monomial occurrences	1990
Ordered square-expansion occurrences	6032
Distinct collected monomials	3403
Largest absolute collected coefficient	2
Total degree	4

Table 13: Literal resource ledger for the deliberately unshared one-step circuit. No auxiliary-minimality claim is made.

The machine-readable `scientific/local-quartic-certificate.json` contains all 3403 collected terms, the full variable order, all 614 gates, all 693 residuals, and all sixteen indicators. A monomial is a list of variable indices with repetitions representing powers; coefficients are exact integers. The independent artifact reader re-expands squared residuals using unordered pairs and compares the resulting sparse polynomial exactly. A second reconstruction begins from the geometric primitive replacements and independently rebuilds indicators, gates, residuals, and every coefficient. Thus the expanded artifact is checked against both its circuit and its intended geometry.

139 Verification and portable reproduction

The release separates proofs from executable evidence. The reviewed source proofs `PROOF.md` and `LOCAL_ALGEBRA.md` are preserved under `scientific/`. This report supplies a connected mathematical presentation. Independent reviews and explicit flip witnesses accompany the source files. No finite regression is offered as an enumeration of all full-shift configurations.

139.1 Independent routes and finite coverage

The direct simulator `component_rule.py` builds occupied components and applies the four replacements. The pointwise evaluator `local_rule.py` imports neither that simulator nor its rule dictionary; it encodes the sixteen occupied-and-empty cylinders independently and exports exact integer masks. Both G and F are evaluated.

The primary suite checks all 4096 supports in a 4×3 rectangle, all 8192 supports in a 13-site line, arbitrary exterior contexts, one-site halo interference, interacting active and malformed templates, births over a full-radius candidate region, large translations, exact orbit phases, first arrivals, both box counts, disjoint traces, and integer-root boundary cases. Its deterministic receipt reports 76680 direct orbit phase steps, 46980 first-arrival checks, and 53601 drift count checks. These are finite checks of the displayed formulas, not substitutes for the all-input proofs.

The quartic artifact reader checks 350 binary local fixtures and their incorrect binary outputs. At the vacuum witness it tests all 614 upward unit perturbations, one per auxiliary. It also compares the sparse identity at 60 nonbinary integer points and obtains all 78 essentiality witnesses. These tests do not exhaust all auxiliary perturbations on all inputs; uniqueness is proved in Theorem [138.1](#). Independent component, drift, cross-implementation, and geometry-to-quartic reconstruction checks provide separate routes. Every scientific test used by the release replay uses explicit exception checks, so optimization does not disable its checks. Normal and `-O` runs must agree with the exact expected receipts, including JSON value types.

139.2 Integrity before execution and isolated outputs

The portable verifier first compares the exact required payload file set and independently pinned SHA-256 values before executing any packaged scientific code. It then runs in fresh external copies. Adjacent-output scripts are executed only in those copies. Expected receipts are compared recursively with exact types, so a Boolean is not accepted in place of an integer. The verifier checks that release bytes, file modes, and modification times are unchanged after the run. The release checksum list and archive checksum provide an additional external integrity layer; executing the verifier still presumes trust in that bootstrap file.

The README supplies exact commands and requirements. Scientific replay needs Python 3.10 or later and only the standard library. PDF rebuild needs `pdflatex` and the listed TeX packages; figure regeneration additionally needs Matplotlib. Supplied vector PDF figures allow a report rebuild without Matplotlib. Build products and replay receipts belong outside the release. Bounded tamper regressions test changed code, missing required files, altered certificate content, and strict typed comparisons; they do not run the scientific suite on tampered payloads.

The portable package contains no Python bytecode caches, private absolute paths, third-party source caches, or files from earlier releases. Those earlier releases are cited, not modified. The exact source and receipt lineage is recorded in `verification/source-lineage.json`.

[write] **The release in this report.** These are the delivered names. `build.sh`, `reproduce.sh`, `replay.py`, `tamper_regression.py` and `figures/make_figures.py` are shipped as `code/21-planar-shuttle-*`, the programs of `scientific/code/` as `code/21-planar-shuttle-code-*` and those of `scientific/audit/` as `code/21-planar-shuttle-audit-*`; the receipts, review bindings and certificates of `scientific/audit/` as `data/21-planar-shuttle-audit-*`, the two certificates of `scientific/` (among them `local-quartic-certificate.json`), the figure data and the `verification/` records as `data/21-planar-shuttle-*`; the two figures as `figures/21-planar-shuttle-rule.pdf` and `figures/21-planar-shuttle-orbits.pdf`; and `scientific/PROOF.md`, `LOCAL_ALGEBRA.md` and the four audit notes as `21-planar-shuttle-*.md` at the report root. Not shipped: the manuscript (printed here), its PDF, README and checksum ledger. Several delivered receipts are byte copies of one another (normal, optimized and expected copies); they are shipped as delivered, because the replay compares them by name. At placement on Windows every replay step passed in normal and optimized mode after two workarounds for harness defects, which the README states: `replay.py` starts its children with `-I`, which ignores `PYTHONUTF8`, so two audits fail to print non-ASCII text on a cp1252 console unless run with `-X utf8`; and several writers emit CRLF line ends on Windows, so the byte comparison of the rebuilt local quartic certificate fails although its content equals the delivered file.

[write] **The programme’s review.** The Hilbert’s-tenth programme reviewed this archive in `review_binary_planar_four_particle31.md`, with its checker and receipt, in `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/` (commit 16f50dcc6, 3 October 2026): “PASS within the stated scope; no error found”. It read the proof notes in full, authenticated the archive and seven members, and, from the four printed replacement rules alone, rebuilt the sixteen indicators, the 32 recognition cylinders of G and F , the 614 product-chain gates and the 693 residuals; their expansion matches all 3403 saved terms, of degree four. It confirms the six noncancelling degree-45 leaders, the 78 essential cells and exact radius six for these rules, the phase intervals including $k = 7$, both missing lines, the cutoff $N \geq k + 1$, and both drift asymptotics, and adds 483 phase states, 183 first arrivals and 261 box counts for $k = 7, 8, 11$. It executed none of the archive’s Python, and it notes that the 614 auxiliaries are not a complete evaluation cost of the quartic and that the rule gives no universal simulation; the repository’s universal operation bounds are unchanged.

140 Consequences limits and next questions

This example realizes two distinct counting regimes with an ordinary binary mass-four orbit: a quadratic stationary-frame wedge count with sharp coefficient $1/2$, and a linear drifted count with a square-root deficit of coefficient four. The deficit records skipped heights of four disjoint traces, rather than weighted-site multiplicities. The two removed wedge lines and the finite horizontal cutoff are essential to the exact formulas.

The construction proves existence for the specified initial family, not a classification of arbitrary malformed orbits. Malformed inputs are covered by the conservation and locality proof; they are not claimed to enter a shuttle. Neither G nor F is asserted reversible or universal. No smallest mass, smallest radius among constructions, minimal number of essential cells, optimal Boolean degree, or optimal auxiliary count is asserted.

The local quartic relates 78 input bits to one next-step output. Its unique real auxiliary tuple follows from acyclic multiplication chains and bit equations. It does not imply an unbounded-time finite-fold representation for arbitrary recursively enumerable sets. Report 30 separately proves that,

for a fixed promised conservative CA and fixed mass-at-most-four input, untimed reachability of a canonical *complete configuration* has an effective ordinary quartic with one natural witness tuple, optionally including first occurrence time [source 20, Part VII]. That theorem uses the full classified timed charts and whole-orbit injectivity in the expanding case. Its hypotheses include this example, but its general certificate is not the 614-auxiliary local polynomial here. Site or partial-pattern recurrence cannot be substituted for complete-target injectivity, and no real-witness conclusion is inherited from that theorem.

Natural next questions remain tightly connected to this construction:

1. Can a binary planar four-particle wedge be realized with a smaller centered radius or fewer essential cells while retaining a transparent all-input conservation proof? ([write] Open. Compare source 17's Question 6 for the one-dimensional rule, which stays open too.)
2. Can shared products reduce the 614 local auxiliaries without introducing noncanonical inactive values, and what exact collected polynomial results?
3. Which changes to the two transverse bounces preserve four-trace disjointness after drift, and which change the coefficient of the square-root deficit?

These are questions, not claims settled by the present finite checks or by the existing general classification.

A Part I: a minimal worked use of the compiler

The following code can be run from the archive root. ([write] “The archive root” is the delivered layout, with `code/signal_certificates.py`; in this report the module is shipped as `code/07-collision-geometry-signal_certificates.py`, so the snippet runs in a copy that restores the delivered name (README of this report).) It builds the first-collision tournament and constructs a complete certificate without using an external algebra system.

```
import sys
sys.path.insert(0, "code")
from signal_certificates import Machine, compile_skeleton

machine = Machine(
    {"A": 2, "B": 0, "C": -1, "H": 0, "R": 0},
    {
        frozenset(("A", "B")): ("H",),
        frozenset(("B", "C")): ("R",),
        frozenset(("A", "B", "C")): ("R",),
    },
)
skeleton = (((0, 1),),)
cert = compile_skeleton(machine, ("A", "B", "C"), skeleton)
gaps = (1, 1)
slacks = cert.witnesses(gaps)
assert cert.evaluate(gaps, slacks) == 0
assert not cert.accepts((3, 1)) # An earlier BC collision.
assert not cert.accepts((2, 1)) # A simultaneous ABC collision.
print(cert.to_dict())
```

The order of rows in the exported matrices is lexicographic and need not match the order used in the displayed mathematical formula. Row order changes only the order of slack coordinates, not the represented relation or the uniqueness statement.

B Part I: dependency map and claim status

Claim	Principal ingredients	Status in this package
Exact chronology chamber	Intercept induction; affine endpoint lemma; static identity and order validation	Full manuscript proof; exact tests
Event-sparse row bound	Local adjacency replacement count	Full manuscript proof; every exported ledger checked
Unique-slack quadratic	Integer strict inequalities; sum-of-squares zeros	Full manuscript proof; zero and mutation tests
Canonical first-hit quartic	Disjoint first-hit chambers; natural one-hot selector equation	Full manuscript proof; finite union tests
Causal denominator and coefficient bounds	Integer speed differences; induction on causal depth	Full manuscript proof; finite denominator checks
Small integral realization	Homogeneity; rational vertex; Cramer’s rule; Hadamard bound	Full manuscript proof
Robustness and codimension	Linear margin estimate; rank of equalities	Full manuscript proof
Finite-horizon semilinearity	Exact chamber theorem; classical Presburger and semilinear-set equivalence	Derived theorem using cited literature
Exact quasi-polynomial counts	Chamber homogeneity; face inclusion–exclusion; classical Ehrhart theorem	Derived theorem; worked counts checked
Universal signal-machine existence	Durand-Lose’s published construction	External result; rule table not reimplemented
Fixed-arity unbounded Diophantine existence	Computable finite-prefix simulation; classical MRDP	External existence theorem, not new compiler
Global convex-quadratic barrier	Positive semidefinite quadratic completion; Presburger decidability	Full manuscript proof

[write] The row “Global convex-quadratic barrier” is a special case of Part XI of [11] (Remark 13.3); source 07’s proof is kept as a second route. The rows “Exact chronology chamber”, “Event-sparse row bound”, “Unique-slack quadratic”, “Causal denominator and coefficient bounds” and “Finite-horizon semilinearity” have second proofs in Part II (Section 1.2).

C Part II: replay and build guide

The accompanying archive contains this article in editable LaTeX, the compiled PDF, exact Python scripts, expected receipts, example coefficient dictionaries, and a manifest with file hashes. It contains no third-party source PDFs. The scripts require Python 3.10 or later and only the standard library. Assertions must remain enabled; do not run Python with `-O`.

From the extracted package root run:

```
python3 scripts/run_replay.py
```

The runner copies the executable scripts into a temporary directory, executes the suites there, compares fresh JSON receipts to the stored expectations, and writes a summary under `replay-output`. The stored expected receipts and packaged examples are not overwritten. Use `-output DIR` to select a different result directory. The worked-example receipt also contains its concrete witness and explicit rejection reason for the false final-cut schema.

[write] These are the delivered names. In this report the five scripts are shipped as `code/11-signal-certificates-*.py`, `build.sh` as `code/11-signal-certificates-build.sh`, the two example files and the manifest as `data/11-signal-certificates-*.json`, and the four expected receipts as `data/11-signal-certificates-expected-*_receipt.json`. The runner looks for `scripts/`, `examples/` and `expected_receipts/` in the parent of its own directory and runs the scripts by their delivered names, so it runs only in a copy that restores the delivered layout (see the README); it writes nothing into the shipped files. Of the archive contents listed above, the editable source is printed as this Part rather than shipped separately, and the compiled PDF and the checksum file `CHECKSUMS.sha256` are not shipped (the checksum file was verified, 17 of 17 files, at placement); the manifest is shipped, and its payload list still names the delivered paths and the delivered PDF and article source.

The main executable interfaces are:

- `signal_geometry.py`: exact simulator, full-layer compiler, polynomial evaluation and rational Gaussian rank calculation;
- `signal_sparse.py`: event-coordinate and adjacency-lifetime compiler;
- `verify_additional.py`: separately seeded edge cases and sharp height checks;
- `verify_examples.py`: the article’s explicit examples and convention checks.

The low-level compilers use assertions to reject malformed schema partitions or structurally impossible collision directions. They are reference mathematical implementations, not hardened parsers for untrusted inputs. JSON residual dictionaries use `@` for the constant term, `x0`, `x1`, and so on for input positions, and the listed variable names for natural witnesses. A certificate evaluates to the sum of the squares of those rows. No expanded polynomial is required to replay it.

To rebuild the article run:

```
sh build.sh
```

A working TeX Live installation with `pdflatex`, Latin Modern fonts, AMS packages, `geometry`, `microtype`, `booktabs`, `enumitem` and `hyperref` suffices. The build script uses fixed document metadata and a fixed source-date epoch; exact PDF byte equality across different TeX distributions is not promised. The mathematical replay receipts are exact and version-independent within supported Python semantics.

D Part II: claim and source guide

The source-supported context is limited to the following claims. All certificate theorems and proofs in this article are supplied here rather than attributed to the literature.

- Durand-Lose 2009: finite configurations in Definition 1; integer inputs in Section 2; speeds in Figure 4; cyclic-tag simulation and halting in Section 5 and Theorem 5. Common-denominator rescaling is our inference, proved in (19.1).
- Durand-Lose 2007: linear-register event simulation in Section 4 of the author version, especially its ninth PDF page; rational models in Section 5. No Diophantine theorem is attributed to it.
- Durand-Lose 2006: energy and live-population bounds in Definition 3 and the following paragraph; accumulation qualifications in Sections 2.2 and 6.
- Ginsburg and Spanier 1966: equivalence of Presburger-definable and semilinear subsets of $\mathbb{N}^n$, Theorem 1.3, pp. 288–289.

Primary sources were checked on 2 October 2026. The bibliography links to the papers or their official bibliographic records; copyright-bearing PDFs are not redistributed in this package.

[write] In this report the Durand-Lose 2009 and Ginsburg–Spanier entries are merged with Part I’s entries for the same papers [3, 7]; the merged entries add source 11’s editors and links to Part I’s.

E Provenance of this report

[write] The report was built from two manuscripts that arrived together in the archives of commit 808b53ed8 (2 October 2026) and were placed by commit 798b0c5d4, which staged their programs, data and provenance files byte-identical to the delivery. (Parts III and IV were added later from two manuscripts of batch 79, and Parts V and VI from four manuscripts of batch 80; their provenance is at the end of this appendix.)

Source	Archive (batch 78)	Pin	Contributed	Printed as
07 (base)	Collision_Geometry_Diophantine_Signal_Machines.zip, manuscript 07; main file collision_geometry/article.tex, 29-page PDF	f1edb38f9	the chamber theorem, the event-sparse bound, the unique-slack quadratic, denominators and heights, small realizations, the NP bound, robustness, semilinearity, the first-hit quartic, Ehrhart counts, the tournament, the clock, the barrier, nine research directions	Part I with Appendices A and B
11	Signal_Machine_Diophantine_Certificates.zip, manuscript 11; main file signal-diophantine-release/article.tex, 15-page PDF	none named	the same theorem with event-coordinate witnesses, halting semantics, full column rank, the size ledgers, the full-layer construction, the four-speed height bound, the bounded shuttle, positive witnesses, five questions	Part II with Appendices C and D

Source 07’s pin is the full commit `f1edb38f93aa0a362a34b531a600909f6e74608a`, named in its Section 14 and in its delivered `PROVENANCE.md` (shipped as `07-collision-geometry-PROVENANCE.md`); it is an ancestor of the placement commit, and the audit blob it cites is unchanged at the write. Source 11 names no repository snapshot and cites no repository file; its only repository-related statement, that the 75- and 87-operation constructions are not improved, is true at the write.

Where the merge had to choose.

- *Base and order.* Source 07 is the base (Part I): it allows signed rational speeds without a normalization interface, proves more about the chamber, and is the longer manuscript. Source 11 follows as Part II. Every result, proof, example, remark, research question, appendix and limitation of both sources is printed, in the sources’ order and with their numbering inside each Part.
- *The common theorem.* Its two statements have different inputs and witnesses, so both are printed, each with its proof. The lemmas of Part II that coincide with lemmas of Part I (Lemmas 19.1, 20.1 and 21.1, Proposition 26.1) open with a [write] note naming their Part I counterpart, and source 11’s proofs are kept as second routes. Section 1.2 compares the two routes, including the two shuttles and the two missed-simultaneity examples, which are kept in both Parts because they are different machines.
- *A result already in the collection.* Part I’s Theorem 13.1 and Corollary 13.2 are a special case of Part XI of *canonical-diophantine-certificates*. They are printed with source 07’s statements and proofs, as a second route, and credited in Remark 13.3; credit notes were added to source 07’s abstract and conclusion. Remark 10.3 relates the first-hit quartic to the same Part.
- *Research questions.* Both agendas are printed in their Parts; Table 3 pairs the coinciding ones, and one [write] note re-scopes the part of Part II’s Question 3 that both sources already prove.
- *Front matter and bibliography.* The two title pages are replaced by one; each source’s title, author line, abstract and keywords open its Part. Section 1 is new. The two bibliogra-

phies are merged: source 11's keys DL09 and GS66 are cited as source 07's `durandlose` and `ginsburgspanier`, whose entries now carry source 11's editors and links; entries [11, 12, 13, 14, 15] were added for the cross-references.

- *Typesetting.* Equations are numbered by section in both Parts (source 07 numbered them consecutively). Source 11's macro `\N` is written `\NN` because source 07's `\N` prints $\mathbb{N}_0$; both print the natural numbers $\{0, 1, 2, \dots\}$ as their sources did. No symbol was renamed in the printed text and no normalization changed.
- *Added text.* Every sentence written in the merge is marked [write]: Section 1; the openings of the two Parts; notes in Part I in Sections 2, 5, 6, 7, 10 (and Remark 10.3), 11, 12, 13 (Remark 13.3), 14 (shipped file names, the pin), 15 and 16, and in its abstract; notes in Part II in Sections 17, 18, 19, 20, 21, 24, 25, 26 and 27; notes in the four source appendices; this appendix; and the bibliography entries marked [write]. Labels were added to the first lemma of Part I's Section 5, to one example subsection, the research directions, the conclusion and the second appendix of Part I, and to several sections, one corollary, the questions, the conclusion and the second appendix of Part II.

Files. The report directory holds this text (`article.tex`, `article.pdf`), a README, source 07's `PROVENANCE.md` and source 11's `REPRODUCIBILITY.md` (as `07-collision-geometry-PROVENANCE.md` and `11-signal-certificates-REPRODUCIBILITY.md`), and the two sources' programs in `code/` and data in `data/`, under the prefixes `07-collision-geometry-` and `11-signal-certificates-`. The README maps each delivered name to its shipped name and gives rerun instructions that do not overwrite the shipped files. Source 11's manuscript, both PDFs, source 11's delivered README and both checksum ledgers (source 07's `SHA256SUMS`, 14 of 14 files, and source 11's `CHECKSUMS.sha256`, 17 of 17, both verified at placement) are not shipped; they survive in the archives of commit `808b53ed8`. Source 07's delivered README was the placed `README.md` and has been replaced by the report README.

Parts III and IV (batch 79). [write] Two manuscripts of batch 79 of ProveIt's incoming reports were added on 2 October 2026. Both arrived in the archives of commit `aebfa386e` and were placed by commit `a7ae02511`, which staged their programs, data and provenance files byte-identical to the delivery (36 files of source 12, 35 of source 13) and retired the archives from the drop zone. A corrected code edition of source 13, `Sparse_Lattice_Diophantine_Certificates_corrected.zip`, arrived in batch 80 (commit `4e270aa46`) and was placed by `8a4e64732` (batch 80, cluster K1), which replaced five of source 13's files by their corrected bytes and added three (38 files of source 13 now); nothing printed changes (Section 47).

Source	Archive (batch 79)	Pin	Contributed	Printed as
12	<code>Conservative_Signal_Frontend_Corrected.zip</code> , manuscript 16; main file <code>paper/conservative-signal-diophantine.tex</code> with <code>paper/morita-table.tex</code> , 21-page PDF	none named	the literal universal conservative signal machine, the stack and loader arithmetic, the complete event normal form, the pivot-scaled integer lift, the one-coordinate mode control, the canonical quadratic step and horizon packet with its ledger, the finite mode closure	Part III with Appendices F–H
13	<code>Sparse_Lattice_Diophantine_Certificates.zip</code> , manuscript 19; main file <code>paper/sparse-lattice.tex</code> reading five further files, 27-page PDF; corrected code edition <code>Sparse_Lattice_Diophantine_Certificates_corrected.zip</code> (batch 80, manuscript 07)	none named	the sparse finite-horizon quartic certificate and its ledgers, the orthant-exact option, the corrected Morita–Imai interface, the false observer, the paid loader and first pulse, the doubling example, the diameter barriers, the mass-two decision and semilinearity theorems, the post-elimination quartic, five questions	Part IV

Neither manuscript names a ProveIt commit. Source 12’s delivered `CORRECTION.md` cites the programme’s review commit `9f033fa6e`, an ancestor of the placement, as the origin of its patch. Source 12’s archive is the corrected edition of batch-79 manuscript 07 (`Conservative_Signal_Diophantine_Frontend.zip`, arrival `2a8a39599`): 36 of its 38 files are byte-identical to manuscript 07’s, its article is the same, and its evaluator `code/quadratic_packet.py` is manuscript 07’s with the programme’s patch applied. Manuscript 07 is therefore superseded; nothing of it is shipped, and it survives in its arrival commit. The delivered checksum ledgers verified at placement (manuscript 07, 37 of 37 files; source 12, 44 of 44; source 13, 48 of 48) are not shipped.

Where the batch-79 addition had to choose.

- *Placement and order.* Both manuscripts answer questions of this report (Section [1.11](#)), so they are additions to it rather than a new report; source 12, the signal machine, comes first as Part III, and source 13, its lattice counterpart, follows as Part IV. Every result, proof, example, remark, question, appendix and limitation of both is printed, in the sources’ order.
- *The two editions of source 12.* The corrected edition is printed; its article is identical to the earlier edition’s. The correction concerns only the generic evaluator program and is described in Section [37](#).
- *Author lines.* Source 12’s title page describes it as a “Research note prepared with AI assistance for private review”; as the report is public, Part III’s opening describes it neutrally as an AI-assisted research note, and the delivered wording is recorded here. Source 13’s author line, “Research report prepared with AI assistance for Vladimir”, is quoted in Part IV’s opening.
- *Re-derived results.* Part III’s event lemma and uniform lift, Part IV’s comparison and sorting gadgets, its post-elimination quartic and the finite-search step of its cutoff proposition re-derive results of Parts I–II or of *canonical-diophantine-certificates*. They are printed as delivered,

with [write] notes naming the earlier result, as second routes; no novelty is claimed for them. The selector/copy construction of Part III is credited to Balas and to Jeroslow and Lowe.

- *Front matter.* The title, subtitle and abstract were widened; Section 1 gained Sections 1.7–1.11, and dated [write] notes were added where batch 79 made earlier sentences stale (Sections 1.1, 1.3, 1.4, 1.5 and 1.6, Table 3, Part I’s direction 15.2 and Part II’s Questions 1 and 2). Nothing was numbered anew: the notation of Parts III and IV is an uncaptioned table, so that the earlier tables keep their numbers, and Part III’s three appendices follow this appendix, so that Appendices A–E keep their letters.
- *Labels.* Source 12’s 49 labels are `smc:cs:` plus their delivered names, and source 13’s 53 are `smc:s1:` plus theirs; none was dropped or renamed apart from the prefix. The labels added in batch 79 (the two Parts and the five subsections of Section 1) also use `smc:cs:`, although the subsections cover both Parts.
- *Bibliography.* Source 12’s keys `d12007` and `d12006` are cited as the existing entries [4, 5] for the same papers, and source 13’s `ginsburgspanier` is the existing entry [7]. The sources’ other entries are appended as delivered, marked with their source; the entries for Balas, Jeroslow–Lowe and the programme’s reviews are added and marked [write].
- *Typesetting.* Both sources’ macro `\N` is written `\NN`; it prints the natural numbers $\{0, 1, 2, \dots\}$ as in the sources. Equations are numbered by section. Source 12’s table file is printed inline, two forced page breaks of the sources and source 13’s own table of contents (with its formatting of contents lines) are omitted, and the titles of Part III’s appendices begin with “Part III:”. No symbol was renamed and no normalization changed.
- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on Parts III and IV; Sections 1.7–1.11 and the dated notes listed above; the openings of the two Parts; in Part III, notes after Lemma 34.1, in Section 35 (the uniform lift, the selector/copy credit), after Theorem 36.1, at the end of Section 37 (the shipped release, the correction and the review), and in Appendices F and H; in Part IV, notes in Sections 38 and 39 (Part V of *canonical-diophantine-certificates*), 41 (the review’s check of the sign correction), 43, 46 (the second route), 47 (the shipped release and the review) and 48; this paragraph and the bibliography entries marked [write].
- *Excluded data.* Four regenerable files of the deliveries are not shipped: source 12’s compact closure records `numeric/MODES.jsonl.gz` and `numeric/EVENTS.jsonl.gz` and source 13’s two coefficient-explicit Morita fixtures `replay/core/fixtures/morita_doubling_n0_pulse_T7.json.gz` and `..._n2_pulse_T41.json.gz` (6.84 MB together). The shipped programs regenerate them with identical decompressed content; the README gives the commands and the retrieval from the arrival commit.

Files added in batch 79. The report directory gained source 12’s `CORRECTION.md`, `SOURCE-PROVENANCE.md` and `numeric/PROOF_AND_LEDGER.md` and source 13’s `SOURCE-PROVENANCE.md` at its root, with the prefixes `12-conservative-signal-` and `13-sparse-lattice-`, and the two sources’ programs in `code/` and records in `data/` under the same prefixes. The README maps each delivered name to its shipped name. Not shipped: both manuscripts and PDFs, both delivered READMEs, the checksum ledgers, source 12’s copy of the patch (it is the programme’s file [29]), an in-archive duplicate of source 12’s machine file, source 13’s manifest checker, packager and duplicate Morita audit, the four regenerable files above, and all of manuscript 07.

Parts V and VI (batch 80). [write] Four manuscripts of batch 80 of ProveIt’s incoming reports were added on 2 October 2026. All five archives of the cluster arrived in commit `4e270aa46` and were placed by commit `345a9e44e` (batch 80, cluster K2), which staged 116 files (programs, data, provenance and audit notes, and one figure) byte-identical to the delivery (51 of source 14, 28 of source 15, 9 of source 16, 28 of source 17) and retired the five archives from the drop zone. Every delivered checksum ledger verified at placement (manuscript 10: 65 of 65 files; 02: 37 of 37; 05: 12 of 12; 06: 11 of 11; 03: 32 of 32); the ledgers are not shipped.

Source	Archive (batch 80)	Pin	Contributed	Printed as
14	<code>Three_Mass_Reversible_Computation.zip</code> , manuscript 10; main file <code>three-mass-release/three-mass-report.tex</code> (also as four modular files), 24-page PDF	none named	the three-unit reversible compiler, the residue query, the rational gap primitive, the cleanup, the exact counts and microtimes, both radius-one realizations, the fixed-horizon quadratic certificate with its loader, literal examples, six questions	Part V, Sections 49–63
15	<code>Exact_Targets_Three_Mass_Units.zip</code> , manuscript 02; main file <code>clean-target-release/clean-target-report.tex</code> , 15-page PDF	source 14 by SHA-256	the clean return wrapper, the fresh-entry condition, exact-pair undecidability, the first-target clock, the wrapper counts, the compact certificate and its affine lift, the worked example, six questions	Part V, Sections 64–73
16	<code>Single_Unit_Three_Mass_Decidability (1).zip</code> , manuscript 05 (attribution revision of manuscript 06); main file <code>single-unit-three-mass/single-unit-mass-three.tex</code> , 10-page PDF	none named	the stationary-unit frame, the no-split lemma, the mass-three core, uniform timed Presburger definability, the numerical three-particle bound, the quartic corollary, the source comparison	Part VI, Sections 74–82
17	<code>Four_Mass_Decidability_Package.zip</code> , manuscript 03; main file <code>four-mass-bound/four-mass-decidability.tex</code> , 12-page PDF	none named	the seed library and thresholds, the additive gap counter, the dispatch, the expanding-shuttle normal form, anchored observations, the binary radius-six shuttle, the one-witness timing quartic, seven questions	Part VI, Sections 83–92

No manuscript names a ProveIt commit. Source 15 pins source 14 by SHA-256 (its “frozen companion release 9”: L^AT_EX source `8bfdbf66...`, PDF `1fc9a5...`, checksum ledger `c9efd37d...`), and all three match source 14’s archive. Source 16 cites source 13 by title and date (“*Sparse Diophantine certificates for finite mass lattice dynamics*, 2 October 2026, Section 9”); source 17 cites source 16 as its companion. Manuscript 06 (`Single_Unit_Three_Mass_Decidability.zip`) is the original of source 16: seven of its twelve files are byte-identical to source 16’s, and its article differs only in the abstract’s attribution, the source comparison and two bibliography entries. It is superseded; nothing of it is shipped, and it survives in its arrival commit.

Where the batch-80 addition had to choose.

- *Placement and order.* All four manuscripts answer questions or non-claims of Part IV (Section 1.16), so they are additions to this report rather than a new report; a separate report on conserved-mass thresholds was the alternative. Source 15 extends source 14 and source 17 builds on source 16, so each pair shares a Part: the universality side (Part V) first, next to Part IV’s Morita–Imai interface, and the decidability side (Part VI) second. Every result, proof, example, remark, question and limitation of the four sources is printed, in the sources’ order, except for the re-proofs below, which are pointers.
- *Re-proofs printed as pointers.* Source 14’s proof of the mass-two theorem (its Section 59) and of its bounded-diameter proposition; source 15’s two paragraphs recalling source 14’s completion and clock, its summary of the mass-two argument and its proof of the forward certificate lemma; and source 17’s proofs of its two small-packet lemmas. Each statement is printed, with a bracketed note naming the theorem it repeats; the omitted text survives in the arrival archives. Source 16 is printed in full, with “Same argument as” notes at its lemmas and sections that follow Part IV.
- *The two editions of source 16.* The revision (manuscript 05) is printed; its mathematics is that of manuscript 06, and its attribution is the fuller one.
- *Author lines.* The four author lines (“Mathematical construction and reproducibility report”, “A clean return extension of the reversible weighted CA construction”, “Decidability theorem and executable checks”, “Decidability proof and a binary timing certificate”) name no assistant; the Part openings describe the manuscripts as AI-assisted research manuscripts of ProveIt’s incoming reports.
- *Companions and bibliography.* Source 15’s and source 17’s “companion” entries are replaced by references to sources 14 and 16, and source 16’s by a reference to Part IV’s Section 46. Source 14’s `durandlose2006` is the existing entry [5]; the four sources’ `moritaimai`, `morita1996`, `alhazovimai`, `cooper` and `ginsburgspanier` are the existing entries of those keys; source 15’s `brandt` is source 14’s `brandt2020`; source 17’s `kong` and `kakenreport` are source 16’s `kongthesis` and `kakenfinal`. The other entries are appended as delivered, marked with their source, and the programme’s reviews are added and marked [write].
- *Typesetting.* The four sources’ macro `\N` is written `\NN`; it prints $N = \{0, 1, 2, \dots\}$ as in the sources. Equations are numbered by section; sources 16 and 17 tagged theirs consecutively, and source 16’s untagged reference “(2)” is now a cross-reference. Source 14’s table of contents and forced page break are omitted. Source 17’s figure is the delivered vector PDF (embedded TrueType fonts, no Type 3 fonts), unchanged. No symbol was renamed and no normalization changed.
- *Front matter.* The title, subtitle and abstract were widened; Section 1 gained Sections 1.12–1.16, and dated [write] notes were added where batch 80 made earlier sentences stale (Sections 1, 1.1, 1.3, 1.4, 1.5, 1.6, 1.10 and 1.11, Table 3, the remark after Theorem 44.1, and Part IV’s fourth and fifth questions). Nothing was numbered anew: the new front tables are uncaptioned, the two new Parts follow Part IV, and all appendices keep their letters.
- *Labels.* Source 14’s labels are `smc:tm:` plus their delivered names, source 15’s `smc:ct:`, source 16’s `smc:su:` and source 17’s `smc:fm:`; labels were added to unlabelled sections, one lemma, two theorems, two corollaries, two propositions, one figure, the questions and

one equation. The two Parts and the five front subsections use `smc:tm:` (for Part VI, `smc:su:part`). The labels of the omitted re-proofs (source 14's `eq:affine`, `eq:nearcount`, `eq:tail`; source 15's `eq:affinefar`) are not used.

- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on Parts V and VI, Sections 1.12–1.16 and the dated notes listed above, the openings of the two Parts and of the four sources, the bracketed notes inside the four sources, this paragraph and the bibliography entries marked [write].
- *Excluded data.* Five literal exports of source 14 (`examples/cycle_rule.json`, `merge_rule.json`, `trap_rule.json`, `mixed_certificate.json` and `mixed_radius_one_certificate.json`; 20,752,637 bytes together) are not shipped. The shipped programs regenerate the three rules byte for byte and the two certificates identically apart from Windows line ends; the README gives the commands and the retrieval from the arrival commit.

Files added in batch 80. The report directory gained, at its root, source 14's `SOURCE_PROVENANCE.md`, source 15's `CLEAN-TARGET-THEOREM.md`, `SOURCE_PROVENANCE.md` and `audit/INDEPENDENT-AUDIT.md`, source 16's `REVISION.md`, `SOURCE-PROVENANCE.md` and `independent/AUDIT.md`, and source 17's seven proof-detail, provenance and audit notes, with the prefixes `14-three-mass-`, `15-clean-targets-`, `16-single-unit-` and `17-four-mass-`; the four sources' programs in `code/` and records in `data/` under the same prefixes; and a new directory `figures/` with source 17's article figure. The README maps each delivered name to its shipped name. Not shipped: the four manuscripts (and source 14's four modular $\text{\LaTeX}$ files), the four PDFs and delivery READMEs, the checksum ledgers and the manifest checkers of sources 14 and 15, source 15's five vendored programs (byte copies of source 14's), source 14's byte copy of a source-13 receipt, source 17's raster figure, the five excluded exports, and all of manuscript 06.

Source 18 and Part VII (batch 82). [write] Four manuscripts of batch 82 of ProveIt's incoming reports were added on 3 October 2026. All four archives arrived in commit `db37d18c8` and were placed by commit `49dfa8fd6` (batch 82, cluster M4), which staged 119 files (programs, data, provenance and audit notes, and three figures) byte-identical to the delivery (18 of source 18, 17 of source 19, 26 of source 20, 58 of source 21) and retired the four archives from the drop zone. Every delivered checksum ledger verified at placement (manuscript 19: 25 of 25 files; 06: 20 of 20; 17: 31 of 31; 01: 61 of 61); the ledgers are not shipped.

Source	Archive (batch 82)	Pin	Contributed	Printed as
18	Timed_Four_Mass_Quartic_Certificates.zip, manuscript 19 (the pipeline's addendum 13); main file phase-quartic-release/fixed-input-timed-quartics.tex, 12-page PDF	source 17 by title; its source -freeze.json pins its own code	the chart interface and half-open partition, the constant-term lifting compiler with ledgers and selector elimination, signed external pairs, the binary charts and the eight- and four-witness certificates	Part VI, Sections 93–100
19	Dimension_Independent_Particle_Thresholds_Package.zip, manuscript 06 (Report 29); main file dimension-independent-threshold-release-20261003/report29.tex, 22-page PDF	source 17's L ^A T _E X source (803bf0c4...), source 18's (fecc0ae2...), Reports 16 (7e644e82...) and 26 (a156d7d9...)	the $\mathbb{Z}^d$ decision theorem with contact rays, the nonparallel switch set, thresholds, dispatch, termination and drift cutoff; the conditional boundary; the $\mathbb{Z}^d$ timed quartics; six questions	Part VII, Sections 101–117, Appendices I–J
20	Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip, manuscript 17 (Report 30); main file sparse-orbit-geometry-release-20261003/report30.tex, 24-page PDF	source 19's L ^A T _E X source (1e4b9cb1...)	semilinear geometry and sharp counts, quadratic clocks and inverses, first arrivals, drift, the weighted planar automata, first-visit quartics; five questions	Part VII, Sections 118–128, Appendix K
21	Binary_Planar_Four_Particle_Shuttle_Package.zip, manuscript 01 (Report 31); main file binary-planar-shuttle-release-20261003/report31.tex, 14-page PDF	sources 17, 18 and 20 (source 20's L ^A T _E X source 0723338c...)	the binary planar rule and its full-shift proof, the orbit, wedge, counts and inverses, the exact dependency set and degree, the one-step quartic; three questions	Part VII, Sections 129–140

No manuscript names a ProveIt commit. The pins of sources 19–21 are recorded in their `source-lineage.json` files (shipped as `data/19-mass-four-zd-source-lineage.json`, `data/20-orbit-geometry-source-lineage.json` and `data/21-planar-shuttle-source-lineage.json`) and in source 19's lineage appendix; source 17's pin equals the hash of `four-mass-bound/four-mass-decidability.tex` in its arrival `4e270aa46`, and all pins were verified at placement. The two batch-82 manuscripts that source 19 pins besides, Reports 16 and 26, were placed in the collection's report `five-particle-binary-automata` by `bca6383e9` and `7d2b1b245`; Report 16 is printed there as Part II (commit `e6a410588`).

Where the batch-82 addition had to choose.

- *Placement and order.* Source 18 answers source 17's last question and sources 19–21 extend source 17's theorem and build only on it and on one another, so all four are additions to this report. Source 18 is a one-dimensional addendum to source 17, written before source 19, so it is the third source of Part VI; sources 19, 20 and 21 form Part VII in their dependency order, after Part VI and before the appendices. A new report for the batch-82 particle series was the alternative. Every result, proof, example, remark, question and limitation of the four sources

is printed, except for the one pointer below.

- *Re-proof printed as a pointer.* Source 18’s subsection on the radius-six rule, a restatement of source 17’s rule, its conservation proof and its potential certificate; the place states what it repeats, and the omitted text survives in the arrival archive. Source 19’s Appendix I, which restates source 18’s chart partition and compiler for $\mathbb{Z}^d$, is printed in full with notes. Source 19’s re-proof of source 17’s theorem at $d = 1$ is printed in full, because it is the $\mathbb{Z}^d$ proof, with bracketed notes at each shared step; source 18’s statement of the imported normal form and its remark on the return quartic are printed with notes.
- *Byte copies.* Source 18’s three byte copies of source 17’s files are not shipped twice. The normal, optimized and expected copies of several receipts of sources 20 and 21 are shipped as delivered, because their replay harnesses compare them by name.
- *Pipeline numbers and author lines.* The author lines (“Research addendum 13 / Complete fixed input orbits and exact witness accounting”, “Research report 29”, “Research report 30”, “Research report 31”) name no assistant; the openings describe the manuscripts as AI-assisted research manuscripts of ProveIt’s incoming reports. The pipeline’s report numbers are kept in the text with bracketed translations.
- *Companions and bibliography.* Source 18’s “companion report 12” entry and the “Report” entries of sources 20 and 21 are replaced by references to sources 17–20. Source 18’s `kong`, `imai` and `ginsburg` are the existing `kongthesis`, `kakenfinal` and `ginsburgspanier`; source 19’s `kong` and `alhazov` are `kongthesis` and `alhazovimai`; the pebble-automaton entries of sources 19 and 20 (`delorme`, `lacalle`, `blum`) are merged into one entry each, and the other new entries are appended, marked with their sources. The six programme reviews cited are added and marked [write].
- *Typesetting.* The macro `\N` of the four sources is written `\NN` and prints `N`; sources 19–21 printed $\mathbb{Z}_{\geq 0}$, the same set. Their `\norm` (the maximum norm) is renamed, because Part IV’s `\norm` differs; `\file` and source 18’s `\Lift` keep their definitions. Tables of contents, forced page breaks and source 18’s `\Needspace` are omitted. Source 19’s figure is its inline TikZ drawing; the figures of sources 20 and 21 are the delivered vector PDFs (embedded TrueType fonts, no Type 3 fonts), unchanged. Equations are numbered by section. No other symbol was renamed and no normalization changed.
- *Front matter.* The title, subtitle, author line, abstract and status line were widened, Part VI’s title names source 18, and Section 1 gained Sections 1.17–1.21; dated [write] notes were added where batch 82 made earlier sentences stale or answered questions (Sections 1, 1.1, 1.5, 1.12 with a new row of the threshold table, 1.16, Table 3, the opening of Part VI, and source 17’s seven questions). Nothing was numbered anew: the new front tables are uncaptioned, source 18 and Part VII follow every earlier section, their captioned tables and figures follow every earlier one, and the appendices of sources 19 and 20 follow Part III’s, so that all earlier appendices keep their letters.
- *Labels.* Source 18’s labels are `smc:tq:` plus their delivered names, source 19’s `smc:zd:`, source 20’s `smc:og:` and source 21’s `smc:bp:`; labels were added to unlabelled sections, two corollaries and one theorem, and to the questions. The new Part and the five front subsections use `smc:zd:`.
- *Wording.* Source 21’s “answers the binary-realization question”, “This proves induction from

$n = 0$ ” and the asymptotic of its stationary inverse carry bracketed precisions; source 18’s hypothesis $d \geq 7$ carries a note that it is not sharp.

- *Excluded data.* None: the largest shipped file has 267,097 bytes.
- *Added text.* Every sentence written in this addition is marked [write]: the title-page and abstract text on source 18 and Part VII, Sections 1.17–1.21 and the dated notes listed above, the openings of Part VII and of the four sources, the bracketed notes inside the four sources, this paragraph and the bibliography entries marked [write].

Files added in batch 82. The report directory gained, at its root, source 18’s `verification/independent-audit.md`, source 19’s six proof, audit, review and literature notes, source 20’s nine proof, review and supplement notes and source 21’s two proof notes and four audit notes, with the prefixes `18-timed-quartics-`, `19-mass-four-zd-`, `20-orbit-geometry-` and `21-planar-shuttle-`; the four sources’ programs in `code/` and records in `data/` under the same prefixes; and three figures in `figures/`. The README maps each delivered name to its shipped name. Not shipped: the four manuscripts, PDFs, delivery READMEs and checksum ledgers, source 18’s manifest checker and its three byte copies of source-17 files, and source 20’s two supplement checksum manifests.

F Part III: the complete Morita transition data

In the table, bRq_s means write b , move right, and enter q_s ; bLq_s is analogous. A dash is unspecified. The two words halt and null are designated terminal cells, not transitions. Symbol order is the release’s digit order $b, Y, N, *, \$, 1$, assigned digits 1 through 6.

[write] The delivered article reads this table from a separate file, `paper/morita-table.tex`; it is printed here inline and unchanged. The same table is shipped as data in `data/12-conservative-signal-MORITA_15_6_TABLE.json`.

State	b	Y	N	$*$	$\$$	1
q_0	$\$Lq_2$	$\$Lq_1$	bLq_{11}	–	–	–
q_1	halt	YLq_1	NLq_1	$*Rq_0$	bLq_1	–
q_2	$*Lq_3$	YLq_2	NLq_2	$*Lq_2$	null	–
q_3	bRq_{10}	$1Rq_4$	bRq_5	bRq_8	–	–
q_4	YRq_6	YRq_4	NRq_4	$*Rq_4$	$\$Rq_4$	–
q_5	NRq_6	YRq_5	NRq_5	$*Rq_5$	$\$Rq_5$	–
q_6	bLq_7	–	–	–	–	–
q_7	NLq_3	YLq_7	NLq_7	$*Lq_7$	$\$Lq_7$	YLq_3
q_8	–	YRq_8	NRq_8	$*Rq_8$	$\$Rq_9$	–
q_9	–	YRq_9	NRq_9	$*Rq_9$	YRq_0	–
q_{10}	–	YRq_{10}	NRq_{10}	$*Rq_{10}$	$\$Lq_3$	–
q_{11}	$*Lq_{12}$	YLq_{11}	NLq_{11}	$*Lq_{11}$	$\$Lq_{11}$	–
q_{12}	bRq_{14}	YLq_{12}	NLq_{12}	bRq_{13}	–	–
q_{13}	NRq_0	YRq_{13}	NRq_{13}	$*Rq_{13}$	$\$Rq_{13}$	–
q_{14}	–	YRq_{14}	NRq_{14}	$*Rq_{14}$	$\$Lq_{12}$	–

G Part III: a small complete quadratic packet

Consider the one-dimensional partial-branch presentation of the total map $F(x) = 2x$ on $\mathbb{N}$: branch 0 has guard $x = 0$ and update 0; branch 1 has guard $x > 0$ and update $2x$. The packet has selectors b_0, b_1 , copies z_0, z_1 , and one slack s :

$$P(x, y; b_0, b_1, z_0, z_1, s) = (b_0 + b_1 - 1)^2 + (x - z_0 - z_1)^2 + (y - 2z_1)^2 \\ + z_0^2 + (z_1 - b_1 - s)^2 + b_1 z_0 + b_0 z_1.$$

Here $B = 2$, $d = 1$, $E = 1$, $T = 1$, $W = 0$, so the ledger gives five variables, five squared residuals, and two products. For $x = 0, y = 0$ the unique witness is $(1, 0, 0, 0, 0)$; for $x > 0, y = 2x$ it is $(0, 1, 0, x, x - 1)$. Every other input/output pair has no natural witness. The core replay exhaustively checks 100,000 bounded assignments, including every auxiliary tuple in $\{0, \dots, 4\}^5$ for $0 \leq x < 4$, $0 \leq y < 8$.

The generic event-branch compiler also fully enumerates a three-signal transparent test machine with speeds $2, 0, -2$. Its 19 branches have $E = 20$, $T = 84$, $W = 0$, yielding 160 auxiliary variables, 111 squared affine residuals, and 19 complementarity products. Both pivot and uniform scalings pass 144 valid-germ tests, including 88 legal steps and two simultaneous steps, with every canonical residual exactly zero.

This example illustrates the exact polynomial mechanics without suggesting that a universal branch family has only two branches. For the signal frontend, the same constructor takes the actual finite label-mode and first-collision branch descriptions.

H Part III: the complete finite mode closure and numeric packet

H.1 A finite family covering every encoded run

The naive family of all 114^{18} ordered label words is unnecessary. Starting from the literal initial mode, consider each nonempty subset J of eligible adjacent closing pairs. A pair is eligible only when its exact incoming labels have a defined rule and their speeds are in incoming order. Exclude subsets with consecutive edges, since those would constitute an undefined collision of at least three signals. Replace every selected pair atomically by its speed-sorted output pair and add the resulting mode. Repeating to exhaustion gives the least mode-only closure for this syntactic successor operation.

Every legal event of an encoded run uses one of these nonadjacent eligible subsets, by the complete-collision lemma. Induction therefore places every event mode and branch in the closure. Conversely, every retained branch has exactly the original mode, germ, earliest-collision and rule guards, so it introduces no spurious geometric step. Restricting the integer map to this closed mode family preserves every encoded computation.

The closure is an overapproximation of modes numerically reachable from the specified input family: successive graph edges may be realizable at different gap vectors which cannot be spliced into one orbit. No assertion of reachability of every graph node is made. Nevertheless every branch is individually feasible. If c is its speed-difference vector, choose

$$h_i = \begin{cases} c_i & i \in J, \\ \max(c_i, 0) + 1 & i \notin J, \end{cases} \quad u = q \sum_i h_i.$$

All gaps are strictly positive, its next delay is one, exactly J collides, and the output is an admissible germ. The exporter checks this witness for every branch.

The completed enumeration has 49,700 modes and 80,501 branches: 65,557 single-collision branches and 14,944 branches with two simultaneous disjoint collisions. The maximum graph discovery depth is 321. This depth is a graph statistic, not a computation-time bound or a numeric reachability certificate.

Among the modes, 1,728 have no eligible legal pair. Only 36 of those have no closing edge at all and are collision-free; the other 1,692 are blocked by undefined collisions. Of the 36 collision-free modes, six carry an accepting halt label and 30 carry the null label. The fixed final accepting mode in Section 35 is the target for encoded runs. Neither all graph sinks nor the union of those six label-bearing modes replaces that target.

H.2 Exact guard and witness counts

For each branch the exported guard convention retains one mode equality, $|J| - 1$ nonpivot tie equalities, one strict germ guard for each $c_i \geq 0$, the strict span guard, the strict pivot-gap guard, and all $17 - |J|$ non- J endpoint guards. Some strict guards are redundant; retaining them makes the convention simple and fully reproducible. Nonnegative coordinates are the ambient natural domain, so no weak-gap slacks are introduced.

The resulting exact ledger is

Quantity	Count
Branches B	80,501
Equality guards E	95,445
Strict guards T	2,667,479
Weak guards W	0
Auxiliary variables $19B + T$	4,196,998
All variables including X, Y	4,197,034
Squared affine residuals $37 + E + T$	2,762,961
Complementarity products	80,501

Thus the literal one-step polynomial is fully specified, quadratic, nonnegative on natural arguments, and canonical by Theorem 36.1. For externally fixed K and supplied endpoints, its unrolling has exactly $4,197,016K - 18$ witness variables. This explicit constant is large; the small physical population does not imply a comparably small Diophantine arity.

The untimed affine rows contain 15,323,489 nonzero variable-coefficient entries and one nonzero constant. The maximum absolute affine coefficient has 124 bits. Gap coefficients remain bounded by 24; the larger values arise from finite mode coding. These are sparse affine-circuit statistics, not claims about minimal gate complexity. Adding the optional timestamp coordinate retains the same branch guards and adds one local copy coordinate per branch and two global input/output residuals. Its derived ledger is 4,277,499 auxiliaries and 2,762,963 squared affine residuals per step.

H.3 What the numeric export contains

The mode file gives each mode's 18 label indices, speed differences, exact code and graph index. The compact event file records every branch index, source/target index, canonical pivot and bitmask of the complete set J . Together with the explicit formulas above, these fixed numbers determine every sparse matrix and guard coefficient. The supplied exporter generates the full numeric branch records, including selector, copy and slack indices. A separate circuit file fixes the global variable

order and the polynomial (36.7). All large codes and generated coefficients are serialized as decimal strings.

The shared selector sum is an expression, not another witness variable. Writing its difference with b_r is exactly $\sum_{h \neq r} b_h$, which is nonnegative on all natural assignments. Keeping this expression factored avoids a quadratic-in- B expansion of the complementarity part. The exporter can regenerate every squared affine residual row or answer individual expanded coefficient queries. The verbose branch and full residual-row files were generated and checked during preparation. They are omitted from the compact archive because they are deterministically regenerable from its complete mode/event data and code.

The supplementary incidence calculation gives 1,059,563,096,023 nonzero terms in the full monomial expansion and a maximum absolute coefficient of 248 bits. Those expansion statistics are derived exactly; no trillion-term monomial file was emitted. The actually generated full affine-residual export was checked independently, including all 15,323,489 nonzero variable-coefficient entries.

This is an explicit coefficient description of one fixed step polynomial and each fixed-horizon family. It does not convert the unknown horizon into a fixed number of natural variables, and it does not assert that the mode graph's finite size bounds the length of a universal run.

[write] In this report the machine file, the ledger and the circuit file are shipped as `data/12-conservative-signal-MORITA_18_SIGNAL_MACHINE.json`, `data/12-conservative-signal-LEDGER.json` and `data/12-conservative-signal-POLYNOMIAL_CIRCUIT.json`, and the exporter as `code/12-conservative-signal-compile_packet.py`. The mode file and the compact event file (`numeric/MODES.jsonl.gz` and `numeric/EVENTS.jsonl.gz` in the delivered layout) are not shipped; the exporter's `build` command regenerates them with identical decompressed content (README of this report). The programme's review [29] enumerated the same 49,700 modes and 80,501 branches independently and derived every entry of the ledger above from the closure.

I Part VII: canonical timed quartic certificates (source 19)

The preceding proof supplies more than reachability decidability: it produces exact phase data with a rigid algebraic form. We now apply the constant term lifting compiler already established in Report 13 [source 18, Section 96]. Its one dimensional conclusion is not new here. The finite dimensional extension comes from the proved normal form and a disjoint lexicographic sorting refinement. The complete companion proof and its independent conditional review are included in `scientific/timed/`. [write] [Shipped as `19-mass-four-zd-timed-PROOF.md` and `19-mass-four-zd-timed-REVIEW.md`; with this appendix the fixed-input answer to source 17's Question 7 holds in every finite dimension.]

Theorem I.1 (Fixed input timed charts and quartics). *Fix a CA and input satisfying Theorem 101.1. There is an effective finite family of disjoint charts, each with at most two natural parameters and an integer affine domain, whose time and original frame occupied coordinates are rational polynomials of degree at most two, integer valued on their domains. Occupied sites are strictly lexicographically ordered. The disjoint chart domains map bijectively to the complete timed orbit.*

For every fixed tuple $\mathbf{a} = (a_1, \dots, a_m)$ of nonzero labels, $0 \leq m \leq 4$, there is an effectively obtained integer polynomial

$$C_{\mathbf{a}}(t, X_1, \dots, X_m; w)$$

of ordinary total degree at most four, with natural witnesses w , whose witness fiber is a singleton

exactly when $F^t(c)$ has precisely those lexicographically ordered sites and labels, and is empty otherwise. Here $t \in \mathbb{N}$ and $X_i \in \mathbb{Z}^d$ are external arguments. The witness count depends on the fixed input and rule, not on time or the queried coordinates.

I.1 Half open charts own every time exactly once

[write] [This subsection is source 18's Section 95 in the notation of Section 110.1 (here q_ℓ is source 18's prefix length b_ℓ , and C is its B); new here are the original-frame coordinates in $\mathbb{Z}^d$ and the raw chart count J . It is printed because the counts below use it.] Materialize the finite prefix as one parameter free chart for each $0 \leq t < T_0$. For a translated periodic or independent tail, extend the prefix through all individual transients and choose one common period p . The phase r has the one parameter formula

$$t = T_0 + r + pn, \quad 0 \leq r < p, \quad n \in \mathbb{N},$$

with affine coordinates. Euclidean division determines a unique phase and parameter. The vacuum uses a one clock tail with no occupied coordinates. In the no unit case the same construction already works in the original frame.

For an expanding shuttle, retain the same cycle indexing as in Section 110.1. Let transition ℓ have flight length $h_\ell(n) = a_\ell n + b_\ell$ and fixed encounter prefix length $q_\ell \geq 0$. The flight length is integral for all natural n , so $b_\ell = h_\ell(0)$ and $a_\ell = h_\ell(1) - h_\ell(0)$ are integers. Both $h_\ell(n) > 0$ and $a_\ell > 0$. The cycle start is

$$T(n) = T_0 + A \frac{n(n-1)}{2} + Cn, \quad A, C \in \mathbb{Z}_{>0},$$

and transition ℓ starts at the quadratic integer valued polynomial

$$S_\ell(n) = T(n) + \sum_{h < \ell} (h_h(n) + q_h).$$

The half open cycle intervals $[T(n), T(n+1))$ partition all times from T_0 onward. Each transition owns $[S_\ell(n), S_\ell(n) + h_\ell(n) + q_\ell)$.

If its packet period is p_ℓ , then for each residue $0 \leq r < p_\ell$ the flight chart has parameters $n, j \in \mathbb{N}$, domain and time

$$h_\ell(n) - r - p_\ell j - 1 \geq 0, \quad t = S_\ell(n) + r + p_\ell j. \quad (\text{I.1})$$

It owns only the times strictly before first contact. For $0 \leq s < q_\ell$, a local prefix chart has parameter n and time $S_\ell(n) + h_\ell(n) + s$. If $q_\ell = 0$, contact is the next checkpoint and belongs to the next transition. Thus no boundary is counted twice.

The cycle interval determines n , the transition interval determines ℓ , and division determines the unique flight phase and j or the unique local prefix phase. This proves one chart point per exact time, even if the spatial configuration repeats. Stationary frame coordinates g_v are affine in the chart parameters, and original frame coordinates

$$P_v = g_v + \delta t$$

have degree at most two. If J denotes the raw chart count, the literal construction gives $J = T_0 + p$ for an affine tail and $J = T_0 + \sum_\ell (p_\ell + q_\ell)$ for a shuttle. These input dependent quantities may be enormous.

I.2 Disjoint lexicographic sorting and its cost

For a raw chart with m occupied sites, refine by every permutation π and every list of first differing coordinates $k_a \in \{1, \dots, d\}$ for adjacent sites $v = \pi(a)$, $w = \pi(a + 1)$. Require

$$g_{w,h} - g_{v,h} = 0 \quad (h < k_a), \quad g_{w,k_a} - g_{v,k_a} - 1 \geq 0.$$

The common drift cancels in pair differences, so these conditions are affine. Occupied sites are distinct; hence exactly one order and one first difference list apply at each chart point. Equal labels do not cause multiplicity. Use positive denominator clearing wherever needed.

The sorting factor is 1 when $m \leq 1$, and $m!d^{m-1}$ otherwise. Let B_{ch} be the retained chart count after any fixed label tuple filtering, K its total private parameter slots, M its domain inequalities, and H its domain equalities. Then

$$B_{\text{ch}} \leq 24d^3 J, \quad K \leq 2B_{\text{ch}}, \quad M \leq 4B_{\text{ch}}, \quad H \leq 3(d-1)B_{\text{ch}}. \quad (\text{I.2})$$

The notation B_{ch} is separate from the geometric prefix bound B in the main proof. Incompatible label tuples simply retain no charts.

I.3 The inherited constant term lifting compiler

A slightly more general interface also covers the pattern consequences below. Chart i has private natural parameters z_i , integer polynomial inequalities $A_{ir}(z_i) \geq 0$, integer polynomial equalities $H_{is}(z_i) = 0$, and rational output vector $P_i(z_i) \in \mathbb{Q}^Q$. All degrees are at most two; outputs are integer valued on the domain. Assume the disjoint domains map injectively to the intended external outputs.

Introduce a natural selector e_i , the private parameters, and one natural slack u_{ir} for every inequality. Write

$$E = \sum_i e_i, \quad Z_i = \sum_j z_{ij} + \sum_r u_{ir}, \quad \mathcal{L}_i(f) = f(z_i) - f(0) + f(0)e_i.$$

Only the literal constant is multiplied by the selector, so no quadratic monomial becomes cubic. For each output coordinate choose a positive common denominator L_q across *all* chart polynomials. Define

$$\begin{aligned} R_0 &= E - 1, & R_{ir} &= \mathcal{L}_i(A_{ir}) - u_{ir}, \\ U_{is} &= \mathcal{L}_i(H_{is}), & G_i &= (1 - e_i)Z_i, \\ V_q &= L_q \left(y_q - \sum_i \mathcal{L}_i(P_{iq}) \right). \end{aligned}$$

All residuals have integer coefficients and ordinary degree at most two in the external and witness variables together. Therefore

$$C(y; w) = R_0^2 + \sum_{ir} R_{ir}^2 + \sum_{is} U_{is}^2 + \sum_q V_q^2 + \sum_i G_i^2 \quad (\text{I.3})$$

is an ordinary quartic sum of squares.

At a natural zero, $E = 1$ forces exactly one selector to be 1. Every inactive gate forces $Z_i = 0$, and nonnegativity forces all its parameters and slacks individually to be zero. Its lifted forms vanish.

In the selected branch, the remaining equations enforce precisely its domain, its uniquely determined integer slacks, and its output. Conversely an admissible chart point gives that unique zero, with every inactive private variable zero. Thus the natural zeros are in bijection with admissible chart points over the specified output. Half open ownership and disjoint sorting prove Theorem I.1.

[write] [This is the argument of source 18's Proposition 96.1, whose residuals (96.2)–(96.3) and squared gates (96.5) appear above as U_{is} , V_q and G_i ; source 19 says itself that the one-dimensional conclusion is not new. The programme has read the shipped companion proof 19-mass-four-z d-timed-PROOF.md in full, in a follow-up (original_frame_first_hit30_fixture.md with its independent review review_original_frame_first_hit30_fixture.md, commit 5b4c091e6) [60], and found its half-open chart ownership, common clock, first-difference sorting, constant-term lifts, pooled denominators, uniqueness of the inactive natural witnesses and original-frame padding sound as conditional statements; the general normal form stays an inherited hypothesis.]

There are exactly

$$N_w = B_{\text{ch}} + K + M \leq 7B_{\text{ch}} \leq 168d^3J \quad (\text{I.4})$$

natural witnesses, and $1 + M + H + Q + B_{\text{ch}}$ squared residual slots before deleting identically zero terms. For a fixed m site label tuple, $Q = 1 + dm$. An empty chart family uses $C = 1$ and no witnesses. These counts are deliberately coarse and input dependent.

The polynomial is nonnegative on all real arguments, but its exactness and witness uniqueness are assertions over *natural* witnesses. Replacing them by unrestricted signed difference pairs breaks the inactive branch argument. Pooled denominators are essential: separately scaled branch outputs cannot simply be added without a common scale.

I.4 Whole configurations and signed external coordinates

One polynomial can cover varying site counts and labels by encoding

$$(t, m, a_1, \dots, a_4, X_1, \dots, X_4)$$

with positive integer nonzero label codes, occupied sites sorted, and all unused label and coordinate slots zero. Append those constants to each chart output; then $Q = 6 + 4d$ and the same witness bound applies. Padded zero coordinates remain literally zero and do not acquire the drift δt .

For equality up to translation, subtract the first occupied site. The relative coordinates $P_v - P_1 = g_v - g_1$ are affine, and time remains quadratic. There is no translation witness. The vacuum is handled separately rather than granting it infinitely many arbitrary translation witnesses.

Signed external integer coordinates are valid polynomial inputs. If all external slots must be natural, replace each scalar X by $X^+ - X^-$ and add the squared residual $(X^+ X^-)^2$. The accepted pair is uniquely $(\max(X, 0), \max(-X, 0))$. These are external slots, not extra witnesses. Total degree remains at most four.

I.5 Fixed patterns including prescribed zeros

For integer vectors v, w , there are $2d + 1$ disjoint comparison outcomes: equality, or either sign at their first differing coordinate. Equality uses d coordinate equations; each strict outcome uses earlier coordinate equalities and one inequality with an integer margin of 1. Comparisons of quadratic outputs therefore introduce only degree two domain conditions, within the compiler interface.

For a fixed anchored pattern on s sites, refine every sorted orbit chart by all comparisons of each occupied site with each window site. The full comparison matrix and fixed labels decide its

truth, including zeros. Retain successful matrices and output only t . The refinement is disjoint, and the orbit determines a unique chart point at each time, so the quartic has an empty or singleton natural witness fiber. One may take

$$B_{\text{anch}} \leq B_{\text{ch}}(2d+1)^{4s}, \quad N_{\text{anch}} \leq (7+4s)B_{\text{anch}}.$$

Here B_{ch} refers to all sorted orbit charts before observation filtering.

For a translated pattern with a nonzero prescribed site w_* , every successful placement sends w_* to one of at most four occupied sites carrying the correct label. The candidate shift is $b_i = P_i - w_*$. Refine by the *complete* matrix of comparisons

$$P_v - P_i \quad \text{against} \quad w - w_*$$

for all occupied i, v and all pattern sites w . The drift cancels and these conditions are affine. The matrix determines the success of every candidate. Retain the matrix once if any candidate succeeds, rather than introducing one branch per success. If a placement is also output, choose the least successful occupied site index. This prevents same time placement multiplicity. Bounds are

$$B_{\text{trans}} \leq B_{\text{ch}}(2d+1)^{16s}, \quad N_{\text{trans}} \leq (7+16s)B_{\text{trans}}.$$

A wholly zero translated pattern, including the empty pattern, occurs in every finite support configuration at every time. For occurrence in external natural t alone, the zero polynomial with no witnesses has the single empty witness tuple. There need not be a lexicographically least placement. If a placement is wanted, choose a specified rule that moves its entire window beyond the maximal first support coordinate, with the vacuum handled separately.

These are fixed pattern statements. The chart family and counts may depend on the whole pattern. No uniform arity claim is made for an arbitrary pattern provided as one encoded external input.

I.6 What time uniqueness does not imply

Time is external in every single witness claim. If it is existentially quantified instead, the same construction has one zero for *each occurrence time*. A stationary nonempty configuration already gives an infinite fiber. Thus this appendix does not establish untimed finite foldness, a generic single fold MRDP theorem, a uniform bound over input configurations, or exactness for rational or real witnesses. The explicit chart interface, not bare decidability, is what permits the quartic compiler.

The companion finite checks cover 208 drift restored two dimensional shuttle states, 325 lexicographic refinement cases, 1,215 finite witness cube evaluations, and additional selectors, inactive gates, domains, signed pairs, and pattern placements. Their normal and optimized receipts agree. They test the displayed algebra on fixtures; they do not implement a general CA to chart compiler or prove the dynamical normal form by experiment.

J Part VII: source lineage and audit status (source 19)

The starting proof source is the 3 October 2026 higher dimensional research packet. Its original SHA 256 is

3e8eb41fe43b5b4b61d0e0a5b4bece3b01cf628f834a820ecc369f4da427b30a.

The delivery copy in `scientific/PROOF.md` replaces environment specific provenance paths with report identifiers and updates its introductory delivery description. The literature copy removes the local reading cache note in favor of primary links. `verification/source-lineage.json` records original and delivered hashes. The mathematical exposition in this report follows the corrected proof and makes its determinant formula, dispatch convention, and observation semantics explicit.

The earlier audit corrected two important wording points: compact exits need the actual gap for their later hit test, and a decreasing cycle may exit to a terminal profile as well as to ordinary. Both are incorporated above. The fresh adversarial review, included as `scientific/independent-review.md`, found no missing dynamical case, circular enumeration, or fatal gap in the lower theorem. Its requested zero drift dispatch, common seed anchor convention, and explicit small launch bound are incorporated above. That review does not reprove the inherited upper construction. Source releases were read without modification. The relevant earlier TeX source hashes are:

Report 12 four mass decision precursor

803bf0c4194bb9d0f1942e6ad3eb762c79d7a811bd785742ee63f0c46a775595

[source 17, Part VI: this is the hash of `four-mass-bound/four-mass-decidability.tex` in arrival 4e270aa46]

Report 16 literal reversible source

7e644e82b986903d473a6f99b379f8141e9a71e18fb01ecf3d4e554009563d81

[batch 82, archive `Literal_Universal_Reversible_Source_Package.zip`; printed as Part II of `five-particle-binary-automata`]

Report 26 parallel involution replacement

a156d7d973721f18c896cf71f0e9eedbaa214a613b8c7a77cf4704be90850dab

[batch 82, archive `Two_Parallel_Conservative_Involutions_Package.zip`; placed in `five-particle-binary-automata`, not yet written]

These are proof dependencies for the inherited statements, not redistributed literature. This report does not revise their one dimensional conclusions.

[write] [The delivered files named in this appendix are shipped as `19-mass-four-zd-PROOF.md`, `19-mass-four-zd-literature.md`, `19-mass-four-zd-independent-review.md` and `data/19-mass-four-zd-source-lineage.json`; the README maps every delivered name. The pins of Reports 12, 16 and 26 were verified at placement against the arrival archives. The Hilbert's-tenth programme has since reviewed source 19's proof (note at the end of Section 116) [58].]

K Part VII: canonical first visit quartic certificates (source 20)

The first arrival theorem yields a direct Diophantine consequence that avoids counting repeated visits as different witnesses. We give the construction rather than invoking a generic finite fold or single fold representation theorem. All witnesses below are natural numbers, including zero; the external site coordinates may be signed integers.

Theorem K.1 (One canonical witness per visited site). *For a fixed promised CA and fixed mass at most four input, an ordinary integer polynomial $P(y; w)$ of total degree at most four can be effectively constructed such that*

$$\#\{w \in \mathbb{N}^W : P(y; w) = 0\} = \begin{cases} 1, & y \in S, \\ 0, & y \notin S, \end{cases} \quad y \in \mathbb{Z}^d.$$

The stationary frame first visit time $\tau(y)$ is canonically reconstructible from the unique witness. Alternatively, one can enforce it as an external argument t or move that time argument into one additional canonical natural witness, still at total degree at most four. The polynomial, coefficients, and witness arity depend on this rule and input.

The stronger form with a time argument has a unique natural witness exactly when $y \in S$ and $t = \tau(y)$, and none otherwise. Time is actual elapsed discrete time, not a cycle number. A certificate for all visits followed by existentially forgetting time would not prove this: recurrent sites could retain many or infinitely many time witnesses.

K.1 Disjoint clauses for the spatial domain

More generally, take any effective Presburger set $S \subseteq \mathbb{Z}^d$ with a specified nonnegative integer function f represented by rational quadratic polynomials on finitely many semilinear cells. The first arrival theorem supplies this data for an expanding tail and its prefix. For independent or translated periodic tails the timed occurrence relation is already Presburger; least time selection and Lemma 122.1 give piecewise rational affine $f = \tau$. Finite and empty sets are included.

Apply effective Presburger elimination to all cell formulas. Collect their integer affine comparison forms $\ell_1, \dots, \ell_a$ and congruence expressions g_j modulo fixed positive m_j , $1 \leq j \leq q$. Enumerate the complete outcomes: each ℓ is negative, zero, or positive, and each g_j has exactly one of its m_j residues. Retain an outcome exactly when it belongs to at least one original cell and assign the polynomial of its least indexed containing cell. All such polynomials give f on that outcome. Unsatisfiable outcomes may be deleted.

Converting strict signs by integer unit offsets yields pairwise disjoint branches i of the form

$$A_{ir}(y) \geq 0, \quad H_{is}(y) = 0, \quad L_{iv}(y) \equiv r_{iv} \pmod{m_{iv}}, \quad (\text{K.1})$$

with integer affine forms, $m_{iv} > 0$, and normalized residues. This partitions spatial points, not coefficients of a semilinear generating family. No uniqueness of those generator coefficients is needed.

Let B be the number of branches and let I, H, C be the total occurrences of inequality, equality, and congruence atoms in them. Before redundant atoms are removed, the full outcome construction gives

$$B \leq 3^a \prod_{j=1}^q m_j, \quad I + H = aB, \quad C = qB.$$

The potentially enormous size is not hidden by the constant degree assertion.

K.2 Affine gating with canonical signed quotients

For each branch introduce a natural selector e_i . For each inequality introduce one natural slack u_{ir} , and for each congruence introduce two natural quotient variables q_{iv}^+, q_{iv}^- . Form the residuals

$$\begin{aligned} R_0 &= \sum_i e_i - 1, \\ R_{ir} &= e_i A_{ir}(y) - u_{ir}, \\ S_{is} &= e_i H_{is}(y), \\ Q_{iv} &= e_i (L_{iv}(y) - r_{iv}) - m_{iv} (q_{iv}^+ - q_{iv}^-), \\ K_{iv} &= q_{iv}^+ q_{iv}^-. \end{aligned} \quad (\text{K.2})$$

Let P be the sum of all their squares. Each residual has integer coefficients and total degree at most two, including the external y coordinates, so P has ordinary total degree at most four.

At a natural zero, $R_0 = 0$ selects exactly one branch. On an inactive branch, $R_{ir} = 0$ forces its slack to zero. Its congruence balance forces $q^+ = q^-$, and their zero product forces both to zero. Thus no inactive auxiliary remains free. On the active branch the slack is uniquely $A_{ir}(y)$, the equalities hold, and each congruence fixes an integer quotient

$$q = (L_{iv}(y) - r_{iv})/m_{iv}.$$

The zero product forces its unique natural encoding $(q^+, q^-) = (\max(q, 0), \max(-q, 0))$. The disjoint clauses determine the selector uniquely. Conversely these canonical assignments make every residual vanish for each $y \in S$. A sum of real squares is zero exactly when every residual is zero, proving membership and witness uniqueness.

For $B > 0$ the literal witness and squared residual counts are

$$W_{\text{mem}} = B + I + 2C, \quad R_{\text{mem}} = 1 + I + H + 2C. \quad (\text{K.3})$$

For an empty branch list use $P = 1$ and no witnesses.

[write] [Second route. This canonical semilinear membership quartic is Part IV's compiler of Section 46 (a unique slack for each inequality and the sign-separated Euclidean quotient (46.1)) combined with Part I's selector device (Theorem 10.2); Part IV gates its clauses through a circuit of truth bits, whereas source 20 multiplies each atom by its selector. Part XI of *canonical-diophantine-certificates* gives every semilinear set a single-fold representation of degree two that is nonnegative on the real orthant (Theorem `cdc:of:thm:singlefolds1`; compare Remark 10.3), so degree four is not needed for membership alone. No novelty is claimed for this subsection. New in this appendix are the one shared cycle variable for quadratic first arrivals, the square lifts for general piecewise quadratic time, the untimed first-hit certificates and the original-frame complete-target certificate through whole-orbit injectivity.]

K.3 One shared cycle variable for actual first arrivals

The expanding shuttle application admits a sharper clock implementation than a general quadratic function. Retain on each final tail clause its rational affine functions $n_i(y) = n_*(y)$ and $k_i(y) = T_0 + h_*(y)$. On every prefix first visit singleton, set $n_i = 0$ and k_i equal to its first time. All clauses then share

$$\tau(y) = cn_i(y)^2 + bn_i(y) + k_i(y),$$

with the same c, b , and both affine functions are nonnegative integers on their assigned cells.

Introduce one shared natural variable N . Choose positive integers A, D so that $a_i = An_i$ and $K_i = Dk_i$ are integer affine, and Dc, Db are integers. Add exactly the two squared residuals

$$AN - \sum_i e_i a_i(y), \quad Dt - DcN^2 - DbN - \sum_i e_i K_i(y). \quad (\text{K.4})$$

The unique selector fixes $N = n_i(y)$ and the second residual fixes $t = \tau(y)$. Both residuals have ordinary total degree at most two, so their squares retain degree at most four. Prefix branches work with $N = 0$, without negative offsets or an extra selector. All inactive membership auxiliaries were already forced to zero.

Thus for actual expanding first arrivals, time external requires

$$W = B + I + 2C + 1, \quad R = 3 + I + H + 2C. \quad (\text{K.5})$$

Making time itself a natural witness changes the first count to $B + I + 2C + 2$, with the same residual count. These are counts of variables in the unique witness tuple, not a claim that the tuple has just one variable. Only the extra clock overhead is dimension independent; the clause and atom counts still depend on the full fixed rule and input. Reconstruction of first time from the membership witness alone needs no added variable.

K.4 General piecewise quadratic time outputs

At the unique membership witness, the selected branch already gives the canonical expression $f(y) = \sum_i e_i f_i(y)$. Its total degree may be three, which is harmless for reconstruction but would produce degree six if squared as a residual. To preserve degree four when enforcing time, use shared natural square lifts

$$U_j = y_j^2 \quad (1 \leq j \leq d), \quad U_{jk} = (y_j + y_k)^2 \quad (1 \leq j < k \leq d).$$

Impose these with their quadratic difference residuals. The universal sufficient number is $M = d(d+1)/2$, and each lift is uniquely nonnegative even for signed coordinates. Polarization gives

$$y_j y_k = \frac{1}{2}(U_{jk} - U_j - U_k).$$

Thus every f_i becomes a rational affine expression in (y, U) . Choose one common positive denominator D , including the polarization halves, and write it as $F_i(y, U)/D$ with integer affine F_i . Add the single pooled residual

$$Dt - \sum_i e_i F_i(y, U). \quad (\text{K.6})$$

It has total degree at most two. Adding its square and the square lift residuals to P retains degree at most four and forces exactly $t = f(y)$. All shared lift values are canonical independently of branch selection; none are left as inactive degrees of freedom.

For external time the literal counts are

$$W_{\text{time}} = B + I + 2C + M, \quad R_{\text{time}} = 2 + I + H + 2C + M. \quad (\text{K.7})$$

Moving t to a natural witness adds one to W and leaves R unchanged. For affine cell functions one may take $M = 0$. Unused square lifts can be omitted after specifying a sufficient list.

If all external slots must also be natural, replace each y_j by $Y_j^+ - Y_j^-$ and add the squared product residual $(Y_j^+ Y_j^-)^2$. This accepts exactly the canonical positive and negative part pair. It changes d signed external slots to $2d$ natural slots, adds d residuals, and adds no witnesses. Affine substitution and quadratic products preserve the degree bound.

K.5 First hits of configurations and fixed patterns

The same construction supplies a useful untimed reachability refinement of Report 29 [source 19]. Fix a finite tuple of external observation parameters z , and a time independent observation predicate that is Presburger in z and the finite labeled support of the stationary frame configuration. The schema and tuple length are fixed when compiling the polynomial.

Substituting the finite phase support lists produces a Presburger occurrence relation $\text{Occ}(z, n, h)$ in half open expanding cycles. It may have many witnesses; only its truth is used. A prefix time can satisfy infinitely many z , so the site-only singleton treatment must be replaced. If $P_t(z)$ is the observation predicate evaluated on the actual prefix configuration at time t , assign that time to the Presburger domain

$$P_t(z) \wedge \bigwedge_{0 \leq s < t} \neg P_s(z).$$

These finitely many domains are disjoint and cover all prefix hits. Exclude their union before minimizing (n, h) on the tail. Least cycle and offset are again Presburger functions and become rational affine on effective semilinear cells. Prefix cells use $n_i = 0, k_i = t$; tail cells use $n_i = n_*(z), k_i = T_0 + h_*(z)$. The shared counter construction (K.4) therefore applies unchanged.

Corollary K.2 (Canonical untimed first hit certificates). *For every such fixed schema, its stationary frame first hit domain has an ordinary integer polynomial of degree at most four with exactly one natural witness tuple per accepted external observation and none otherwise. Its first absolute hit time is canonically reconstructible, external, or one additional natural witness. The counts in (K.3) and (K.5) apply to its final disjoint clauses. This includes the following observations.*

1. *Complete finite configuration targets, canonically encoded by a site count $0 \leq m \leq 4$, positive codes for active nonzero labels from the fixed alphabet, and at most four signed coordinate slots, strictly lexicographically sorted, with zero label and coordinate padding beyond m .*
2. *anchors of a fixed finite pattern, including prescribed vacuum symbols. The anchor is external; the pattern's finite offsets and symbols are fixed at compilation.*

For complete targets, cardinality, labels, coordinate equality with all phase support sites, ordering, and padding are finite Presburger conditions. Thus actual untimed stationary frame configuration reachability has a unique natural witness tuple for every reachable canonical target. This does not follow merely by dropping the time argument from an all occurrence certificate.

For a fixed pattern, each nonzero entry requires a matching occupied site at the anchor plus its offset; each zero entry requires absence of all occupied sites there. This is again a finite Presburger predicate. The argument does not cover an arbitrary unbounded external pattern encoding.

There is no quadratic distance growth assertion for general observation parameters. A wholly zero pattern can already hold at infinitely many distant anchors at time zero. The corollary concerns semilinear first hit domains, piecewise quadratic first times, and canonical quartic certificates. As before, it is a fixed rule, fixed input result, and does not by itself assert general expanding original frame reachability under nonzero drift. Complete targets admit the separate corollary below.

K.6 Complete configuration targets in the original frame

For full configurations there is a further argument that removes the frame restriction. It uses exact trajectory injectivity, not a first visit spatial substitution.

Lemma K.3 (An expanding whole configuration orbit is injective). *If the classified tail is expanding, the complete configurations $F^t(c_0)$ are pairwise distinct for all $t \geq 0$, including the initial prefix.*

Proof. At section n the actual two residual unit sites have separation $(x_0 + n\Delta)\nu + z$, with a fixed transverse offset z . Hence the configuration diameter is at least $\|(x_0 + n\Delta)\nu + z\|_\infty$ and tends to

infinity. Translating by $T_n\delta$ leaves this diameter unchanged, so this statement holds in the original frame.

If $F^a(c_0) = F^b(c_0)$ for $a < b$, determinism implies $F^{a+s}(c_0) = F^{b+s}(c_0)$ for every $s \geq 0$. The future then consists of finitely many repeated configurations and has bounded diameter. This contradicts the unbounded section diameters. The argument excludes every repeated pair, even if one or both times precede the eventual tail. $\square$

Corollary K.4 (Original frame untimed configuration certificate). *For every fixed promised CA and input, original frame untimed complete configuration reachability admits an effective ordinary integer polynomial of degree at most four with exactly one natural witness tuple for each reachable canonical target and none otherwise. The first reachability time can be part of that tuple.*

For an expanding tail, the timed chart construction of Report 29 [source 19, Appendix I, whose compiler is source 18's] gives the explicit bound

$$W \leq 7B_{\text{ch}} + 1,$$

where B_{ch} is its retained sorted timed chart count, not the Presburger clause count B in this appendix. No uniform over input arity bound is asserted.

Proof. Report 29's Appendix A [source 19's Appendix I, bound (I.4)] supplies a quartic $C(t, z; w)$ for canonical complete original frame targets z , with exactly one natural witness tuple for each actual time and target pair, and at most $7B_{\text{ch}}$ witness variables. Its half open charts, lexicographic sorting, and inactive branch gates establish that uniqueness. Reclassify the external elapsed time t as one additional natural witness. Lemma K.3 says that a reachable complete target has exactly one occurrence time. Therefore its combined tuple (t, w) is unique, and total degree remains at most four. The time is automatically its first occurrence.

For independent or whole translated periodic tails, the full original frame timed occurrence relation is Presburger: restoring drift preserves the affine phase rays. Select the least occurrence time and apply the disjoint clause construction with a piecewise affine time output. Prefix overlaps and repeated periodic configurations are handled by this first time selection. Finite and empty cases are included. $\square$

This argument is specific to complete configurations. Sites, labels, and pattern observations can recur even when the entire configuration never repeats. The original frame extensions for their first visits therefore do not follow from this corollary. In particular it does not repair a substitution of $y - \delta t$ into a stationary site first visit certificate.

K.7 Application and limits

For expanding shuttles, first remove the finitely many sites first visited in the prefix and then select the lexicographically least cycle and within cycle offset using the Presburger occurrence relation. Equation (122.1) supplies the quadratic cell function. This proves Theorem K.1 for the entire stationary frame set, not just its eventual tail. Affine, periodic, independent, and finite alternatives use their affine first time functions. Vacuum uses the empty polynomial certificate $P = 1$.

If $\delta = 0$, the result directly concerns F . For affine tails, restoring δt preserves affine timed phase rays, so the same construction also works in the original frame. For expanding site and pattern observations, no general original frame first hit corollary is asserted. Complete configurations have the separate argument of Corollary K.4. Substituting $y - \delta t$ into a stationary first visit certificate is

insufficient: a later return to a stationary frame site can create a new original frame site that the first visit certificate omits.

This is a direct elementary construction using disjoint Presburger clauses and explicit square lifting. It is not a generic single fold MRDP theorem, an inference from decidability alone, an assertion for rational or real witnesses, or a uniform over input arity bound. The full supporting note and its independent review are included in the scientific sources. [write] [Shipped as 20-orbit-geometry-first-visit-PR00F.md, 20-orbit-geometry-first-visit-independent-review.md, 20-orbit-geometry-original-frame-PR00F.md and 20-orbit-geometry-original-frame-REVIEW.md.] Finite algebra fixtures check the formula bookkeeping, inactive branch zeros, signed quotients, time lift, and degree counts; they are not a general Presburger elimination or CA classification implementation.

[write] **The programme's intake.** The Hilbert's-tenth programme read the shipped original-frame proof and first-visit companion in full, and the central parts of the geometry proof, in review_original_frame_first_hit30_intake.md (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/, commit c875bad40, 3 October 2026) [59]. It found no defect in the step of Corollary K.4 that turns the timed certificate into an untimed one through whole-orbit injectivity (Lemma K.3), nor in the separate treatment of affine and periodic tails. It is an intake, not a review of the compiler: it did not review the inherited timed certificate of source 19's Appendix I (whose companion proof it names as the next audit), and it stresses that the witness and residual counts here are not gate counts, that the bound $W \leq 7B_{\text{ch}} + 1$, with $B_{\text{ch}} \leq 24d^3J$ (source 19's (I.2)), depends on the whole automaton and input, and that the shared cycle variable is a lead for a paid comparison, not an established saving. It executed no archived code. In a follow-up (original_frame_first_hit30_fixture.md with its independent review review_original_frame_first_hit30_fixture.md, commit 5b4c091e6) [60], the programme then read source 19's timed proof in full and found no gap in it, conditional on the normal form, and, for one drifted orbit of the weighted planar automaton of Section 123 composed with a unit vertical shift, emitted complete first-hit quartics for complete original-frame targets, the smallest with 35 paid operations (11 multiplications, 24 additions), three natural witnesses, eight squared rows and exact degree four. That is one fixed-input certificate in a decidable class, not a compiler, an arity bound over inputs or a change of the universal bounds.

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